SURFING

WACE HUMAN BIOLOGY

Unit 1 The Functioning Human Body

• Kerri Humphreys •

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Science Press

© Science Press 2017
First published 2017

Science Press
Bag 7023 Marrickville NSW 1475 Australia
Tel: (02) 9516 1122 Fax: (02) 9550 1915
sales@sciencepress.com.au
www.sciencepress.com.au

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Introduction

Each book in the *Surfing* series contains a summary, with occasional more detailed sections, of all the mandatory parts of the syllabus, along with questions and answers.

All types of questions – multiple choice, short response, structured response and free response – are provided. Questions are written in exam style so that you will become familiar with the concepts of the topic and answering questions in the required way.

Answers to all questions are included.

A topic test at the end of the book contains an extensive set of summary questions. These cover every aspect of the topic, and are useful for revision and exam practice.

Words To Watch

account, account for State reasons for, report on, give an account of, narrate a series of events or transactions.

analyse Interpret data to reach conclusions.

annotate Add brief notes to a diagram or graph.

apply Put to use in a particular situation.

assess Make a judgement about the value of something.

calculate Find a numerical answer.

clarify Make clear or plain.

classify Arrange into classes, groups or categories.

comment Give a judgement based on a given statement or result of a calculation.

compare Estimate, measure or note how things are similar or different.

construct Represent or develop in graphical form.

contrast Show how things are different or opposite.

create Originate or bring into existence.

deduce Reach a conclusion from given information.

define Give the precise meaning of a word, phrase or physical quantity.

demonstrate Show by example.

derive Manipulate a mathematical relationship(s) to give a new equation or relationship.

describe Give a detailed account.

design Produce a plan, simulation or model.

determine Find the only possible answer.

discuss Talk or write about a topic, taking into account different issues or ideas.

distinguish Give differences between two or more different items.

draw Represent by means of pencil lines.

estimate Find an approximate value for an unknown quantity.

evaluate Assess the implications and limitations.

examine Inquire into.

explain Make something clear or easy to understand.

extract Choose relevant and/or appropriate details.

extrapolate Infer from what is known.

hypothesise Suggest an explanation for a group of facts or phenomena.

identify Recognise and name.

interpret Draw meaning from.

investigate Plan, inquire into and draw conclusions about.

justify Support an argument or conclusion.

label Add labels to a diagram.

list Give a sequence of names or other brief answers.

measure Find a value for a quantity.

outline Give a brief account or summary.

plan Use strategies to develop a series of steps or processes.

predict Give an expected result.

propose Put forward a plan or suggestion for consideration or action.

recall Present remembered ideas, facts or experiences.

relate Tell or report about happenings, events or circumstances.

represent Use words, images or symbols to convey meaning.

select Choose in preference to another or others.

sequence Arrange in order.

show Give the steps in a calculation or derivation.

sketch Make a quick, rough drawing of something.

solve Work out the answer to a problem.

state Give a specific name, value or other brief answer.

suggest Put forward an idea for consideration.

summarise Give a brief statement of the main points.

synthesise Combine various elements to make a whole.

1 Assumed Knowledge

1. The cell is the basic unit of life. What structural features of cells are possessed by all living things?
2. Draw a fully labelled diagram of a human epithelial cheek cell as seen under a light microscope.
3. Identify the following parts of a light microscope and use by a person.

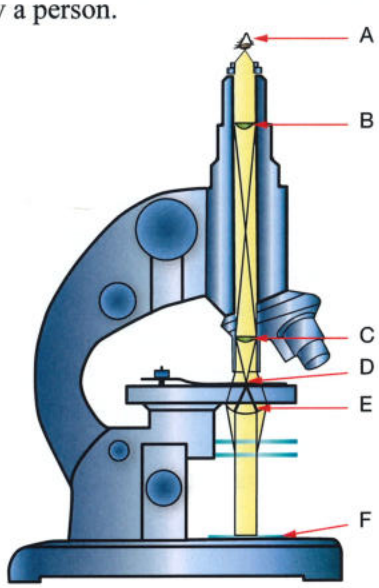


Figure 1.1 Light microscope.

4. Describe one safety precaution you should follow while using a light microscope.
5. What is the function of the nucleus of a cell?
6. What is the function of the cell membrane?
7. What is cytoplasm?
8. Define protoplasm.
9. Define cytosol.
10. Outline the function of the circulatory system.
11. A common prefix used in human biology is 'vas' or 'vaso'. What does this mean and give an example.
12. Outline the function of blood.
13. Name the three types of blood vessels.
14. Outline the function of the heart.
15. Identify the components of the human circulatory system.
16. Define respiration.
17. In humans, what structures make up the respiratory system?
18. Outline the function of the respiratory system.
19. What are lymph vessels?
20. Define metabolism.
21. Name the four basic groups of organic compounds.
22. What are inorganic compounds?
23. What is the function of the digestive system?
24. What are the two main uses of food?
25. Define an enzyme.
26. To what does the prefix 'hepat-' refer?
27. Distinguish between anabolism and catabolism.
28. Figure 1.2 shows the human digestive tract. Identify each part.

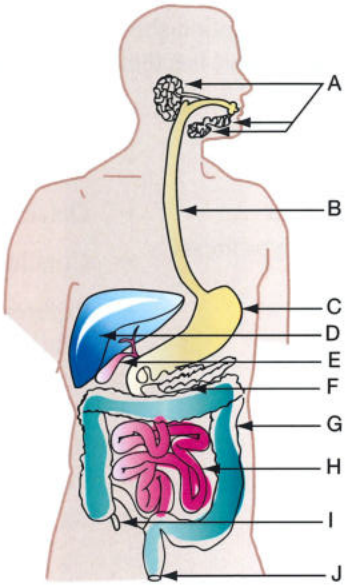


Figure 1.2 Human digestive tract.

29. Construct a table to show each of the following parts of the digestive system, to outline its structure and its main function.
    (a) Mouth.
    (b) Oesophagus.
    (c) Stomach.
    (d) Small intestine.
    (e) Large intestine.
    (f) Anus.
30. The diagram shows a simple model of digestion.

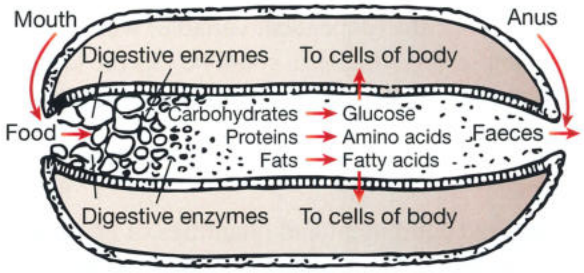


Figure 1.3 Simple model of digestion.

    (a) From this model identify what foods are broken down into:
        (i) Glucose.
        (ii) Amino acids.
        (iii) Fatty acids.
    (b) Explain why scientists make simple models such as this simple model of digestion.
31. What is the function of the excretory system?
32. Outline the function of the excretory system.
33. What is the function of the muscular system?
34. Why is movement of the whole body or parts of the body needed?
35. Describe muscle tissue.

2 Writing a Practical Report

Each practical investigation should be recorded in a scientific report. A standard report has the following aspects.

- Title
- Aim
- Introduction or rationale or hypothesis
- Equipment
- Variables
- Method
- Risk assessment
- Results
- Discussion
- Conclusion
- References or Bibliography

A practical report is written in the **third person**, **past tense**, so that making a slide of onion skin would be written as 'a slide of onion skin was made'.

Aim

The aim is a brief statement of the purpose of the investigation. The aim usually begins with 'To', e.g. 'To investigate how activities affect the heart and breathing rate in humans.'

Introduction

An introduction is needed if there are scientific theories, laws, or relationships which are being used or tested to achieve the aim. If an hypothesis is being tested it needs to be included in the introduction. An **hypothesis** can be written '**As the** ..... (independent variable) ..... , **the** (dependent variable) **will** .....'. For example, an hypothesis could be 'As the level of physical exercise increases, the breathing rate increases.'

Equipment

All required equipment and quantities of equipment should be listed. If the equipment needs to be assembled into a specific set-up, a diagram of the set-up should be drawn. The diagram should be a suitable size, drawn in pencil, with labels and follow the standard rules for scientific diagrams. The choice of equipment contributes to showing the validity of the method.

Variables

When designing an investigation it is important to identify the independent variable and the dependent variable and the factors that have been controlled to make sure the method is valid. For example, when testing if increasing the amount of physical exercise causes an increase in the breathing rate, the independent variable is the amount of physical exercise and the dependent variable is the breathing rate.

When carrying out this experiment you would need to control the age of the people, the type of physical exercise, the physical conditions of the person exercising, e.g. temperature, humidity, terrain/floor on which they are exercising, when the person had last eaten, physical fitness level of people involved.

However, not all biology investigations involve dependent/independent variables, e.g. studying the structure of red blood cells using a light microscope would not need this section in the practical report.

Method

The method is usually written as a series of dot points showing the logical sequence of steps that were followed as the experiment was carried out. The degree of detail to put into the method is usually left to common sense. The way the results are measured and recorded needs to be included and the number of multiple trials needs to be stated. The method tests the aim and shows the validity of the experiment and the number of repeated trials contributes to the reliability of the experiment.

Risk assessment

All relevant risks need to be identified with a reason why it is a risk. A safety precaution for each risk then needs to be stated to show how each risk can be avoided. The treatment for each situation also needs to be stated.

Results

Collected data needs to be presented in the most suitable format. In many instances data can be summarised in a table, chart or graph. The results section can include calculations, e.g. working out an average and the manipulation of data.

Discussion

The discussion is very important and needs to have an evaluation of the investigation. There should be a statement to indicate if the results supported the hypothesis (if there was an hypothesis) and if the data shows any trend or pattern. The results need to be explained, stating any assumptions made about the system, or identifying sources of error and/or any limitations of the data. The results can be compared with known data and suggestions made about improvements and ideas for further investigations.

Conclusion

The conclusion needs to be concise and relate back to the aim. If an hypothesis was being tested there should be a statement saying the results supported/did not support the hypothesis. The results should *not* be repeated and there should *not* be another discussion.

References

Sources of information and help need to be acknowledged using correct formats. Sources can be written material, e.g. books, periodicals and journals, web pages and/or advice from help bodies. Assistance from individuals with an expertise that helped in some way with the investigation can also be acknowledged in this section.

QUESTIONS

1. A Human Biology student measured the heart rate of a test person by putting a finger on the neck of the test person and counting the number of beats in 60 seconds. The test person jumped on the spot for 2 minutes and the heartbeat of the test person was measured again in the same way.
   (a) Suggest an hypothesis that could be investigated for these observations.
   (b) For the hypothesis you suggested in part (a), identify the dependent variable and the independent variable.
   (c) If the students carried out an investigation to test their hypothesis, what factors would they need to control and keep constant?
2. Some Human Biology students studied red blood cells and white blood cells using the high power objective of a light microscope. They used a minigrid scale to work out the diameter of these blood cells.
   (a) Write a suitable aim for this experiment.
   (b) Suggest why the students needed to use the high power objective to observe red blood cells and white blood cells.
   (c) List the equipment the students would need to carry out this investigation.
   (d) Identify a safety precaution the students would have needed to follow to carry out this investigation.
3. The diagram shows a graph drawn by students in their results section of their practical report.

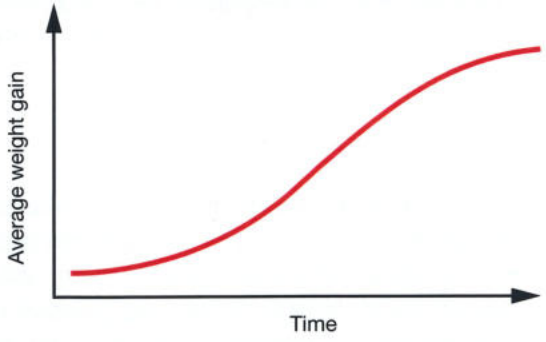


Figure 2.1 Graph of student results.

For these results what is the dependent variable and the independent variable?

4. A spirometer is used to measure human lung capacity measuring the volume of air inspired and expired by the lungs.

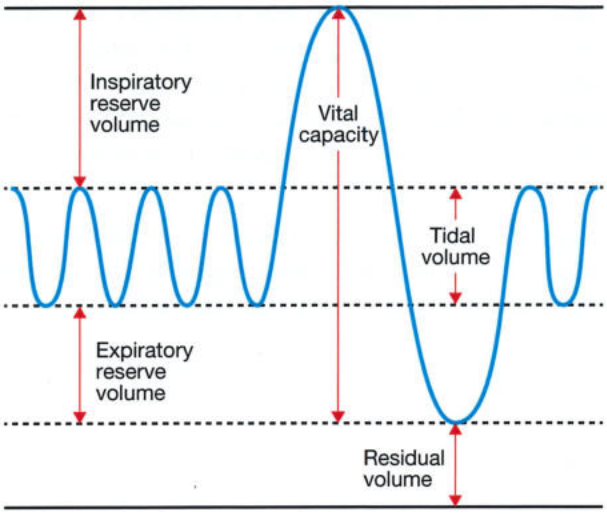


Figure 2.2 Spirometer data.

Some students used a spirometer when rested and sitting down recording the result as normal breathing rate per minute. Other students used the spirometer by breathing in as deeply as possible and then resuming normal breathing. Other students breathed out as far as possible and then resumed normal breathing. The results were tabulated and average vital lung capacity was deduced from the data. Determine the validity of this experiment.

5. The diagram shows the process of scientific method.

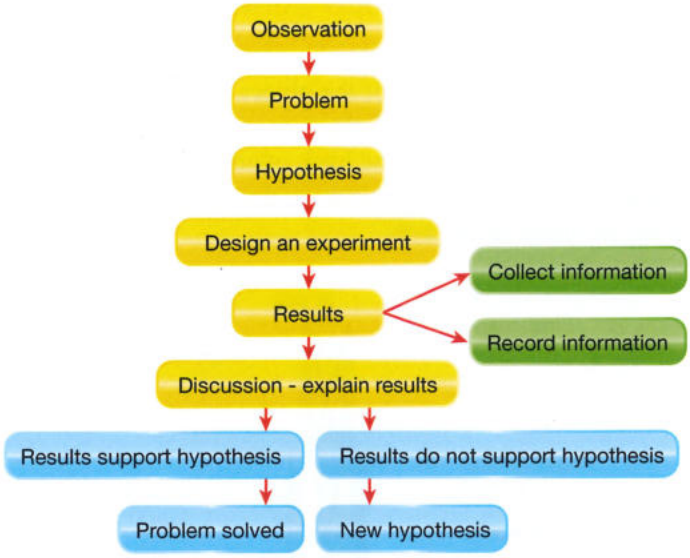


Figure 2.3 Scientific method.

Explain how this process relates to writing practical reports.

3 The Modern Light Microscope

The compound light microscope operates on the main principle that an objective lens with a very short focal length can form a highly magnified real image of the object. Visible light passes through the specimen and then a series of lenses. The resolution of the microscope is limited by the shortest wavelength of light used to view the specimen.

Images from a light microscope can be captured with a camera to produce a **photomicrograph**. Digital images can be shown directly on a computer screen.

Table 3.1 Features of the modern light microscope.

Feature	Light microscope
Magnification	Effective up to 1000×.
Resolution	Up to 0.2 μm.
Stains	Allows the use of many different coloured stains to identify substances, structures and provide contrast for easier viewing.
Living specimen	The light microscope allows viewing of living specimen and processes occurring within a cell or within an organism.
Mounting	The specimen is mounted on a glass slide in air.
Focusing	By glass lenses.
Energy source for viewing	A beam of light is passed through the specimen.

How a light microscope works

The objective lens is brought close to the specimen to create an enlarged image of the object. The image is inverted. In most modern light microscopes the eyepiece is a compound lens near the back of an eyepiece tube. Light travels from the light source up the microscope to form an image at the eye.

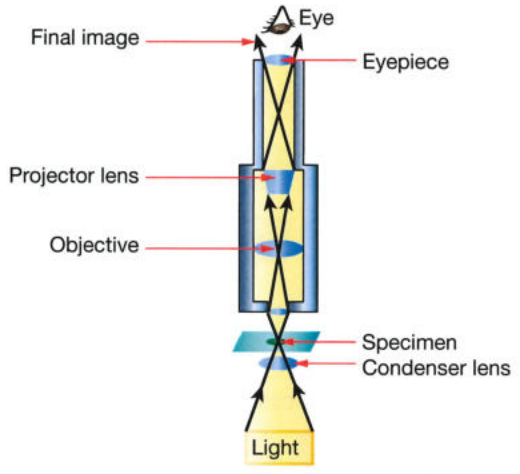


Figure 3.1 How light travels through a light microscope.

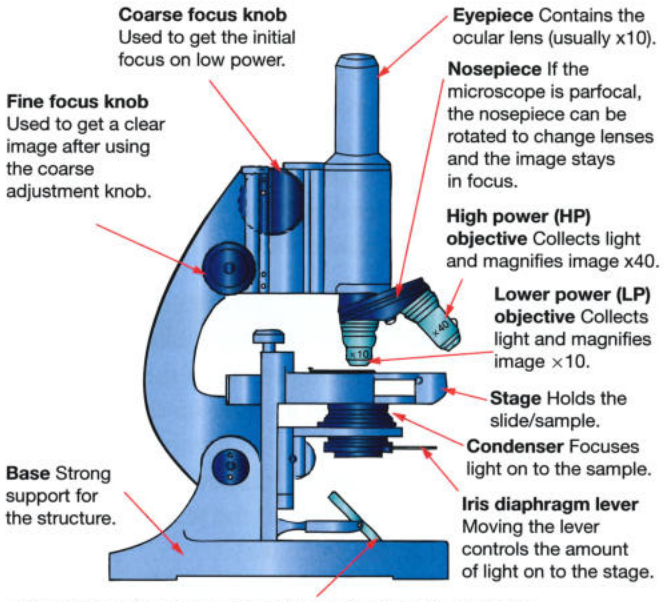


Figure 3.2 Features of the modern light microscope.

Advantages of a light microscope

The main advantages of light microscopes are that:

- Living cells can be observed.
- Coloured stains can be used.
- The specimens are easy to prepare.
- The microscopes are relatively inexpensive (compared to the cost of an electron microscope).
- Their size means they are relatively easy to store.

Disadvantages of a light microscope

The main disadvantages of the light microscope are:

- Its limited magnification (effective magnification begins to reduce after 1000×).
- Its limited resolution.

During the 20th century many different illumination techniques and other developments have increased the detection power of the light microscope for observing living cells.

Phase contrast microscopes

The phase contrast microscope uses interference rather than absorption of light to increase the contrast in unstained cells by amplifying variations in density within the cell. It improves our ability to study living, unpigmented cells in biological and medical research. Many dyes and stains stop chemical processes in cells which means the phase contrast microscope has improved our ability to see detail in living cells, e.g. the process of cell division. Frits Zernike was awarded with the Nobel Prize in Physics, 1953 for the development of phase contrast illumination.

Fluorescence microscopes

In the fluorescence microscope the specimen is illuminated through objective lenses with a narrow set of light wavelengths. The specimen either fluoresces in its natural form, e.g. chlorophyll or has been treated with fluorescing chemicals or antibodies. The fluorescent substances absorb UV light and emit visible light so that the fluorescence shows the location of specific molecules in the cell.

Other illumination techniques

Other illumination techniques include:

- **Bright field** – passes light through the specimen and contrast comes from the absorbance of light in the specimen. If the specimen is unstained or unpigmented there is little contrast.
- **Cross-polarised** – contrast occurs when polarised light is rotated through the sample.
- **Confocal** – is a type of fluorescence microscopy using optical sectioning by scanning lasers of fluorescently-stained specimens.

Stains and dyes used with the light microscope

There are many stains that are used to highlight structures being viewed under a light microscope.

- **Gram staining** – uses several stains, e.g. crystal violet, iodine, fuchsin or safranin to stain cell cells to differentiate bacteria into Gram positive (purple/blue colour occurs) and Gram negative (pink/red colour occurs). This is usually the first step in identifying bacteria.
- **DAPI** – is a fluorescent nuclear stain that shows a blue fluorescence when bound to DNA and viewed with ultraviolet light.
- **Eosin** – is used as a counterstain with haematoxylin in H&E staining and gives a pink/red colour to cytoplasmic material, red blood cells and cell membranes.

QUESTIONS

1. Outline the basic principle behind the operation of a light microscope.
2. When using a light microscope identify the function of:
   (a) The condenser.
   (b) The iris diaphragm.
3. What restricts the resolution of the light microscope?
4. Construct a table to compare the advantages and disadvantages of a light microscope.
5. What is a photomicrograph?
6. Outline the benefit of the phase contrast microscope.
7. Outline the benefit of fluorescence microscopy.
8. The table shows some objects and their sizes.

Object	Size (μm)
Frog egg	1100
Paramecium	100
Plant epithelial cell	60
Human ovum	10
Red blood cell	7 to 8
Mitochondrion	1.1
Nanobes	0.025

   Identify which objects could be seen with the naked eye and/or light microscope.
9. Use an example to show how the use of stains and the light microscope have increased our understanding of processes in cells.
10. Most modern light microscopes in schools have at least two different objectives. Name objectives you have used and state the purpose of each objective.
11. What is a parfocal microscope?
12. The diagram shows red blood cells as seen and as drawn by a Human Biology student using a light microscope.

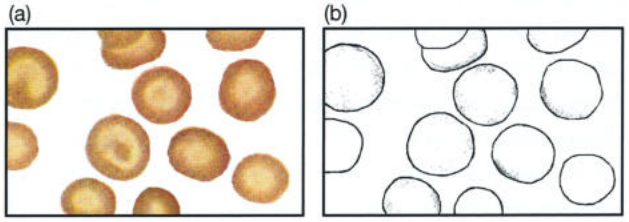


Figure 3.3 Red blood cells seen using a light microscope. (a) Seen under a light microscope. (b) As drawn by a student.

   Red blood cells are usually about 6 to 8 micrometres in diameter.
   Which objective would the student most likely have been using to view these cells as seen in the diagram? Explain your answer.
13. Explain why the Gram stain is used.
14. What is the correct order of the movement of light when an object is viewed under a light microscope?
   (A) Light → condenser → objective → eyepiece → eye
   (B) Light → objective → condenser → eyepiece → eye
   (C) Light → objective → eyepiece → condenser → eye
   (D) Eye → eyepiece → objective → condenser → light
15. What is the limit of resolution of a light microscope?
   (A) 0.2 centimetres
   (B) 0.2 millimetres
   (C) 0.2 micrometres
   (D) 0.2 nanometres

4 The Electron Microscope

The electron microscope provides greater detail about cell structure. The electron microscope sends a stream of electrons through a vacuum. The electron beam is focused by electromagnets, magnified by an objective lens and projected onto a fluorescent screen or photographic film. Since the beam of electrons has a much shorter wavelength than visible light resolution is greatly improved, e.g. approximately 0.002 nm, although in practical situations it can be limited to 2 nm. Typically they provide a resolution of 0.5 nm (400 times better than a light microscope) and magnify up to 500 000 times.

Transmission electron microscope

The transmission electron microscope (TEM) uses the same basic principles as the light microscope with a beam of electrons passing through the specimen instead of a beam of light. The TEM is used to study the internal ultrastructure of cells. The specimen is stained with heavy metal atoms which attach to particular cellular structures, preserved by a chemical fixative, embedded in plastic then cut into exceedingly thin slices (50 to 100 nm). You can see objects to the order of several nanometres (10^{-9} m) increasing the capacity for medical, biological and materials research. Many organelles were discovered using the TEM, e.g. ribosomes. One of the main disadvantages of the TEM is that the method of specimen preparation kills the cells and the specimen is viewed in a vacuum. This means living cells cannot be viewed. Specimen preparation also produces artefacts and structural features that do not exist in living cells.

Table 4.1 Features of the electron microscope.

Feature	Electron microscope
Magnification	Up to 300 000×
Resolution	Approximately 0.0005 μm
Stains	Heavy metal stains identify substances, structures and provide contrast for easier viewing. No colour stains.
Living specimen	The electron microscope does not allow viewing of living cells.
Mounting	Metal background in a vacuum chamber.
Focusing	By electromagnetic lenses.
Energy source for viewing	A beam of electrons is passed through the specimen.

Scanning electron microscope

The scanning electron microscope (SEM) emits an electron beam which is rapidly passed back and forth over the surface of the specimen. Surface variations alter the pattern of the scattering of the electrons and the pattern is recorded, amplified and transmitted to a TV monitor. This gives a three-dimensional detailed view of the surface of the specimen. The image has great depth of field.

QUESTIONS

1. Outline why the electron microscope has better resolution than the light microscope.
2. Outline the basic principle behind the operation of an electron microscope.
3. Construct a table to compare the advantages and disadvantages of a light microscope.
4. Discuss the importance of the transmission electron microscope.
5. Discuss the importance of the scanning electron microscope.
6. Identify the type of microscope that was used to view each of the following.

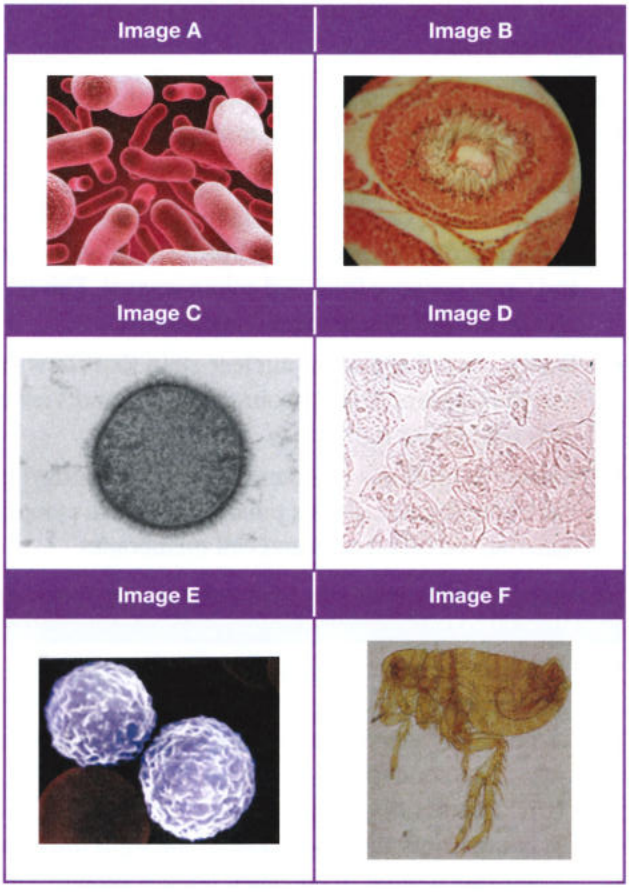


Figure 4.1 Images from different types of microscopes.

7. What is the limit of resolution of a transmission electron microscope?
   (A) 0.5 centimetres (B) 0.5 millimetres
   (C) 0.5 micrometres (D) 0.5 nanometres

5 The Stereo Microscope

The **stereo microscope** has two ocular lens so that the left eye and the right eye each view the object from different angles. This sends two images to the brain. The brain then produces a three-dimensional image of the object. Thus the microscope gives a stereoscopic view or 'stereo' view of an object.

Some compound microscopes have double eyepieces which make viewing easier but give both eyes the same image and do not provide the three-dimensional images provided by stereo microscopes.

To use a stereo microscope the eyepieces need to be adjusted. There is variation in the distance between the pupils of the eyes of each person and this means that the position of the eyepieces of the stereo microscope need to be corrected to suit the requirements of each person.

Different parts of a specimen come into focus when focusing up and down while using a stereo microscope. Organisms are three-dimensional and the stereo microscope allows viewing of the different levels of the specimen. For larger specimens every feature cannot be seen clearly at the same time.

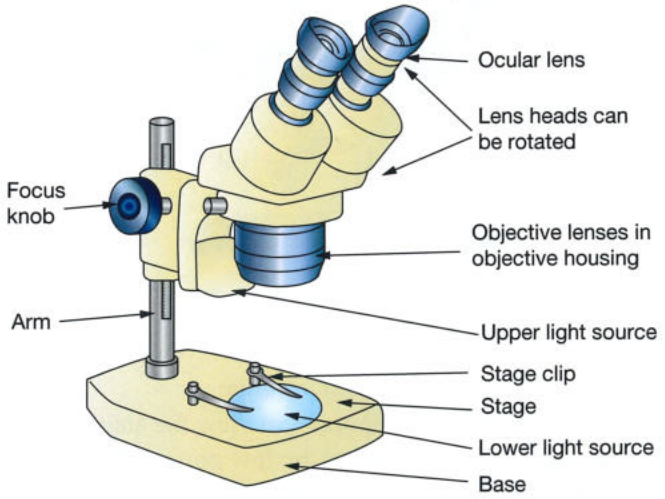


Figure 5.1 Stereo microscope.

Resolution, illumination and magnification of a stereo microscope

A stereo microscope usually has poor resolution as the lenses are a distance away from the object.

Most stereo microscopes use reflected light from an upper and/or lower light source. It is important when observing live organisms, e.g. worms and insects not to leave the light source shining on the organism for too long. The heat can kill the organism.

Most stereo microscopes have magnifications from 10× to 40×.

Uses of a stereo microscope

A stereo microscope is often used when dissecting small organisms and is very useful when you need to carefully observe the external features of a specimen.

If a camera and/or computer is attached to the stereo microscope the image can be recorded and displayed on a screen. Since the camera is only connected to one eyepiece the result will be a 2-D image.

QUESTIONS

1. What is the distinguishing feature of a stereo microscope?
2. When are you likely to use a stereo microscope?
3. The diagram shows a side view of a stereo microscope.

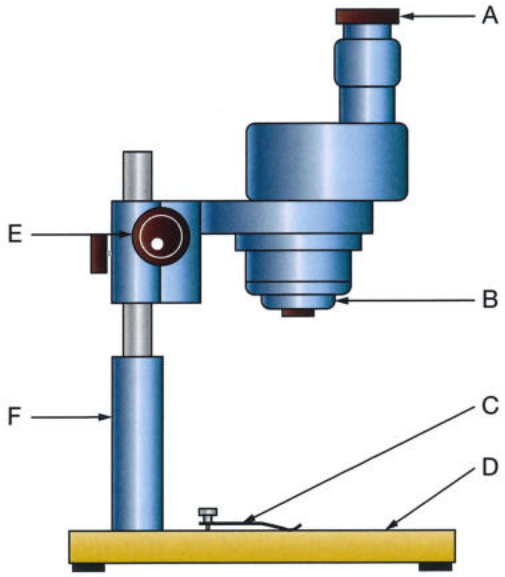


Figure 5.2 Side view of a stereo microscope.

Identify the parts of the microscope labelled, A, B, C, D, E and F.

4. Explain how a compound microscope with double eyepieces may not provide a three-dimensional image to form a 'stereo microscope'.
5. Explain why many stereo microscopes have poor resolution.
6. Compare the magnification of a typical stereo microscope with a typical monocular microscope found in schools.
7. When studying crystals, e.g. salt crystals under a stereo microscope, explain why it is advisable to put a piece of black paper beneath the crystal.
8. Explain why some parts of an organism may appear out of focus while studying a specimen with a stereo microscope.
9. Which of the following would *not* be suitable to be observed using a stereo microscope?
   (A) Red blood cells. (B) Honey bee.
   (C) Eucalypt leaf. (D) Salt crystal.

6 Experiment – The Light Microscope

Biology students need to be able to use a monocular light microscope. The light microscopes found in schools are compound microscopes with magnifying lenses used in series to provide an enlarged image of a specimen. Light is transmitted through the specimen to reach the eye of the observer. The specimen needs to be very thin to allow light to pass through. A simple light microscope like the one used by van Leeuwenhoek has one lens.

The eyepiece lens on most microscopes is 10× and some microscopes have interchangeable eyepieces, e.g. 5× or 10×. There is usually a choice of at least two objective lenses, for example 10× and 40×. The overall magnification is calculated by multiplying the degree of magnification of the objective lens by the degree of magnification of the eyepiece lens. The 100× objective is an oil immersion lens and can only be used with a special oil between the slide and the lens.

Many school light microscopes are **parfocal** which means that a specimen viewed under low power will automatically be in focus or near focus when the nosepiece is rotated from low power to high power.

Rules for using a light microscope

There are several important **rules** to follow when using a light microscope.

1. You must always use two hands when carrying a microscope. One should hold the arm and the other hand needs to be beneath the base.
2. Never touch the lenses with your fingers.
3. When you have finished using your microscope the nosepiece needs to be rotated so the low power objective is facing the stage and then lower the objective down to the stage.
4. Never leave a slide on the stage.
5. Place the dust cover over the microscope and return the microscope to its proper location.
6. If you made a wet mount slide, make sure your work area is clean and dry and all glass slides and cover slips are returned to their proper location at the end of the experiment.

Setting up a microscope

1. Place the microscope on the desk or bench with the arm *towards* you. This means that the stage is facing *away* from you.
2. Turn the coarse adjustment knob and raise the nosepiece.
3. Rotate the nosepiece so that the low power objective is facing the stage.
4. Look through the eyepiece and adjust the mirror and the diaphragm so that there is a bright circle of light visible.
5. Place the slide on the stage. Stage clips can be used.
6. Look at the stage from the side and lower the low power objective using the coarse adjustment knob. Stop before the objective touches the slide.
7. Look through the eyepiece and slowly turn the coarse adjustment knob until the image comes into focus.
8. Use the fine adjustment knob to bring the image into sharp focus.

QUESTIONS

1. Identify each part of the compound light microscope shown in Figure 6.1.

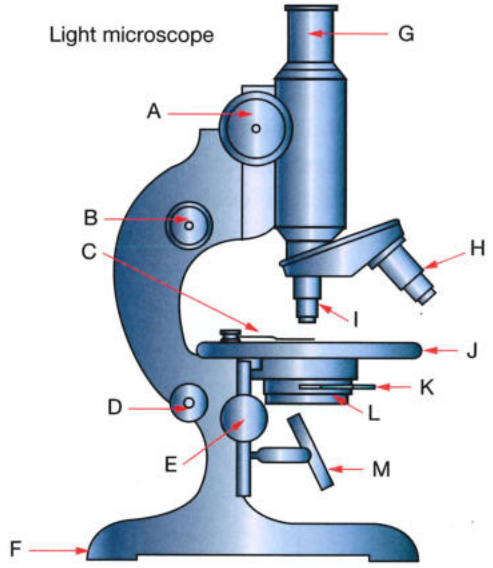


Figure 6.1 Compound light microscope.

2. Explain why it is necessary to always use two hands when carrying a light microscope.
3. Explain why it is important to view the stage from the side when you lower the low power objective and then always move the objective lens away from the microscope slide.
4. If an eyepiece has a magnification of 10×, what is the overall magnification for each of these objective lenses?
   (a) Low power objective 10×.
   (b) High power objective 40×.
5. Outline the function of the iris diaphragm.
6. Explain why it is usually necessary to adjust the amount of light after changing objective lenses.
7. How does a stereo microscope vary from a light compound microscope?
8. Explain why it is important not to touch the lens with your fingers and discuss what you would do if you found a fingerprint on the lens.

7 Experiment – Using a Light Microscope

The use of a light microscope is determined by its magnification and resolution. **Resolution** refers to the minimum distance between two points to distinguish them as two points – it refers to the clarity of the image. **Magnification** in microscopy refers to the ratio of the object's actual size to its image. Most light microscopes cannot resolve detail finer than 0.2 μm and can effectively magnify objects up to 1000×. Many light microscopes used by students in schools are only equipped with lenses that allow up to 400× magnification.

Magnification is limited by the resolving power of the microscope – increasing magnification will only give a bigger blur and not more detail. The resolution is determined by the shortest wavelength of light being used to illuminate the specimen.

When using a light microscope it is important to determine the diameter of the field of view under low power. A minigrid slide that has a 1 cm square of graph paper gives you the ability to estimate the diameter of the field of view. Figure 7.1 shows a section of minigrid with a central mm square divided into 0.1 mm divisions.

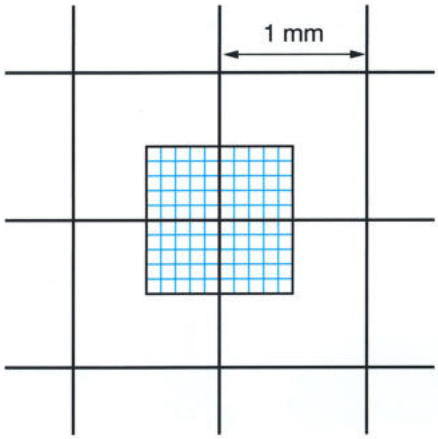


Figure 7.1 Section of a minigrid with the central area subdivided into 0.1 mm divisions.

By placing the minigrid with a line crossing the diameter of your field of view and the mini-minigrid on this line, you can estimate the diameter of the field of view in millimetres (mm). The measurement can easily be converted into micrometres (μm).

Depth of field in microscopy can be shown by placing three different strands of hair over each other on a slide and making a wet mount. To determine the position of the three strands you need to focus up and down with the fine adjustment knob. This skill is used to determine positions of organelles or structures in thicker sections of tissue or in large unicellular organisms.

QUESTIONS

1. Distinguish between magnification and resolution.
2. Figure 7.2 shows a minigrid under lower power of a light microscope using an eyepiece 10× and a low power objective 10×.

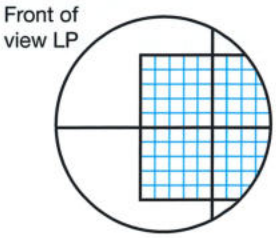


Figure 7.2 Minigrid under a light microscope.

   (a) Estimate the diameter of the field of view in millimetres and micrometres.
   (b) If the high power objective (40×) was used on this microscope, what would be the diameter of the field under high power?
3. Figure 7.3 shows the appearance of the letter 'e' as seen under low power of a light microscope.

Figure 7.3 Letter 'e' as seen under the microscope.

   Discuss why viewing the letter 'e' under the microscope is used to show important aspects of the viewed image of an object.
4. Describe how you would determine the position of different organelles in a large unicellular organism being viewed in a wet mount under a light microscope.
5. Figure 7.4 shows a plant root hair as seen under a light microscope that has a minigrid.

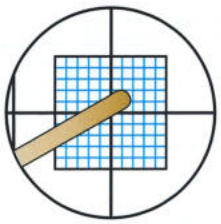


Figure 7.4 Root hair and minigrid under a light microscope.

   What is the diameter of the root hair in millimetres and in micrometres?
6. Describe how you would record your observations when carrying out an experiment using a light microscope.
7. If you are trying to focus a microscope and have raised the body tube more than 2 cm above the slide, what should you do next?

8 Cells, Tissues, Organs and Systems

Multicellular organisms consist of many cells and many require specialised cells to carry out particular jobs. Specialisation allows a more efficient supply of nutrients, removal of wastes, and the other processes essential to life.

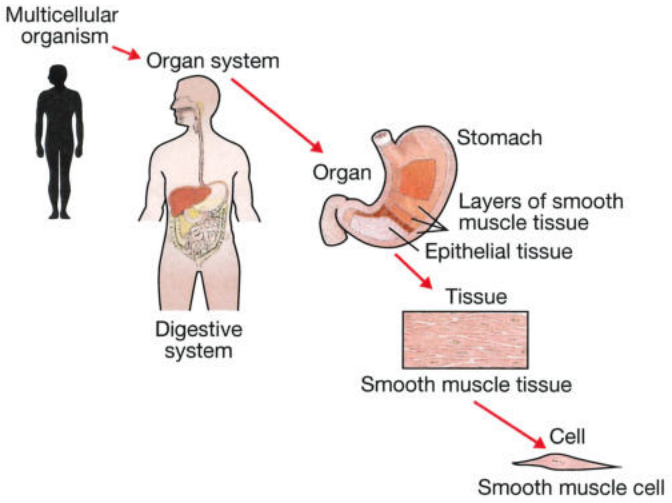


Figure 8.1 Levels of organisation.

The **cell** is the basic unit of life. Many cells are specialised to carry out particular tasks, e.g. red blood cells carry oxygen. This means that specialised cells often have a distinct shape and special kinds of chemical reactions occur in their cytoplasm.

A **tissue** is a group of cells with similar structure and function. For example, muscle tissue is made of many aligned muscle cells which are long cells able to shorten to produce movement. There are four main types of tissues – epithelial tissue, connective tissue, muscle tissue and nervous tissue. Tissue regeneration can occur when some of the original cells are present and can replicate by mitosis.

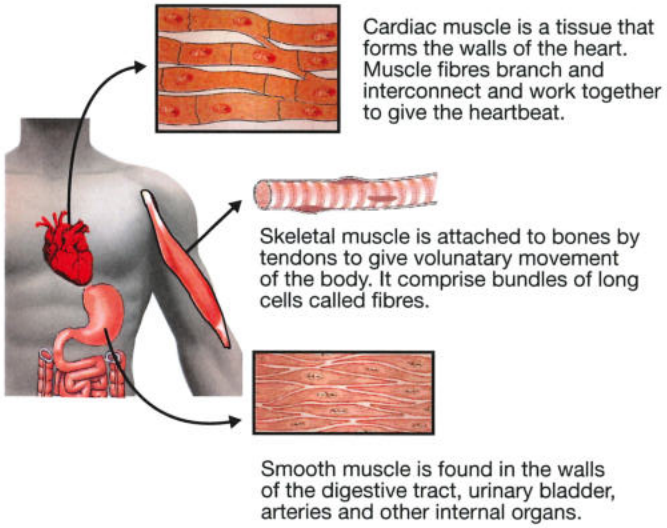


Figure 8.2 Muscle tissue.

Muscle tissue generates motion and generates force. It can also generate heat and help maintain body temperature in endotherms. The three different types of muscle tissue are found in different organs and different systems.

An **organ** is a group of tissues grouped together to make a structure with a special function. For example, the stomach is an organ which has an epithelial lining, gland tissue and muscle layer supplied with blood vessels and nerves.

A **system** is usually a group of organs whose function is closely related, for example the stomach and intestines.

QUESTIONS

1. Define the following terms.
   (a) Cell
   (b) Tissue
   (c) Organ
   (d) System
2. Give an example of each of the following.
   (a) Organ
   (b) Tissue
   (c) Specialised cell
3. Describe how the different types of muscle tissue are in different systems but they have a similar general function.
4. The kidneys, certain nerves, bladder, urethra and ureters cooperate to remove chemical wastes from the body. Identify the level of organisation for this cooperation.
   (A) Tissue
   (B) Organ
   (C) System
   (D) Cell
5. The diagram shows two types of cells.

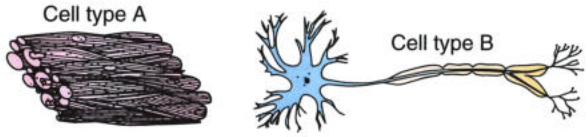


Figure 8.3 Two types of cells.

What is the importance of cells having different shapes?
   (A) More cells can fit into the same space.
   (B) Different cell shape is related to different function.
   (C) Cell shape indicates different species.
   (D) Different cell shape shows evolutionary relationships.
6. What is the correct order from smallest to largest level of organisation?
   (A) System → organ → tissue → cell
   (B) Organ → system → tissue → cell
   (C) Cell → tissue → system → organ
   (D) Cell → tissue → organ → system

9 Differentiation and Specialised Cells

Differentiation is a process when a young, relatively unspecialised cell develops into a more mature state to become a more specialised cell. When cells differentiate they are no longer similar to the parent cell in structure or function. Specialised cells carry out particular functions leading to a division of labour which means particular tasks can be carried out more efficiently.

Cell specialisation occurs when there is **differential gene expression**. The nucleus of each nucleated cell (except gametes which are haploid and have half the number of chromosomes) has a full set of chromosomes. When a specific gene is 'switched on' the cell can change structure and shape so that it becomes specialised to carry out a particular function. Differentiated cells have genes that are switched on that make them specialists in making tissue-specific proteins. For example, muscle cells are specialised to produce the proteins myosin and actin which are needed for muscle contraction. In the human body there are over 200 recognisably different types of cells.

Tissues are groups of cells carrying out a particular function. A tissue may have only one kind of cell or several kinds of cells. There are four main types of mammalian tissue.

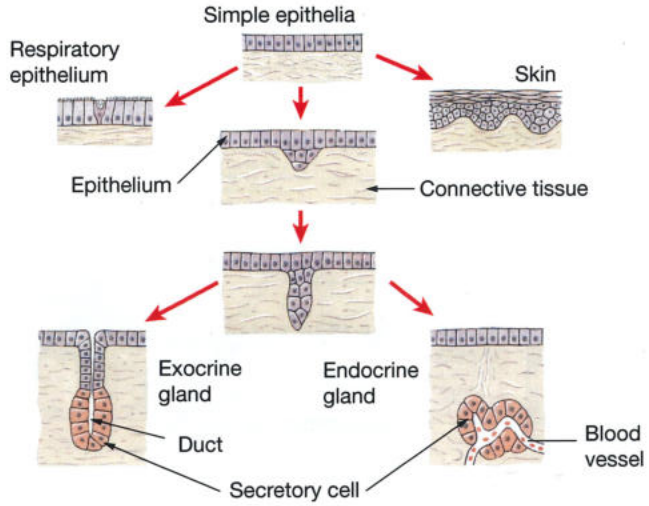


Figure 9.1 Epithelial tissue.

Epithelial tissue

Epithelial tissue (epithelia) covers many surfaces and linings of body cavities. Endothelia lines internal cavities. Epithelial tissue acts as a protective layer and as a barrier against infection by microbes or water loss. Epithelial tissue is found in the outer part of the skin and the linings of the digestive tract, the lungs, the blood vessels, the body cavity and various ducts.

Epithelial cells group together to form sheets which can be strengthened with bundles of filaments running from cell to cell. Simple epithelia can be squamous (flat), cubical or columnar with some having microvilli on one side. Microvilli increase the surface area of the cell to aid the movement of substances into and out of the cell, e.g. in the small intestine and the walls of kidney tubules. Epithelium tissue is usually separated from the underlying tissue by an extracellular fibrous **basement membrane**. The basement membrane is usually made of a network of white, wavy, non-elastic collagen fibres.

Epithelia can differentiate to form the secretory cells of glands or become sensory cells, e.g. olfactory cells in the nose. The gland cells can be unicellular gland cells or a section of the epithelial tissue can invaginate to form a multicellular gland such as an endocrine gland (ductless gland) or an exocrine gland (has duct).

Connective tissue

Connective tissue is found in many areas of the body and includes bone, cartilage, blood, tendons and ligaments. A large proportion of connective tissue is an intercellular matrix that contains a network of protein fibres in a semiliquid ground substance. The matrix is secreted by connective tissue cells and can be liquid, semisolid or solid. Blood and lymph have liquid matrixes. Connective tissue contains **collagenous fibres** which are non-elastic and do not easily tear when pulled lengthwise. **Elastic fibres** which can stretch and return to their original length and **reticular fibres** which are thin and branched crosslink to form a fine network.

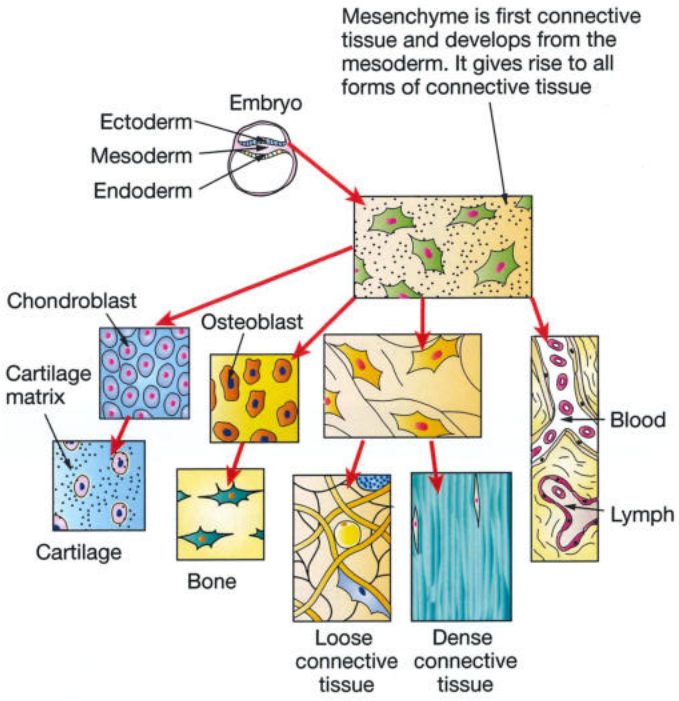


Figure 9.2 Connective tissue.

Muscle tissue

Muscle tissue is specialised to convert chemical energy in ATP into mechanical energy for movement. Smooth muscle forms the walls of many internal cavities, e.g. intestines, blood vessels and bladder. Striated muscle is connected to bones either directly or via a tendon to give voluntary motion. The heart is made of cardiac muscle.

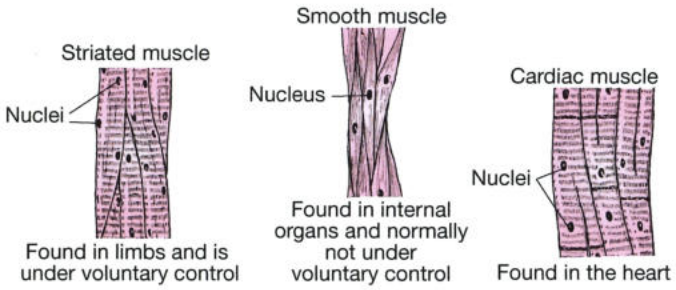


Figure 9.3 Muscle tissue.

Nervous tissue

Nervous tissue consists of specialised cells called neurons (nerve cells) and glial cells. The stem cells in the embryonic brain produce both neurons and glia. A neuron consists of dendrites which pick up the signal and transmit the impulse from their tips to the rest of the neuron. There are several types of neurons, e.g. in the retina there are bipolar cells and ganglion cells and in the brain the neurons of the cerebellum are different to the neurons of the cerebral cortex. Motor neurons take messages from the central nervous system to muscles or glands. Sensory neurons take messages from a sensory receptor to the central nervous system. Supporting glial cells help neurons function correctly.

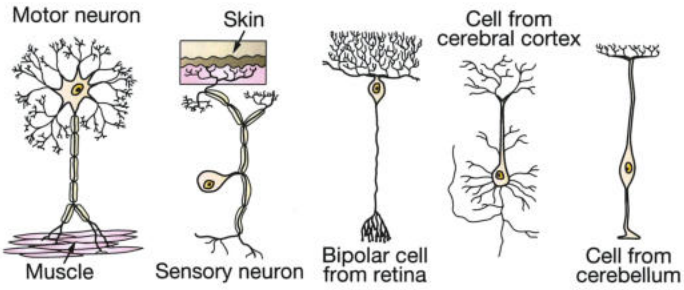


Figure 9.4 Different types of neurons.

QUESTIONS

1. Define differentiation.
2. Explain why multicellular organisms need specialised cells.
3. What is meant by differential gene expression?
4. Identify some proteins made by muscle cells that give these cells their particular ability to contact and relax and outline why this is an important substance for muscle cells.
5. Are all cells in a tissue identical? Use an example to explain your answer.
6. Identify the four main types of mammalian tissue.
7. Describe where epithelial tissue is found.
8. What are the three basic shapes of epithelial cells?
9. Explain the presence of microvilli on epithelial cells.
10. What is the basement membrane?
11. Distinguish between an exocrine gland and an endocrine gland.
12. Where is connective tissue found in the body?
13. Outline the function of muscle cells.
14. Identify the function of the three types of muscle cells.
15. Name the types of cells found in nervous tissue.
16. Use examples to show the range of different specialised nerve cells.
17. The diagram shows a type of specialised cell.

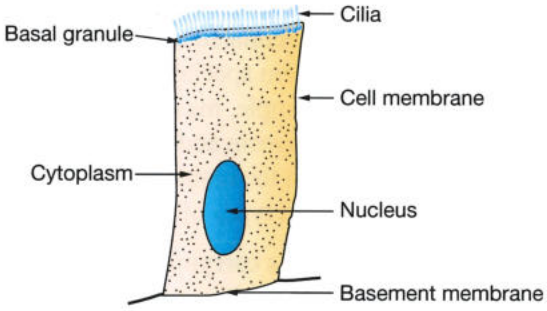


Figure 9.5 A specialised cell.

What is the name of this type of cell?
(A) Columnar epithelial cell.
(B) Muscle cell.
(C) Neuron.
(D) Squamous epithelial cell.

18. The diagram shows a type of tissue.

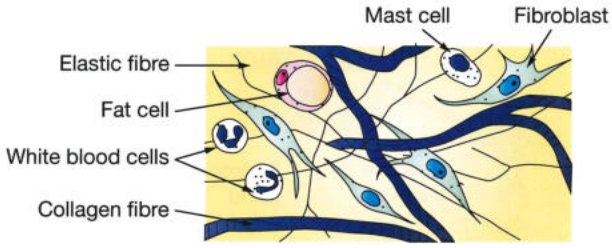


Figure 9.6 A specialised tissue.

What is the name of this type of tissue?
(A) Nervous tissue.
(B) Dense connective tissue.
(C) Loose connective tissue.
(D) Epithelial tissue.

19. What type of cells can become specialised to form gland cells?
(A) Neurons.
(B) Epithelial cells.
(C) Muscle cells.
(D) Connective tissue cells.

10 The Light Microscope and Cell Organelles

Technological improvements in microscopy have allowed scientists to increase their knowledge of cells and the structures in cells. Staining techniques cause different organic compounds to react with coloured dyes to identify specific chemicals and also make certain structures stand out.

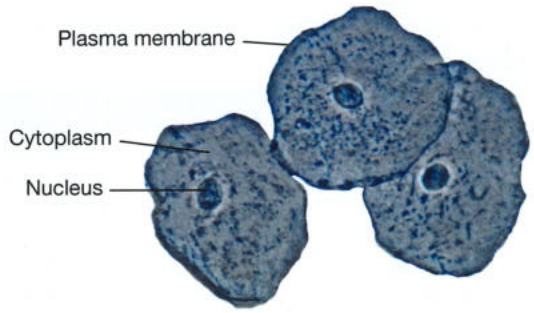


Figure 10.1 Stained human epithelial cheek cells under light microscope (x400).

Plasma membrane

The cell membrane or plasma membrane forms a barrier around the cell contents and is semipermeable, i.e. it controls the movement of substances into and out of the cell. It is very thin and often hard to see under a light microscope.

Cytoplasm

All material between the cell membrane and the nuclear membrane, including the cell membrane, make up the cytoplasm. When viewed under the light microscope, the cytoplasm appears as a thick clear fluid with particles, and living cells show **cytoplasmic streaming**, which is the constant movement of the particles in the cytoplasm.

The **cytosol** is the semifluid part of the cytoplasm.

Vacuoles

Vacuoles are membrane spaces whose function depends on the type of cell, e.g. for food storage, water balance. Human cells have small vacuoles.

Nucleus

The nucleus is surrounded by a double membrane containing hereditary information in the form of DNA in chromosomes. The nuclear membrane has pores which control movement into and out of the nucleus. The information coded in the chromosomes controls cell metabolism and the development of the organism. The nucleus stains readily with dyes. The nucleus will stain yellow with iodine solution and will stain blue with methylene blue.

Nucleolus

The nucleolus is located in the nucleus and is the site of manufacture of ribosomal RNA and ribosome subunits. Ribosomal RNA plays a major role in protein synthesis. It is a dark staining area of the nucleus.

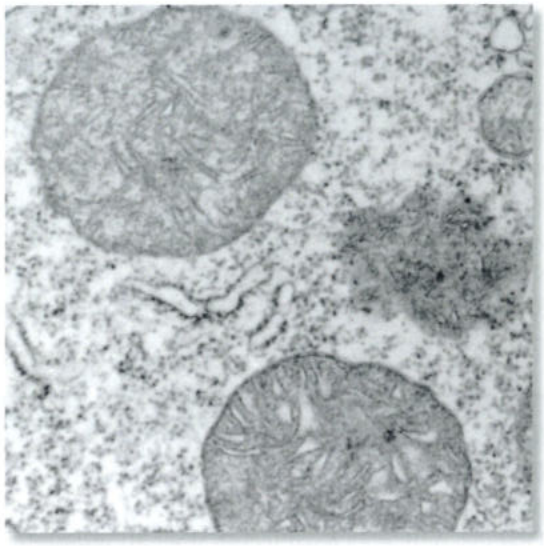


Figure 10.2 Mitochondria.

Mitochondria

Mitochondria have a double membrane with the inner layer being highly folded to increase the surface area to increase the amount of respiration able to be carried out. The respiration releases energy and mitochondria are the 'power house' of the cell. Mitochondria have their own DNA and ribosomes and are thought to have evolved from bacteria which were engulfed by other cells. Mitochondria are just visible under the light microscope at the highest magnifications.

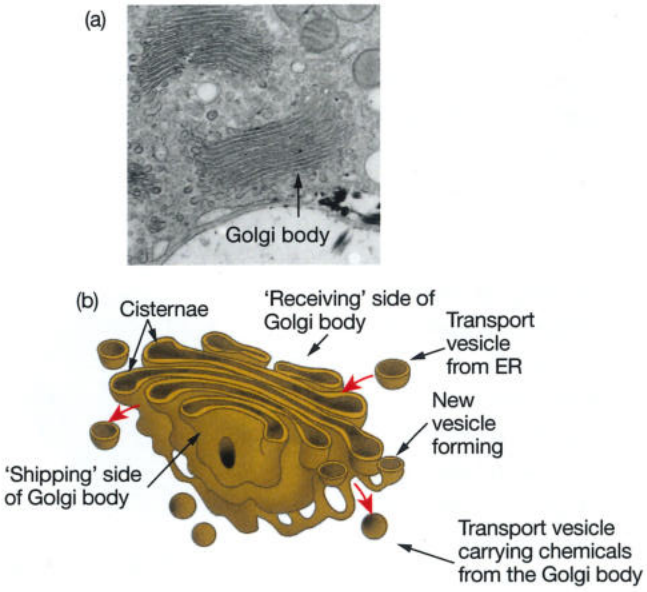


Figure 10.3 Golgi body.

Golgi bodies

Golgi bodies (or Golgi apparatus) were first observed under the light microscope when Camillo Golgi used a silver stain on nerve cells. At first scientists were not sure whether the mark was a product of the staining technique or an organelle. The invention of the electron microscope 60 years later confirmed that the Golgi bodies are organelles consisting of stacks of membrane sacs with many small vesicles at the perimeter. They receive, sort, store and secrete materials, e.g. proteins. They have a large surface membrane on which reactions can occur and the materials are packaged into the vesicles which move across the cell, fuse with the cell membrane and then are released from the cell. **Exocytosis** is the secretion of macromolecules from a cell by the fusion of vesicles with the plasma membrane. Many Golgi bodies are involved in exocytosis.

QUESTIONS

1. How have staining techniques assisted our understanding of cell structure?
2. Discuss why it is hard to see the cell membrane under a light microscope.
3. Outline the function of the cell membrane.
4. Is cytoplasm a single substance? Explain your answer.
5. A Human Biology student was asked to draw a human cheek cell as seen under a light microscope at school. The following diagram is his drawing.

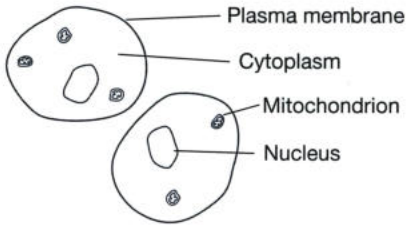


Figure 10.4 Human cheek cell under HP (×400).

Explain why this drawing would not receive full marks when being assessed.

6. Suggest why very active cells have many mitochondria present in the cytoplasm.
7. What stains can be used in schools that will show the presence of the nucleus?
8. The diagram shows a photomicrograph of a human cell. Would this photomicrograph have been taken using a light microscope? Explain your reasoning.

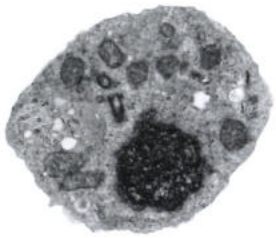


Figure 10.5 Photomicrograph.

9. How did our understanding of mitochondria follow advances in technology?
10. Name the scientist who first identified Golgi bodies.
11. Suggest why cells in the salivary glands should have many Golgi bodies.
12. What is exocytosis?
13. Suggest a reason for cytoplasmic streaming.
14. Which organelles contain DNA?
15. When viewing cells under the light microscope and you change from a low power objective to a high power objective, why do you often need to adjust the iris diaphragm?
16. Which of the following matches organelle with its description?

	Golgi body	Nucleus	Mitochondrion
(A)	Modifies, packages and stores substances for secretion.	Assists water balance.	Controls all activities in cell.
(B)	Releases energy in respiration.	Controls all activities in cell.	Assists water balance.
(C)	Controls all activities in cell.	Modifies, packages and stores substances for secretion.	Releases energy in respiration.
(D)	Modifies, packages and stores substances for secretion.	Controls all activities in cell.	Releases energy in respiration.

17. Which of the following most accurately describes the plasma membrane?
    (A) Impermeable to all molecules.
    (B) Permeable to some molecules and impermeable to others.
    (C) Impermeable to small lipid soluble molecules.
    (D) Permeable to all molecules.
18. The diagram shows a light microscope.

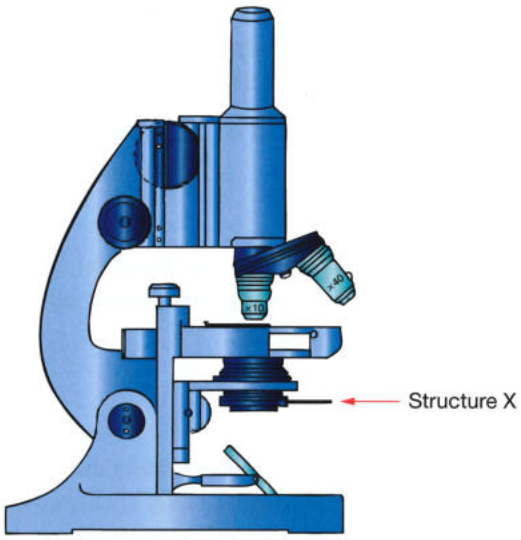


Figure 10.6 Light microscope.

Identify structure X.
(A) Iris diaphragm.
(B) Condenser.
(C) Light source.
(D) High power objective.

11 Experiment – Human Cells

Most animal cells range in size from 10 to 100 μm. Figure 11.1 shows a typical animal cell as seen under the light microscope. While many plant cells have a regular shape, animals cells have many varied shapes and some animal cells, e.g. phagocytes can change their shape when they engulf foreign particles.

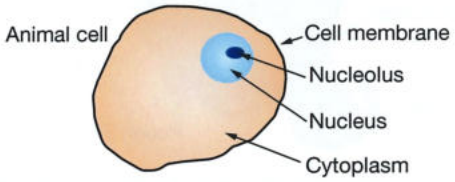


Figure 11.1 Typical animal cell.

Animal cells have a cell membrane (plasma membrane) which controls which substances can enter or leave the cell. Most animal cells have a nucleus – the cells of some specialised tissues lose their nucleus, e.g. mature red blood cells do not have a nucleus. The nucleus controls all cellular activities and contains genetic information. The nucleolus is also visible under a light microscope. The nucleolus is involved in the production of ribosomes and proteins. The region between the nucleus and the plasma membrane is the cytoplasm and within the animal cytoplasm there are a variety of organelles suspended in cytosol.

Some animal cells have small vacuoles. Animal cells do not have a large central vacuole like plants.

There are several organelles present in animal cells, visible under a light microscope.

Organelles visible under a light microscope

Mitochondria are visible under the light microscope in both plant and animal cells. Mitochondria are the sites of cellular respiration. The number of mitochondria present in a cell is related to the metabolic activity of the cell. Since mitochondria release energy in the form of ATP for use in the cell, cells that are highly active have high numbers of mitochondria, e.g. muscle cells.

The **Golgi bodies** are visible under the light microscope, being first seen by Camillo Golgi when he stained nerve cells with silver. For many years some people believed the Golgi body was an artefact of the staining technique and it was not until the electron microscope was invented that its structure was finally described.

The **centrosome** is an organelle that is not surrounded by a membrane. It is involved in cell division and the formation of the spindle. In animal cells there is a pair of centrioles at the centre of the centrosome. The centrioles are not essential for the formation of the spindle as a laser microbeam can be used to destroy the centrioles and the cell will still form a spindle.

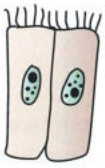
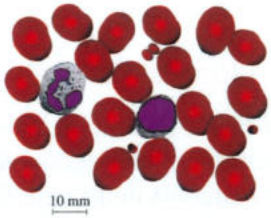
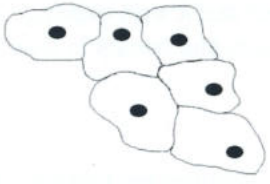
QUESTIONS

1. Outline the function of each of the following cell structures.
   (a) Mitochondria.
   (b) Centrosome.
   (c) Cell membrane.
   (d) Nucleus.
2. Figure 11.2 shows stained human cheek cells under HP. Draw a biological diagram of these cells.

Figure 11.2 Human cheek cells.

3. Explain how an animal cell differs from a plant cell.
4. Identify each of the following types of animal cells.

Table 11.1 Different types of animal cells.

Human cell A	Human cell B
Human cell C	**Human cell D**
	
Human cell E	**Human cell F**
	

5. Which type of animal cell does *not* have a nucleus?
   (A) Neuron.
   (B) Muscle cell.
   (C) Epithelial cell.
   (D) Mature red blood cell.

12 The Electron Microscope and Cell Organelles

The invention of the electron microscope allowed cell organelles (or little organs) to be seen in more detail, or seen for the first time.

Lysosomes

Lysosomes are small membrane-bound sacs which contain enzymes. The enzymes break down large molecules such as fats, proteins and polysaccharides and recycle the unwanted material inside the cell.

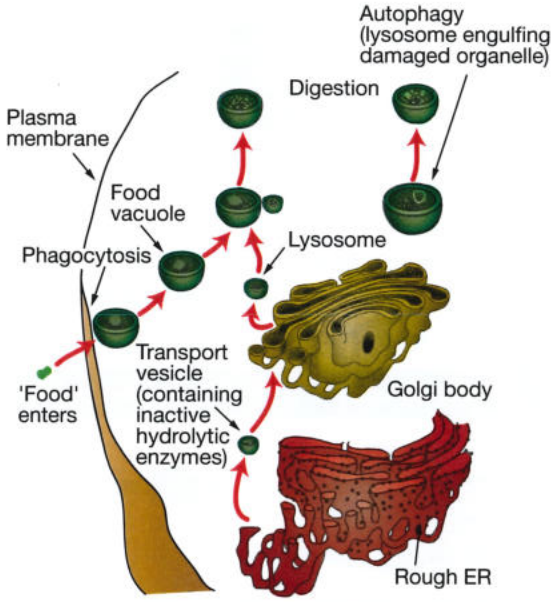


Figure 12.1 Lysosomes digest material taken into the cell and recycle unwanted material inside the cell.

Endoplasmic reticulum

The endoplasmic reticulum is a large network of membranes throughout the cell. It is involved in transport of materials in the cell. Rough endoplasmic reticulum has ribosomes on its surface. The ribosomes are the site of protein synthesis. Smooth endoplasmic reticulum has no ribosomes and is involved in the production of enzymes that assist with making various chemicals, e.g. steroids, lipids and assists in inactivating some drugs, e.g. alcohol.

Ribosomes

Ribosomes are particles made of ribosomal RNA and protein and they carry out protein synthesis. There are free ribosomes in the cytosol and bound ribosomes attached to the endoplasmic reticulum. The free ribosomes mainly produce proteins that function within the cytosol. As all ribosomes are structurally identical they can alternate between these two roles.

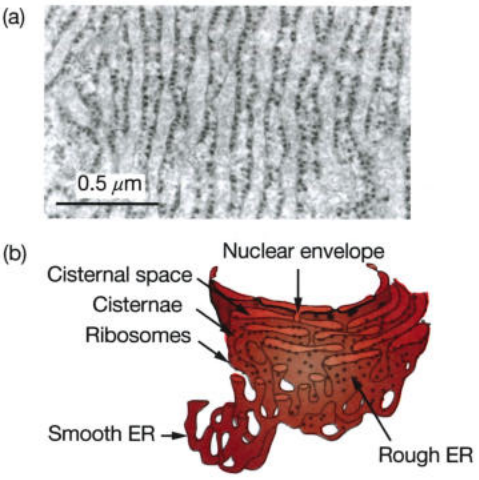


Figure 12.2 Endoplasmic reticulum. (a) Electron micrograph. (b) A system of tubes and flattened sacs within the cell. Ribosomes cover the rough endoplasmic reticulum and are the site where protein synthesis takes place.

QUESTIONS

1. Outline the function of lysosomes.
2. Identify the dark spots attached to endoplasmic reticulum and outline their function.
3. The following electron micrograph shows a cell.

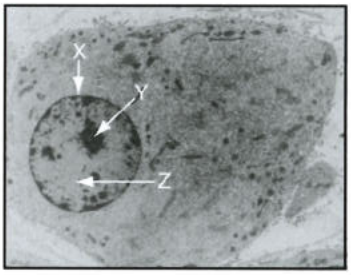


Figure 12.3 Electron micrograph.

Identify the parts labelled X, Y and Z.

4. Explain why lysosomes are needed in a cell.
5. State two advantages and two disadvantages of the light microscope compared with the electron microscope.
6. Explain how the structure of endoplasmic reticulum is related to its function.
7. Which of the following organelles are found in both plant and animal cells?
   (A) Nucleus, nucleolus, cell wall, very large vacuoles.
   (B) Endoplasmic reticulum, chloroplasts, nucleoli, ribosomes.
   (C) Nuclear membrane, mitochondria, endoplasmic reticulum, lysosomes.
   (D) Ribosomes, cell membrane, cell wall, mitochondria.
8. Which of these organelles can only be seen under an electron microscope?
   (A) Mitochondria. (B) Nucleus.
   (C) Golgi body. (D) Ribosome.

13 Mitochondria

Mitochondria (sing. mitochondrion) are organelles with a double membrane found in plant and animal cells (in nearly all eukaryotic cells). Figure 13.1 shows a mitochondrion and Table 13.1 summarises the structures found in a mitochondrion and their function.

Table 13.1 Parts of a mitochondrion.

Feature	Structure	Function
Outer membrane	Smooth phospholipid bilayer with scattered integral proteins.	Protects the organelle. Controls the substances that enter/leave the mitochondrion. Freely permeable to small molecules.
Inner membrane	Convoluted phospholipid layer, infolded to form cristae. Contains proteins with different functions. Folding greatly increases surface area of inner membrane.	Large surface area assists steps of respiration that occur on the inner membrane. The proteins in the inner membrane are involved in reactions and assist in the transport of specific molecules in/out matrix.
Intermembrane space	Region between the outer membrane and inner membrane.	Concentration of small molecules is the same as the matrix while large molecules that need active transport, e.g. cytochrome c, can have a different concentration to matrix.
Cristae	Inner membrane invaginates and forms compartments. Has large surface area. Cells with high demand for ATP, e.g. muscle cells have higher numbers of cristae.	Surface area of the cristae is needed for cellular reactions. Increased surface area means greater ability to carry out reactions.
Matrix	Region enclosed by inner membrane. Has many enzymes involved in respiration.	Is the site of ATP synthesis and many other reactions.
Mitochondrial DNA (mtDNA)	Contains DNA, usually circular and attached to the inner membrane.	Can manufacture own RNA and proteins, e.g. for the inner membrane.
Ribosomes	Consist of rRNA and protein molecules.	Site of protein synthesis.

Mitochondria are oval shaped and often appear as circular if the section has cut across the mitochondrion in cross-section. They are approximately 1 to 10 μm long.

Figure 13.1 Mitochondrion.

The outer membrane is a regular membrane that protects the organelle and the inner membrane is highly folded to form cristae. The cristae increase the surface area of the inner membrane and extend into the matrix. The enzyme that makes ATP is built into the inner membrane. The space between the two membranes is called the intermembrane space. The intermembrane space has a high concentration of hydrogen ions (H^+) which is highly important in maintaining chemical reactions. The matrix is an aqueous solution containing enzymes and many different molecules. Some of the steps in cellular respiration are catalysed by enzymes in the matrix. Mitochondria have their own DNA which is usually circular and is found in the matrix. Mitochondria also have ribosomes for protein synthesis. Figure 13.2 shows a 3-D image of a mitochondrion.

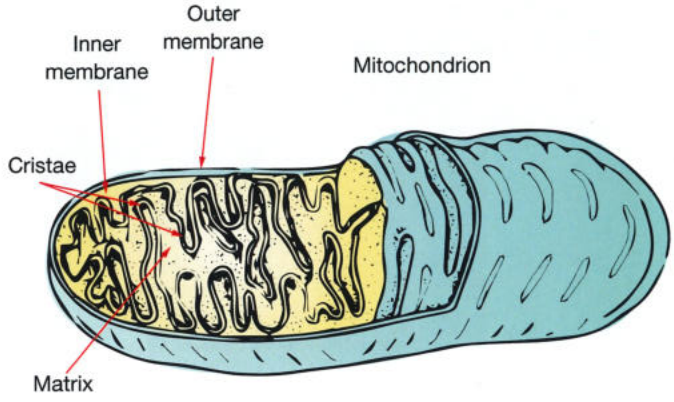


Figure 13.2 Stylised 3-D image of a mitochondrion.

QUESTIONS

1. Draw a generalised labelled diagram to show the structure of a mitochondrion.
2. Describe the structure of a mitochondrion.
3. Explain how the structure of a mitochondrion is suited to its function.
4. Figure 13.3 shows dividing mitochondria.

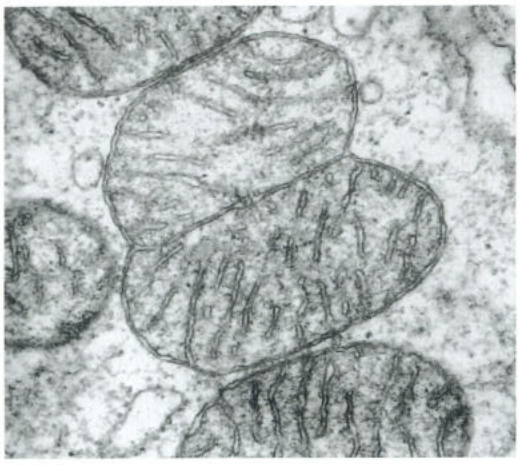


Figure 13.3 Mitochondrion dividing.

Explain why it is important for a cell to produce more mitochondria.

14 Golgi Bodies

A Golgi body is a stack of flattened membrane-bound sacs called cisternae. The cisternae form an extensive network in the cell. The membrane of each sac surrounds a separate internal space that is not continuous with the cytosol. The membranes on the opposite sides of a Golgi body are quite different in thickness and composition with the side receiving the materials called the cis face and the side releasing the materials the trans face. The vesicles around the Golgi body are involved in transferring materials between parts of the Golgi body and other structures.

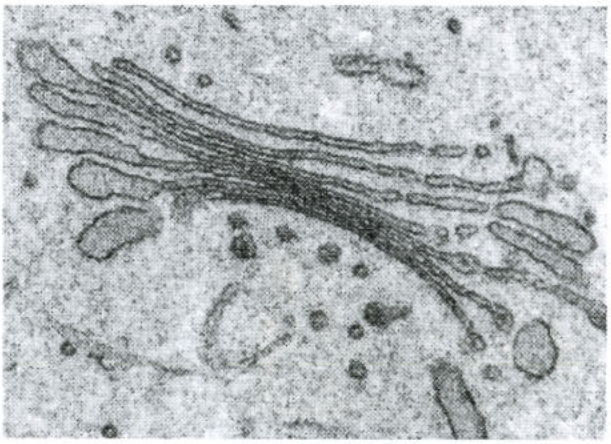
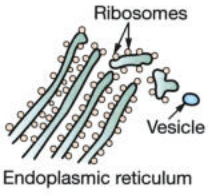
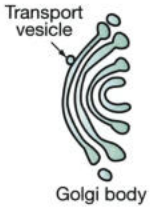
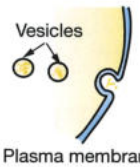


Figure 14.1 Golgi body.

Table 14.1 How Golgi bodies are involved in secretion from a cell.

Organelle	Diagram	What happens
Rough endoplasmic reticulum		**Step 1** – Ribosomes on rough endoplasmic reticulum produce polypeptide chains. The new protein enters the lumen of the rER and is sectioned off to form a vesicle. The transport vesicle moves to the Golgi body.
Golgi body		**Step 2** – The transport vesicle from rER joins with the Golgi body and its contents are added to the lumen. The Golgi body modifies the substance. New vesicles are formed and bud off. Domes of these transport vesicles contain secretions that leave the cell.
Plasma membrane		**Step 3** – Transport vesicles from the Golgi body travel to the plasma membrane. The membrane fuses with the vesicle and the substance is secreted from the cell.

The cis face is usually found near the rough endoplasmic reticulum (rER) and is the convex side of the stack faces. Vesicles containing the proteins formed by the ribosomes on the endoplasmic reticulum fuse with the cis face fuse with the cisternae and release their contents into the lumen inside the sac. As the substance moves from the cis face to the trans face, it is modified by enzymes so that the Golgi bodies are capable of producing a large variety of carbohydrates and proteins. Vesicles are pinched off from the trans face and transport the Golgi products to specific organelles within the cell or to the plasma membrane where they are secreted from the cell.

Most Golgi bodies consist of 5 or 6 cisternae. Within the Golgi body the proteins from the ribosomes may be cut into smaller proteins or carbohydrates may be attached to make glycoproteins.

Cytoplasmic vesicles are classified as either secretory vesicles which contain proteins to be exported or storage vesicles which do not leave the cytoplasm. Lysosomes are the major storage vesicles of the cell. Endocrine cells secrete hormones. Endocrine cells contain large numbers of secretory vesicles formed by the Golgi bodies. When the cell receives a signal for secretion the secretory vesicle releases its contents by exocytosis into the extracellular space.

QUESTIONS

1. Describe the structure of Golgi bodies.
2. Outline the function of Golgi bodies.
3. Discuss the importance of the cis and trans face of Golgi bodies to the function of the organelle.
4. Draw a flow diagram to show how Golgi bodies are involved in secretion.
5. Describe cytoplasmic vesicles.
6. Identify a type of cell that is likely to have a large number of Golgi bodies and explain why they need such a large number of these organelles.
7. Figure 14.2 shows a Golgi body.

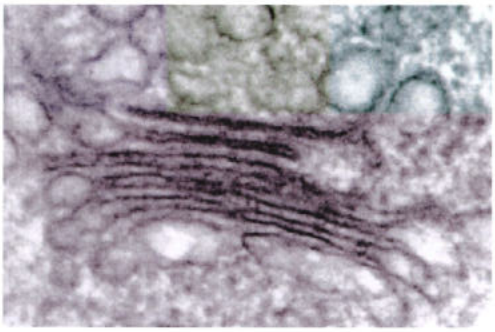


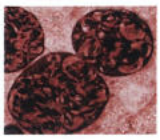
Figure 14.2 Golgi body.

Draw a line diagram of this organelle and discuss how you can distinguish the cis and trans faces.

15 Cell Organelles Summary

All cells have a **cell membrane**. The **cytoplasm** is the entire contents of the cell inside the cell excluding the nucleus. The semifluid part of the cytoplasm is the **cytosol**. Cell organelles are specialised structures found in the cell.

Table 15.1 Summary of cell organelles.

Organelle	Its structure	Its function
Mitochondria	Mitochondria are oval shaped with a double membrane. The inner membrane is highly folded (folds called cristae) to increase surface area for site of chemical reactions.	It is the site of aerobic respiration and where most ATP (adenosine triphosphate) is generated.
Golgi body	Is a single membrane structure with sacs stacked like pancakes. Vesicles contain macromolecules, e.g. enzymes, hormones and pinch off from the stack.	Modification of lipids and proteins and storage and packaging of materials for export from the cell.
Lysosome 	A membrane bound organelle containing enzymes and are produced in Golgi bodies.	Pick up and break down unwanted materials or foreign invaders, e.g. old organelles, bacteria, food.
Endoplasmic reticulum	Network of internal membranes, tubules and vesicles which increase in number as the cell becomes more active in protein synthesis.	It is involved in the transport of materials in the cell, the site of synthesis of many proteins and lipids and connects the cell membrane with the nuclear membrane.
Ribosome Black dot on ER	Very small, spherical granules made of RNA and protein often attached to endoplasmic reticulum.	It is the site of polypeptide and protein synthesis.
Vacuole	A fluid-filled membrane bound sac. In humans they are usually small.	Variety of functions, e.g. storage, digestion or water removal.
Microfilaments	Small strands of protein.	Provide structural support and maintain characteristic shape of cell.
Microtubules	Contractile protein structures in the cytoplasm.	Involved in movement of cell and organelles within the cell. Movement of cilia on the free border of some cells.

QUESTIONS

1. Define cytosol.
2. List the parts of the cell that are involved in the membrane system of the cell.
3. Identify the organelles involved in collecting or releasing energy.
4. Explain why cells that have a high rate of protein synthesis have a very large number of ribosomes present in the cytosol.
5. In plant and animal cells, the endoplasmic reticulum (ER) makes up more than half the total membrane of the cell. The tubules (cisternae) of ER have an internal space called the ER lumen (cisternal space). The ER membrane connects with the nuclear membrane so that the ER lumen is continuous with the space between the two membranes of the nuclear membrane.
   Suggest why there is a need to connect the nuclear membrane to the membrane system in the cytosol.
6. Name the form of energy produced by cellular respiration.
7. What is the function of vacuoles?
8. Which of the following organelles is involved in the breakdown of unwanted materials?
   (A) Golgi bodies. (B) Mitochondria.
   (C) Ribosomes. (D) Lysosomes.
9. The diagram shows a section of a cell.

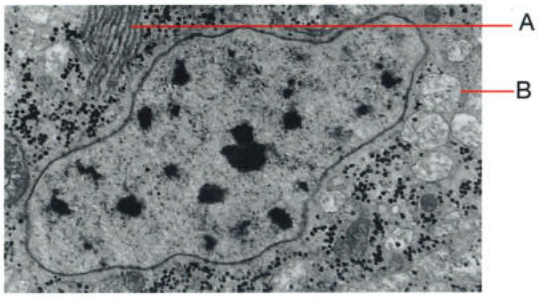


Figure 15.1 Section of a cell.

What are structures A and B?

	Structure A	Structure B
(A)	Endoplasmic reticulum	Nucleus
(B)	Golgi body	Lysosome
(C)	Endoplasmic reticulum	Lysosome
(D)	Golgi body	Mitochondria

10. Which of the following organelles is involved in the packaging and export of materials from a cell?
    (A) Golgi bodies.
    (B) Mitochondria.
    (C) Ribosomes.
    (D) Lysosomes.

16 Chemicals in Cells

Living organisms use organic and inorganic compounds. **Organic compounds** contain carbon and were originally made by living things, e.g. carbohydrates, lipids, proteins and nucleic acids. **Inorganic compounds** are compounds that do not contain carbon as the main element, except for carbonates, e.g. water, mineral salts, and trace elements such as calcium and manganese.

Carbohydrates

Carbohydrates contain the elements carbon, hydrogen and oxygen. They are used in respiration, storage of energy and some of the larger polysaccharides are used for structural support. Glucose is one of the most simple forms and is a monosaccharide (single sugar). Glucose is formed in photosynthesis and is the most common energy source in plants and animals in respiration. Disaccharides (double sugars) include sucrose, lactose and maltose. Most of the sugars transported in phloem in plants is in the form of sucrose.

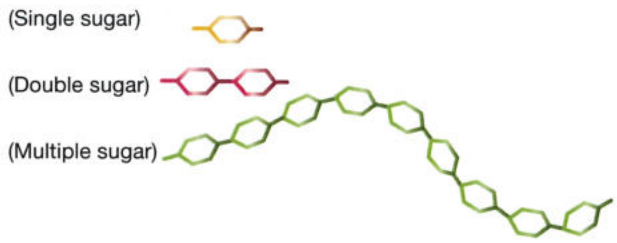


Figure 16.1 Carbohydrates.

Proteins

Proteins contain the elements carbon, hydrogen, oxygen and nitrogen and often sulfur atoms. They are used in the structure, storage, movement and transport of cells. The basic unit of proteins is the **amino acid** and these bond together in long chains joined by peptide bonds to form polypeptides. Different proteins have amino acids arranged in different combinations, numbers and sequences. They have been named amino acids as all contain an amino group ($-NH_2$) and an acid group ($-COOH$). There are over 20 amino acids and they include alanine, arginine, asparagine, leucine, serine, proline, valine and glutamic acid. One group of proteins is the enzymes, which control the rate of chemical reactions in the cell. DNA determines the actual sequence of amino acids in any protein.

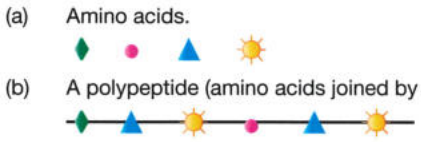


Figure 16.2 Proteins.

Lipids

Lipids contain the elements carbon, hydrogen and oxygen, and store energy. It is a more energy-efficient form of storage compared with carbohydrates but the energy is more difficult to release for cell use. Lipids break down into fatty acids and glycerol. At room temperature, oils are liquid lipids, while fats are solid lipids. Most animal fats contain saturated fatty acids, while lipids in plants and fish usually contain unsaturated fatty acids. Cholesterol and other steroids are lipids.

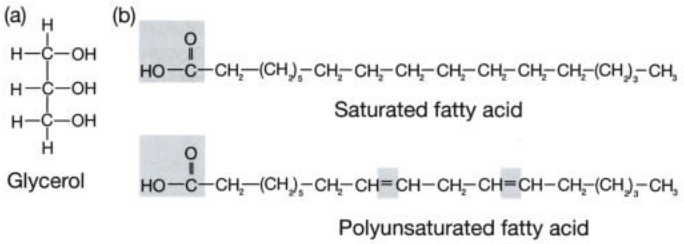


Figure 16.3 Structure of lipids.

Nucleic acids

Nucleic acids include DNA (deoxyribose nucleic acid) and RNA (ribose nucleic acid) and are the largest and most complex of the organic molecules. The subunit is the **nucleotide**, which each consist of a sugar, phosphate and a nitrogenous base. The sequence of nucleotides provides instructions to direct the production of polypeptides, passing down genetic information and controlling the functioning of the cell and organism.

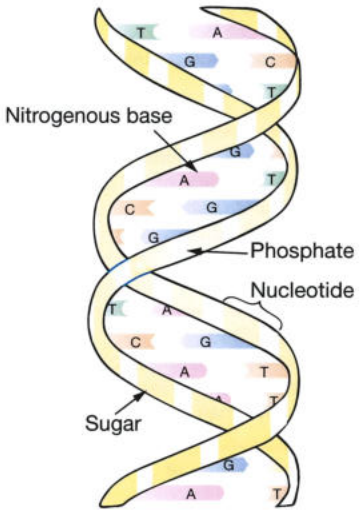


Figure 16.4 DNA.

QUESTIONS

1. Distinguish between organic and inorganic compounds.
2. Identify the four main groups of organic compounds.
3. Name four inorganic compounds.

4. Name the basic unit found in each of the following groups.
   (a) Carbohydrates. (b) Proteins.
   (c) Lipids. (d) Nucleic acids.
5. Identify the components that give amino acids their name.
6. Which of the following is the largest and most complex organic molecule?
   (A) Sucrose. (B) Glycerol.
   (C) DNA. (D) Alanine.
7. In 1944 F Sanger, a British chemist, began to work on the structure of the protein insulin. In 1954 he published his results, showing that insulin contained 51 basic units in two chains linked together. He later received a Nobel prize in chemistry for his work. What type of 'basic unit' is present in insulin?
   (A) Glucose. (B) Fatty acids.
   (C) Nucleotides. (D) Amino acids.
8. Which of the following correctly matches the group of compounds and their function?

	Lipids	Proteins	Nucleic acids
(A)	Energy storage	Genetic information	Respiration
(B)	Genetic information	Cell structure	Respiration
(C)	Cell structure	Respiration	Genetic information
(D)	Energy storage	Cell structure	Genetic information

9. Which organic compounds do *not* contain nitrogen?
   (A) Carbohydrates only.
   (B) Proteins.
   (C) Nucleic acids.
   (D) Carbohydrates and lipids.
10. Which of the following only contains lipids?
   (A) Butter, grease, cholesterol, cod liver oil.
   (B) Cellulose, lard, wax, peanut oil.
   (C) Glucose, sucrose, fructose, maltose.
   (D) Alanine, guanine, proline, valine.
11. The diagram shows the formation of a bond between two amino acids.

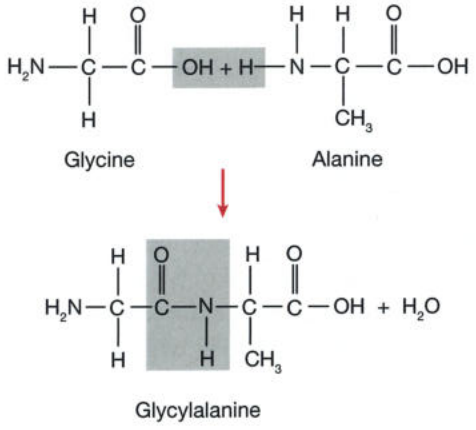


Figure 16.5 Bond forming between two amino acids.

What is the name of this bond?
   (A) Ionic bond.
   (B) Peptide bond.
   (C) Hydrogen bond.
   (D) Metallic bond.
12. 'Protein' means 'of first importance'. Why would scientists give this name to proteins?
   (A) They are needed to provide energy for cell metabolism.
   (B) They store energy for the cell.
   (C) They provide structure.
   (D) They contain genetic information.
13. The diagram shows how glucose and fructose join to form sucrose.

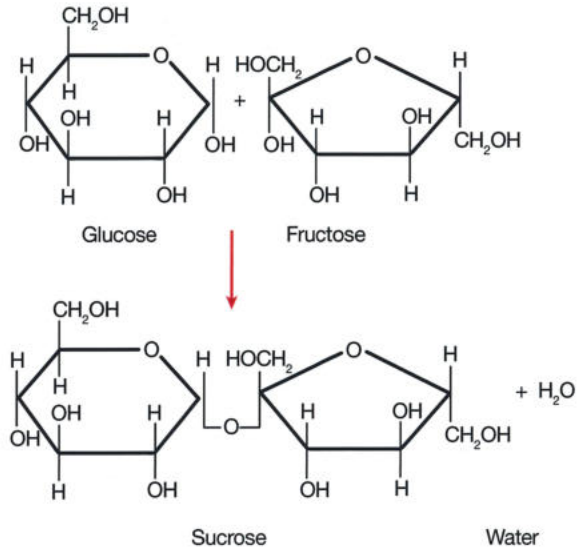


Figure 16.6 Formation of sucrose.

How would glucose, fructose and sucrose be classified?

	Glucose	Fructose	Sucrose
(A)	Monosaccharide	Polysaccharide	Disaccharide
(B)	Monosaccharide	Monosaccharide	Disaccharide
(C)	Disaccharide	Disaccharide	Polysaccharide
(D)	Disaccharide	Disaccharide	Polysaccharide

14. Humans have a layer of lipids under the skin. What is the function of this layer?
   (A) Insulation.
   (B) Structural support.
   (C) Store genetic information.
   (D) Internal transportation of materials.
15. When cells need to store glucose they form polysaccharides. What is the most common polysaccharide in humans?
   (A) Cellulose.
   (B) Starch.
   (C) Glycogen.
   (D) Valine.

17 Experiment – Substances in Tissues

Some Human Biology students decided they would use tests to identify the following range of substances in tissues – glucose, starch, lipids, proteins, chloride ions and lignin. They collected information about distinguishing tests for specific substances and the risk assessment for those chemicals which are considered dangerous.

Table 17.1 Distinguishing tests.

Test	What it shows
Iodine solution	Is used to test for starch. The yellow-brown iodine solution will turn blue-black with starch.
Tes-tape	Is used to test for glucose. Tes-tape paper is yellow and when it contacts glucose it will change to differing shades of green depending on the concentration of sugar.
Brown paper	Lipids make brown paper turn from opaque to translucent.
Biuret test	Is used to test for the presence of peptide linkage. Proteins cause a change from pale blue to pink/violet reaction. Biuret solution needs to be freshly made using equal amounts of copper sulfate solution and sodium hydroxide solution.
Silver nitrate solution	Is used to test for the presence of chloride ions. A precipitate of white silver chloride forms.
Toluidine blue	Stains nucleic acids blue and used to stain chromosomes.

Table 17.2 Risk assessment.

Medium risk		
Name	**Nature of hazard**	**Risk management**
Biuret solution	Toxic if ingested; corrosive due to sodium hydroxide.	Wear safety glasses and gloves. Wear safety glasses and gloves.
Toluidine blue	Toxic if ingested.	Do not ingest.
Silver nitrate	Toxic if ingested.	Do not ingest.
Iodine solution	Toxic if ingested.	Do not ingest.

QUESTIONS

1. Outline the standard safety precautions which should always be followed when carrying out experiments in a laboratory.
2. Describe how you would perform a first-hand investigation to identify the presence of starch in a range of foodstuffs.
3. A student wished to investigate the presence of lipids in a range of biscuits. He collected a range of biscuits, e.g. dry biscuits, sweet biscuits, wafers, crackers; ground up each with a mortar and pestle and then rubbed a sample of each on brown paper. He wore rubber gloves while holding the brown paper.
   (a) Suggest why he wore rubber gloves while holding the brown paper.
   (b) How could the student check the reliability of his results?
4. Biuret solution is used to test for proteins.
   (a) Name the substances in biuret solution.
   (b) Outline the safety precautions needed when using Biuret solution and explain why they are needed.
   (c) Name four foodstuffs you would expect to show a positive result with Biuret solution.
5. Describe how you would perform a first-hand investigation to identify the presence of nucleic acids in skin cells.
6. Proteins are highly important for life and make up more than 50% of the dry weight of most cells. Identify the main functions of proteins.
7. Many plants contain starch. Starch is a major part of the diet of many people.
   (a) Explain why people need starch.
   (b) Describe where starch is stored in plants.
   (c) Identify the distinguishing test for starch.
8. If you wrapped some food in brown paper and noticed after a while that the paper had turned translucent, what could you deduce about the food?
   (A) The food included fish and chips.
   (B) The level of wrapping had allowed some microbes into the package.
   (C) The combination of starch and protein had caused the reaction in the paper.
   (D) The food contained lipids.
9. Which of the following would all show a positive test with iodine solution?
   (A) Onion, bread, mince meat, peanut butter.
   (B) Potato, bread, cake, flour.
   (C) Bread, egg yolk, apple, egg white.
   (D) Liver, fish, chicken, egg white.

18 Development of the Model of the Cell Membrane

In the 19th century scientists realised that there had to be some form of semipermeable barrier around cells that separated the internal contents of the cell from its environment. Several observations and experiments had provided some information about this barrier. The cell membrane is not usually visible under normal light microscopes which means that detail of its structure was not easily seen before the invention of the electron microscope in the 20th century.

It was obvious that animal cells needed to have some kind of boundary and experiments with plant cells also showed that there was a barrier around the cytoplasm beside the cell wall. The plant cell wall is easily seen under the light microscope and 19th century scientists found that the cytoplasm of a plant cell placed in strong solutions, e.g. salt solution, pulled away from the cell wall showing the cytoplasm was surrounded by some kind of membrane.

Moritz Traube

Moritz Traube (1826-1894) realised that cell membranes were molecular sieves and produced artificial semipermeable membranes to explain the physical-chemical theory of plant cell growth. Traube created artificial cells by putting droplets of glue in tannic acid. He suggested that the barrier around cells was formed by an interfacial reaction of the cell protoplasm with the extracellular fluid.

Quincke

In 1888 Quincke suggested the existence of a thin lipid layer at the boundary of the protoplasm. In his studies he showed how a cell forms a spherical shape in water and if a cell is broken into two it will form two smaller spherical shapes in water. This behaviour is similar to the behaviour of oil in water and led to his suggestion that the cell membrane was a fluid layer of fat less than 100 nm thick.

Ernest Overton

Overton found that the ability of a substance to pass through the membrane was related to its chemical nature. Non-polar substances could quickly pass through the membrane and the speed with which a substance penetrated depended on the fat solubility of the substance.

Hugo Fricke

In 1923 Fricke conducted experiments to measure the capacitance of the red blood cell membrane. He assumed the membrane to be an oil film and calculated that the cell membrane was 3.3 nm thick. This was the first indication that the membrane had molecular dimensions. Although his measurements were accurate his assumption that the cell membrane was a single molecular layer meant he misinterpreted the data.

Evert Gorter and F Grendel

In 1925 Gorter and Grendel removed the lipids from a known number of red blood cells and calculated the total cell area. They found that the total lipids compressed into a monolayer that was twice the total cell area. From this they suggested the bilayer concept of the cell membrane.

JF Danielli

In 1940 Danielli theorised that the cell membrane consisted of inner and outer layers of protein with two layers of lipid between them. He calculated the total width of the membrane to be 8.0 nm.

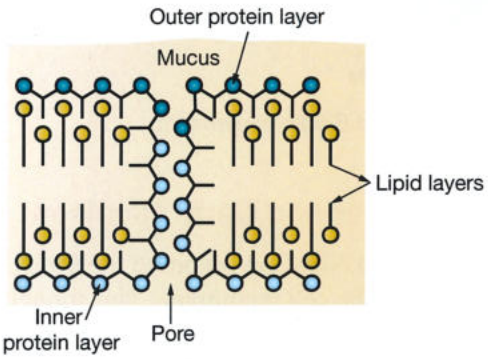


Figure 18.1 Danielli model of the cell membrane.

J David Robertson

Robertson took electron micrographs of cell membranes that showed the cell membrane consisted of two electron-dense layers separated by a wider lighter area. The measured total thickness of the membrane was 7.5 nm. He suggested the **unit membrane** which is a lipoprotein sandwich with lipids arranged between two layers of protein. The two dark electron-dense areas were the headgroups and associated proteins and the two lipid monolayers were lying in opposite orientation.

Mueller and Rudin

Mueller and Rudin suggested the **BLM** which can stand for either the 'black lipid membrane' or the bimolecular lipid membrane'. They 'painted' a solution of lipid in organic solvent across an aperture to create an artificial bilayer. They then determined different properties of the membrane which showed correlation with natural cell membranes, e.g. lateral fluidity, high electrical resistance, and self-healing after puncture.

Frye and Edidin

In 1970 Frye and Edidin used fluorescent dye-labelling techniques to conclusively demonstrate the notion of fluidity of the cell membrane. In their experiments they fused human and mouse cells in culture to produce human-mouse cell hybrids and used antibodies labelled with fluorescent dyes to identify the proteins of mouse or human origin. They found that immediately after fusion the human and mouse proteins were to be found in separate halves of the hybrid cells. Then after a short time and incubation at 37°C the human and mouse proteins were intermixed over the cell surface. They thus showed that the proteins in the cell membrane could move from place to place on the cell membrane.

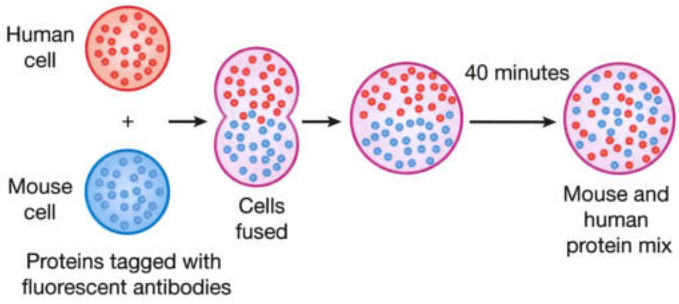


Figure 18.2 Frye and Edidin experiment.

Jonathan Singer and Garth Nicolson

In 1972 Singer and Nicolson proposed the **fluid mosaic model** of the cell membrane. This model kept the lipid bilayer and suggested that the proteins were macromolecules embedded in the bilayer going partway or completely across the membrane. They distinguished between **peripheral proteins** which dissociate from the membrane when the membrane is treated with polar reagents, e.g. extreme pH or high salt and **integral membrane proteins** which are inserted in the membrane and can only be dissociated with reagents that are detergents which displace the membrane lipids.

More recent research

More recent research has shown the presence of a carbohydrate layer on the outer surface of the cell membrane known as the **glycocalyx**. The glycocalyx consists of specific carbohydrates that form complexes with membrane proteins (**glycoproteins**) and membrane lipids (**glycolipids**). The glycocalyx is involved in the protection of the cell surface and different complexes act as markers for cell-cell interactions, e.g. interactions between leucocytes (white blood cells) with endothelial cells of blood vessels which can lead to the inflammatory response if the endothelial cells are injured. There is also ongoing research into the structure of channel proteins in the membrane. There is also research into the structure of other membranes relating structure to function.

A study of different membranes has shown that all membranes are not identical with membrane thickness varying from 5 to 10 nm depending on the location of the membrane. Different membranes have different proteins and carbohydrates or similar proteins and carbohydrates but in different proportions, e.g. the cell membrane has more cholesterol than other membranes and the mitochondrial membrane has more proteins than other membranes.

QUESTIONS

1. Explain why 19th century scientists knew that the cell membrane existed but did not clearly describe its structure.
2. Explain why 19th century scientists knew that plant cells had a barrier around the cell contents besides the cell wall.
3. Construct a table to summarise the work of Traube, Quincke, Overton, Fricke, Gorter and Grendel, Danielli, Robertson, Mueller and Rudin, Frye and Edidin and Singer and Nicolson.
4. In the Singer and Nicolson proposal, what is the difference between peripheral and integral membrane proteins?
5. What is the glycocalyx?
6. The diagram shows a model of the cell membrane that was proposed in 1935.

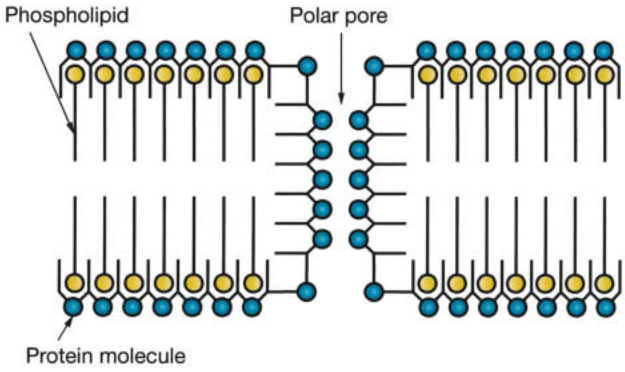


Figure 18.3 Model of cell membrane proposed in 1935.

Who proposed this model?
(A) Taube.
(B) Danielli.
(C) Mueller and Rudin.
(D) Singer and Nicolson.

7. What is the name of the current model of the cell membrane?
(A) Lipid monolayer model.
(B) Glycocalyx model.
(C) Unit membrane model.
(D) Fluid mosaic model.

19 The Current Plasma Membrane Model

The cell membrane forms a barrier around the outside of the cell, containing the cell contents. It is **semipermeable**, i.e. compounds such as water may easily enter and leave the cell but larger molecules such as polysaccharides cannot.

Fluid mosaic model

In 1972 Jonathan Singer and Garth Nicolson proposed the fluid mosaic model to represent the structure of the cell membrane. Phospholipids form a bilayer with the hydrophobic (water hating) end of the lipid facing the middle of the membrane and the hydrophilic (water loving) end facing towards the watery contents of the cell.

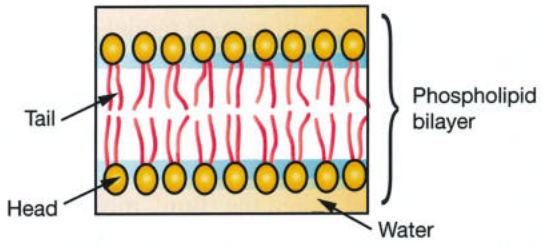


Figure 19.1 Lipid bilayer.

There is a range of protein molecules throughout the phospholipid bilayer. Different proteins serve different functions and are found on specific types of cell membranes. Carbohydrates are present on the surface of the cell membrane and are important in cell-cell recognition. This is important in the immune response and the rejection of foreign cells, e.g. transplanted tissues and organs. Some carbohydrates bond to the lipids forming **glycolipids**. Most carbohydrates bond with the proteins forming **glycoproteins**.

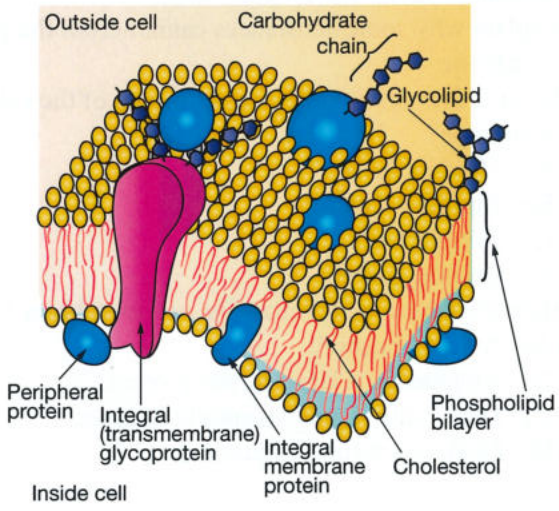


Figure 19.2 Fluid mosaic model.

Most of the phospholipids and some of the proteins can 'drift' and move their position and hence their location is 'fluid'. The model also explains why the membrane is semipermeable and how different molecules pass through the membrane.

Movement of phospholipids

Phospholipids move laterally within the membrane. The movement is very fast which means a particular phospholipid can move about 2 μm in one second.

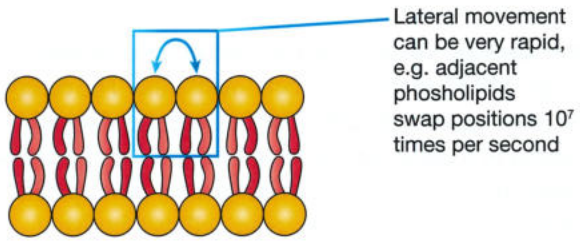


Figure 19.3 Lipid bilayer.

Unsaturated hydrocarbon tails of phospholipids often have 'kinks' that reduce tight packing of the molecules and increase membrane fluidity. Many organisms regulate the amount of unsaturated fatty acids in their membranes to compensate for temperature changes. Lipids become a solid at low temperatures (like cooking grease) and at low temperatures membranes with high amounts of unsaturated hydrocarbons have a greater capacity to stay fluid and not crystallise.

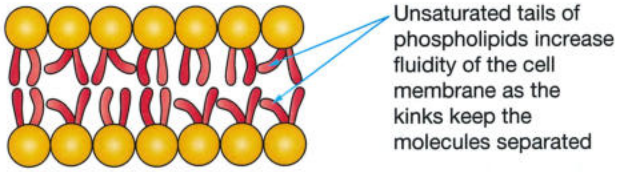


Figure 19.4 Unsaturated tails of phospholipids increase fluidity of the cell membrane.

Membrane proteins

Embedded in the phospholipid bilayer there are large protein molecules. **Integral membrane proteins** (IMPs) are permanently attached to the membrane. Some IMPs extend from one side of the membrane to the other side and are called transmembrane proteins. Some proteins extend only partway across the bilipid layer or the protein can be attached to the surface and part of the integral protein (peripheral protein). The proteins have different compositions and different functions.

- **Transport proteins** – assist the movement of ions, small molecules or macromolecules in or out of the cell.

- **Membrane receptor proteins** – communicate with the environment, e.g. they can receive a signal from a hormone, neurotransmitter, cytokine or growth factor which triggers the cell to carry out a specific internal response, e.g. signals the nucleus to begin gene expression to start the synthesis of a particular protein.
- **Membrane enzymes** – some enzymes that modify molecules needed near the cell surface are found on either side of the cell membrane.

Signal transduction

Signal transduction occurs when an extracellular molecule activates an IMP receptor in the cell membrane. The receptor sends the message into the cell. The message may signal the activation of a particular gene to start gene expression to make a particular protein or the message may alter specific chemical reactions in the cell and alter cell metabolism.

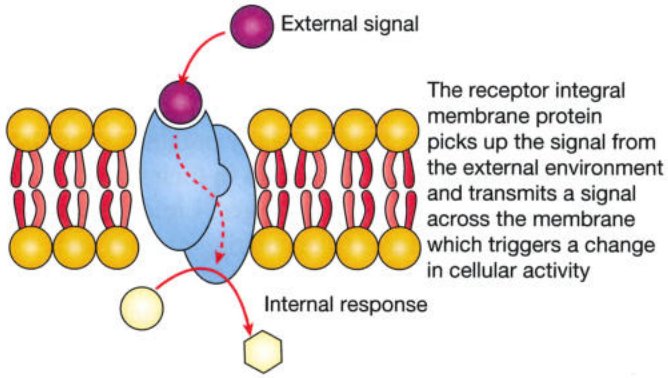


Figure 19.5 Fluid mosaic model.

Diffusion across the cell membrane

Molecules soluble in lipids, e.g. oxygen and carbon dioxide, dissolve in the phospholipid bilayer and diffuse across the membrane. Molecules not soluble in lipids, e.g. water, also diffuse across the cell membrane through small transitory openings made by the movement of the fluid lipids. **Aquaporins** are integral membrane proteins that form pores in the membrane that act as water channels allowing the fast movement of water in or out of the cell.

Facilitated diffusion across the cell membrane

Many compounds need to combine with a particular 'carrier' molecule to cross the membrane. As the carrier molecule moves across the membrane, it transports the molecule with it, e.g. large water-soluble molecules such as amino acids and simple sugars combine with transport integral proteins. If energy is used to move a chemical against a concentration gradient, it becomes active transport across the membrane.

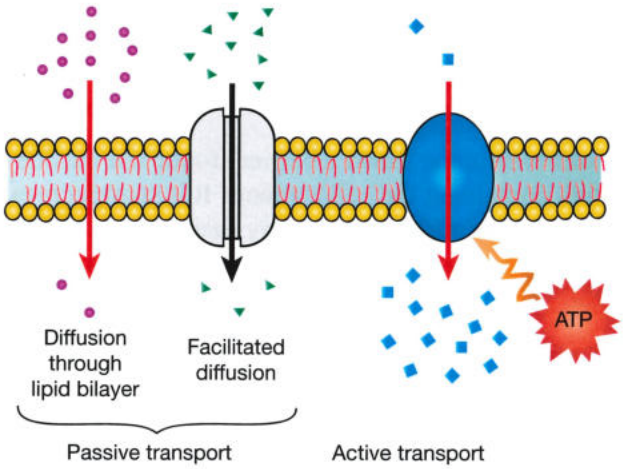


Figure 19.6 Transport across the cell membrane.

QUESTIONS

1. What is meant by semipermeable?
2. Who proposed the fluid mosaic model of the cell membrane?
3. Describe the structure of the cell membrane.
4. What is the function of carbohydrates on the surface of the cell membrane?
5. Why is the model called the 'fluid' model?
6. What is the importance of the 'kinks in the unsaturated hydrocarbon tails of phospholipids?
7. Outline what happens to lipids at low and high temperatures.
8. Outline the benefits of lipids having high amounts of unsaturated hydrocarbons.
9. What are integral membrane proteins?
10. Construct a table to summarise the action of three types of proteins associated with the cell membrane.
11. Describe how water crosses the cell membrane.
12. Describe how water-soluble amino acids cross the cell membrane.
13. When is active transport involved in the movement of substances across the cell membrane?
14. Explain why some substances cannot cross the cell membrane.
15. What is the name of the current model of the cell membrane?
    (A) Fluid mosaic.
    (B) Biprotein layer.
    (C) Trilipid layer.
    (D) Drifting protein.
16. How does most water cross the cell membrane?
    (A) Active transport.
    (B) Attached to a large carrier protein.
    (C) Dissolving in the phospholipid layer.
    (D) Diffusion through aquaporins.

20 Passive Transport

Passive transport is the movement of particles that does not require an input of energy. **Diffusion** is the movement of particles from an area of high concentration of particles to an area of low concentration of particles. This leads to the random spreading out of particles. **Osmosis** is a type of diffusion. In biology, osmosis refers to the movement of water, across a semipermeable membrane, from an area of high water concentration to an area of low water concentration.

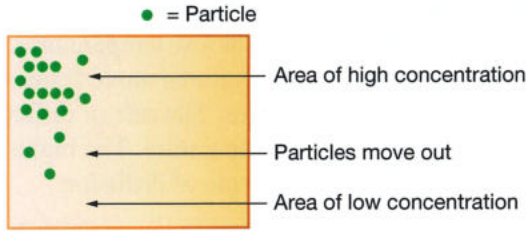


Figure 20.1 Diffusion.

The rate of diffusion depends on the concentration difference – the greater the difference, the faster the rate of diffusion. The difference in the number of particles compared to the number of particles in another area is called the concentration gradient. Substances will move down a **concentration gradient** by **passive transport**.

Substances can move *against* a concentration gradient, however, energy is needed to force the particles to move to a place of higher concentration of those particles and this movement is called **active transport**.

When a semipermeable membrane separates two solutions, only the small molecules can pass through the membrane, e.g. water moves across by osmosis.

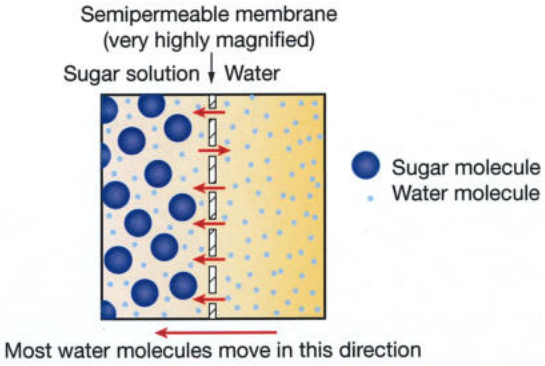


Figure 20.2 Semipermeable membrane.

If water and a sucrose solution are separated by a semipermeable membrane, the water molecules can move across the membrane, but the sucrose molecules cannot. The osmotic pressure wants to equal out the concentrations, so the water will move into the sugar solution, trying to equalise the concentrations, causing the volumes to change.

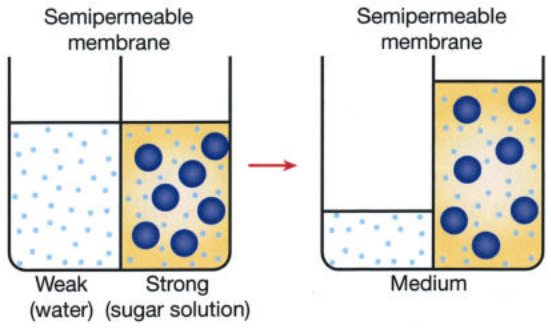


Figure 20.3 Osmosis.

When comparing two solutions, **hypertonic** is used to describe the more concentrated solution and **hypotonic** describes the less concentrated solution. If the two solutions have the same concentration they are called isotonic.

QUESTIONS

1. Define diffusion.
2. Define osmosis.
3. How does the concentration gradient influence the rate of diffusion?
4. How is active transport related to the concentration gradient?
5. What is the difference between hypertonic, hypotonic and isotonic solutions?
6. In what ways are the processes of diffusion and osmosis similar and different?
7. If a human red blood cell was placed in distilled water, explain what would happen to the cell.
8. If a strong sucrose solution is placed one side of a semipermeable membrane and a weak sucrose solution is placed the other side of the membrane, what would you expect to happen?
   (A) The sucrose will move from the weak sucrose solution to the strong sucrose solution.
   (B) The sucrose will move from the strong sucrose solution to the weak sucrose solution.
   (C) The water will move from the weak sucrose solution to the strong sucrose solution.
   (D) The water will move from the strong sucrose solution to the weak sucrose solution.
9. Figure 20.4 shows the movement of particles.

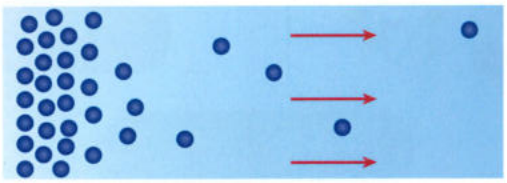


Figure 20.4 Movement of particles.

What is the best name for this movement?
(A) Osmosis. (B) Diffusion.
(C) Dissolving. (D) Active transport.

21 Diffusion and Osmosis

Diffusion is the movement of particles from an area of high concentration of particles to an area of low concentration of particles. Diffusion is due to the random movement of particles. The particles will spread out until they evenly fill the available space.

Net diffusion stops when equilibrium is reached. At equilibrium the molecules continue to move but there is no net change in concentration.

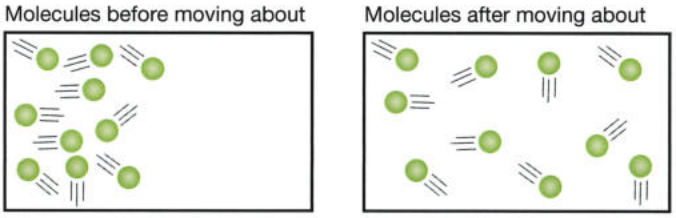


Figure 21.1 Diffusion.

Diffusion is passive transport as energy does not need to be provided for the movement to occur unlike active transport which requires an input of energy. In diffusion the particles move down a concentration gradient from an area of high concentration to an area of low concentration while in active transport the particles move against the concentration gradient from an area of low concentration to an area of high concentration.

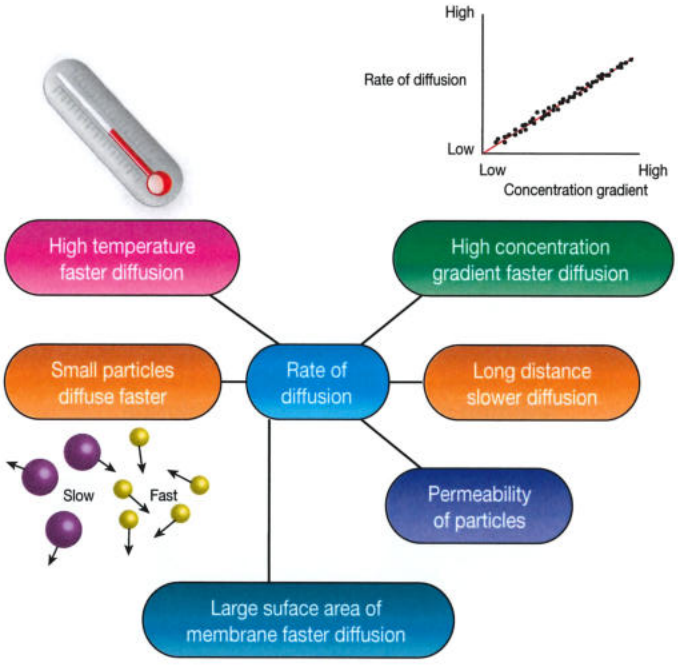


Figure 21.2 Factors affecting the rate of diffusion.

Factors affecting the rate of diffusion

The rate of diffusion depends on the following.

- **Concentration gradient** – the difference in the number of particles in one area to the number of particles in another area forms a concentration gradient between the two areas. The higher the concentration gradient the faster the rate of diffusion. In diffusion particles move **down** a concentration gradient.
- **Size of the molecule** – the size of a particle and its mass is inversely proportional to the rate of diffusion. The smaller the particle the faster the rate of diffusion.
- **Temperature** – particles at high temperatures have more energy than particles at low temperatures. The particles at high temperatures move faster than particles at low temperatures. The rate of diffusion is directly proportional to temperature. The higher the temperature, the faster the rate of diffusion.
- **Diffusion distance** – the distance the particle has to travel will affect the time it takes for particles to spread to another area. The further the distance, the slower the rate of diffusion.
- **Permeability of the particle** – permeability is the ability of the particle to pass through. **Solubility** refers to the amount of a substance that can be dissolved in a given amount of solvent. Substances that dissolve in lipids can more easily pass through a cell membrane than substances that dissolve in water. The rate of diffusion across a membrane is related to the permeability of the particle. Substances that can move across a membrane by simple diffusion include carbon dioxide, oxygen and ethanol. The membrane is not permeable to ions.
- **Surface area of the membrane over which the molecule moves** – if diffusion occurs across a membrane then the greater the surface area of the membrane, the faster the rate of diffusion.

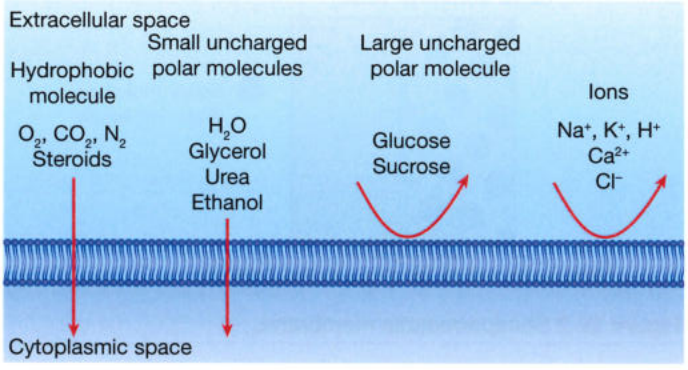


Figure 21.3 Permeability across a membrane.

Osmosis

Osmosis is a type of diffusion. It is the movement of water from an area of high concentration of water to an area of low concentration of water across a semipermeable membrane. (*Note*: In chemistry osmosis refers to the net movement of any solvent, not just water.)

The process of osmosis is highly important in osmoregulation. **Osmoregulation** is the active regulation of the osmotic pressure of body fluids so they do not become too dilute or concentrated in particular areas. **Osmotic pressure** is the pressure needed to stop the flow of water across a semipermeable membrane. The **water potential** is the physical property that predicts the direction in which water will flow and is determined by the solute concentration and the applied pressure.

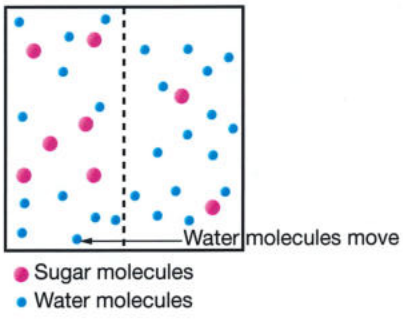


Figure 21.4 Osmosis.

QUESTIONS

1. Define diffusion.
2. Why does diffusion occur?
3. What is meant by a concentration gradient?
4. During diffusion, how do the particles move with respect to the concentration gradient?
5. How does the size of the particle affect the rate of diffusion?
6. How does temperature affect the rate of diffusion?
7. What is the relationship between the rate of diffusion and how far the particle has to travel?
8. Define permeability.
9. Define solubility.
10. If two molecules, one water soluble and one lipid soluble had equal size, which one would more easily pass through a cell membrane?
11. For two lipid soluble molecules, would a small molecule or a large molecule have the fastest rate of diffusion?
12. Identify two substances that enter the human body by diffusion.
13. Define osmoregulation.
14. What is meant by the water potential?
15. Define osmotic pressure.
16. In an aqueous solution of glucose, what is the solvent and what is the solute?
17. Glucose has the formula $C_6H_{12}O_6$ while sucrose has the formula $C_{12}H_{22}O_{11}$. Explain why when carrying out experiments using dialysis tubing to show that some substances can cross a membrane while other substances cannot cross the membrane you need to use an aqueous sucrose solution and not an aqueous glucose solution.
18. The diagram shows a human epithelial cell in a container filled with distilled water.

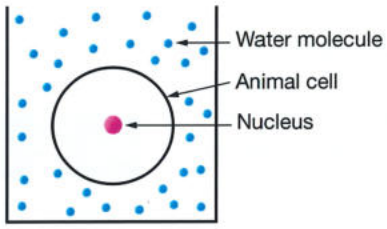


Figure 21.5 Human cell in a container of water.

What would you expect to happen to the epithelial cell? Explain your answer.

19. The kidney can change the permeability of the collecting duct. How does this assist excretion for the body?
20. At the arterial end the capillary blood pressure (hydrostatic pressure) is about 5 kPa while at the venous end of the capillaries the hydrostatic pressure is reduced to about 2 kPa. The osmotic pressure in the capillaries is 3 kPa. How does this affect water movement?
21. Which of the following does *not* cross the cell membrane by simple diffusion?
    (A) Carbon dioxide. (B) Glucose.
    (C) Ethanol. (D) Oxygen.
22. The diagram shows the results of an experiment using dialysis tubing.

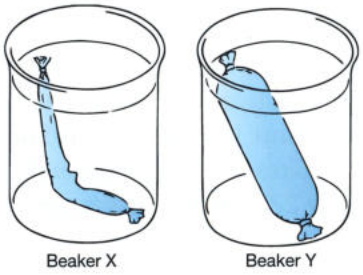


Figure 21.6 Experiment using dialysis tubing.

Pure distilled water and sucrose solution were used in setting up the equipment. Which of the following best accounts for the results of this experiment?

(A) Sucrose solution was in bag Y with distilled water outside and water moved out of the bag by osmosis.
(B) Sucrose solution was in bag X with distilled water outside and water moved out of the bag by osmosis.
(C) Water was in bag Y with sucrose solution outside and water moved into the bag by osmosis.
(D) Water was in bag X with sucrose solution outside and water moved out of the bag by osmosis.

22 Active Transport

Active transport involves the movement of substances across a membrane against a concentration gradient. This process requires energy usually in the form of ATP which is supplied by respiration.

Active transport and the cell membrane

Transporter proteins (pump proteins) in cell membranes carry out active transport. The transporter proteins are all carrier proteins. Figure 22.1 shows active transport across a membrane.

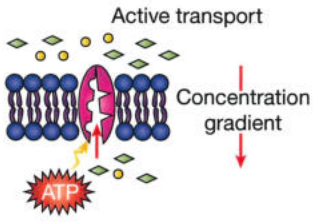


Figure 22.1 Active transport across a membrane.

There are many different transporter proteins, e.g. plant roots absorb potassium and other ions by active transport and neurons have a sodium/potassium pump which pumps potassium ions in and sodium ions out by active transport as part of a nerve impulse. Active transport is also involved in moving ions against an electrochemical gradient. An electrogenic pump is a transporter protein that generates voltage across a membrane, e.g. the sodium/potassium pump in neurons.

Endocytosis

Cytosis is the passage of large 'packages' of material across a membrane. The materials are large and cannot move across by diffusion or with the assistance of carrier or channel proteins. Endocytosis is a process where a cell takes in macromolecules by forming vesicles from the plasma membrane. If the substance is liquid, it is called **pinocytosis** while if the substance is solid it is called **phagocytosis**. Endocytosis requires an energy input and is a form of active transport. Figure 22.2 shows the two types of endocytosis.

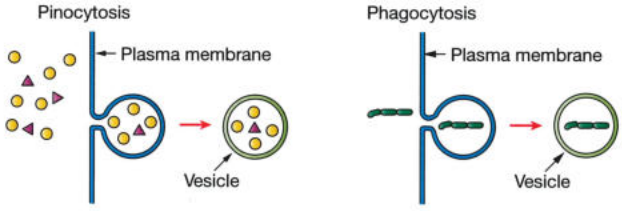


Figure 22.2 Endocytosis.

Exocytosis

Exocytosis is the process when vesicles inside the cell fuse with the cell membrane and the contents of the vesicle are secreted from the cell. Exocytosis requires an energy input and is a form of active transport. Figure 22.3 shows exocytosis.

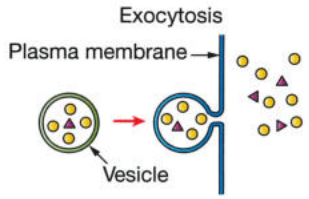


Figure 22.3 Exocytosis.

QUESTIONS

1. Define active transport.
2. What is an electrogenic pump?
3. Define cytosis.
4. Distinguish between endocytosis and exocytosis.
5. Distinguish between pinocytosis and phagocytosis.
6. Construct a table, with a column for a diagram, to compare the different types of transport across a membrane.
7. Figure 22.4 shows a biological process that occurs in cells.

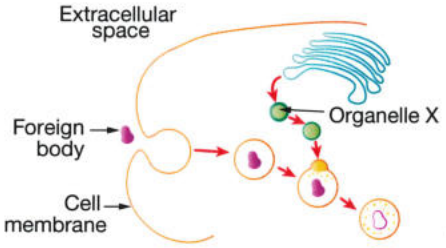


Figure 22.4 A biological process.

   (a) Identify the process that is happening to the foreign body.
   (b) Describe this process.
   (c) Identify organelle X and describe what it is doing.
8. Figure 22.5 shows a sodium-potassium pump.

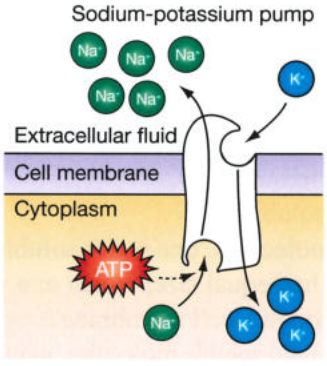


Figure 22.5 Sodium-potassium pump.

   Describe what is happening in this process.
9. Which cells produce ATP?
   (A) Only cells with a nucleus.
   (B) Only plant cells and animal cells.
   (C) Only animal cells.
   (D) All living cells.

23 Experiment – Membranes, Diffusion and Osmosis

Some Human Biology students wished to perform a first-hand investigation to show how a semipermeable membrane allows osmosis and diffusion of some substances to occur but prevents the movement of other substances.

They collected four gas jars, four 20 cm lengths of dialysis tubing, string, a measuring cylinder, sucrose solution, distilled water, starch solution, iodine solution, glucose solution and parafilm. They set up the following apparatus.

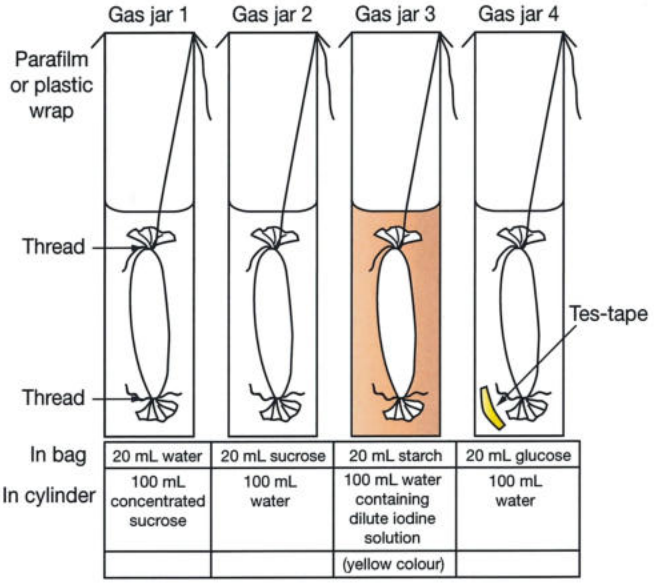


Figure 23.1 Set-up for experiment.

Once the equipment was set up and each jar was suitably labelled, the students noticed that the Tes-tape in jar 4 began to turn green before the end of the lesson. They recorded the 'fullness' of each bag and then left the equipment overnight in the laboratory. The next day they recorded any changes in the 'fullness' of each bag and any colour changes. Their results are in the following table.

Table 23.1 Student results.

	Initial appearance	Final appearance
Gas jar 1	Bag nearly full.	Bag more empty with shrivelled appearance.
Gas jar 2	Bag nearly full.	Bag extended and very full.
Gas jar 3	Bag nearly full, water outside a pale yellow colour.	Bag full and blue-black colour and water outside colourless.
Gas jar 4	Bag nearly full, during lesson Tes-tape turned green.	Bag nearly full, Tes-tape reverted to yellow.

QUESTIONS

1. What is meant by a semipermeable membrane?
2. In this experiment what was acting as a semipermeable membrane?
3. What happened to the iodine molecules?
4. Compare and explain the results of gas jar 1 and gas jar 2.
5. Explain the results in gas jar 4.
6. Identify the substances that were able to pass through the dialysis tubing.
7. Identify the substances that were *not* able to pass through the dialysis tubing.
8. When the overall size of molecules has been measured, it is found that starch molecules are very large as they are a polysaccharide made of many glucose units; sucrose is not as large as it is a disaccharide made of two monosaccharides joined together; glucose is a monosaccharide and smaller; and water and iodine molecules are very small. Propose a hypothesis to relate the size of molecules to the results of this experiment.
9. In this experiment, dialysis tubing was used to represent a cell membrane. Explain why, or why not, this was a suitable substitute to show the nature of a selectively permeable membrane.
10. Outline how this experiment showed the difference between diffusion and osmosis.
11. The diagram below shows what happened when two identical human epithelial cells were each placed in different solutions.

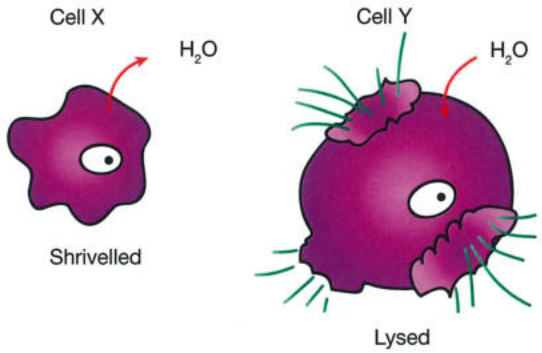


Figure 23.2 Human epitheliall cells placed in solutions.

What type of solutions would have caused these results?

	Solution for cell X	Solution for cell Y
(A)	Sucrose	Distilled water
(B)	Distilled water	Sucrose
(C)	Distilled water	Glucose
(D)	Sucrose	Glucose

24 Experiment – Surface Area to Volume Ratio and Diffusion

Some Human Biology students wished to carry out an experiment to show how the size of a cell influenced its ability to survive.

Student planning

The students worked out that they would have to measure the surface area of the cell as this determines the amount of material that can be moved in and out of the cell. They also worked out that they would have to work out the volume of the cell as this determines the amount of material the cell would need to survive.

The students decided that they would use cube blocks of jelly to represent cells. This would allow easy calculations as:

- Surface area of cube = (side × side) × 6
- Volume of cube = side × side × side

To represent the 'cells' they decided they would use a jelly that contained the alkali sodium hydroxide and had also been coloured with the indicator phenolphthalein. Phenolphthalein is pink in alkali and colourless in acid.

The students cut three cubes – a large cube with 3 cm sides, a medium size cube with 2 cm sides and a small cube with 1 cm sides.

The students then investigated the risk assessment and recorded the following table.

Table 24.1 Risk assessment.

Medium risk		
Name	**Nature of hazard**	**Risk management**
1 mol L^{-1} Hydrochloric acid	Corrosive to skin and clothing	Wear gloves, lab coat and safety glasses and use with awareness and care
Jelly containing sodium hydroxide	Corrosive to skin and clothing	Wear gloves, lab coat and safety glasses and use with awareness and care
Knives and scissors	Sharp objects can cause injury	Use appropriately and with care

Each cube was placed in hydrochloric acid for exactly three minutes and as the acid diffused into the cube the indicator changed colour showing the depth of diffusion. The cubes were removed from the acid and placed on paper towelling.

The cubes were then cut in half and the depth of colour change measured and recorded in a table with the results from other groups. The average depth of colour change was calculated for the class. The students also drew diagrams to show the appearance of each cube.

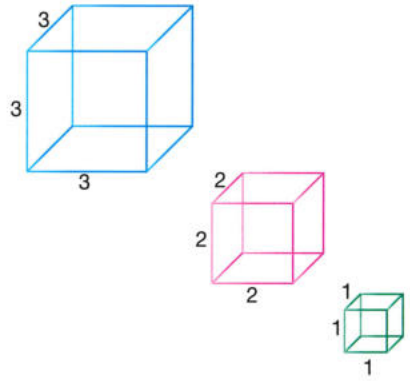


Figure 24.1 Three cubes.

Table 24.2 Surface area to volume ratio before starting.

Cube	Surface area	Volume	Ratio SA : vol	Reduced ratio
Small	1 × 1 × 6 = 6	1 × 1 × 1 = 1	6 : 1	6 : 1
Medium	2 × 2 × 6 = 24	2 × 2 × 2 = 8	24 : 8	3 : 1
Large	3 × 3 × 6 = 54	3 × 3 × 3 = 27	54 : 27	2 : 1

The students found that the acid had diffused in 2 mm in each side of each cube. This meant that the new width of the *pink* part of the cubes were 2.6 cm for the large cube, 1.6 cm for the medium cube and 0.6 cm for the small cube.

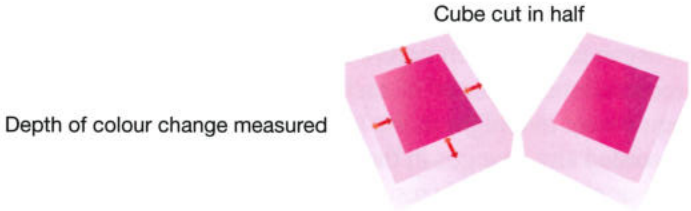


Figure 24.2 Measuring the depth of colour change.

To understand the significance of their results, the students decided to calculate the percentage of diffusion into the cell. To do this they followed the following calculation.

$$\text{Percentage of diffusion into cell} = \frac{\text{original volume} - \text{volume after acid}}{\text{original volume}} \times \frac{100}{1}\%$$

After carrying out these calculations, the students realised that the smallest cell had the greatest percentage of diffusion into the cell. They then decided that cell size would influence its ability to survive, with a smaller cell having a greater chance of survival than a larger cell.

QUESTIONS

1. Discuss why the students decided they needed to measure the surface area of the cell.
2. Discuss why the students decided they needed to measure the volume of the cell.
3. When the students began their experiment, their 'cells' were pink. Explain why the jelly had this colour.
4. Describe two safety precautions you need to follow while conducting this experiment.
5. Explain why the students chose different size cubes, rather than another shape.
6. What was the depth of diffusion of acid into each cube and what does this show about the rate of diffusion?
7. Draw up a table similar to Table 24.2 and calculate the reduced surface area to volume ratio of the new size of each pink cube.
8. Determine the percentage of diffusion for each of the cubes to show why the students decided that the smallest cube had the greatest percentage of diffusion.
9. Explain why the students decided that cell size would influence the ability of a cell to survive with a smaller cell having a greater chance of survival than a larger cell.
10. Surface area to volume ratio also influences the ability to gain and lose heat. Predict which cube would most easily lose and gain heat and explain your reasoning.
11. Many cells continue to grow in size, until they reach certain dimensions and then they divide, split into two and the cycle begins again. Suggest why cells divide when they reach a certain size.
12. The table shows the sizes of different types of human cells.

Table 24.3 Different shapes.

Cell type	Volume (μm^3)
Oocyte	4 000 000
Fat cell	600 000
Osteoblast	4000
Neutrophil	300
Lymphocyte	130
Red blood cell	100
Spermatozoon	30

Which of the following correctly links cell size and volume with function?

(A) Neutrophil is large to contain more Golgi bodies to process insulin for secretion.
(B) Red blood cell is large to contain more haemoglobin to transport more oxygen.
(C) Spermatozoon is large to assist mobility to swim to egg.
(D) Oocyte is large as it contains large amounts of cytoplasm to nourish early development.

13. Which of the following shapes would have the most suitable cross-section shape for a cell that needed highly efficient diffusion of substances in and out of its body?
(A) Circle
(B) Square
(C) Rectangle
(D) Triangle
14. Which of the following shapes would have the smallest surface area to volume ratio?

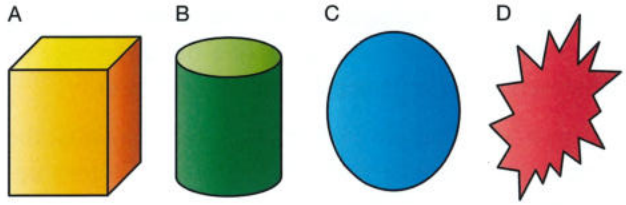


Figure 24.3 Different shapes.

The following information is to be used for the next TWO questions.
There are four shapes:
Shape A Cube with 1 cm sides
Shape B Cube with 2 cm sides
Shape C Box with 2 sides 1 cm × 1 cm and 4 sides 1 cm × 2 cm
Shape D Box with 2 sides 2 cm × 2 cm and 4 sides 1 cm × 2 cm

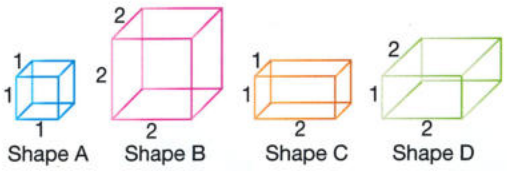


Figure 24.4 Four shapes.

15. Which of these shapes has the largest surface area?
(A) Shape A
(B) Shape B
(C) Shape C
(D) Shape D
16. Which shape has the largest surface area to volume ratio?
(A) Shape A
(B) Shape B
(C) Shape C
(D) Shape D
17. At which stage of the cell cycle would a cell have the largest size?
(A) G_1 stage
(B) S stage
(C) G_2 stage
(D) End of M stage

25 Experiment – Material Exchange and Concentration Gradient

The material exchange across a membrane depends on the concentration gradient across the membrane.

If the particles are spread evenly throughout a space there is a state of **equilibrium**. If there is a difference in the concentration of particles on either side of the membrane then the rate of diffusion is directly proportional to the concentration gradient – the bigger the difference in concentration, the faster the rate of diffusion.

A graph with the rate of diffusion plotted against the concentration gradient shows the linear relationship between these two factors.

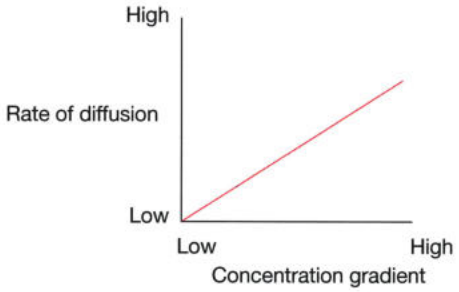


Figure 25.1 Rate of diffusion versus concentration gradient.

If you compare the rate of diffusion with the speed of a child going down a slippery dip, the steeper the slope the faster the speed of the child.

Figure 25.2 Rate of diffusion is like the slope of a slippery dip.

QUESTIONS

1. Define equilibrium.
2. What is the relationship between the rate of diffusion and the concentration gradient?
3. Some senior biology students wished to investigate how the concentration of a sugar solution affected the loss of water from cubes of potato.
   One of the students wrote the following notes in his practical report.

Method

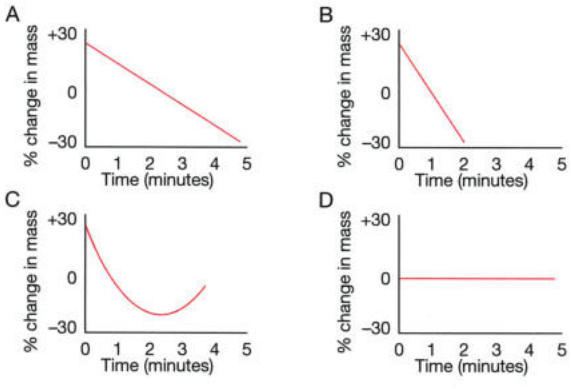
1. 24 cubes of potato each measuring 1 cm × 1 cm × 1cm were cut up and weighed.
2. 6 beakers were set up containing 0 mol L^{-1}, 0.2 mol L^{-1}, 0.4 mol L^{-1}, 0.6 mol L^{-1}, 0.8 mol L^{-1} and 1.0 mol L^{-1} sucrose sugar solutions respectively.
3. 4 potato cubes were added to each beaker.
4. The potato cubes were removed from the beakers at the same time and weighed.
5. For each cube the percent change in mass was calculated and then the average percent change in mass was calculated for one cube/beaker.

Results

Sugar solution in beaker	Change in mass cube/beaker
0	+20%
0.2 mol L^{-1}	+8%
0.4 mol L^{-1}	–4%
0.6 mol L^{-1}	–16%
0.8 mol L^{-1}	–22%
1.0 mol L^{-1}	–26%

(a) Write a suitable hypothesis for this experiment.
(b) Graph the student results for this experiment.
(c) What is the significance of the graph crossing the *x*-axis?
(d) Explain why the potato in a 0 mol L^{-1} sugar solution had a +20% change in mass of cube/beaker.
(e) Explain why the potato in a 1.0 mol L^{-1} sugar solution had a −26% change in mass of cube/beaker.
(f) Identify the relationship between change in mass of the potato cube and the molarity of the sugar solution.

4. Define diffusion.
5. Suggest how temperature affects the rate of diffusion.
6. Some students carried out an experiment to investigate what happens to a cylinder of carrot when it is placed in different concentrations of sucrose sugar solution. Which of the following graphs would correspond to a carrot cylinder placed in a solution with a high concentration of sugar?



26 Material Exchange and the Nature of the Material

The physical and chemical properties of a substance determine how the substance moves across membranes and how the substance travels around the bodies of multicellular organisms.

The **physical properties** of a substance do not change the chemical nature of the substance and can be observed and measured without changing the substance. Physical properties include boiling point, melting point, density, permeability, size of particles, molecular weight, molecular surface area and molecular volume.

The **chemical properties** of a substance do change the chemical nature of a substance and cannot be observed without changing the substance. Chemical properties include heat of combustion, reactivity with water, pH and flammability.

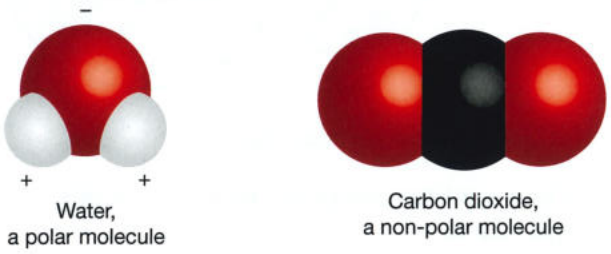


Figure 26.1 Polar and non-polar molecules.

Non-polar molecules

Non-polar molecules do not have oppositely charged ends. Non-polar molecules are permeable and can pass through the cell membrane as they can dissolve in the non-polar fatty chains of the membrane phospholipids.

Polar molecules

Polar molecules are molecules that have a separation of electric charge with sections of the molecule having positive and negative charges (electric dipole) or multipole expansions. Water is a polar molecule with a bent shape that has a negative charge in the middle and positive charges on the ends. Polar molecules and charged ions cannot freely diffuse across cell membranes. Polar molecules and ions move across cell membranes through either protein or ion channels.

Integral membrane proteins (IMPs) are proteins that are permanently attached to the membrane. IMPs form ion channels and ion transporters that control the flow of ions into and out of cells. This also helps regulate cell volume. There are different types of transport membrane proteins and each moves specific materials across the membrane.

Aquaporins are IMPs that form water channels to allow the rapid movement of water into/out of a cell across the cell membrane. Some transporters bind to the substrate molecule and undergo changes that ensure only specific molecules can cross using that channel. Glucose and amino acids move down a concentration gradient using transport channels in facilitated diffusion. The carrier proteins change their conformation as they take the glucose or amino acids across the cell membrane.

Size and mass of molecule

Gases such as oxygen (O_2) and carbon dioxide (CO_2) and small, uncharged polar molecules (e.g. urea and ethanol) can cross membranes by simple diffusion. The rate of diffusion for these molecules depends on the concentration gradient and how hydrophobic the molecule is (how much the substance in unable to dissolve in water). Macromolecules cross the cell membrane either by active transport or by facilitated diffusion through IMPs.

QUESTIONS

1. Distinguish between the chemical and physical properties of a substance.
2. Construct a table to list examples of physical and chemical properties.
3. (a) What is the difference between polar and non-polar molecules?
   (b) Give an example of a polar and a non-polar molecule.
4. What are integral membrane proteins?
5. The diagram shows the structure of glucose ($C_6H_{12}O_6$).

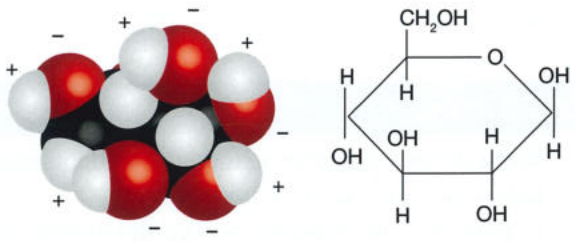


Figure 26.2 Structure of glucose.

   (a) Is glucose a polar or non-polar molecule?
   (b) Will glucose dissolve in water?
   (c) How does glucose cross the cell membrane?
6. Which of the following types of substances can easily cross the cell membrane by simple diffusion?
   (A) Large macromolecules.
   (B) Sugars such as glucose.
   (C) Proteins.
   (D) Small lipid-soluble molecules.

27 Experiment – Surface Area and Rate of Reaction

Some Human Biology students wished to perform a first-hand investigation into the relationship between surface area and the rate of reaction. They were given the following information.

- Soluble aspirin is meant to easily dissolve in water.
- Surface area can be increased by grinding a block of a substance into a powder.
- The rate at which soluble aspirin dissolves is related to the rate of diffusion of the aspirin particles into the water.

The students worked out how they would conduct the experiment and requested the following list of equipment.

- Soluble aspirin
- Mortar and pestle
- Stopwatch
- Safety goggles
- 100 mL measuring cylinder
- Distilled water
- Two 250 mL beakers
- Safety gloves

The students recorded the following results for their experiment.

Table 27.1 Student results.

Contents of beaker	Time to dissolve (seconds)
Whole tablet	120
Ground tablet	38

QUESTIONS

1. Write down a possible method the students could follow to carry out their investigation.
2. Explain why the students needed the mortar and pestle.
3. Describe one safety procedure that needed to be followed.
4. When designing this experiment, identify two variables that the students would have kept constant and explain why it was important to keep these variables constant.
5. Discuss the student results.
6. Write a valid conclusion for this investigation.
7. Discuss the validity of the conclusion you made in Question 6.
8. Describe one aspect of the method you describe in Question 1 that could lead to inaccuracy in the results.
9. Suggest one way to improve the method to reduce this inaccuracy.
10. Outline why many structures in the body require a large surface area.
11. Discuss the reliability of the data that would be gathered in this experiment.
12. Outline one strategy the students would have to follow to work effectively as a group.
13. The diagram shows a piece of apparatus used in an experiment to investigate the relationship between surface area and rate of reaction.

Figure 27.1 Scientific apparatus.

Why would this equipment be needed in this experiment?

(A) To make sure both aspirin tablets are the same size.
(B) To decrease the surface area of one tablet.
(C) To increase the surface area of one tablet.
(D) To prevent temperature changes in the room influencing the results.

14. In an experiment, what is meant by the reliability of the results?

(A) The same results will occur if the experiment is repeated.
(B) The experiment was carried out following scientific procedures.
(C) The method is able to properly investigate the aim.
(D) The variables had been properly controlled and the results are correct.

15. Students carrying out an investigation into the rate of reaction ground up an aspirin tablet and timed how long it took to dissolve and compared this rate with the time a whole tablet took to dissolve under the same conditions. What result would you expect?

(A) Whole tablet dissolved more quickly.
(B) Ground tablet dissolved more quickly.
(C) They both took the same length of time to dissolve.
(D) Not enough data has been given to know.

28 Human Organ Systems

In most multicellular organisms different tissues are organised into organs which have distinct functions. Organ systems are groups of interacting organs. The human body can be divided into 11 main organ systems. Table 28.1 summarises the major functions of each of these systems.

Table 28.1 Major functions of the human organ systems.

Feature	Structure
Skeletal system	Support, protection of soft tissue, storage of minerals and blood formation.
Muscular system	Support, locomotion and heat production.
Digestive system	Breakdown and processing of food so nutrients can be absorbed.
Cardiovascular system	Transport of substances, e.g. nutrients, waste products, hormones and heat around body.
Respiratory system	Delivers oxygen to respiratory surface for gas exchange and removes carbon dioxide.
Nervous system	Detects changes in the environment and responds to the stimuli by coordinating other organs.
Urinary system	Removes nitrogenous wastes and other substances, e.g. water, salts in osmoregulation.
Lymphatic system	Defence against disease and infection.
Endocrine system	Production of hormones to control body changes and processes.
Integumentary system	Protection from the environment and temperature control (skin).
Reproductive systems	Productive of gametes and hormones for sexual reproduction.

Types of tissues

The organ systems are comprised of different tissues. Two types of tissue you should know are epithelial tissue and connective tissue. **Epithelial tissue** includes epithelia and glands. The function of epithelial tissue is to provide physical protection, control the substances entering or leaving the body, provide sensations as many contain sensory receptors and produce specialised secretions. Different types of epithelial cells include squamous epithelia, glandular epithelia and columnar epithelia which can be ciliated. **Connective tissue** is found throughout the body and is never exposed to the external environment. Connective tissue provides a structural framework, protection, stores energy, acts in defence and transports fluids and dissolved materials around the body.

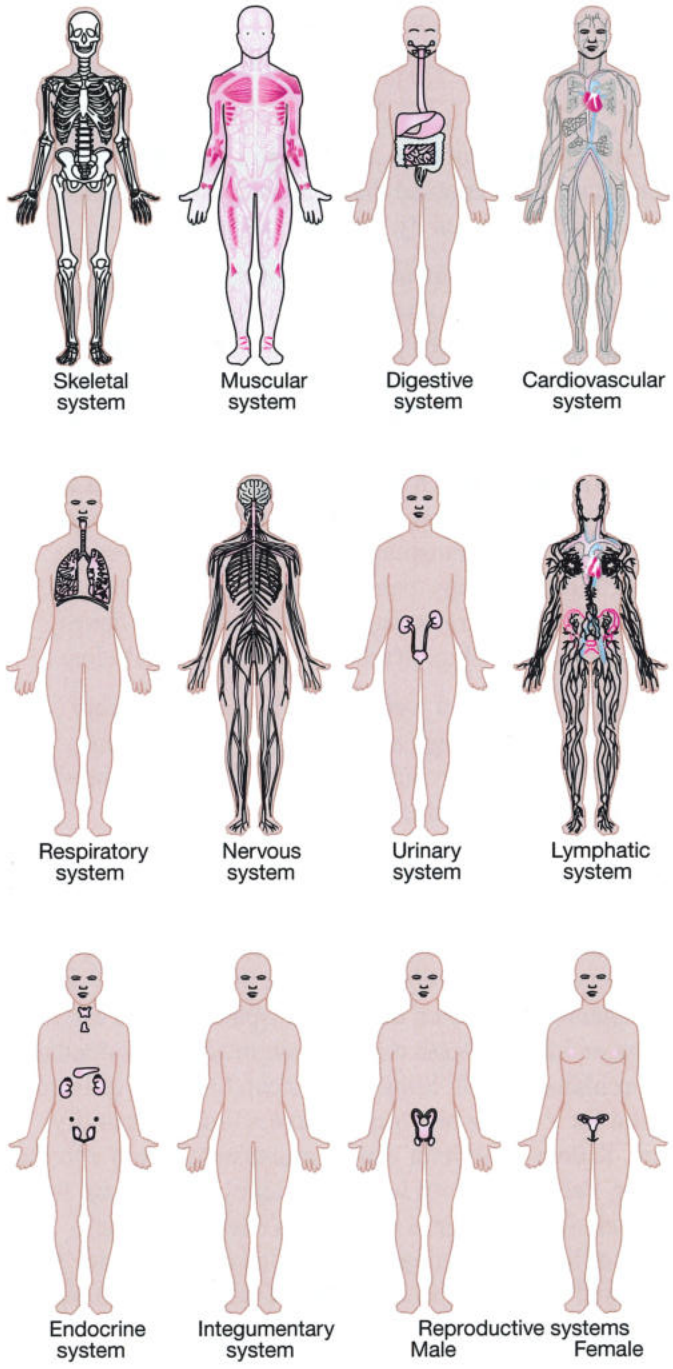


Figure 28.1 Human organ systems.

QUESTIONS

1. Define an organ system.
2. Construct a table to list the main components of each of the organ systems.
3. Which organ system(s) are involved in the process of locomotion?
   (A) Muscular system.
   (B) Skeletal system.
   (C) Nervous system.
   (D) All of the above.

29 Organ and Tissue Transplantation

The development of new technological and new surgical techniques have enabled diseased or damaged organs and tissues to be replaced by transplantation.

A **tissue transplantation** is the moving of a tissue from one body to another or from a donor site to another location on the patient's own body. An **organ transplantation** is the moving of an organ from one body to another. An **autograft** is a transplanted tissue or organ from the patient's own body to another area of their body. An **allograft** is a transplanted tissue or organ from the body of one person to the body of another person. Successful organ transplants have occurred for hearts, livers, kidneys, lungs, pancreas, intestines and thymus. Successful tissue transplants have occurred for bones, tendons, cornea, skin, heart valves, nerves and veins.

In Australia there are waiting lists for transplantable organs for heart, lungs, kidneys, liver, intestine and pancreas. A person is put on a waiting list when they have end-stage organ failure.

Transplant donors

Australia has a high rate of success for transplantations in both survival of the recipient and in the number of organs transplanted from each donor. Transplant donors can be living or 'dead'. Death can be brain dead and/or dead due to circulatory death. Some tissues can be preserved and stored for a time while organs have a limited usability time. Kidneys and liver can be transplanted from a living donor. People are born with two kidneys so one can be donated and the liver is regenerative which means a piece of healthy liver can be removed and the donor's liver will regenerate in a number of weeks back to full functionality.

Ethical issues have arisen about the definition of death, when, how and who can give consent for an organ to be transplanted and the payment for the donation. The demand of transplantation has led to illegal organ and tissue trafficking, forced donation and 'transplantation tourism'. In some countries it is easier, cheaper and quicker to get a transplant but this may involve the exploitation of the poor and needy and the violation of their human rights.

Tissue rejection

The ability to transplant different healthy organs and tissues to replace diseased or damaged parts has involved a range of new developments. The development of **immunosuppressive drugs** has decreased the rejection by transplant recipients.

Rejection occurs when the **major histocompatibility complex** (MHC) present on the transplanted organ/tissue is recognised by the patient's white blood cells, e.g. T cells as non-self. MHC is a set of cell surface molecules that act as antigens. Only identical twins have the same set of MHC molecules which means that rejection will not occur if the transplant is from one identical twin to another or if the transplant is an autograft. In all other transplants attempts are made to match the donor and recipient MHC molecules as closely as possible. If the transplant is to a very young child there is less need to be totally donor-recipient compatible as the immune system of young children, e.g. under 24 months old is not fully developed.

Immunosuppressive therapy involves the use of immunosuppressive drugs and other drugs to suppress the immune system, e.g. glucocorticoids are given after the transplant operation to prevent acute transplant rejection and graft-versus-host disease. When a person is on immunosuppressive therapy they are more susceptible to infection and cancer.

QUESTIONS

1. Define tissue transplantation.
2. Define organ transplantation.
3. What is an autograft?
4. Construct a table to show the successful transplants for tissues and for organs.
5. Distinguish between an autograft and an allograft.
6. When is a person put on a waiting list for a transplantable organ?
7. What waiting lists for transplantable organs are there in Australia?
8. Identify the source of donor transplants.
9. The diagram shows a MHC molecule attached to the cell membrane.

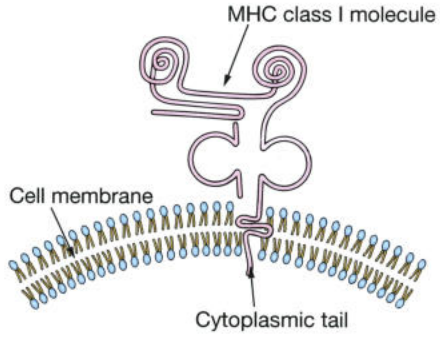


Figure 29.1 MHC class I molecule.

Discuss why MHC molecules are involved in transplant rejection.

10. Outline some ethical issues that have arisen due to organ and tissue transplantation.
11. Explain why there are health risks for people on immunosuppressive therapy.

30 Bioartificial Organs

There is a demand for bioartificial tissues and organs as an alternative to donor organs and tissues.

The need for bioartificial organs

- The supply of donor organs and tissues is limited causing some people to wait months, even years on a waiting list for transplantation.
- Donor tissues and organs can be rejected by the recipient's body.
- Transplantation involves immunosuppressive therapy which makes the recipient more susceptible to infection and cancer.
- There are ethical issues about the definition of death, when, how and who can give consent for an organ to be transplanted and the payment for the donation.
- There has been a growth in illegal organ and tissue transplantation, forced donation and 'transplantation tourism'.
- Longer life expectancy and more people surviving to older ages has increased the need for transplant donations.

Producing bioartificial organs

Scientists use knowledge of the structure and functions of organs and tissues to design a scaffold to create a new functional tissue. **Regenerative medicine** is the process of replacing or regenerating human cells, tissues or organs to give the recipient normal functioning of that part.

Some replacement parts are not cellular, e.g. in a hip replacement the 'ball' of the ball-and-socket joint is replaced by a metal head that is set in the femur (thighbone) and a plastic cup is inserted to become the 'socket'.

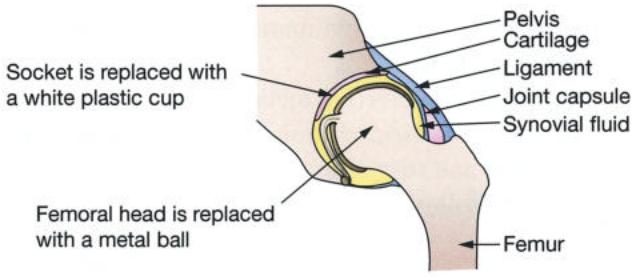


Figure 30.1 Hip joint replacement.

New techniques are developing artificial tissues and organs that are partly or completely cellular. In these replacement parts living cells or tissues are embedded in natural or synthetic scaffolds and the tissue is grown in cell culture in the laboratory. The cells can come from the patient who will be the recipient of the bioartificial part which means there is no tissue rejection.

Tissue engineering has been successful in growing and transplanting joint-cartilage and skin replacements.

If a patient's own cells are too badly damaged or diseased, tissues can be grown from a stem cell bank. Bioengineered stem cell-based tracheas which contain both artificial and biological material have been successfully transplanted into recipients. The stem cells can be harvested from the patient's own bone marrow which means there is a reduced risk of tissue rejection.

Tissue engineering

Tissue engineering involves the use of cells, engineering and material methods and suitable biochemical and physiochemical factors to improve or replace biological functions. Tissue engineering has produced artificial pancreas, artificial bladders, artificial skin, lab-grown cartilage and artificial bone.

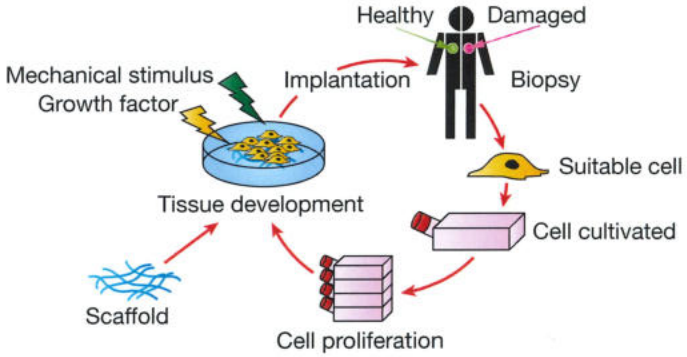


Figure 30.2 Tissue engineering.

Developments in tissue engineering could lead to a future where tissues and organs using a patient's own cells could be produced within a suitable framework to replace damaged or diseased body parts. This would remove the need for donor organs and tissues, remove waiting lists and be a more efficient and more effective way of improving the quality of life for many people.

QUESTIONS

1. Explain why there is a need for bioartificial organs.
2. What is regenerative medicine?
3. Outline what can happen in a hip replacement.
4. What is tissue engineering?
5. Explain why there is interest in the developments in tissue engineering.
6. Explain how the use of scientific knowledge about the structure and function of organs is influenced by social, economic, cultural and ethical considerations.
7. Why is a scaffold used in tissue engineering?
   (A) To support a 3-D tissue formation.
   (B) Allow cell attachment and migration.
   (C) Deliver and retain cells and biochemical factors.
   (D) All of the above.

31 Metabolism

Metabolism refers to the sum of all chemical reactions in an organism. **Metabolic pathways** are series of chemical reactions that either build complex macromolecules or break down complex molecules into simpler molecules.

A **catabolic pathway** is a series of reactions that release energy by breaking down complex molecules into simpler molecules, e.g. cellular respiration. Catabolic reactions are often called 'downhill' reactions as energy is released.

An **anabolic pathway** is a series of reactions that build complex macromolecules from simpler molecules, e.g. synthesis of proteins from amino acids. Anabolic reactions are often called 'uphill' reactions as energy is needed to drive the reaction.

Enzyme control

Each step in a metabolic pathway is controlled by a specific enzyme. An **enzyme** is an organic catalyst that controls the rate of a specific chemical reaction.

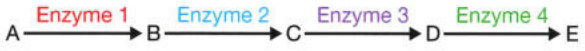


Figure 31.1 Metabolic pathway.

Enzymes lower the activation energy of the chemical reactions that they catalyse.

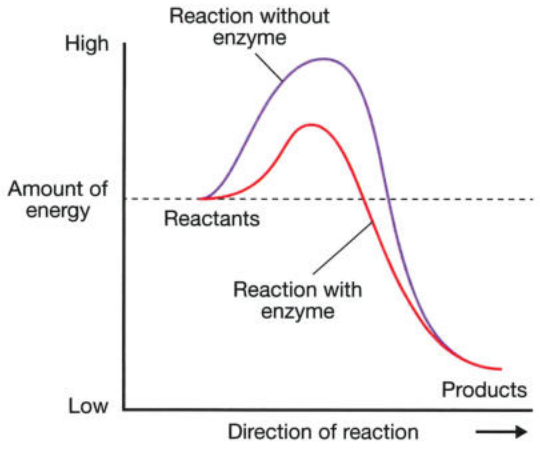


Figure 31.2 Activation energy.

The graph in Figure 31.2 shows exothermic reactions (releases energy by light or heat) – energy is released with the energy levels of the reactants higher than the energy level of the products. The energy required for the reaction to proceed without an enzyme is higher than with an enzyme present.

Each enzyme has an optimal temperature and an optimal pH at which it is most active.

Types of cellular work

There are three main types of work done by cells.

Transport work – substances are moved across a membrane by active transport against the concentration gradient.

Chemical work – energy is used to drive specific chemical reactions, e.g. biosynthesis of macromolecules.

Mechanical work – involving the transfer of energy, e.g. beating of cilia, muscle cells change shape in contraction, chromosomes move in mitosis.

Adenosine triphosphate (ATP)

ATP is the 'energy currency' of a cell. It is used to drive transport and mechanical work.

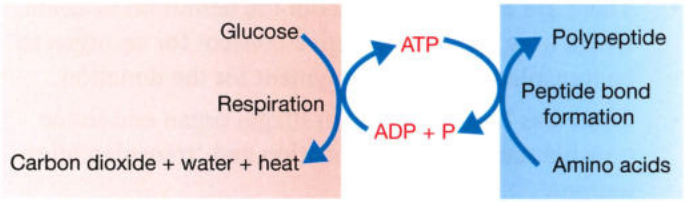


Figure 31.3 ATP/ADP cycle.

QUESTIONS

1. Define metabolism.
2. What is meant by a metabolic pathway?
3. Define a catabolic pathway.
4. Define an anabolic pathway.
5. Explain why catabolic and anabolic reactions are often called 'downhill' and 'uphill' reactions respectively.
6. Define an enzyme.
7. Explain how enzymes affect the activation energy of a reaction.
8. Define an exothermic reaction.
9. Construct a table to summarise the three types of work done by cells.
10. How is ATP involved in metabolism?
11. Draw a word equation to show the linking of ATP formation and respiration.
12. Study the following linked reactions.

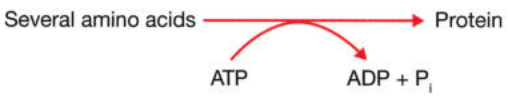


Figure 31.4 Linked reactions

What is shown in this diagram?

(A) ADP becomes a part of protein structure
(B) ATP becomes a part of protein structure
(C) ADP provides energy in protein synthesis
(D) ATP provides energy in protein synthesis.

32 Respiration

Respiration is a series of chemical reactions which releases energy from complex carbohydrates. All living things respire. **Aerobic respiration** is the complete breakdown of molecules such as sugars, using oxygen to form carbon dioxide and water. **Anaerobic respiration** occurs when there is no oxygen available, or the cell has insufficient oxygen. The products of anaerobic respiration depend upon the situation. For example, yeast cells respire anaerobically and form carbon dioxide and ethanol – a fact used by beer makers and wine makers. In animals, long periods of exercise and activity can lead to a shortage of oxygen in the muscles and the cells can begin to respire anaerobically, forming lactic acid – this leads to a 'cramp'.

General equation for aerobic respiration

Glucose + oxygen → carbon dioxide + water

$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$

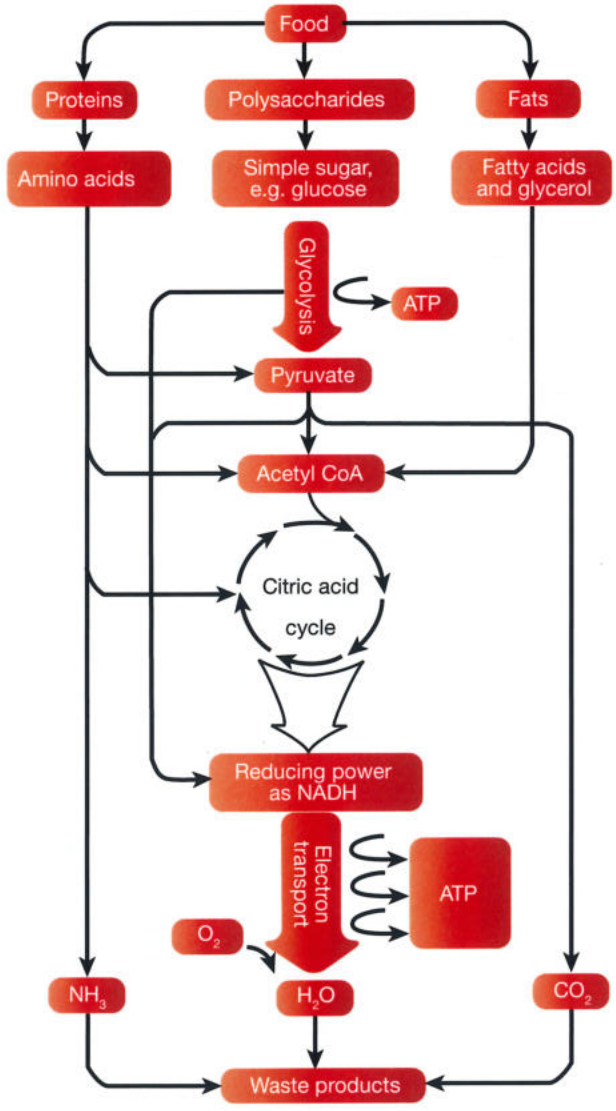


Figure 32.1 Respiration.

Stages of respiration

The general equation for aerobic respiration shows the reactants and the final products; however, the process does not occur in one step. It is a series of controlled chemical reactions with about 50 different stages, each catalysed by a different enzyme. An **enzyme** is a chemical made by living things and its function is to control the rate of a specific chemical reaction that occurs in the body.

Respiration can be divided into two main stages. The first stage breaks down glucose, a 6-carbon sugar, into two 3-carbon molecules called pyruvate. A small amount of energy is released at this stage. The second stage is the breakdown of the 3-carbon molecule into a 1-carbon molecule, carbon dioxide. This stage uses oxygen and releases a much larger amount of energy.

QUESTIONS

1. Define respiration.
2. Which organisms respire?
3. Distinguish between aerobic and anaerobic respiration.
4. Use an example to show the products of anaerobic respiration and how humans apply this knowledge.
5. Write the general equation for aerobic cellular respiration.
6. In what way could the general equation for aerobic respiration be misleading?
7. What is an enzyme?
8. Outline why enzymes are important in respiration.
9. Outline the two main stages of aerobic respiration.
10. Which of the following correctly shows the general equation for aerobic respiration?
    (A) Glucose + carbon dioxide → oxygen + water
    (B) Water + oxygen → carbon dioxide + glucose
    (C) Glucose + oxygen → carbon dioxide + water
    (D) Carbon dioxide + water → oxygen + glucose
11. Identify the process which provides the energy for life for activities such as movement, growth, repair and other vital functions.
    (A) Respiration.
    (B) Phagocytosis.
    (C) Mitosis.
    (D) Neutralisation.
12. The scientist Joseph Priestley (1733-1804) conducted experiments in which he placed a mouse in a sealed jar and added the gas released when oxides of mercury were heated. This gas enabled the mouse to live twice as long as in normal air. Identify this gas.
    (A) Carbon dioxide.
    (B) Oxygen.
    (C) Helium.
    (D) Hydrogen.

33 Reactions in Respiration

Cellular respiration is a series of oxidation-reduction (redox) reactions. **Oxidation** involves the loss of electrons from a substance in a redox reaction and **reduction** is the addition of electrons to a substance involved in a redox reaction. An oxidised substance has less energy than a reduced substance. The amount of energy in a molecule is also related to the size of the molecule with larger molecules containing more energy than smaller molecules.

ATP (**adenosine triphosphate**) is an organic compound that stores a large amount of energy and is the 'energy currency' of the cell. It is produced in respiration.

There are three main steps in cellular respiration – glycolysis, the Krebs cycle and the electron transport system.

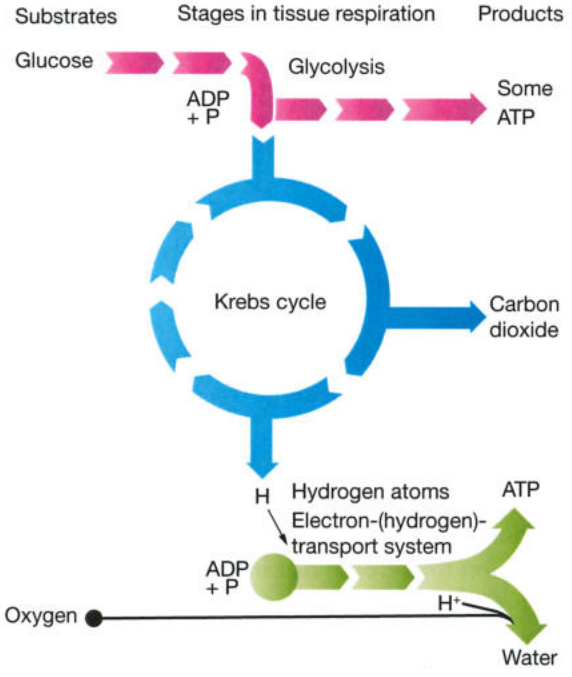


Figure 33.1 Stages in cellular respiration.

Glycolysis

Glycolysis is named by *glycol* + *lysis* – splitting of sugar and is a series of reactions where glucose (six-carbon sugar) is broken down into pyruvate (three-carbon molecule). Glycolysis does not need oxygen to occur and occurs in all living organisms beginning fermentation and aerobic respiration.

The overall reaction for glycolysis is:

$$\text{Glucose} + 2\text{ADP} + 2\text{NAD}^+ + 2\text{P}_i \rightarrow 2 \text{ pyruvate} + 2\text{ATP} + 2\text{NADH} + 2\text{H}^+$$

Each reaction in glycolysis is catalysed by a specific enzyme and there is a net synthesis of two molecules of ATP for each split glucose molecule. Phosphorylation occurs during glycolysis. **Phosphorylation** is the addition of a phosphate group to a molecule.

Table 33.1 Glycolysis.

Step in glycolysis	What is happening
Glucose; ATP → ADP; Hexokinase	Glucose enters the cell and an enzyme transfers a phosphate group from ATP to glucose in **phosphorylation**.
Glucose-6-phosphate; Phosphoglucoisomerase	An enzyme causes conversion to the isomer.
Fructose-6-phosphate; ATP → ADP; Phosphofructokinase	An enzyme transfers a phosphate group from ATP to the sugar in **phosphorylation**.
Fructose-1, 6-biphosphate; Aldolase; Dihydroxyacetone phosphate ↔ Glyceraldehyde -3-phosphate	An enzyme splits sugar by **lysis** into two 3-carbon sugars that are isomers of each other. An enzyme converts the not used isomer into the form that is used in the next step.
Glyceraldehyde-3-phosphate; 2NAD+ → 2NADH + 2H+; Triose phosphate dehyrogenase; 2Pi	Triose sugars are dehydrogenated and inorganic phosphate added. In the redox reaction hydrogen is used to reduce 2 molecules of NAD^+ to form NADH + H^+ for each triose.
1, 3-Biphosphoglycerate; 2 ATP → 2ADP; Phosphoglycerokinase	A phosphate group is transferred to ADP to produce ATP. Since 2 ATP were used earlier and 2 more molecules have been synthesised, the energy debt has been repaid and the reaction is energy equal for use/synthesis of ATP.
3-Phosphoglycerate; Phosphoglyceromutase	An enzyme relocates the last phosphate group for the next reaction.
2-Phosphoglycerate; $2H_2O$; Elonase	An enzyme causes dehydration and water is formed. The bond with the last phosphate becomes unstable.
Phosphoenolpyruvate; 2ADP → 2 ATP; Pyruvate kinase; Pyruvate	An enzyme catalyses the formation a molecule of ATP and a molecule of pyruvate in **phosphorylation**. Since there are 2 molecules of phosphoenolpyruvate, 2ATP are produced per glucose.

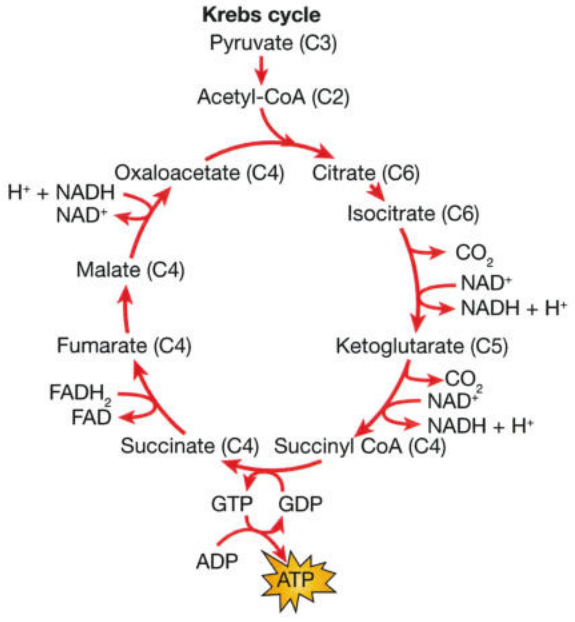


Figure 33.2 Krebs cycle.

The Krebs cycle

The Krebs cycle was worked out in the 1930s by Sir Hans Krebs. It is also known as the citric acid cycle or the tricarboxylic acid (TCA) cycle. It is a series of reactions that occur in the mitochondria with each step catalysed by a specific enzyme. The enzymes for the Krebs cycle are in the matrix of the mitochondria.

Pyruvate formed by glycolysis enters the mitochondrion by active transport and is converted into acetyl coenzyme A (**acetyl-CoA**). The acetyl-CoA is then degraded to carbon dioxide and water with the release of ATP. Since the breakdown of one glucose molecule produces two acetyl-CoA molecules the cycle must turn twice to process each glucose.

Electron transport pathway

The electron transport chain is a series of electron carrier molecules, e.g. membrane proteins that move electrons during redox reactions in order for the energy to be used to make ATP. Hydrogens or their electrons are transferred from one electron acceptor molecule to the next. The final electron (and hydrogen) acceptor is molecular oxygen to produce water as a by-product. The released energy is used to make ATP by chemiosmosis.

Chemiosmosis is an energy-coupling mechanism that uses stored energy in the form of a hydrogen ion (proton) gradient across a membrane to power cellular activities. In cellular respiration the synthesis of ATP is driven by the electrochemical gradient across the inner membranes of the mitochondria.

The enzymes and carriers of the electron transport chain are attached to the inner membrane of the mitochondria.

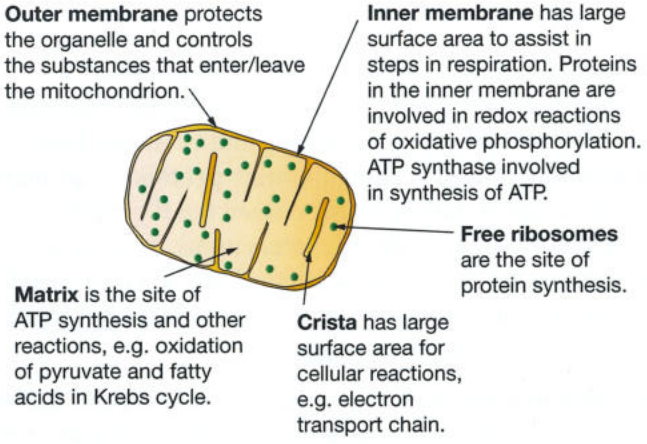


Figure 33.3 Mitochondrion.

NAD^+ (nicotinamide adenine dinucleotide) is an electron acceptor that acts as an oxidising agent in respiration. It is reduced to NADH. Each NADH formed in respiration has stored energy that can be used by the electron transport chain to produce more ATP in oxidative phosphorylation.

Anaerobic respiration

When there is no oxygen, glycolysis still occurs in the cytoplasm but the Krebs cycle and electron transport pathway are blocked. The pyruvate formed by glycolysis can become the hydrogen acceptor which leads to lactic acid fermentation, e.g. in vertebrate muscles that are being worked hard. In plants and yeasts anaerobic conditions lead to alcoholic fermentation where the pyruvate is the hydrogen acceptor to form ethanol.

QUESTIONS

1. Distinguish between oxidation and reduction in chemical reactions.
2. Compare the energy of an oxidised substance and a reduced substance.
3. Would a glucose molecule ($C_6H_{12}O_6$) have more or less energy than a methane molecule (CH_4)? Explain your reasoning.
4. Would a molecule of methane (CH_4) have more or less energy than a molecule of carbon dioxide (CO_2)? Explain your reasoning.
5. Name the three main steps in cellular respiration.
6. What is the meaning of the word 'glycolysis'?
7. What happens in glycolysis?
8. Explain why glycolysis can occur in both aerobic and anaerobic conditions.
9. How much ATP is synthesised per glucose molecule in glycolysis?
10. What is phosphorylation?
11. Where does the Krebs cycle occur?
12. What is chemiosmosis?
13. In a mitochondrion identify what reactions occur:
    (a) On the inner membrane.
    (b) In the matrix.
14. In which step of cellular respiration is glucose broken down into two pyruvate molecules?
    (A) Glycolysis.
    (B) Krebs cycle.
    (C) Electron transport chain.
    (D) Chemiosmosis.
15. Which molecule is the main 'energy currency' of a cell?
    (A) Adenosine diphosphate.
    (B) Adenosine triphosphate.
    (C) Adenine.
    (D) Acetyl-CoA.
16. What happens during phosphorylation?
    (A) Phosphate is broken down into phosphorus and oxygen.
    (B) Phospholipids are broken down with a phosphate being removed.
    (C) A phosphate is added to a molecule.
    (D) Hydrogen is added to a phosphate.
17. Which of the following can act as an electron acceptor?
    (A) CO_2 (B) $C_6H_{12}O_6$
    (C) NADH (D) NAD^+

34 Protein Structure

Proteins make up more than 50% of dry mass of most cells and are involved in nearly every aspect of life. They are macromolecules that always contain the elements carbon, hydrogen, oxygen and nitrogen. Proteins are formed on ribosomes and are made from a linear sequence of amino acids.

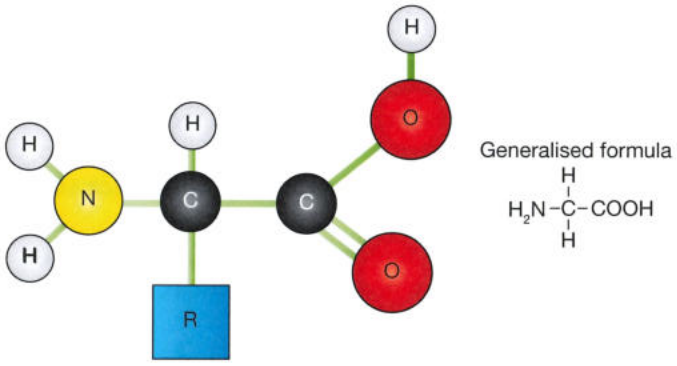


Figure 34.1 Generalised structure of an amino acid.

Amino acids

Amino acids are crystalline solids that dissolve in water but are insoluble in organic solvents. Each amino acid has an amino group (NH_2), a carboxyl group (COOH) and a side chain (R). The different chemical properties of different amino acids, e.g. polar (hydrophile) or non-polar (hyrodphobe) depend on the properties of R. The simplest amino acid is glycine where R = H.

There are over 150 types of amino acids that occur in cells but only 20 are used to make proteins.

Amino acids can be joined together by **peptide bonds**. Two amino acids joined together form a **dipeptide** while three amino acids joined together form a **tripeptide**. Hundreds or thousands of amino acids can be joined together to make a **polypeptide**.

Biuret solution (containing equal amounts of 1% copper sulfate solution and 5% potassium hydroxide solution) is used to test for peptide linkage. The biuret solution is a pale blue and a positive test for peptide linkage (protein) is a change to a purple colour.

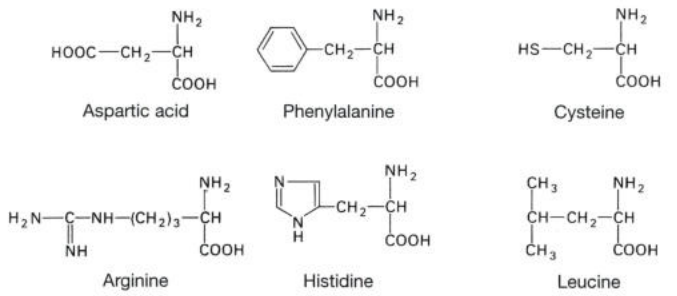


Figure 34.2 Structures of some amino acids.

Primary structure

The primary structure of a protein is the unique linear sequence of amino acids in its molecule and is determined by genetic information. The order of the amino acids is extremely important as it influences higher levels of organisation in the protein and its biological function.

Secondary structure

Secondary structures are coils or folds caused by hydrogen bonds between repeating CO and NH groups. Secondary structures include the helix and beta pleated sheet. Some spirals are 'right handed' (alpha helix) and others are 'left handed' (beta helix). Pleated sheets make up the basis of many globular proteins and fibrous proteins. If the coil of amino acids remains extended and threadlike it forms a **fibrous protein**. Keratin is a fibrous protein that mainly consists of alpha helices and is found in hair, wool, claws, beaks, feather, horns and nails. Silk protein is a fibrous protein that mainly consists of beta sheet form.

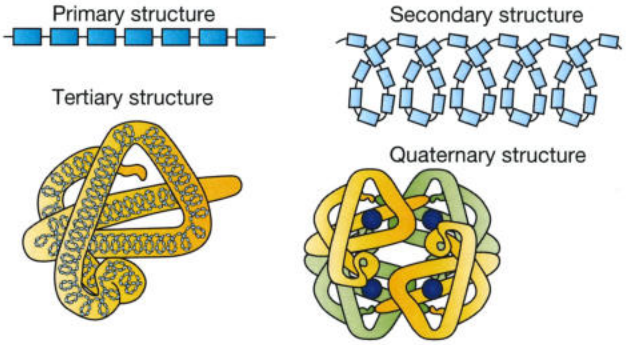


Figure 34.3 Protein structure.

Tertiary structure

Tertiary structure is formed by the folding of the secondary structure into a complex and compacted shape. The long coils of amino acids loop and twist and fold back on themselves to form **globular proteins**. The shape is determined by interactions between the R groups of the amino acids, hydrophobic interactions, hydrophilic interactions and hydrogen bonding. Three types of bonds hold the tertiary configuration in shape – hydrogen bonds, ionic bonds and disulfide bonds.

Disulfide bridges are strong covalent links between sulfur containing amino acids. Sulfur is often in the form of –SH groups and the most common amino acid with this –SH group is cysteine. If there are two cysteine amino acids distant from each other in the chain but folded to lie next to each other, their –SH groups with link to form a disulfide bridge –S–S– which will stabilise the structure. The folding of tertiary structures can provide structural strength, e.g. microtubules are globular proteins, or can produce an 'active site', e.g. enzymes are globular proteins.

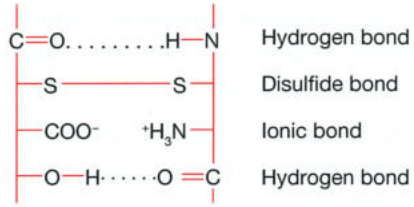


Figure 34.4 Bonds in proteins.

Quaternary structure

Quaternary structure involves the interaction between several polypeptide chains that aggregate into one functional macromolecule. The individual subunits are held together by hydrogen bonds rather than covalent linkages, though disulfide bonds occur. The hydrogen bonds are weak and can be easily broken by physical or chemical changes in the environment. If many bonds are broken, e.g. by excessive heat, pH, the globular structure cannot be maintained and the protein is **denatured**.

Conjugated proteins contain non-protein material and the non-protein part is called the prosthetic group. Many important proteins use quaternary structure to carry out a large variety of biological functions, e.g. collagen fibres and haemoglobin. Collagen is a fibrous protein with helical subunits twisted to form a triple helix which gives great strength to the long fibres. In the haemoglobin molecule of vertebrate blood, the protein component consists of four linked polypeptide chains – two alpha chains and two beta chains. Haemoglobin also contains four haem groups which are the prosthetic group of the protein. Conjugated proteins include chlorophyll and many enzymes in the electron transport chain.

QUESTIONS

1. What is a protein?
2. (a) Name the main parts of an amino acid.
   (b) Draw the generalised structure of an amino acid.
3. What determines the different properties of an amino acid?
4. Identify the simplest amino acid.
5. Distinguish between a dipeptide, a tripeptide and a polypeptide.
6. (a) What is the distinguishing test for protein?
   (b) What does this test actually detect?
7. What is meant by:
   (a) The primary structure of a protein?
   (b) The secondary structure of a protein?
   (c) The tertiary structure of a protein?
   (d) The quaternary structure of a protein?
8. Describe the types of coils/folds of secondary structures of proteins.
9. Construct a table to distinguish between a fibrous protein and a globular protein.
10. Identify the three types of bonds that hold the tertiary structure of proteins in shape.
11. Outline the significance of disulfide bonds in protein structure.
12. (a) What happens when a protein is denatured?
    (b) List some factors that can cause denaturation of a protein.
13. The diagram shows one polypeptide chain of a particular protein.

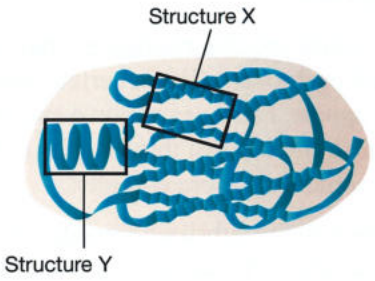


Figure 34.5 One polypeptide chain of a particular protein.

(a) Discuss how this diagram shows the primary, secondary and tertiary structure of this protein.
(b) Identify structure X and structure Y on the diagram.
(c) Is this a fibrous or globular protein? Explain your reasoning.
(d) If this protein consists of two of these polypeptide subunits, draw a possible appearance of this protein and identify the level of structure that has been formed.

14. The diagram shows what happened to a particular protein when it was heated to a high temperature.

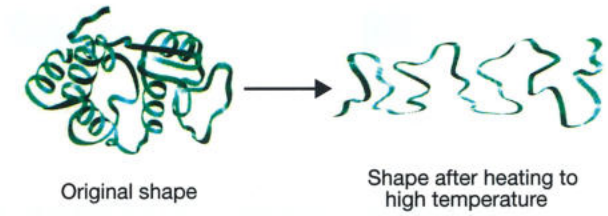


Figure 34.6 A protein is heated.

What has happened to this protein?
(A) Renaturation. (B) Denaturation.
(C) Melting. (D) Mutation.

15. What are the main components of all amino acids?
    (A) A phosphate group, a pentose sugar and a nitrogen base.
    (B) A carboxyl group (COOH) and an unbranched hydrocarbon chain($CH_3(CH_2)_n$–)
    (C) An amine group (NH_3), a carboxyl group (COO) and a side chain (R).
    (D) An amine group (NH_2), a carboxyl group (COOH) and a side chain (R).
16. What is the simplest amino acid?
    (A) Glycine (B) Leucine
    (C) Cysteine (D) Histidine

35 Enzymes

Enzymes are biological catalysts, that is, they are organic macromolecules that speed up reactions without being consumed by the reaction. Most enzymes are proteins. Examples of enzymes include pepsin which is in gastric juice for the breakdown of protein in the stomach and trypsin which is in the small intestine and also involved in protein digestion.

Enzymes control specific reactions, often working in 'teams' with each controlling a certain step until the final product(s) appear. Enzymes act on a specific substrate to either break them down into simpler substances or to synthesise something more complex.

Enzymes bind to the substrate forming an enzyme-substrate complex. The active site of an enzyme can distinguish its substrate from closely related molecules and this makes enzymes highly specific for different substrates. This binding is often called the '**lock and key mechanism**'. The **active site** on the enzyme is often a surface groove which perfectly fits the substrate and brings the active sites of the chemicals involved into alignment so that the chemical reaction can occur more quickly. The active site is not a rigid structure and the enzyme can slightly change shape to make a tight fit with the substrate. The 'lock and key' model was modified into the '**induced fit**' model as information was gathered about enzyme activity.

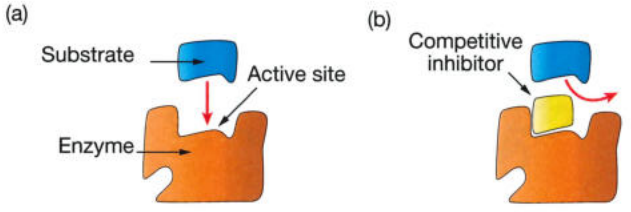


Figure 35.1 Lock and key mechanism of an enzyme and its substrate.

The activity of enzymes is affected by the amount of substrate present, temperature, pH, presence of coenzyme or cofactor, the presence of inhibitors and heavy metals and the availability of energy.

If there is an increase in substrate, the rate of the reaction will increase until all the active sites of the enzyme are occupied. This is the maximum rate at which the reaction can occur, according to substrate concentration and is known as the **saturation point**.

The induced fit mechanism explains why enzymes are denatured at high temperatures – their surface shape alters so that the substrate can no longer bind with the enzyme and reactivity is lost. The hydrogen bonds in proteins are broken by high temperatures and this disrupts their three-dimensional shape.

Coenzymes are non-protein molecules which need to be present before an enzyme can function. They take part in the reaction and are changed by it, e.g. vitamins and ATP.

Cofactors are mineral ions that need to be present to help activate the enzyme. They are not changed by the reaction, e.g. potassium ions and magnesium ions.

QUESTIONS

1. Identify the role of enzymes in living organisms.
2. Outline the chemical composition of enzymes.
3. What is the 'substrate'?
4. Draw a simple diagram to explain enzyme specificity on a substrate.
5. (a) Discuss why the enzyme-substrate binding is often referred to as the 'lock and key' mechanism.
   (b) Discuss why the 'lock and key' model has been replaced by the 'induced fit' model.
6. List the factors that influence the activity of enzymes.
7. Name one enzyme and state how its activity is influenced by each of the following.
   (a) Increased temperature.
   (b) Change in pH.
   (c) Change in substrate concentration.
8. Define saturation point.
9. Explain why enzymes denature at high temperatures.
10. Which of the following best describe enzymes?
    (A) Control all the chemical reactions in the body's metabolism.
    (B) Speed up reactions and are consumed by the reaction.
    (C) Decrease the rate of the reaction.
    (D) Enable a reaction to occur which would not normally proceed without the enzyme.
11. The enzyme ATP synthase is found on the inner membrane of a mitochondrion and is responsible for ATP synthesis. The antibiotic oligomycin inhibits ATP synthase. Which of the following would *not* inhibit the functioning of this enzyme?
    (A) Decreasing the temperature.
    (B) Lowering pH.
    (C) Addition of heavy metals.
    (D) Increasing concentration of ADP.
12. In 1894 Emil Fischer, a German chemist, proposed the lock and key model for enzymes. What does this model propose?
    (A) The overall shape of an enzyme is the same as the overall shape of the substrate.
    (B) Enzymes attach to the substrate by fitting into its shape.
    (C) Enzymes are inserted and added to the substrate molecule.
    (D) The substrate and enzyme alter their shapes so that they tightly fit together.

36 Enzymes and Reactions

All enzymes are specific and will only catalyse a particular reaction. The lock-and key model of enzyme activity was modified to the induced-fit model to more accurately explain how the active site changes shape to more closely fit the substrate during the reaction. Invertase is an enzyme that will only catalyse the hydrolysis of sucrose (cane sugar) to glucose and fructose. It does not work on any other sugar. The enzyme maltase splits the sugar maltose into glucose units.

Enzyme inhibitors attach to the enzyme and block its functioning. **Competitive inhibitors** block the substrate from entering the active site. **Non-competitive inhibitors** do not bind at the active site but their presence attached to another part of the enzyme still affects the functioning ability of the enzyme.

Graphs can be drawn to show how enzyme activity varies with substrate concentration, increase in temperature and change in pH.

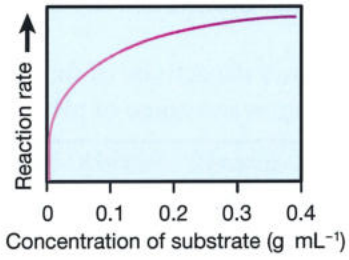


Figure 36.1 Enzyme activity and substrate concentration.

The graph for enzyme activity and substrate concentration shows an initial rapid increase with increase in substrate but the rate levels off at higher concentrations due to other limiting factors, e.g. amount of enzyme present.

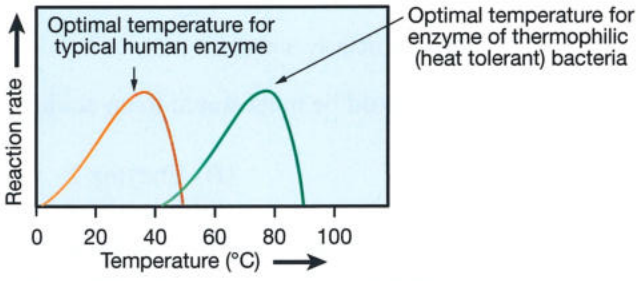


Figure 36.2 Enzyme activity with increase in temperature.

Increased temperature increases the movement of the molecules so that the substrate molecules collide more frequently with the active sites of the enzymes. The graph shows a gradual increase in enzyme activity until an optimum temperature is reached. After this temperature, the enzyme begins to denature. An enzyme which has partially denatured will regain its shape after cooling, but an enzyme which has completely denatured cannot correctly function again.

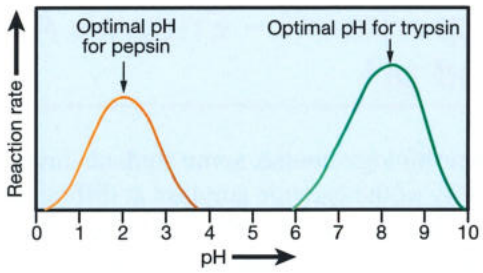


Figure 36.3 Enzyme activity with change in pH.

Each enzyme functions best at its own optimum pH. Extreme pH will denature enzymes.

QUESTIONS

1. Describe the 'induced fit' model and why it replaced the 'lock and key' model.
2. Distinguish between competitive inhibitors and non-competitive inhibitors.
3. For a reaction to occur the molecules in the system must overcome the energy barrier. This barrier is called the activation energy. The graph shows the energy needed for a reaction to occur, without an enzyme.

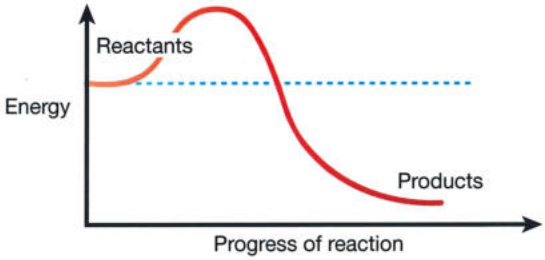


Figure 36.4 Graph of energy for a reaction without an enzyme.

   (a) Use a diagram to show how an enzyme would assist this reaction to occur more easily.
   (b) The enzyme carbonic anhydrase catalyses the interconversion of carbonic acid with carbon dioxide and water with each molecule of carbonic anhydrase able to catalyse the conversion of 600 000 molecules of carbon dioxide per second. Suggest how the rate of a catalysed reaction compares with an uncatalysed reaction.
4. Use an example to show the specificity of enzymes.
5. Why do small amounts of enzymes have a huge metabolic impact?
   (A) They change the pH of interstitial fluid, halting other reactions.
   (B) They can be used over and over again.
   (C) They are excreted by the kidneys after they have catalysed the reaction.
   (D) They transform into phospholipids and cholesterol after the reaction.

37 Experiment – Enzyme Activity and pH

During their biology course, some students investigated the reactivity of the enzyme amylase at different pH levels. Amylase is found in saliva and converts starch into smaller sugar units. The class was divided into groups so that each group tested one pH level and each group repeated their experiment five times.

Each group made up a beaker of test solution which contained 5 mL starch solution with 1 mL of their designated pH solution. Before starting the reaction, they had prepared a tray which had 1 drop of iodine solution in each indentation and each had a labelled time.

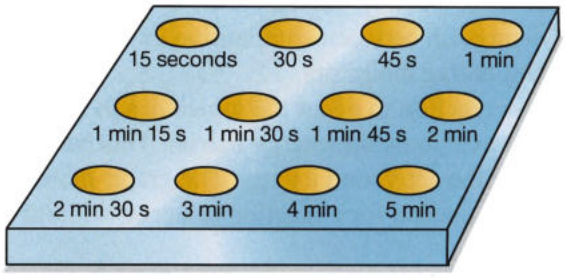


Figure 37.1 Tray with one drop of iodine per indentation and each labelled with a time.

At time 'zero' they added 0.5 mL amylase solution to their beaker. At each successive time, as indicated on the tray, they removed a few drops of the reacting solution with a pipette and immediately added this to the iodine in the correct time indentation on their tray.

When starch is added to iodine it forms a blue-black compound. Students noted the time when the added solution did *not* turn blue-black – the iodine stayed yellow-brown. If the solution still turned blue-black after 5 minutes, the result was recorded as 'no reaction'.

The table shows the average time from the five trials, recorded by each group.

pH	Time when iodine stayed yellow-brown	$\frac{1}{\text{time}}$
1	No reaction	0
3	No reaction	0
5	1 minute	
6	30 seconds	
7	15 seconds	
8.5	30 seconds	
9.5	1 minute 15 seconds	
11	3 minutes	
13	No reaction	0

QUESTIONS

1. Complete the table by calculating $\frac{1}{\text{time}}$ in s^{-1} for each pH to two significant figures.
2. Plot these results showing pH versus $\frac{1}{\text{time}}$.
3. Explain why students repeated the experiment and took the average time.
4. Explain why it was advisable to plot pH versus $\frac{1}{\text{time}}$ rather than pH versus time.
5. Using the results from the graph, state the optimum pH for the activity of this amylase.
6. Propose a reason why this should be the optimum pH for an amylase in the mouth.
7. Name one other factor besides pH which could influence the rate of this reaction.
8. Acidophiles are organisms that can live in highly acidic environments, e.g. below pH = 2 and alkaliphiles are organisms that can live in highly alkaline environments, e.g. above pH = 9. Explain why the scientific community is interested in investigating the enzyme activity of acidophiles and alkaliphiles.
9. The graph shows the activity of four different enzymes acting over a range of pH.

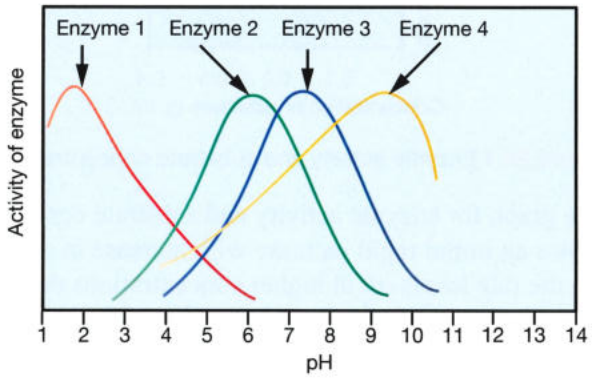


Figure 37.2 Enzyme activity and pH.

Which enzyme would be most suited to an acidic environment?

(A) Enzyme 1. (B) Enzyme 2.
(C) Enzyme 3. (D) Enzyme 4.

10. *Lactobacillus acidophilus* is a species of bacteria that occurs naturally in the human gastrointestinal tract and is believed to have probiotic characteristics. It is used in the production of many dairy products including the production of some types of yoghurt. What would be the optimum conditions for the growth of this bacteria?

(A) Low pH and 20°C.
(B) Low pH and 37°C.
(C) High pH and 20°C.
(D) High pH and 37°C.

38 Experiment – Enzyme Activity and Temperature

A Human Biology student wished to show that enzymes will not function if they have been boiled. She chose the enzyme rennin which is found in the gastric juice of mammals and causes milk proteins to clump together, taking them out of solution. She set up the following apparatus.

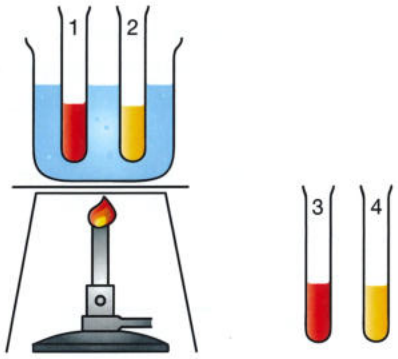


Figure 38.1 Apparatus to show boiling enzymes stops their action.

QUESTIONS

1. Suggest the most suitable contents for each tube.
2. What is the control?
3. Explain why boiling causes rennin to cease functioning.
4. Amylase is an enzyme present in saliva and helps break down starch in food products. Describe an experiment you could perform to show the action of amylase.
5. Many enzymes are destroyed by high temperatures, e.g. above 60°C, while low temperatures reduce the rate of reaction but do not destroy the enzyme. Suggest a use for this property of enzymes.
6. Human body temperature can fall due to a variety of reasons, e.g. wearing wet clothing for a long time, being exposed to cold weather. Low body temperature can also occur as part of the symptoms of particular diseases, e.g. diabetes, Addison's disease, infection, kidney failure, liver failure and hypothyroidism.
   (a) What symptoms would you expect to observe in a person with a low core temperature?
   (b) How does low core temperature affect enzyme activity?
   (c) Hypothermia can occur when human core temperature is less than 35.0°C. What is happening to the body during hypothermia?
7. Which of the following graphs correctly shows the relationship between enzyme activity and temperature, given that the enzyme involved has an optimum temperature of 38°C?

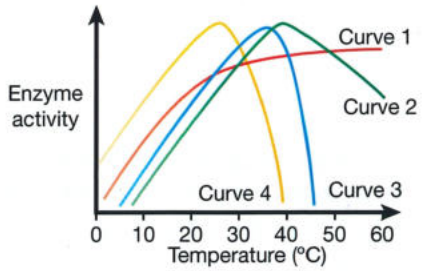


Figure 38.2 Enzyme activity and temperature.

(A) Curve 1 (B) Curve 2
(C) Curve 3 (D) Curve 4

8. Hyperthermia has been classified as body temperature greater than 38.3°C, with temperatures over 40°C life threatening. If a person has heat stroke, what enzyme activity would be expected?
   (A) Increased amount of enzymes produced.
   (B) Increased enzyme activity.
   (C) Reduced enzyme activity.
   (D) Completely normal functioning of enzymes.
9. Sucrose is a double sugar or disaccharide which can be split by the enzyme sucrase into a monosaccharide. Sucrose intolerance or congenital sucrose-isomaltase deficiency (CSID) is caused by a lack of proper sucrose. Which of the following graphs does *not* correctly show the relationship between enzyme activity and the condition?

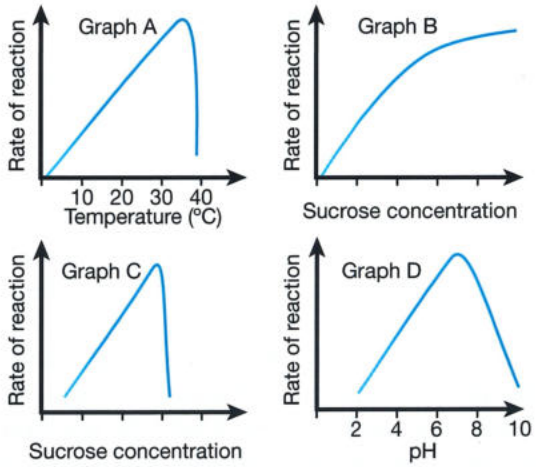


Figure 38.3 Enzyme activity under different conditions.

(A) Graph A (B) Graph B
(C) Graph C (D) Graph D

10. The induced fit mechanism for enzyme action explains why enzymes lose their activity with significant changes in pH and temperature as the changes cause a:
    (A) change in the shape of the enzyme and active sites no longer 'fit'.
    (B) coiling of the substrate so that the enzyme is surrounded.
    (C) union of the enzyme and substrate into a fixed locked complex.
    (D) melting of the enzyme onto the substrate forming a macromolecule.

39 Experiment – Enzyme Activity and Substrate Concentration

The **substrate** is the substance on which the enzyme acts and the **concentration** of the substrate refers to how many molecules of the substrate are present in the solution or mixture.

Some Human Biology students wished to investigate how the concentration of the substrate influences the rate of the reaction. They decided to use rennin as it is easily available as junket tablets. Rennin is an enzyme secreted by the walls of the stomach. It acts on the milk protein casein causing it to clump into a semisolid curd.

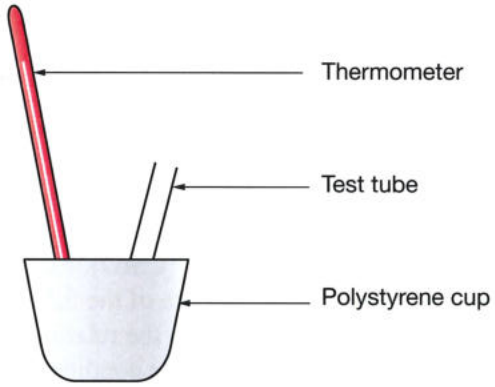


Figure 39.1 Equipment used by students.

They set up seven water baths – polystyrene cups – and kept the temperature of each at approximately 37°C by adding hot or cold water as needed. Next, they crushed a junket tablet with a spoon in 10 mL distilled water in a small beaker. To change the concentration of the substrate they decided to set up a series of test tubes, adding milk and water to each test tube so that each had a different concentration of milk. The total volume of milk/water in each test tube was 3 mL. When they were ready to start, they recorded the time and added 3 drops of the junket solution to each test tube. Each tube was examined by tilting gently and the time recorded when clotting occurred or its appearance after 10 minutes.

Student results:

Test tube	Contents of test tube	Time to clot
1	3 mL milk	45 sec
2	2.5 mL milk + 0.5 mL water	1 minute
3	2 mL milk + 1 mL water	1 min 19 sec
4	1.5 mL milk + 1.5 mL water	2 min 3 sec
5	1 mL milk + 2 mL water	Few lumps
6	0.5 mL milk + 2.5 mL water	No clotting
7	3 mL water	No clotting

QUESTIONS

1. Define substrate and substrate concentration.
2. Explain why the temperature of the water baths had to be kept at approximately 37°C.
3. (a) What is rennin?
   (b) Describe the function of rennin.
   (c) What is junket?
4. Explain how the students changed the concentration of the substrate.
5. Which milk concentration clotted in the fastest time?
6. (a) Plot the student results on a graph, using time to clot on the *x*-axis and milk concentration on the *y*-axis.
   (b) Describe the shape of the curve of your graph.
   (c) The following graph shows the usual graphical presentation of enzyme activity and substrate concentration.

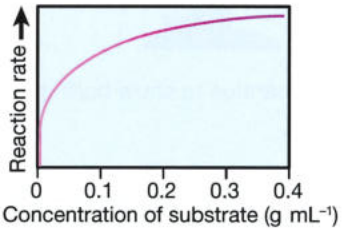


Figure 39.2 Effect of substrate concentration on enzyme activity.

   Compare the graph you have drawn with this graph.
7. What conclusion could the students draw from this experiment?
8. Describe two safety precautions you would need to follow while doing this experiment.
9. Why do many snake venoms and the 'nerve gases' developed during World War II prevent the transmission of messages along nerve pathways?
   (A) They cause the amino acids in the enzyme to deaminate.
   (B) They cause enzymes to break apart by splitting the bonds between each monosaccharide unit.
   (C) They stimulate the formation of polysaccharides around each enzyme.
   (D) They bind with the active site of enzymes and inhibit enzyme functioning.
10. The enzyme carboxypeptidase-A is a zinc-containing protein containing 307 amino acids. Which of the following relates to its function as a catalyst?
   (A) The bonds in carboxypeptidase-A are broken down in the reaction.
   (B) Each of the 307 amino acids are separated in the reaction.
   (C) It can only be used once to change the speed of the reaction.
   (D) Only small amounts are needed to speed up the reaction.

40 Enzymes and Inhibitors

Each enzyme reacts with a specific substrate. When the enzyme joins with the substrate it forms an **enzyme-substrate complex**. Each enzyme catalyses a specific reaction and only a specific part of the enzyme actually binds with the substrate – the **active site**.

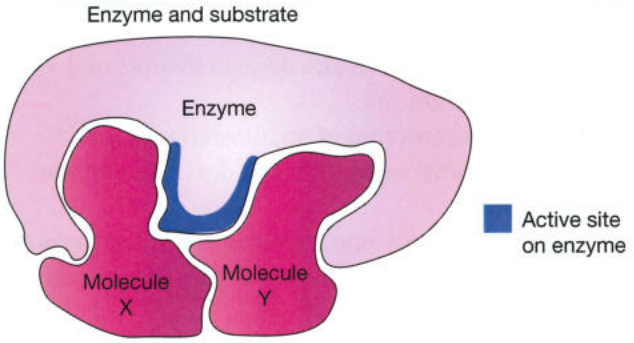


Figure 40.1 Active site on an enzyme.

The specificity of an enzyme is linked with the fit between the shape of the substrate and the shape of the active site.

Inhibitors can stop enzymes functioning and thus inhibit a specific reaction from occurring. Some chemicals cause **irreversible inhibition**, e.g. when there is a strong covalent bond formed between the enzyme and the inhibitor and this bond cannot be broken. Toxins, poisons and pesticides can be irreversible inhibitors, e.g. DDT, parathion and sarin. While other chemicals cause **reversible inhibition** as they form weak bonds with the enzyme which can be broken and the enzyme is able to resume normal functioning.

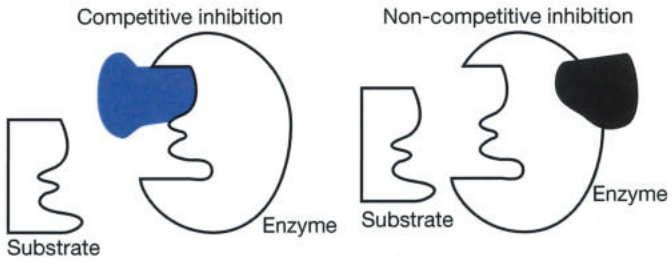


Figure 40.2 Types of inhibitors.

Types of inhibitors

There are two main types of inhibitors – competitive inhibitors and non-competitive inhibitors.

Competitive inhibitors are chemicals which bind to the enzyme with weak bonds and prevent the binding of the substrate to the enzyme. The inhibitor competes for the active site with the substrate.

Non-competitive inhibitors are chemicals which reduce the ability of the enzyme to bind with the substrate without actually competing for the active site. The inhibitor binds with the enzyme and can alter the shape of the enzyme so that even if the enzyme and substrate link in some way to form an enzyme-substrate complex the active site is less effective in forming the product.

QUESTIONS

1. Define active site.
2. How is the active site related to the specificity of an enzyme?
3. Distinguish between irreversible inhibition and reversible inhibition.
4. Explain how competitive inhibition prevents an enzyme from functioning.
5. Explain how non-competitive inhibition prevents an enzyme from functioning.
6. The enzyme acetylcholinesterase breaks down the neurotransmitter acetylcholine. Acetylcholine passes the nerve message from the end of one neuron across the synapse to the dendrites of the next neuron in the nerve pathway and is also involved in activating skeletal muscles.

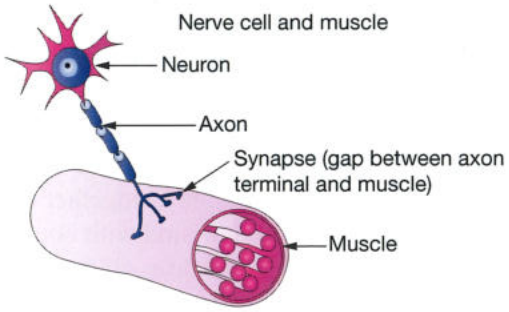


Figure 40.3 Acetylcholine is a neurotransmitter that activates skeletal muscle.

An acetylcholinesterase inhibitor (or anticholinesterase) stops the enzyme from working and increases the level and duration of the neurotransmitter. Sarin is an irreversible acetylcholinesterase inhibitor that has been declared a weapon of mass destruction and was used in two terrorist attacks in 1994 and 1995.

(a) What is acetylcholine?
(b) What is acetylcholinesterase?
(c) Explain why an acetylcholinesterase inhibitor will cause prolonged muscle contraction.
(d) Sarin affects lung muscles. Outline why people with sarin poisoning die of asphyxiation.

7. What do we call a substance that blocks the active site of an enzyme?
   (A) Non-competitive inhibitor.
   (B) Competitive inhibitor.
   (C) Reversible inhibitor.
   (D) Irreversible inhibitor.

41 Allosteric Activation and Inhibition

Allosteric regulation occurs when the functioning of an enzyme or protein is controlled by the binding of a regulatory molecule to a separate site. **Allosteric activators** can speed up the reaction while **allosteric inhibitors** slow down the reaction.

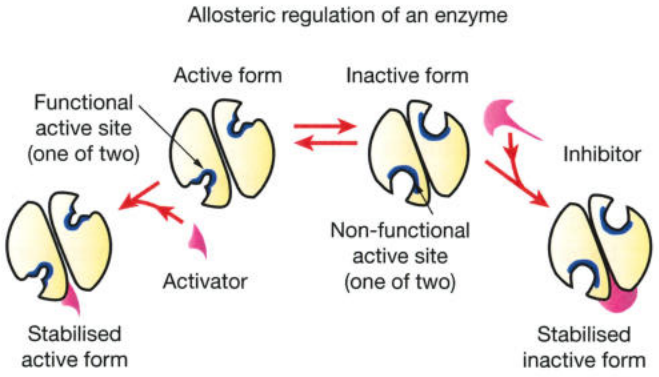


Figure 41.1 Actions of activators and Inhibitors.

Allosteric regulated enzymes usually consist of two or more subunits which each have an active site. The active site is separated from the **regulatory site** (or allosteric site). The enzyme has an **active form** and an **inactive form**. The activator binds to the enzyme causing the active site to become functional. The inhibitor binds with the enzyme causing the active site to become non-functional. The way the subunits fit together means that only one activator is needed to bind with one subunit and the change in shape then causes all the active sites of the other subunits to change shape so that all become active. Similarly only one inhibitor is needed to bind with one subunit and the change in shape affects all the other subunits so that all active sites become non-functional.

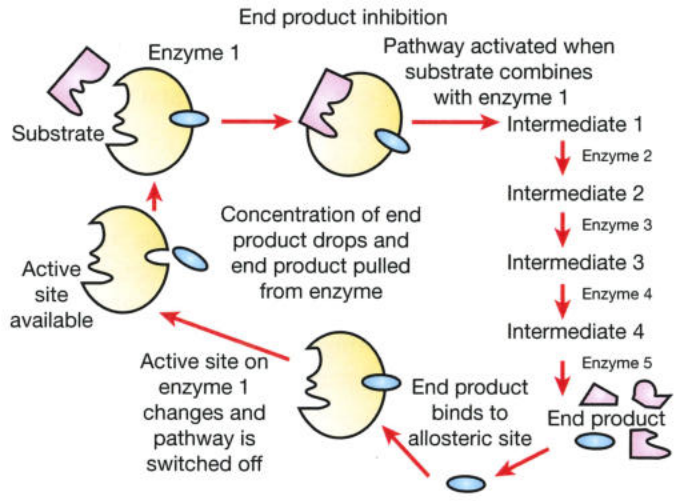


Figure 41.2 End product Inhibition.

End product inhibition

End product inhibition occurs in a multistep pathway where the end product binds to an enzyme that is early in the chain of reactions. This inhibits the series of reactions and prevents the production of more end product. It is a type of feedback mechanism.

QUESTIONS

1. Define allosteric regulation.
2. Distinguish between allosteric activators and allosteric inhibitors.
3. What is the structure of an allosteric enzyme?
4. Distinguish between the regulatory site and the active site.
5. Explain why only one activator is needed to cause all subunits of an allosteric enzyme to become functional.
6. Explain why only one inhibitor is needed to cause all subunits of an allosteric enzyme to become non-functional.
7. Explain what is meant by end product inhibition.
8. Caspases are a group of enzymes that are involved in protein digestion. They are important in cell death and inflammation and are in most human cells killing the cell once activated. There have been several research studies to find allosteric regulators that will act as inhibitors of caspases. It has been found that some viruses have genes for caspase inhibitors and are able to extend the life of the host cell so that the virus can reproduce inside its host. Discuss why scientists are interested in the virus genes for caspase inhibitors.
9. The flow chart shows a system where each step is catalysed by a different enzyme.

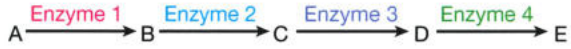


Figure 41.3 Series of reactions.

If there is end product inhibition of this series of reactions, what is the most likely substance to inhibit the activity of enzyme 1?
(A) Product B.
(B) Enzyme 3.
(C) Enzyme 4.
(D) Product E.

10. What is the name of the site on an enzyme molecule that allows a substance other than the normal substrate to bind to the molecule and to change the shape of the molecules and the activity of the enzyme?
(A) Active site.
(B) Allosteric site.
(C) Feedback site.
(D) Regulation site.

42 Enzyme Deficiencies

The information about the structure of different proteins in the body, e.g. the structure of enzymes is found in the nucleic acids DNA and RNA. A **protein** is one or more polypeptides folded and coiled into specific shapes. A **gene** has the code for a polypeptide. In gene expression cells will switch genes on and off to make the polypeptides. There are **structural genes** which have the code for the sequence of amino acids in a polypeptide and there are **regulator genes** that control the activity of structural genes.

Enzyme production

There are enzymes that are produced continuously under all conditions. These enzymes are needed for basic growth and survival, e.g. the enzymes involved in respiration. There are also enzymes that are only produced under certain conditions. **Intracellular enzymes** are made in cells and stay within the cell. **Extracellular enzymes** are made in the cell, packaged and secreted from the cell to work elsewhere, e.g. digestive enzymes.

Figure 42.1 Lactose intolerance means the person cannot digest milk sugar.

Lactose intolerance

Lactose is the disaccharide sugar found in the breast milk of mammals. The enzyme **lactase** digests lactose. Human babies normally produce lactase to be able to drink breast milk. In some human population, e.g. many European populations which have raised cattle and consumed milk products for centuries, the lactase is produced throughout their whole life. In other human populations the production of lactase stops during childhood or adolescence and the gene for producing lactase is permanently switched off. An autosomal recessive gene causes the lactase gene to be switched off. If a person without lactase consumes milk, they will not be able to digest the milk so that when the milk reaches the large intestine it is fermented by intestinal bacteria causing diarrhoea and vomiting. These people are called lactose-intolerant – they cannot consume lactose.

Phenylketonuria

Phenylketonuria (**PKU**) is a genetic disease caused by a mutation on chromosome 12 in the gene that makes the enzyme phenylalanine hydroxylase (PAH). This enzyme breaks down the amino acid phenylalanine into the amino acid tyrosine. Without the enzyme PAH, phenylalanine cannot be broken down in the body and builds up in the blood and tissues and is converted into phenylpyruvic acid. Toxic levels of phenylalanine and phenylpyruvic acid affect the nervous system affecting brain development, causing intellectual disability, seizures, behavioural problems and heart problems.

The occurrence of PKU varies with different populations around the world. In Australia newborns are tested for PKU. If the baby has PKU the parents are advised about the special diet the child must follow to reduce the intake of phenylalanine. Phenylalanine is an essential amino acid found in red meat, chicken, fish, eggs, nuts, cheese, milk and other dairy products. Supplementary tablets need to be taken, e.g. the amino acid tyrosine needs to be taken as tyrosine is normally made from phenylalanine in the body. Treatment has improved the quality of life for people born with PKU.

QUESTIONS

1. Where is the information about the structure of different proteins found in the body?
2. What is a protein?
3. What is the relationship between a gene and a polypeptide?
4. Outline the difference between structural genes and regulator genes.
5. Distinguish between intracellular enzymes and extracellular enzymes.
6. What is lactose?
7. Name the enzyme that digests lactose.
8. Explain why the people in some human populations produce lactase throughout their whole life while people in other populations stop producing lactase in childhood or adolescence.
9. Outline what happens if a person who does not produce lactase eats dairy products.
10. What is PKU and how is it related to an enzyme deficiency?
11. Outline how treatment has improved the quality of life for people born with PKU.
12. Where is the information about enzyme production?
    (A) In the cytoplasm of cells.
    (B) In the genetic code.
    (C) In the brain.
    (D) In the extracellular fluid.

43 The Respiratory System

A respiratory system is needed to facilitate the exchange of materials between the internal and external environments of the organism. The respiratory system allows gas exchange. '**Gas exchange**' is the movement of oxygen and carbon dioxide in different directions across a membrane. The structure and function of the respiratory system at the cell, tissue and organ levels allows gas exchange.

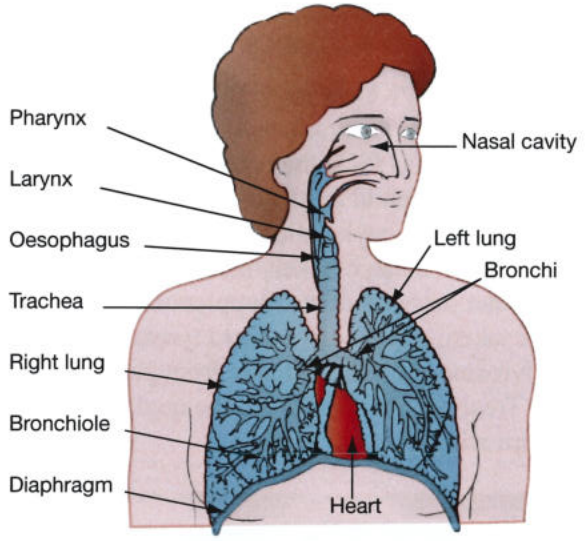


Figure 43.1 Human respiratory system. When we breathe in, the epiglottis closes off the oesophagus and air passes down narrower and narrower tubes until it reaches the alveoli.

Respiration

Individual cells obtain energy by the process of **respiration**. During aerobic respiration organic molecules such as glucose are combined with oxygen to release energy, and carbon dioxide and water are formed as waste products. The word equation for respiration can be written as follows.

Glucose + oxygen → water + carbon dioxide + energy

For respiration to occur, organisms need to be able to supply oxygen to the cells and remove carbon dioxide.

Respiratory surfaces

Respiratory surfaces where gas exchange occurs require several features.

- Respiratory surfaces must be moist so oxygen and carbon dioxide can dissolve so they can diffuse across the membrane.
- Respiratory surfaces need a large surface area to allow maximum diffusion.
- Respiratory surfaces need a rich blood supply to remove the oxygen once it has been absorbed so more oxygen can then be absorbed.
- Respiratory surfaces need to be thin to reduce the amount of diffusion needed.

The human respiratory system consists of the lungs and passages involved with the intake, expulsion and exchange of oxygen and carbon dioxide. The lungs are located in the thoracic cavity and air enters through the nostrils.

Organs of the respiratory system

The organs of the respiratory system include – nose, pharynx, larynx, trachea, two bronchi, bronchioles, two lungs with their pleura and muscles of respiration (intercostal muscles and diaphragm).

Respiratory system pathway

Nose – In humans, air enters the **nostrils** and the **nasal cavity**. The nasal cavity is divided into two equal passages divided by a **septum**. The nose is lined with a **ciliated epithelium** which has mucus-secreting **goblet cells**. In the nose the air is warmed, moistened and filtered. Hairs near the nostrils trap larger particles and smaller particles are trapped in the mucus that also protects the lining of the nose from chemical and physical injury. Beating of the cilia forces the mucus with trapped inhaled foreign particles back up the respiratory tract.

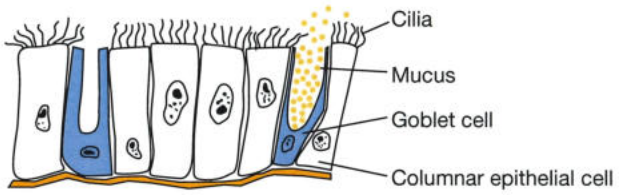


Figure 43.2 Ciliated epithelium and goblet cells.

Pharynx – Is behind the nose, mouth and larynx and is a tube around 12 to 14 cm long. It has three layers of tissue – mucous membrane lining, fibrous tissue and muscle tissue and is also involved in warming and humidifying the air.

Larynx (voice box) – Extends from the root of the tongue to the top of the trachea. The larynx is involved in pitch, volume and resonance of sound production. When swallowing the larynx moved upwards and the epiglottis closes over the larynx.

Epiglottis – Is a small flap of tissue that prevents food entering the trachea.

Trachea (windpipe) – Is a tube with C shaped rings of cartilage. It has three layers of tissue – outer layer is fibrous and elastic tissue that enclose the cartilages, the middle layer has the cartilage and bands of smooth muscle that wind around the trachea with blood and lymph vessels and autonomic nerves and the inner lining is columnar epithelium with mucus-secreting goblet cells.

Bronchi – The trachea splits into two **bronchi**, which each enter a lung and branch into smaller tubes called **bronchioles**. The right bronchus is wider and shorter (about 2.5 cm) and more vertical than the left bronchus (about 5 cm long). The bronchi are lined with ciliated columnar epithelium.

Alveoli – Bronchioles end in air sacs called **alveoli** which have a thin epithelium surrounded by a network of capillaries for gas exchange. Defence in the alveoli relies on lymphocytes, plasma cells and macrophages in lung tissue.

The oxygen diffuses across the wall of the alveoli into the blood in the capillary and carbon dioxide diffuses out of the bloodstream into the alveoli. In the capillary the oxygen attaches to the haemoglobin of a red blood cell, forming oxyhaemoglobin, and is transported to the heart in the pulmonary vein and hence to a needy cell that requires oxygen.

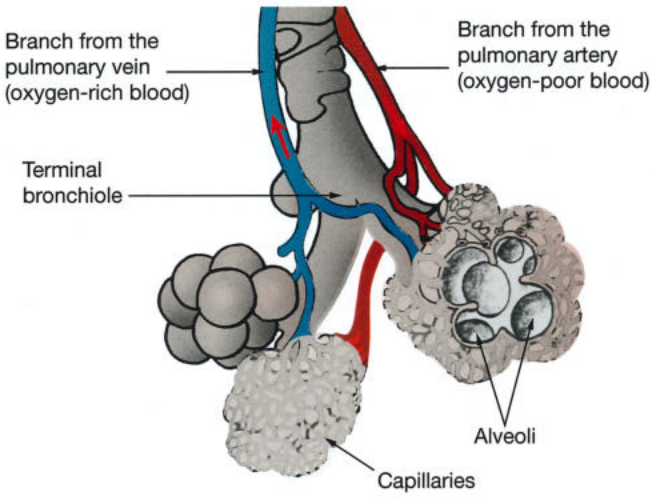


Figure 43.3 Part of a respiratory system.

Breathing

Breathing is the process of getting air into and out of the lungs. The lungs are two large, spongy organs found either side of the heart. Each lung is in a **pleural cavity** which is surrounded by a **pleural membrane** and contains a thin layer of fluid. Human lungs can extract up to 20 per cent of the oxygen in the air that enters the lungs. The breathing mechanism is due to change of air pressure in the lungs relative to the environmental air pressure.

Inspiration happens when there is a fall in air pressure in the lungs, e.g. when the diaphragm contracts and pulls the lungs downwards and **expiration** occurs when there is a rise in air pressure in the lungs, e.g. when the lungs shrink and the inspiratory muscles relax.

QUESTIONS

1. How do cells obtain energy?
2. What happens during aerobic respiration?
3. Write the word equation for respiration.
4. What is meant by 'gas exchange'?
5. Outline the four necessary characteristics of a respiratory surface.
6. Outline the structure of the respiratory system of mammals.
7. List the organs of the respiratory system.
8. Outline the structure and function of the nose.
9. Suggest why nose bleeds can involve a large loss of blood.
10. Outline the function of goblet cells in the lining of the respiratory tract.
11. Draw a flow chart to show the passage of air into the human body.
12. Where does oxygen enter the bloodstream in humans?
13. What happens to oxygen once it enters the bloodstream?
14. What is the difference between breathing and respiration?
15. Respiratory pigments combine reversibly with oxygen. What is the respiratory pigment in human blood and outline why it is advantageous for humans to have this pigment in their blood.
16. In humans the right lung is divided into three lobes and the left lung is divided into two lobes. Suggest why there is a difference in the number of lobes of human right and left lungs.
17. What is meant by gas exchange?
    (A) Oxygen taken in and carbon dioxide released.
    (B) Oxygen released and carbon dioxide taken in.
    (C) The movement of oxygen and carbon dioxide in different directions across a membrane.
    (D) Gases moving in and out of blood.
18. The diagram shows a section of the lower respiratory tract. Which part is a bronchiole?
    (A) Part A
    (B) Part B
    (C) Part C
    (D) Part D

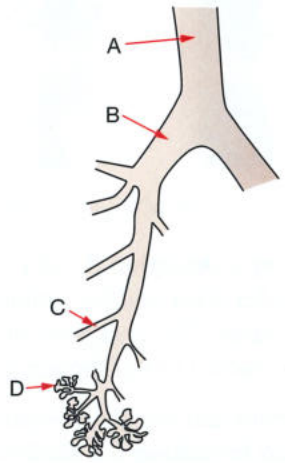


Figure 43.4 Part of a respiratory system.

44 Gas Exchange, Breathing and Ventilation

The efficient exchange of gases in the lungs is maintained by the actions of breathing, blood flow and the structure of alveoli.

Ventilation is any process that increases the contact between a respiratory surface and the respiratory medium. **Gas exchange** is the process where oxygen is absorbed and carbon dioxide is excreted across a membrane. **Breathing** is often called pulmonary ventilation as it is the physical movement of air in and out of the bronchial tree. Breathing is the process of inhalation and exhalation to ventilate the lungs.

How breathing causes ventilation

Air flow in the lungs is tidal with air entering and leaving the lungs going along the same pathway.

During **inhalation** the diaphragm contracts and moves downwards increasing the volume of the thoracic cavity. The **diaphragm** is a dome-shaped sheet of muscle that separates the thoracic and abdominal cavities. During inhalation the **intercostal (rib) muscles** contract and the ribs move upwards also increasing the volume of the thoracic cavity. The lungs adhere to the walls of the thoracic cavity so that an increase in the volume of the thoracic cavity causes an increase in the space inside each lung. The increase in volume of the lung drops the air pressure inside the lung compared to the outside air pressure. The difference in air pressure forces air into the lungs.

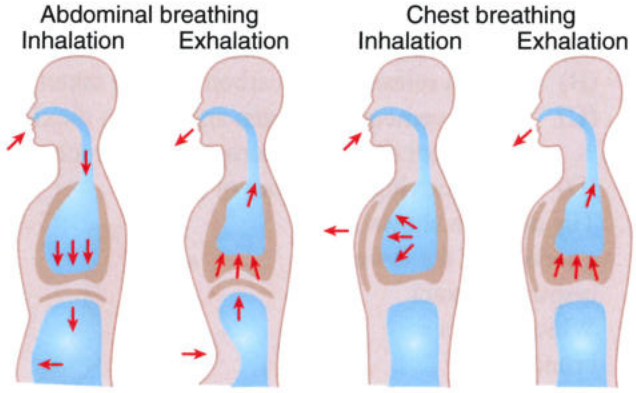


Figure 44.1 How breathing occurs.

During **exhalation** the diaphragm and rib intercostal muscles relax and the volume of the chest cavity decreases. This increases the pressure inside the lungs and air is forced out of the lungs.

Humans can use chest breathing or abdominal breathing. **Chest breathing** is a shallow form of breathing that predominantly relies on the intercostal muscles to raise and lower the ribs. Chest breathing is used in the early stages of childbirth and in the 'iron lungs' of artificial respiration.

Abdominal breathing predominantly relies on the diaphragm which contracts to enlarge the chest cavity during inhalation and relaxes during exhalation.

Lung capacity

Air flow in the lungs is tidal with air entering and leaving the lungs going along the same pathway. **Lung capacity** is the volume of air in the lungs after maximum inhalation. When investigating the quantity of air respired there are several terms that need to be understood.

- **Tidal volume** is the volume of air inhaled/exhaled with each breath.
- **Vital capacity** is the maximum tidal volume that a person can achieve with forced breathing. Vital capacity is greater than tidal volume.
- **Residual volume** is the volume of air that remains in the lungs after forced exhalation.
- **Ventilation rate** is the number of inhalations or exhalations per minute.

Expired air has only about one third of its oxygen removed.

Table 44.1 Composition of inhaled air and exhaled air.

Component	Oxygen (%)	Carbon dioxide (%)	Nitrogen (%)
Inhaled air (atmospheric air)	20.9	0.04	79
Exhaled air (alveolar air)	14.0	5.60	79

Control of breathing

Thoracic ventilation is controlled by the respiratory centre of the brain located in the **medulla oblongata**. There are two sections – the inspiratory centre and the expiratory centre – which connect to the intercostal muscles and diaphragm by cranial nerves. Breathing is an involuntary reflex action which automatically adjusts to meet the body's needs.

QUESTIONS

1. Distinguish between ventilation and gas exchange.
2. Discuss why breathing is described as a type of ventilation.
3. What is the diaphragm?
4. Construct a table to compare inhalation and exhalation.
5. Distinguish between chest breathing and abdominal breathing.
6. What controls breathing?
7. What is the difference between tidal volume and vital capacity of the lungs?
8. Explain why the composition of exhaled air is different to the composition of inhaled air.

45 The Circulatory System

A circulatory system is needed to transport materials around the body, e.g. glucose, hormones, amino acids; blood cells, such as. red blood cells and white blood cells; and heat throughout their body. The human circulatory system is a **closed circulatory system** with the circulating fluid always contained in a set of blood vessels. The transport of materials is facilitated by the structure and function of the circulatory system at the cell, tissue and organ levels.

The circulatory system consists of the heart, blood vessels and blood.

Double circulation

The human circulatory system is a **double circulation** system as blood goes through the heart twice in each circuit. It involves the **pulmonary circulation** and the **systemic circulation**. Blood from the body enters the right side of the heart and is pumped to the lungs where it is oxygenated. The right side receives oxygen-poor blood. From the lungs the blood is then returned to the left side of the heart which pumps the blood out the aorta. The left side of the heart receives oxygenated blood. From the left side of the heart the blood then travels through arteries which connect to capillaries and then to veins which return the blood to the heart.

The closed circulatory system is able to meet high energy needs allowing humans to be very active and have high metabolic rates requiring glucose and oxygen for respiration.

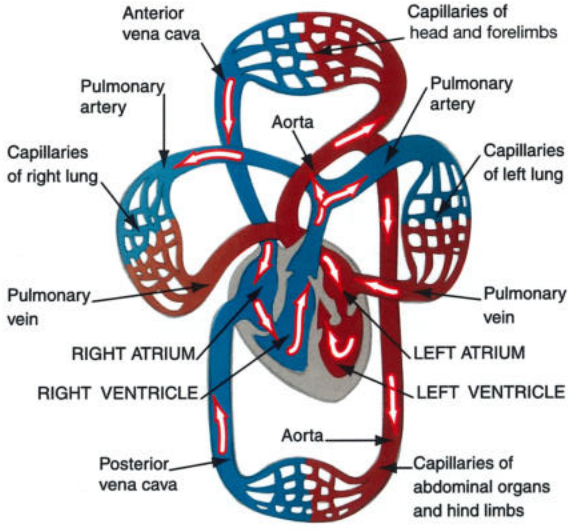


Figure 45.1 Human circulatory system.

A closed system requires a strong heart to pump blood through thin tubes with high pressure to overcome the resistance to flow and the exchange of materials occurs across the wall of capillaries. In a closed system the circulating fluid and the interstitial fluid are separated allowing the blood to have special properties.

This means that the lymph (interstitial fluid) does not have red blood cells while the blood does contain red blood cells which have the respiratory pigment haemoglobin that combines with oxygen to form oxyhaemoglobin. Haemoglobin greatly increases the ability of blood to carry oxygen.

Heart

The heart is a four-chambered organ that pumps blood through blood vessels. It is located behind the breastbone and is about the size of a clenched fist.

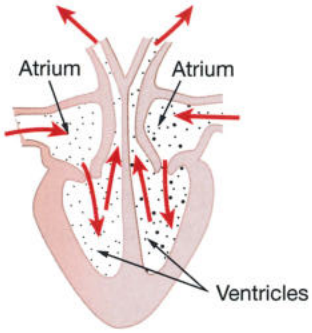


Figure 45.2 Human heart.

The heart consists of two atria which are receiving chambers and have relatively thin walls and two ventricles which have thicker walls and force blood into arteries when the ventricle wall contract. **Valves** which are flaps of connective tissue prevent backflow of blood. A **heart murmur** involves blood squeezing backwards.

Blood vessels

Blood vessels include arteries and arterioles, veins and venules and capillaries. Most diffusion of materials in and out of the blood vessels occurs in the thin walled capillaries.

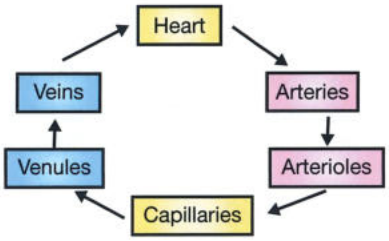


Figure 45.3 Human blood vessels

Arteries transport blood away from the heart and veins transport blood to the heart. All blood vessels except capillaries have smooth muscle fibres which are connected to nerves of the autonomic nervous system. Nerve stimulation causes **vasoconstriction** of small arteries and arterioles reducing blood flow while reduced nerve stimulation causes the smooth muscle to relax leading to **vasodilation** and increased blood flow. The diameter of large arteries varies according to the amount of blood they contain.

Blood

Blood is a connective tissue that makes up about 7% of the body weight for the average man. The proportion is less in women and greater in children. Blood consists of plasma and three types of blood cells – red blood cells (erythrocytes), white blood cells (leucocytes) and platelets (thrombocytes). Plasma is mainly water – around 90% water and dissolved substances such as plasma proteins, hormones, enzymes, gases, nutrients such as glucose, amino acids, fatty acids, glycerol and vitamins and inorganic salts such as sodium chloride, sodium hydrogen carbonate, potassium, magnesium, copper, iodine, calcium and iron.

Homeostasis maintains blood volume and the concentration of its components within narrow limits. Any fluid that seeps from the blood vessels into the extracellular fluid, called the **interstitial fluid**, is collected by the lymphatic system and then returned to the bloodstream.

Separating blood from the body and blood from the lungs means that the aorta contains high levels of oxygen allowing maintenance of body temperature and a high metabolic rate.

QUESTIONS

1. Outline the need for a circulatory system.
2. List some materials transported in the human circulatory system.
3. What is interstitial fluid?
4. Explain why the human circulatory system is called a double system.
5. What happens to any fluid that seeps out of the blood vessels?
6. Which side of the heart receives oxygenated blood and which side of the heart receives deoxygenated blood?
7. Describe the pathway of blood in the double system.
8. Explain why humans need a strong heart.
9. What are the main components of the circulatory system?
10. Describe the human heart.
11. Discuss why a separation of circulating fluid and interstitial fluid can be an advantage.
12. What are valves and what is their function?
13. Draw a table to show three differences between the three main types of blood vessels.
14. What is a heart murmur?
15. Which blood vessels have smooth muscle?
16. Explain how vasodilation and vasoconstriction occur.
17. What causes the change in the diameter of large arteries?
18. What is blood?
19. List some of the components of plasma.
20. (a) What is homeostasis?
    (b) How is the circulatory system involved in homeostasis?
21. What is the correct pathway of blood through the circulatory system?
    (A) Artery → capillary → arteriole → venule → vein
    (B) Artery → arteriole → capillary → venule → vein
    (C) Artery → capillary → vein → arteriole → venule
    (D) Artery → arteriole → capillary → vein → venule
22. The diagram shows the human circulatory system.

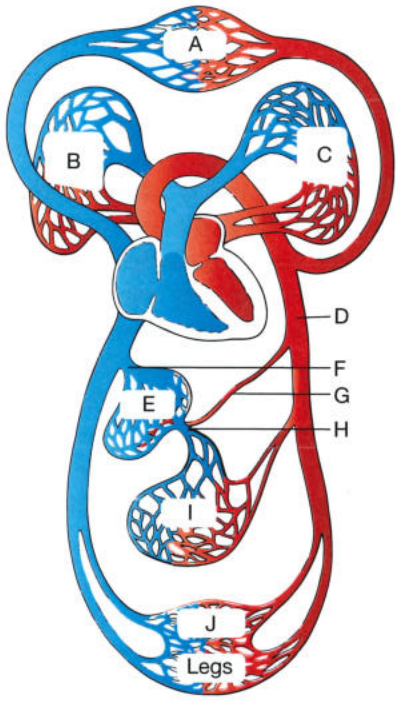


Figure 45.4 Human circulatory system.

    Which labelled part(s) would have capillaries that are picking up oxygen?
    (A) Part A.
    (B) Parts A and B.
    (C) Parts B and C.
    (D) Parts J and K.
23. What is the name of the inflow chambers of the heart?
    (A) Atria.
    (B) Ventricles.
    (C) Aorta.
    (D) Sinusoids.
24. What is the function of the systemic circulation?
    (A) Pump blood to the lungs from the right side of the heart.
    (B) Pump blood to the lungs from the left side of the heart.
    (C) Supply all the body organs and return deoxygenated blood to the left side of the heart.
    (D) Supply all the body organs and return deoxygenated blood to the right side of the heart.

46 The Cardiac Cycle

The **cardiac cycle** is one complete sequence of filling and pumping of the heart. It is a series of events that occur from the beginning of one heartbeat to the beginning of the next heartbeat. The events in the cardiac cycle maintain constant circulation of blood.

In each circuit of the body a blood cell will go twice through the heart – once through the left side and once through the right side due to the presence of a separate circulation for the lungs (pulmonary circulation).

The human heart is a **myogenic heart** which means that the pacemaker is specialised muscle tissue located in the heart.

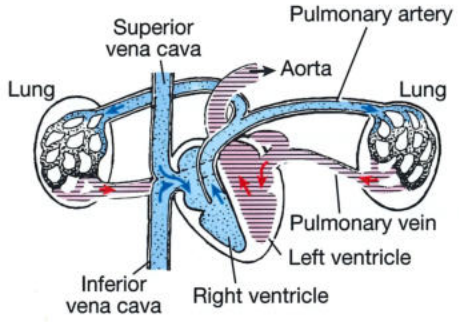


Figure 46.1 Pulmonary circulation.

The pathway through the pulmonary system is – vena cava → right atrium → right ventricle → pulmonary artery → lungs → pulmonary vein → left atrium → left ventricle → aorta.

Stages of the cardiac cycle

There are three main stages in the cardiac cycle – the atrial systole, the ventricular systole and the complete cardiac diastole.

The **systole** is the contraction phase and **diastole** is the relaxation phase.

The **cardiac output** is the volume of blood each ventricle pumps per minute and is determined by the **heart rate** which is the rate of contraction or number of beats per minute and the **stroke volume** which is the amount of blood pumped by a ventricle in a single contraction.

Atrial systole – Heartbeat is initiated by a group of specialised muscle cells in the right atrium called the **sinoatrial node**. The sinoatrial (SA) node is a region of the heart composed of specialised muscle tissue that sets the rate and timing at which all cardiac muscle cells contract (the pacemaker). In humans the SA node is near the point where the superior vena cava enters the heart. The signal from the SA node stimulates contraction as it propagates through the walls of the atria and then the walls of the ventricles. Because cardiac muscle cells are electrically coupled impulses from the SA spread rapidly.

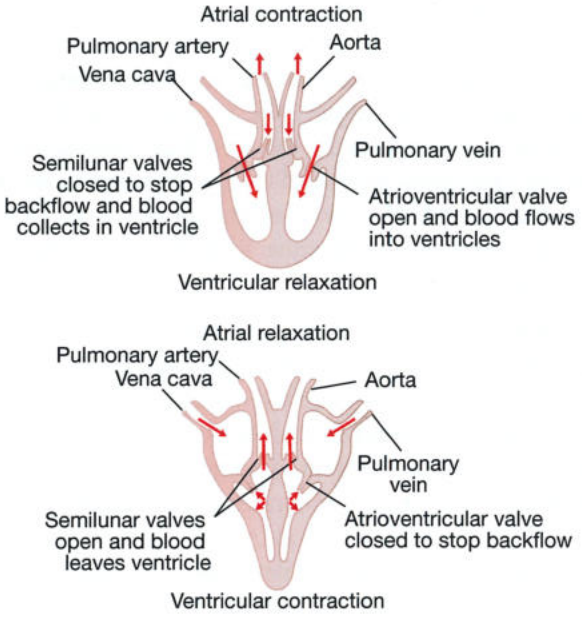


Figure 46.2 Changes in shape of the heart during contraction.

Ventricular systole – The atrioventricular node is a region of muscle tissue between the right atrium and the right ventricle where electrical impulses are delayed for about 0.1 seconds before spreading to the ventricles and causing them to contract.

Stages of heartbeat	
Pacemaker (SA node)	The pacemaker (sinoatrial SA node) sets the rate and timing for contraction of cardiac muscle cells. It is in the wall of the right atrium and generates an electrical impulse. It goes to the atrioventricular (AV) node.
Atrioventricular (AV) node	The signal is delayed at the AV node to make sure atria are completely empty.
Bundle branches; Heart apex	Bundle branches and Purkinje fibres take the signal to heart apex.
Purkinje fibres	Signal spreads through walls of ventricles and ventricles empty.

Figure 46.3 Stages of heartbeat.

Controlling the cardiac cycle

Factors that affect heart rate include the following.

Autonomic nervous system – Autonomic nerves in the cardiovascular centre in the medulla oblongata of the hindbrain affect heart rate. The heart rate centre has two pairs of nerves that connect the medulla oblongata to the pacemaker.

One pair of nerves go through the spinal cord and accelerate the heart. The other pair form part of the vagus nerves and decelerate heart rate. **Parasympathetic nerves** (vagus nerves) supply the SA and VA nodes and atrial muscle and reduce the rate at which impulses are produced decreasing heartbeat rate. **Sympathetic nerves** supply the SA and VA nodes and the myocardium of the atria and ventricles and increase heartbeat rate. These are antagonistic nerves.

Circulating chemicals – Hormones adrenaline (epinephrine) and noradrenaline secreted from the adrenal medulla increase heartbeat rate by stimulating cardiac muscle and increasing the excitability of the body. Adrenaline is a catecholamine (monoamine) hormone that mediates 'fight or flight' responses to short term stress causing the release of glucose into the blood from the liver so that the glucose can be used in the cellular respiration required for vigorous physical activity. The faster heart rate causes the faster transport of oxygen and nutrients to the muscles for respiration needed to perform the physical activity and the faster flow removes wastes, e.g. carbon dioxide more quickly from the tissues. Adrenaline is also produced at the ends of sympathetic nerve fibres as a neurotransmitter.

Temperature – Heart rate is influenced by temperature with higher temperatures accelerating heartbeat and lower temperatures decelerating heartbeat.

Position – When a person is upright the heart rate is usually faster than when lying down.

Exercise – Lower pH, e.g. due to increased exercise and the release of carbon dioxide speeds up breathing and heartbeat. Active muscles need more blood than resting muscles needing faster heart rate and selective vasodilation.

Age and gender – Heart rate is faster in women than in men and faster in babies and small children than in older children and adults.

Emotional state – Heart rate is faster when a person is feeling fear, anxiety or excitement.

Graphing the cardiac cycle

Graphs can be drawn to show the three major pressure waves in the atria – waves a, c and v.

Wave a is caused by atrial contraction, **wave c** occurs when the ventricles begin to contract and there is slight backflow pressure and **wave v** occurs at the end of the ventricular contraction from slow blood flow into the atria from veins while the A-V valves are closed. Ventricular pressure rises slightly during atrial contraction. When the A-V valves close ventricular contraction begins and ventricular pressure rises during isovolumetric contraction. When the ventricular pressure rises above 80 mmHg the semilunar valves open and blood flows out of the ventricles in ejection. The ventricles empty and ventricular pressure decreases.

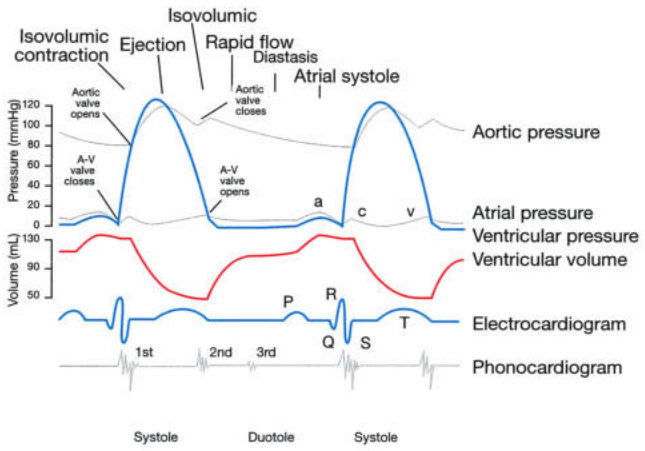


Figure 46.4 Pressure changes during the cardiac cycle.

Venous return

Cardiac output is affected by venous return which in turn is affected by the **position of the body**, e.g. gravity aids return from the head and neck when standing or sitting, by the **skeletal muscle pump**, e.g. contraction of skeletal muscles around the deep veins puts pressure on the veins to keep blood flowing up the legs and by the **respiratory pump** as inhalation creates negative pressure in the thorax helping blood flow towards the heart.

QUESTIONS

1. What is the cardiac cycle?
2. Draw a flow chart to show the pathway for pulmonary circulation.
3. Distinguish between systole and diastole.
4. Identify the three main stages of the cardiac cycle.
5. (a) What is meant by cardiac output?
   (b) How is cardiac output determined?
7. The coronary arteries supply the heart with blood. What would happen if the coronary arteries become blocked?
8. Construct a table to summarise the function of the chambers and valves of the heart.
9. Why is the human heart classified as a myogenic heart?
10. The diagram shows an electrocardiogram.

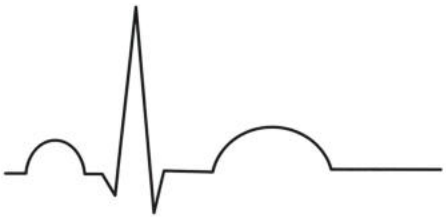


Figure 46.5 Pressure changes during the cardiac cycle.

   (a) What is an electrocardiogram (ECG)?
   (b) Copy the diagram and explain what is happening at each section of the ECG.
11. Construct a table to summarise the different factors that can affect heart rate.

47 Blood Vessels

In humans, blood flows in an internal closed circulatory system. There are several advantages of a closed circulatory system.

- Higher blood pressures are possible, allowing blood to be returned more quickly to the heart.
- Blood is more specialised for transport, e.g. higher oxygen-carrying capacity, and for defence.

There are three main types of blood vessels: **arteries**, which carry blood away from the heart; **veins**, which carry blood toward the heart; and **capillaries**, which connect arteries to veins.

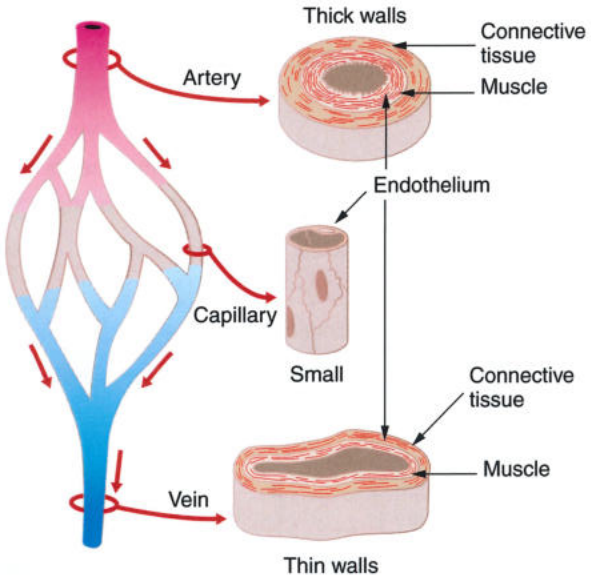


Figure 47.1 Blood vessels.

QUESTIONS

1. Discuss why arteries and veins are defined by the direction in which they carry blood rather than by the blood composition.
2. (a) The diagram shows three blood vessels.

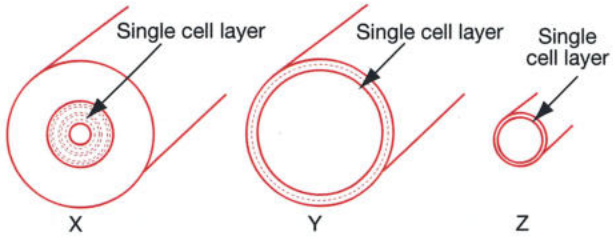


Figure 47.2 Three different types of blood vessels.

Identify each blood vessel, giving a reason for your decision.

(b) Choose one feature of veins not shown in the diagram and explain how this feature is related to the function of veins.

3. Many capillaries are only 6 to 8 μm wide, which means that many red blood cells distort into a bell shape.

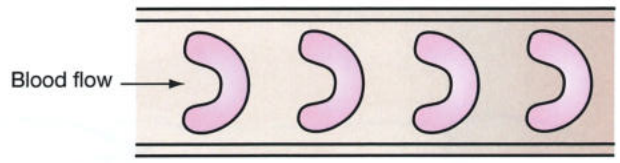


Figure 47.3 Red blood cells in a capillary.

Discuss the advantages of this behaviour.

4. The diagram shows the heart, lungs and associated blood vessels.

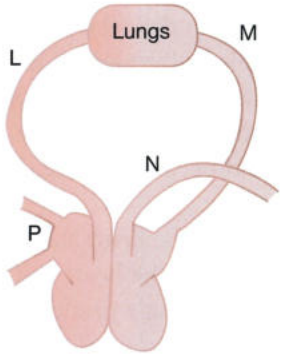


Figure 47.4 Heart, lungs and major blood vessels.

(a) Identify the blood vessel labelled N.
(b) Draw a labelled cross-section of the blood vessel N.
(c) Choose one feature you have labelled and explain why it is necessary for the efficient functioning of this blood vessel.
(d) Of the labelled vessels L, M and P, which would have a structure most similar to that of N? Explain your reasoning.
(e) Describe one difference in the chemical composition of blood in vessels labelled N and P. Explain why this difference occurs.

5. The diagram shows the structure and functioning of valves.

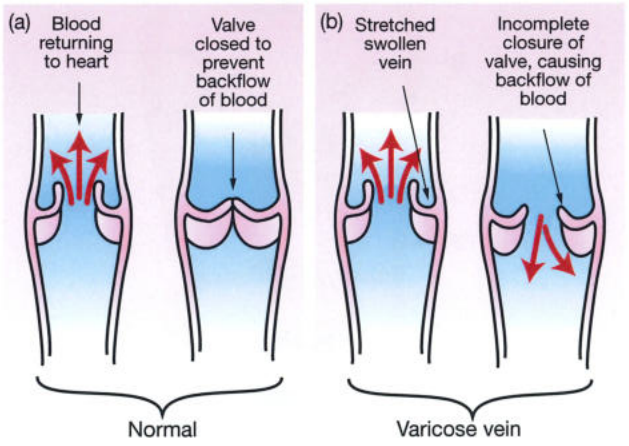


Figure 47.5 Valves.

Discuss why veins have valves, but arteries do not.

6. The diagram shows a section of skeletal muscle with a vein.

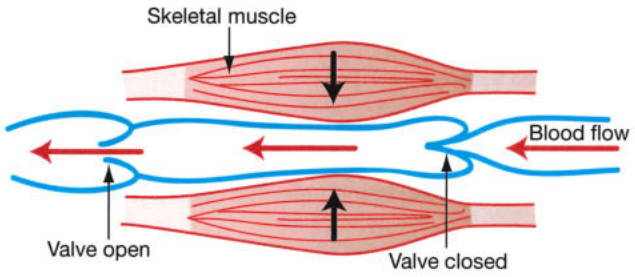


Figure 47.6 Skeletal muscle and vein.

Use this diagram to describe two features of the movement of blood through veins.

7. Which of the following blood vessels would be carrying the most carbon dioxide?
   (A) Major vein between the heart and the rest of the body.
   (B) Vein between the heart and lungs.
   (C) Artery between the heart and lungs.
   (D) Capillary in muscle tissue.

8. Which of the following most accurately describes the functions of arteries, veins and capillaries?

	Arteries	Veins	Capillaries
(A)	Carries blood to heart	Carries blood from heart	Site of gas exchange
(B)	Carries blood from heart	Site of gas exchange	Connects arteries and veins
(C)	Carries blood from heart	Carries blood to heart	Site of gas exchange
(D)	Carries blood to heart	Site of gas exchange	Carries blood from heart

9. The diagram shows a modified scheme of the circulatory system of humans.

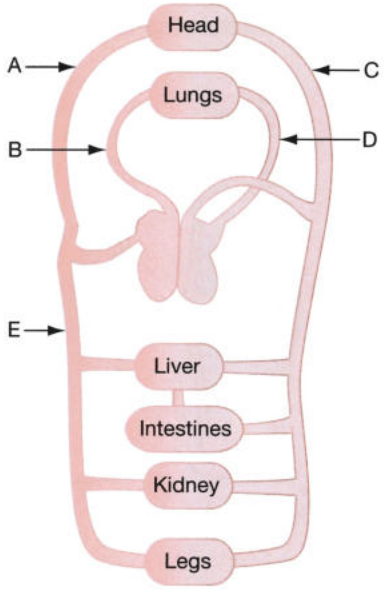


Figure 47.7 Circulatory system of humans.

Which blood vessel would have the highest concentration of carbon dioxide?
(A) A (C) C (B) B (D) D

10. The diagram shows the part labelled X (coeliac artery) delivering blood to the intestines and the part labelled Y (hepatic portal vein) taking blood from the intestines to the liver.

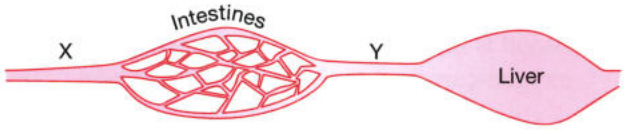


Figure 47.8 Blood flow to and from the intestines.

What would you expect to be carried in the blood in Y?
(A) Oxygen.
(B) Fatty acids and glycerol.
(C) Glucose.
(D) Urea.

11. The diagram compares the wall components of an artery, an arteriole, a capillary, a venule and a vein.

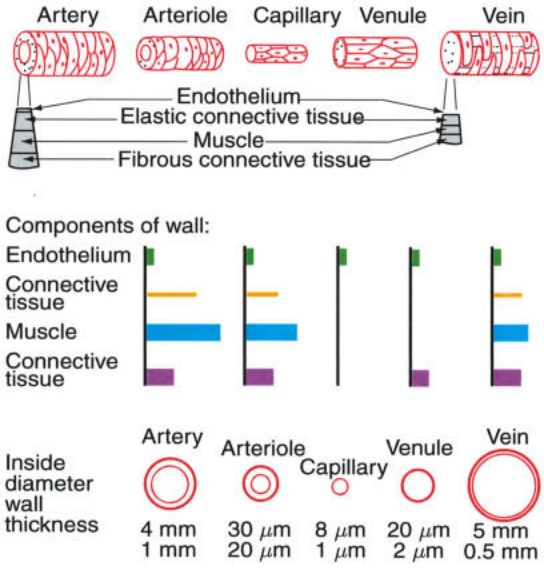


Figure 47.9 Wall structure in blood vessels.

Which of the following statements can be made from this data?
(A) The muscle layer is the same thickness in all blood vessels.
(B) The endothelium is thickest in veins.
(C) Elastic connective tissue is thickest in veins.
(D) Capillaries only have an endothelium layer.

12. Where is the portal circulation?
   (A) Venous blood from the capillaries of the abdominal part of the digestive system to the liver.
   (B) Arterial blood from the capillaries of the abdominal part of the digestive system to the liver.
   (C) Arterial blood to the liver.
   (D) Venous blood to the lungs.

48 Case Study – Coronary Occlusion

An **occlusion** is a blockage or closing of a blood vessel or hollow organ. If an artery is completely blocked the tissues supplied with oxygen and nutrients will quickly degenerate and die from **ischaemia** (lack of blood flow to a tissue) which leads to **infarction** (tissue damage due to inadequate flood supply). The amount of tissue damage depends on the size of the blocked artery, the amount and type of tissue involved in the blockage and the extent of the collateral circulation affected by the blockage.

If a coronary artery is blocked, there will be a restricted supply of oxygen and glucose for cellular respiration and the heart muscle will not be able to function properly. Myocardial infarction occurs. This can lead to heart attacks.

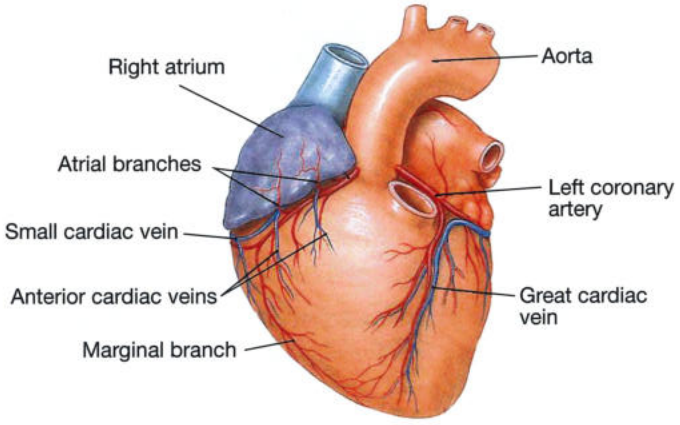


Figure 48.1 Coronary vessels.

A coronary occlusion is defined as the complete or partial blockage of blood flow in a coronary artery which are the arteries that supply the heart with blood delivering oxygen and nutrient to the cardiac muscle cells.

Cause

There are several causes for a coronary occlusion with the most common cause being a gradual build-up of plaque containing cholesterol (atherosclerosis) on the artery wall. This makes the arteries narrow and hardened with little elasticity. The formation of a blood clot on plaque (coronary thrombosis), a temporary spasm of muscles in an artery wall or inflammation of a coronary artery wall will also cause an occlusion.

Risk factors

- The build-up of plaque is associated with certain contributing risk factors.
- Smoking.
- High blood pressure.
- Certain chronic diseases.
- Overweight.
- Diet – high animal fat and high low density lipoproteins (LDL) with less high density lipoproteins (HDL).
- Family history.

Consequence

A coronary occlusion can lead to death of the area of cardiac muscle served by that artery (myocardial infarct); a heart working harder can cause pain (angina) and/or a heart attack. Without treatment the occlusion can cause death.

Treatment

Treatment depends on the precise nature and site of the occlusion. Medication, e.g. nitrate group such as glyceryl trinitrate can increase the diameter of the artery and increase the flow of blood containing the required oxygen and nutrients. Or calcium antagonists such as nifedipine which causes the muscle in the coronary artery wall to relax can be administered. Angioplasty is another option with the artery stretched open and a balloon introduced in surgery or bypass surgery can be carried out, e.g. using vein graft or using a nearby artery.

QUESTIONS

1. What is an occlusion?
2. Define ischaemia.
3. Define infarction.
4. The diagram shows the human heart.

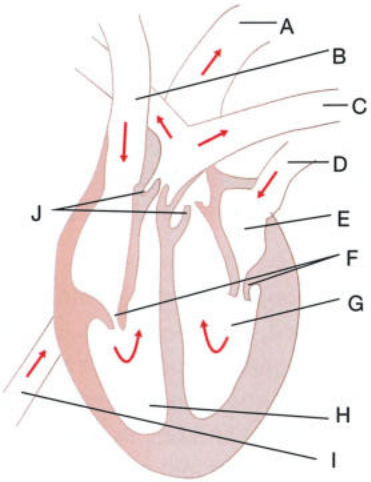


Figure 48.2 Human heart.

Label each part of the heart.

5. Define a coronary occlusion.
6. Construct a table to summarise the cause, risk factors, consequence and treatment for coronary occlusion.

49 Blood

Human blood can be divided into two parts – the plasma and the cellular components. The cellular components include red blood cells, white blood cells and platelets.

Plasma

Plasma is a straw coloured liquid that is about 90% water and carries many substances in solution, e.g. urea, carbon dioxide, salts, amino acids, hormones, vitamins, antibodies, clotting factors etc. The **water** in plasma travels as water molecules. Most **carbon dioxide**, about 70%, travels in the plasma in the form of hydrogen carbonate ions (HCO_3^-); about 23% combines with haemoglobin, forming carbaminohaemoglobin; and the rest, about 7%, dissolves directly in the plasma. **Salts** travel in the plasma as either positive or negative ions, e.g. sodium, potassium, calcium, magnesium and chlorine. In mammals **nitrogenous wastes** usually travel as urea. Many **lipids** are water insoluble and can only travel in blood when they are coated with proteins, becoming lipoproteins, and travel as high density lipoproteins (HDL) or low density lipoproteins (LDL). **Other products of digestion** travel as separate molecules, e.g. glucose and amino acids.

Erythrocytes

Erythrocytes or **red blood cells** (RBCs) are biconcave discs which contain the iron pigment haemoglobin. They have no nucleus, are made in bone marrow, remain in the blood for a short time (about 120 days) and are destroyed in the spleen and liver. RBCs are about 7 to 8 μm in diameter and 1 to 2 μm thick. In areas of high oxygen concentration, e.g. lungs, **oxygen** combines with haemoglobin to form oxyhaemoglobin and the oxygen is then efficiently transported through the blood vessels to areas of low oxygen concentration, e.g. body tissues, where it dissociates and diffuses into the tissues to be used in respiration.

The adaptive advantage of haemoglobin is that more oxygen can be carried, which allows a higher rate of respiration and the generation of more heat. If humans go to areas of high altitude where there is decreased oxygen levels, heartbeat and respiration rates will increase quickly. Mountain sickness, caused by quickly moving to high altitude, is indicated by breathlessness, heart irregularities, headache, nausea and oedema. If low oxygen levels persist, the bone marrow starts producing more RBCs so that there are more circulating RBCs to pick up oxygen. Indigenous people in these areas often have an increased number of capillaries in various tissues to assist diffusion to all cells.

Leucocytes

There are several types of **leucocytes** or **white blood cells** (WBCs) but all are involved in defence against infection. They are the largest blood cells and are less numerous than RBCs accounting for around 1% of blood volume. Leucocytes can move out of the blood vessels into the interstitial fluid. There are two main types of leucocytes – **granulocytes**, e.g. neutrophils, eosinophils and basophils and **agranulocytes**, e.g. monocytes and lymphocytes.

Phagocytes surround and ingest foreign substances such as bacteria in a process called phagocytosis.
B lymphocytes make antibodies.

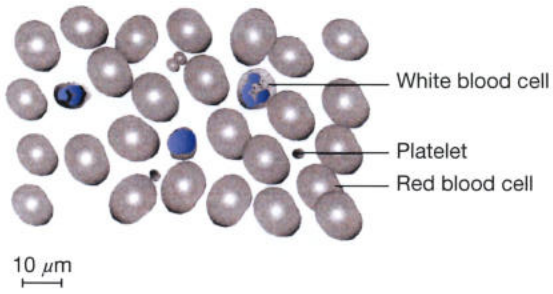


Figure 49.1 Components in blood.

Thrombocytes

Thrombocytes or **platelets** are fragments of cells and are important in clotting the blood. They are non-nucleated discs around 2 to 4 μm in diameter and are formed in red bone marrow. The life span of a platelet is 8 to 11 days and if not used in blood clotting are destroyed by macrophages, e.g. in the spleen.

QUESTIONS

1. (a) Identify the form in which oxygen is carried in mammalian blood.
   (b) The ability of oxygen to combine with oxygen depends on the partial pressure of oxygen and several other factors. The graph shows the oxygen dissociation curve for haemoglobin.

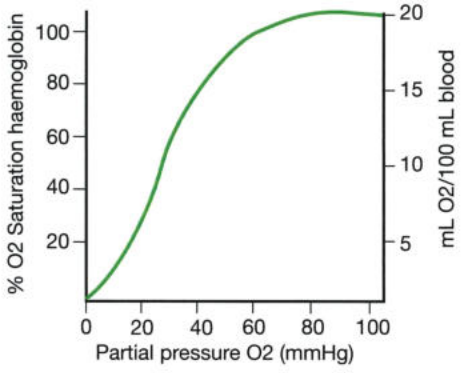


Figure 49.2 Oxygen dissociation curve for haemoglobin.

(i) The partial pressure in tissues at rest is usually, around 40 mmHg. What is the per cent oxygen saturation of haemoglobin in tissues at rest?

(ii) Explain this value in terms of oxygen loading/unloading.

(iii) On the graph, mark in the section which would correlate with the lungs.

(c) Explain why Chinese immigrants when they arrived in Tibet had high rates of in utero deaths.

2. The chemical composition of blood that leaves the liver may be very different from the blood that entered via the hepatic portal vein. Choose one substance and explain how and why its concentration changes due to the action of the liver.

3. Trace one change in the chemical composition of the blood that occurs when it flows through each of the following organs. Explain why this change occurs.
(a) Lungs. (b) Kidneys.

4. In humans, identify the form in which the majority of carbon dioxide is carried in blood.
(A) Combined with haemoglobin to form carbaminohaemoglobin.
(B) As hydrogen carbonate ions in the plasma.
(C) Dissolved directly in the plasma.
(D) Bonded to white blood cells.

5. Identify one of the main differences between the blood in capillaries and venules in body tissues with arterial blood.
(A) Arterial blood has higher oxygen levels.
(B) Arterial blood has higher carbon dioxide levels.
(C) Arterial blood has lower pH.
(D) Arterial blood has higher concentration of hydrogen carbonate ions.

6. Which of the following explains why gas exchange occurs in the mammalian lung?
(A) Partial pressure of oxygen is greater in the capillary than in the alveoli.
(B) Partial pressure of oxygen is greater in the alveoli than in the capillary.
(C) Partial pressure of nitrogen is greater in the capillary than in the alveoli.
(D) Partial pressure of carbon dioxide is greater in the alveoli than in the capillary.

7. Which of the following outlines one major adaptive advantage of haemoglobin?
(A) Allows the blood to carry more nitrogenous wastes.
(B) Changes osmotic pressure in blood for faster diffusion.
(C) Forms a permanent bond with oxygen forming a very stable compound.
(D) Allows blood to carry more oxygen.

8. As blood moves around the body, where would you expect to find the greatest concentration of urea?
(A) Renal artery. (B) Renal vein.
(C) Hepatic vein. (D) Vena cava.

9. Why is oxygen needed by living cells?
(A) To convert glucose to pyruvate which is the main form of energy in the body.
(B) To control the amount of carbon dioxide distributed in the plasma.
(C) To release large amounts of energy in the form of ADP.
(D) To release large amounts of energy in the form of ATP.

10. The diagram shows the structure of a villus with fatty acids and glycerol which are the end product in the digestion of fats.

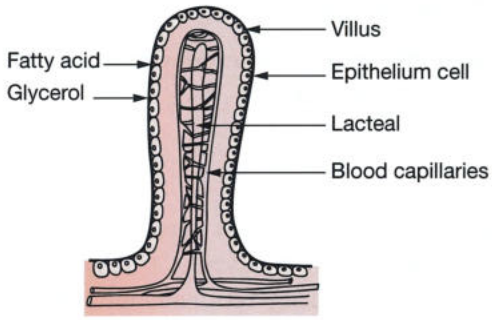


Figure 49.3 Structure of a villus.

What happens to the fatty acids and glycerol?
(A) They diffuse across the epithelium and enter the capillaries in the same form to travel as fatty acids and glycerol.
(B) They diffuse across the epithelium and are resynthesised in the lacteal to travel as chylomicron.
(C) They are absorbed by epithelial cells and transported to the lacteal to travel as fatty acids and glycerol.
(D) They are absorbed by epithelial cells, recombined to form chylomicron and transported to the lacteal.

11. Which of the following is a physiological response of humans to decreased oxygen concentrations at high altitudes?
(A) Decrease in respiration and heart rate.
(B) Using oxygen breathing masks.
(C) Formation of more red blood cells.
(D) Increase in body weight to change body mass index.

12. Identify the equation which summarises why oxygen is needed by living cells.
(A) Glucose + oxygen → carbon dioxide + water
(B) Carbon dioxide + water → glucose + oxygen
(C) Starch + oxygen → glucose + water
(D) Glucose + water → oxygen + carbon dioxide

50 Experiment – Scaled Diagram of Blood Cells

Some Human Biology students wanted to use a light microscope and a prepared slide of blood to estimate the size of red and white blood cells.

Choosing equipment

The students decided it was first necessary to work out the field of view under low power and then high power of the light microscope being used for the experiment.

They decided to use a prepared minigrid slide. This slide is like a transparent piece of graph paper with a 1 cm × 1 cm square on a microscope slide with lines drawn at 1 mm intervals.

The centre of this square has a 'mini' minigrid with lines drawn at 0.1 mm intervals.

Method

The students used a school light microscope and focused under low power, which involved a '×10' ocular lens and a '×10' objective lens. They estimated the field of view as shown in the following diagram.

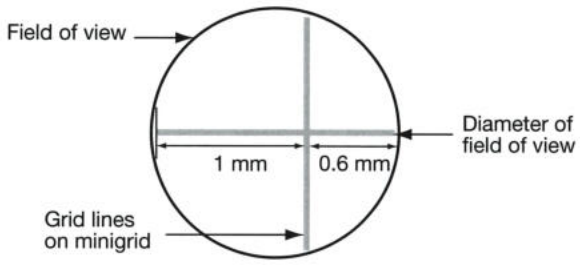


Figure 50.1 View of minigrid under low power.

To change to high power, the students rotated the nosepiece of the microscope to the '×40' objective lens.

The students then placed the prepared slide of blood on the microscope stage and estimated the number of red blood cells (RBCs) which would fit across the diameter of the field of view.

Results

The students estimated that 46 RBCs would fit across the diameter of the high power field of view.

To estimate the size of a white blood cell (WBC), the students compared the size of a WBC with a RBC and, knowing the scale for the field of view, estimated the size of the WBC. They found a WBC with a horseshoe shaped nucleus and, since it was slightly larger than the RBC, estimated that the diameter of the WBC was 10 μm.

QUESTIONS

1. Use Figure 50.1, 'View of minigrid under low power', to estimate the diameter of the field of view of the microscope used by the students in this experiment.
2. (a) When the students changed to high power, what was the new diameter of their field of view in millimetres?
   (b) Convert the diameter of high power field of view into micrometres.
3. Using the student results that 42 red blood cells would fit across the diameter of the high power field of view, what is the estimated diameter of one RBC?
4. Compare this result with the usually accepted diameter of RBCs and account for any differences.
5. The following diagram shows the cellular components of blood and their relative sizes.

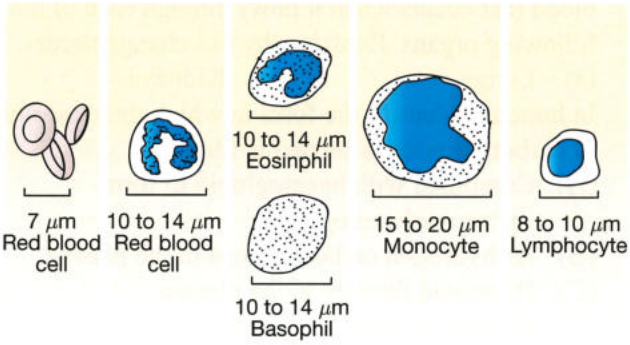


Figure 50.2 Cellular components of blood.

   (a) Use these diagram to identify the type of WBC measured by the students.
   (b) Discuss how the students would have distinguished between the RBCs and WBCs under the microscope.
6. The diagram shows a scaled drawing of a red blood cell and a white blood cell.

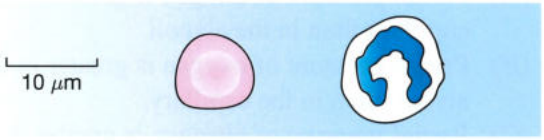


Figure 50.3 Scaled drawing of red blood cell and white blood cell.

Measure the diameter of each of these cells and identify which of the following is the closest correct answer.

	Red blood cell	White blood cell
(A)	11 μm	8 μm
(B)	8 μm	11 μm
(C)	16 μm	22 μm
(D)	16 μm	22 μm

51 Haemoglobin

Haemoglobin is a globular protein consisting of a globin (protein) and an iron-containing substance (haem). It is synthesised inside developing red blood cells in red bone marrow.

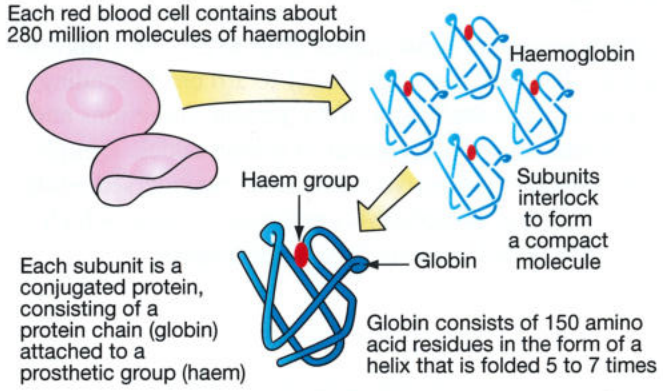


Figure 51.1 Haemoglobin.

Measuring haemoglobin

The measurement of haemoglobin content is a routine procedure in clinical practice. The **haemoglobin value** is the weight of haemoglobin in whole blood and is measured in grams per 100 mL, e.g. normal values for haemoglobin (Hb) for males is 13 to 18 g/100 mL and for females 11.5 to 16.5 g/100 mL.

The **mean cell haemoglobin** (MCH) is the average amount of haemoglobin in each cell measured in pictograms (pg = 10^{-12} gram).

The **mean cell haemoglobin concentration** is the amount of haemoglobin in 100 mL of red cells.

Function of haemoglobin

Haemoglobin has several functions. The main function of haemoglobin is the transport of oxygen from areas of high oxygen concentration to areas of low oxygen concentration. However, it also is involved in the transport of some carbon dioxide and it helps to act as a buffer in the blood – preventing dramatic changes in pH which would seriously disrupt metabolism.

Haemoglobin acts as a buffer by picking up most of the hydrogen ions (H^+) when the carbon dioxide reacts with water to form carbonic acid. The carbonic acid breaks apart into hydrogen ions (H^+) and hydrogen carbonate ions (HCO_3^-). The reactions can be summarised as:

$$CO_2 + H_2O \rightarrow H_2CO_3 \rightarrow H^+ + HCO_3^-$$

Because HCO_3^- affects the pH of a solution, the transport of carbon dioxide in venous blood changes the pH of blood – the pH of blood in veins is 7.34 while the pH of blood in arteries is 7.4.

Haemoglobin is a globular protein made of 4 polypeptide chains which are folded to enclose a haem group. A haem group is a subunit with an iron(II) ion at the centre. The iron ion combines with the oxygen to form oxyhaemoglobin. There is cooperation between the 4 subunits of the haemoglobin when loading and unloading oxygen. When one oxygen molecule combines with one of the haem groups, there is a slight change in the shape of the molecule which increases the affinity of haemoglobin for more oxygen.

QUESTIONS

1. What is haemoglobin?
2. Where is haemoglobin synthesised?
3. What is meant by the haemoglobin value?
4. State the three functions of haemoglobin.
5. Haemoglobin has a far greater affinity for carbon monoxide than for oxygen. Carbon monoxide is found in coal gas used for heating and cooking and in car exhausts. Which of the following explains why death occurs from carbon monoxide poisoning?
   (A) Insufficient oxygen carried to the tissues.
   (B) White cells detecting changed appearance of haemoglobin.
   (C) Increased affinity of haemoglobin for oxygen.
   (D) Decreased ability of plasma to carry carbon dioxide.
6. The graph shows the oxygen dissociation curve for foetal and maternal human haemoglobin.

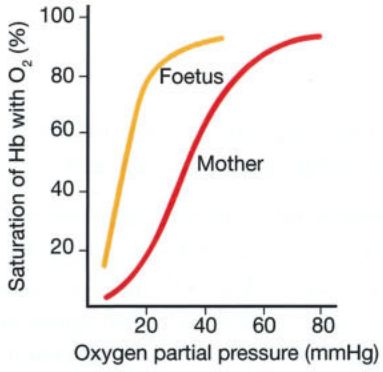


Figure 51.2 Oxygen dissociation curve for foetal and maternal human haemoglobin.

Which of the following is shown by these curves?
(A) Foetal haemoglobin loads at higher oxygen partial pressure.
(B) Maternal haemoglobin has a higher affinity for oxygen than foetal haemoglobin.
(C) Maternal haemoglobin is more alkaline.
(D) Foetal haemoglobin loads at lower oxygen partial pressure.

52 The Lymphatic System

The lymphatic system can be considered to be a part of the vascular system as it removes interstitial fluid from tissues and takes it to the heart paralleling veins. This is a one-way movement of fluid from the tissues to the blood circulation. The lymphatic system also plays an important role in the immune response.

When tissue fluid is absorbed into lymph capillaries it is called **lymph**. Since proteins and large particulate matter cannot be absorbed directly into blood capillaries, the lymphatic system is important as it moves these substances away from the tissue spaces.

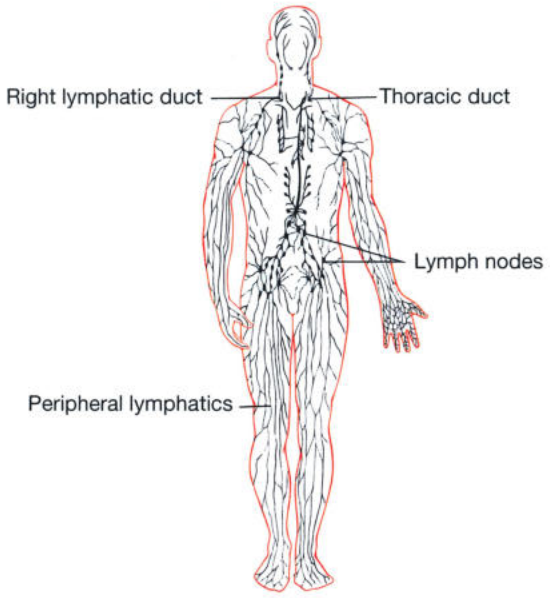


Figure 52.1 Lymphatic system.

Composition of lymphatic system

The organs in the lymphatic system include lymphatic vessels, lymph nodes, spleen and thymus.

Lymphatic vessels are veins and capillaries which carry lymph and lymphocytes from peripheral tissues to the cardiovascular system. Lymphatic drainage from the lower limbs, pelvis, abdomen, left upper side of the thorax, neck and head drains into the **thoracic duct** while lymphatic drainage from the right upper limb and right side of the thorax, neck and head drains into the smaller **right lymphatic duct**. Compared to blood capillaries lymphatic capillaries are larger in diameter and more irregular. The walls of lymph veins are thinner than the walls of blood vessel veins with valves in both types of veins preventing backflow. Unlike blood vessels which form a continuous circuit in the body the lymph capillaries begin at blind ends in the tissues. Interstitial fluid seeps into the lymph capillaries which connect to larger and larger ducts.

The walls of the lymph capillaries are held in place to the surrounding connective tissue by fibres that keep the capillary open. Gaps between the cells allow interstitial fluid and proteins and particulate matter such as cell debris and bacteria to be pushed into the lymph capillary by bulk flow. In mammals lymph is moved by the contractions of body muscles with the valves stopping backflow.

Lymph nodes are bean shaped structures with a fibrous capsule that vary in size from 0.1 to 2.0 cm depending on location and the health of the person. Infections and cancer can cause enlargement of lymph nodes. Lymph nodes contain connective tissue with interspersed white blood cells, e.g. lymphocytes and macrophages which filter, capture and destroy foreign pathogens.

Function of lymphatic system

The lymphatic system has several functions.

- Tissue fluid is collected and returned to the bloodstream.
- Transports fats absorbed from the digestive tract to the bloodstream.
- The walls of lymph vessels are permeable to proteins so that blood proteins lost by leakage from blood capillaries are returned to the blood in the lymph. This maintains normal osmotic balance between blood and tissue which is highly important as without the return of the proteins a person would die within about 24 hours.
- Lymph nodes filter tissue debris and invading pathogens from the lymph and phagocytes destroy and remove these substances.
- Non-digestible particles, e.g. dust and soot that phagocytes cannot remove are stored in lymph nodes.

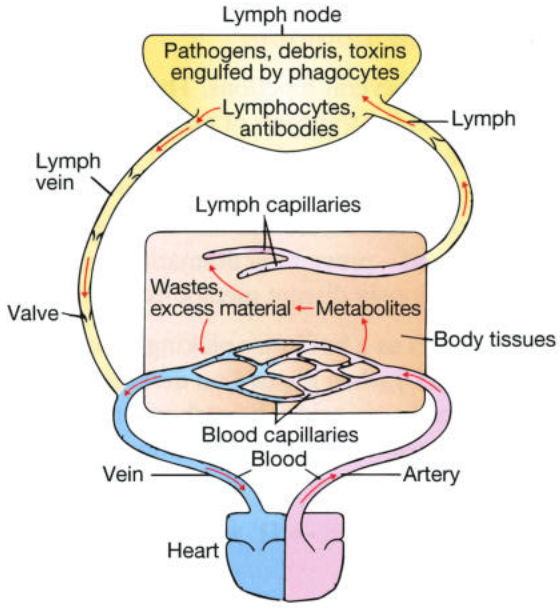


Figure 52.2 Lymphatic circulation.

Lymphatic circulation begins at blind-end lymph capillaries in tissues and ends when lymph empties into venous blood circulation via the thoracic ducts.

Malfunction of lymphatic system

If major lymph vessels become blocked, the protein concentration in tissue fluid can rise and upset the osmotic balance in the tissues. This will cause less fluid to be reabsorbed by blood capillaries leading to excessive swelling in the tissue and severe oedema.

QUESTIONS

1. Define lymph.
2. Explain why movement in the lymphatic system is a one-way movement.
3. Construct a table to compare lymph vessels with blood vessels.
4. Identify the organ components of the lymphatic system.
5. Construct a table to summarise how the lymphatic system interacts with the cardiovascular system, the digestive system and the immune system.
6. The diagram shows a lymph valve.

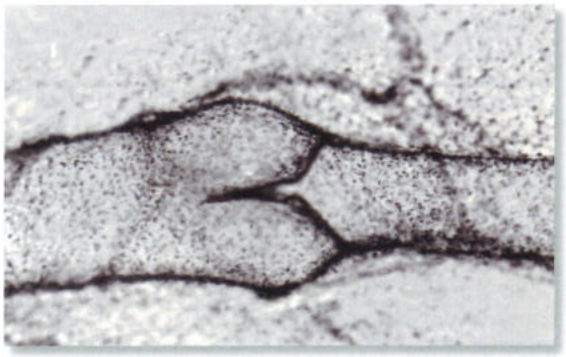


Figure 52.3 Lymph valve.

Where are lymph valves and what is their function?

7. In mammals blood flow in arteries relies on the beating of the heart to maintain blood pressure. What assists the flow of lymph in the lymphatic system?
8. Describe the size and shape of lymph nodes.
9. List the main functions of the lymphatic system.
10. Explain why the recycling of blood proteins is an important function of the lymphatic system.
11. (a) Explain why an immobilised limb will frequently swell in a condition called oedema.
    (b) What treatment is recommended to prevent oedema occurring?
12. Explain how the release of histamines in the inflammation response affects the flow of lymph.
13. Elephantiasis is a chronic condition caused by parasites blocking the lymph vessels in the lower legs. Explain why this condition causes massive enlargement of the legs.
14. The diagram shows the movement of lymph through a lymph node.

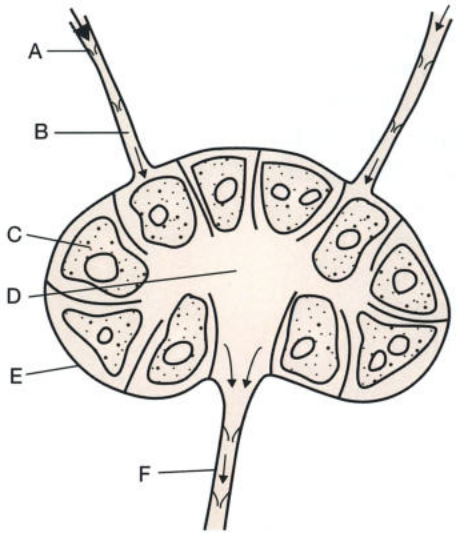


Figure 52.4 Movement through a lymph node.

Identify parts A, B, C, D, E and F.

Use the following diagram for the next TWO questions.

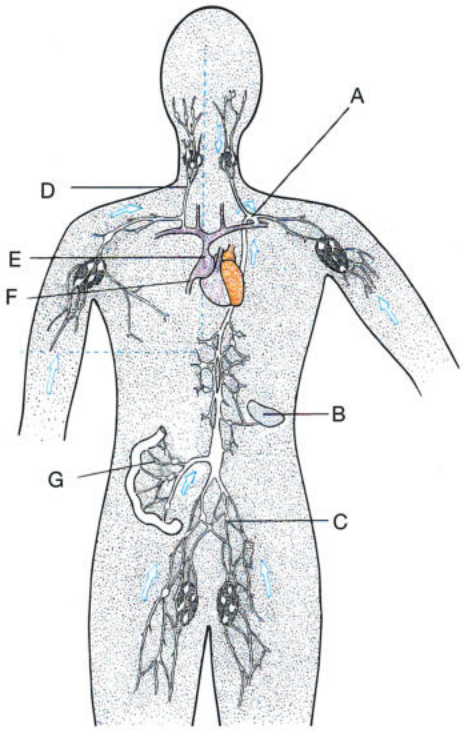


Figure 52.5 Lymphatic system.

15. Which lymph vessel receives lymph from the lower limbs, pelvis, abdomen, left upper side of the thorax, neck and head?
    (A) Part A (B) Part B (C) Part C (D) Part D
16. Peyer's patches have large lymphoid follicles that aid defence against microbes in food. Which shows the location of Peyer's patches?
    (A) Part A (B) Part B (C) Part C (D) Part G

53 Play – In the Blood

Dramatic personae:
Robert Red Bloodcell
Rupert Red Bloodcell
Willie White Bloodcell
Wally White Bloodcell
Owen Oxygen
Bertie Bacteria
Betty Bacteria
Pauline Platelet
Penny Platelet
Cain Sugar
Hannah Hormone

SCENE 1 – END OF A LONG BONE IN A LEG

(Robert and Rupert Red Bloodcell are talking in the end of a long bone)

Robert I've just been made in the marrow and I love my shape; what do you call this?

(Rupert studies Robert)

Rupert I'd call it disc shape and personally I think that the thin section in the middle looks a bit pale. I prefer cubes to donut shapes.

Robert You mean you'd prefer to be square than this lovely round disc?

Rupert Yes. Though I do like my haemoglobin. It gives me a nice colour and my size is just perfect. How big are we?

Robert We're about 8 μm in diameter. Just perfect for fitting through little capillaries. My shape and size is much better than those large white blood cells.

Robert Stop being so vain, it's time to swim and see where this blood vessel goes.

Rupert Okay. Oh no! I just lost my nucleus. What does this mean?

Robert It means you're going to die and can't reproduce!

Rupert If that's the case, then let's enjoy ourselves before we go to that big red spleen in the abdomen!

Willie Excuse me, excuse me, out of the way. I'm on my way to an accident. There's a cut in the foot.

Wally Hurry up, move your red corpuscle, the white warriors are coming through and we have priority passage.

(Robert and Rupert watch the two white blood cells pushing their way along the blood vessel)

Rupert Let's follow those white blood cells. I love a good fight. It sounds as though there's trouble in the foot. Do you think they need us?

Robert I want to go to the lungs as I'm feeling a little blue, so I'll meet you later. How about we meet at the Hepatic Portal Vein for afternoon coffee and biscuits?

Rupert Sounds good. See you later at the Hepatic Portal Vein. By the way, where is this vein?

Robert The Hepatic Portal Vein takes digested food, like glucose and amino acids, to the liver.

Rupert Oh THAT vein! I'll meet you later and tell you all about the fight in the foot.

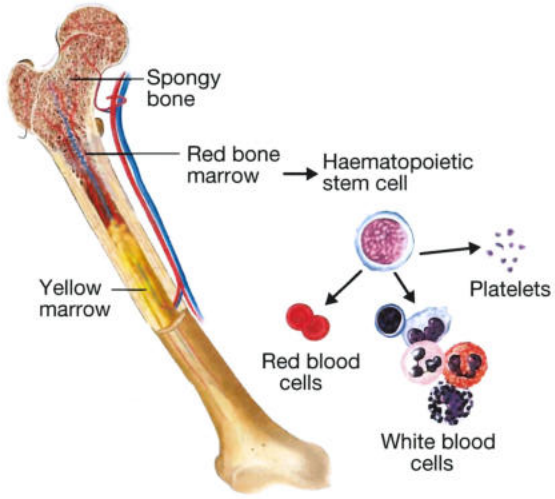


Figure 53.1 End of a long bone in a leg.

Questions Scene 1

1. Where do red blood cells form?
2. Describe the shape and size of red cells.
3. Why do red cells have a limited life span?
4. Why would a red blood cell be blue?
5. Explain why they go to the Hepatic Portal Vein for afternoon tea.

SCENE 2 – THE FOOT

(Bacteria are attacking through a cut in the foot)

Bertie Charge the foot, mates! We've got to infect quickly!

Betty I'll give this foot gangrene, and then it'll have to be amputated. It's fun being a bacterium.

Willie Stop! In the name of the immune system! I'm a law and order officer of the white cell squadron!

Wally I'll get Betty and engulf her!

Betty No you won't! You oversized phagocyte!

Wally Phagocytes are specialised white blood cells. We are the elite and we engulf horrible bacteria, like you.

Betty No! Go away! Don't come near me!

Bertie Keep fighting, Betty! Don't give in! Bacteria can win if we stick together.

Willie I'll get Bertie!

(Willie and Wally, the white blood cells, fight Betty and Bertie Bacteria. A savage battle follows, with valiant efforts made by both sides)

Betty I may be the smallest cell around this place but I could've taken over the foot.

Wally But I'm here to see that you don't, even if I die also.

Betty I've killed 100 skin cells, so I can die happy.

Wally And I'm dying too, so what does that make us?

Willy Pus. Dead cells, bacteria and white cells make pus.

Pauline Hurry up Penny. If we don't hurry up and get to work, this foot is going to bleed to death.

Penny I'm ready to clot! What about all that pus in the way?

Pauline That's not our problem. Our job is to clot the blood.

Rupert If you two platelets are going to clot the blood, I'm leaving. Your union may allow scabs, but I'm not working at this site. This is no place for a red blood cell.

Penny If this gash doesn't seal up soon it'll need a toe-truck instead of a band-aid.

Pauline I love being a platelet and clotting blood. It's a pity that Rupert escaped. He would have made a nice part of this scab.

Questions Scene 2

1. What is the function of phagocytes?
2. What happens when bacteria invades the human body?
3. What is the function of platelets?
4. What forms 'pus'?

SCENE 3 – THE HEPATIC PORTAL VEIN

(Rupert arrives at the Hepatic Portal Vein)

Rupert Robert, are you here? I'm here for afternoon tea.

Robert Here I am. I've just been to the lungs.

Rupert What happened in the lungs?

Robert I picked up Owen Oxygen. He attached to my haemoglobin.

Owen Hello Rupert. Pleased to meet you.

Rupert Welcome to our afternoon tea party. Where's the food?

Cain I'm Cain the Sugar molecule. What do you mean afternoon tea?

Robert We heard there was free food here in the afternoon.

Cain Well, I used to be starch in a potato chip, and then I was broken down into glucose in the digestive system.

Then I diffused across the wall of a villus in the small intestine, but I'm not food for you. I'm ready for a reaction in respiration.

Rupert Why not? A reaction sounds good. Owen Oxygen and I had a reaction in the lungs on my way here. It was fun picking up oxygen.

Robert Sounds like a hitchhiker to me.

Rupert Don't be so boring. Another reaction sounds fun.

Cain The reaction is called respiration and sugar is the most important reactant.

Owen I beg your pardon, but it's the oxygen that's the most important reactant.

Rupert How am I involved?

Cain You're the transport but we can't have the reaction here. This is not the right place.

Robert The right place? What's the matter with this place?

Owen What's the matter with the Hepatic Portal Vein?

Rupert Is this vein too wide?

Cain Yes it is. Respiration is needed in every living cell and we have to go to a needy cell for our reaction.

Owen We'll have to dissociate.

Robert You want to WHAT?

Owen Dissociate. When the oxygen concentration gets low, oxygen dissociates from the haemoglobin. How long will it take to get to a needy cell?

Cain It only takes 45 seconds for a red blood cell to complete one circuit of the circulatory system.

Rupert Well if that's the case, let's get going. Are there any side effects I should know about?

Cain You'll lose your oxygen.

Rupert What! That's not fair. I went all the way to the lungs to find Owen.

Robert You can get more oxygen in the alveoli.

Rupert But it won't be Owen.

(They all travel together for awhile and then suddenly Owen breaks away from Rupert)

Rupert What's happening?

Owen Help! Help! The diffusion gradient is too strong.

Cain Goodbye Rupert, I'm taking your oxygen and I'm diffusing out of this capillary into the tissues.

Owen Help me!

Rupert Come back here, thief!

Cain Ha! Ha! You're too late. I've got your oxygen.

Robert That glucose just diffused away and you lost Owen! He was such a nice chap.

Rupert Cain's a THIEF! They should lock him up in a dark cell.

Robert Their reaction is probably very boring.

Rupert He said I'd be part of the reaction. Why should they be the only ones to release energy?

Robert Well, if we meet again, we can pay him back with some mean trick of our own.

Questions Scene 3

1. How does sugar get into the circulatory system?
2. What is the chemical reaction? Write the word equation for this reaction.
3. Where does the chemical reaction occur?
4. What happens to red blood cells in the lungs?
5. How long does it take for blood to complete one circuit of the system?

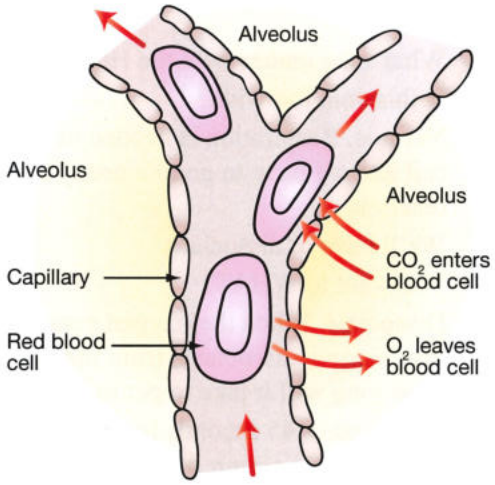


Figure 16.1 A capillary – red blood cells flow in single file.

SCENE 4 – A SMALL CAPILLARY

(The two friends Rupert and Robert meet again)

Robert Have you found that sweet sugar?

Rupert Sweet? That Cain Sugar stole Owen away from me. Cain was not sweet.

Robert They both diffused through the wall of the capillary, but we can't seem to get out that way.

Rupert I don't want to go out there anyway. There's lymph out there and it has no colour at all. Very boring.

Robert Does this mean we're stuck between these walls forever?

Hannah Hello red men. What's going on here?

Rupert Nothing.

Robert He had a friend called Owen and was looking forward to a special reaction and then Owen diffused off with Cain.

Rupert It's true. I lost my oxygen and now I'm blue.

Hannah Well I'm a hormone, and I can't help you personally, but I'll keep you company on the journey.

Robert Most kind of you. And which hormone are you in particular?

Hannah Actually, I'm insulin. I was made in the pancreas and my job is to stop the formation of new glucose.

Rupert and **Robert** What!!!!

Rupert Did you say STOP the formation of glucose?

Hannah Yes. I make the liver and muscles change the glucose from the intestine into glycogen for storage.

Robert Fantastic!

Rupert It certainly is.

Robert We were conned and led up the garden path, or hepatic vein to be precise, by a certain sugar molecule.

Rupert And we would just love to escort a hormone that's going to put sugar into storage for a long time.

Hannah Well I'm glad to be a help. Just don't go into insulin shock with the lack of sugar.

Robert We won't. We're so glad you came along to cheer up our short lives.

Hannah Do you want to go to the liver with me?

Rupert Can you change Cain into glycogen?

Hannah Of course I can. Cain will be trapped in the liver until the body needs more glucose.

Robert Fantastic revenge!

Rupert Let's go friends, my revenge will be sweet.

Questions Scene 4

1. How did the glucose molecules move through the wall?
2. What makes red blood cells become 'blue'?
3. Where is insulin made?
4. What is the function of insulin?
5. When do people suffer 'insulin shock'?
6. What is found in blood?

54 Donated Blood and Its Products

Blood is a type of connective tissue that contains cells in a liquid matrix (plasma). An average adult has approximately 5 to 6 litres of blood which makes up about 7% of the body weight for an adult male. The proportion is usually less in women and higher in children.

Functions of blood

Blood is needed for the transport of nutrients from the alimentary tract and oxygen from the lungs to cells and for the transport of wastes to excretory organs. It also carries heat for thermoregulation from active tissues to less active tissues, hormones for coordination functions in osmotic regulation and stabilising pH, and transports substances for defence, e.g. antibodies and clotting factors.

Donated blood

The Australian Red Cross collects donated blood from a donor. Donated blood can be either directed, standard or autologous donation.

Directed blood is given by friends or family and is donated for a specific person.

Standard donation involves the removal of 450 mL of blood for a non-specific purpose and usually takes 15 minutes.

Autologous donation is when a patient's own blood may be collected for future transfusions.

There needs to be a constant supply of donated blood as platelets only have a shelf life of up to 5 days, red cells have a shelf life of up to 42 days and plasma can only be stored for up to one year.

Most donated blood is used to assist people with major medical conditions such as heart disease, cancer, e.g. leukaemia, haemophilia or diseases of specific organs, e.g. stomach, bowel, liver or kidney. Products of donated blood are also used to assist burn and accident victims or used in emergency surgery.

Regulations and restrictions for donated blood

There are several regulations concerning the eligibility of people allowed to donate blood, e.g. age, recent illness, past medical history and lifestyle issues. Donors must be aged between 16 and 70 years old, weigh at least 45 kg and have normal temperature and blood pressure.

If you have just had your ears pierced with clean single use equipment only your donated plasma can be used for the 24 hours after the ear piercing procedure while if you have had other body piercing using clean single use equipment only the plasma portion can be used for 4 months after the procedure. After these times full donated blood can be used.

If you have travelled to an overseas area that has a malaria risk, you can only donate the plasma portion for 4 months after returning to Australia. From 4 months to 12 months after returning the Australian Red Cross will carry out a malaria test on the blood and if the test is negative the blood can be used for transfusion.

Donated blood is tested for blood type, Rh group and red cell antibodies. Blood is also tested for infectious diseases – HIV 1 and 2; hepatitis B core antibody; hepatitis C virus; human T cell lymphotropic virus, type 1(HTLV); hepatitis B surface antigens; and syphilis.

If a donor has an abnormal result for any of these tests, they are contacted and confidentially advised about their health issue.

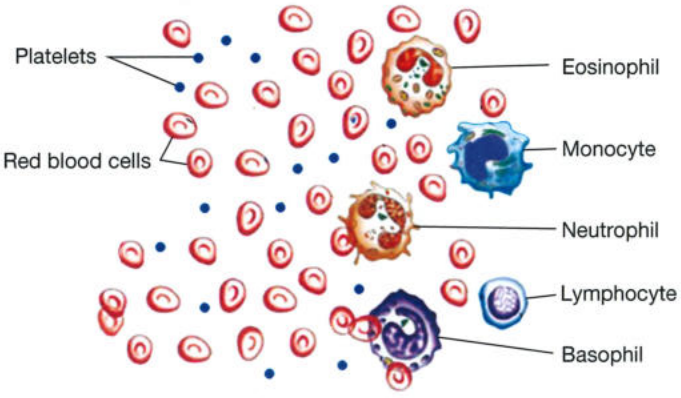


Figure 54.1 Cellular components in blood.

Components of donated blood

Donated blood is centrifuged at different speeds to collect the different components – red blood cells, platelets and plasma.

Red blood cells make up around 40% by volume of blood and the red haemoglobin combines with oxygen to form oxyhaemoglobin. Haemoglobin transports oxygen from the lungs to needy cells. Red blood cells are refrigerated and have a shelf life of up to 42 days. They are used to treat people with anaemia, accident victims, during surgery and for those people whose red blood cells do not function correctly.

Platelets are fragments that assist in the clotting of blood. If the platelet count is too low, people may suffer from spontaneous bleeding. Platelets are needed for people with diseases such as leukaemia or people having chemotherapy. Platelets have a shelf life of 5 days.

Plasma is the straw coloured liquid that is up to 90% water and carries many substances in solution, e.g. urea, carbon dioxide, salts, amino acids, hormones, vitamins and antibodies. Extraction of different products from plasma assists a variety of medical conditions. Fresh frozen plasma is used to treat people who have clotting problems, e.g. after a liver transplant or after trauma. Donated plasma is also used to obtain intragam, immunoglobulins, anti-D, albumex and prothrombinex. Plasma has a shelf life of 12 months.

Packaging of donated blood

Donated blood is packaged into several main products, including whole blood, packaged RBC, platelet concentrates, plasma, cryoprecipitated AHF, and plasma fractions such as stable plasma protein solution (SPPS), immune serum globulin and hyperimmune globulins.

Packaged red blood cells are extracted from whole blood by removing the liquid plasma, e.g. whole blood is centrifuged and because red blood cells weigh the most they will settle to the bottom. Red blood cells contain the iron-rich pigment haemoglobin which carries oxygen in the blood. The red blood cells make up approximately 47% of blood volume in an adult male. Packaged RBCs are used to increase the haemoglobin levels of a patient while not increasing the blood volume. People suffering from chronic anaemia, kidney failure or have acute blood loss from surgery or trauma receive packaged RBCs.

Cryoprecipitated AHF is removed from the plasma by freezing and then slowly thawing the plasma. Cryoprecipitate from fresh frozen plasma contains factor VIII, fibrinogen, von Willebrand factor and factor XIII and is used for liver transplants and people who have clotting problems related to deficiencies of these proteins. Patients with haemophilia or inherited coagulation abnormalities use cryoprecipitated AHF to control or prevent bleeding.

Plasma is collected by a process called **apheresis** where a separating machine takes blood from the arm and separates out and collects the plasma or platelets and returns the rest back into the arm. Fresh frozen plasma (FFP) is used in blood transfusions for patients who are bleeding or at risk of bleeding, e.g. for patients with cardiac bypass, liver disease and during massive transfusions.

QUESTIONS

1. What type of tissue is blood?
2. How much blood is in the average adult?
3. List some of the functions of blood.
4. Distinguish between directed donation, standard donation and autologous donation.
5. Outline some of the regulations and restrictions for blood donation.
6. Why does there need to be a constant supply of donated blood?
7. What are the main uses of donated blood?
8. Identify the main components of blood.
9. What is apheresis?
10. (a) Identify four products extracted from donated blood.
    (b) Describe one of these fractions and how it is extracted.
    (c) Explain the use of the fraction you chose in part (b).
11. (a) Name some infectious diseases for which blood is tested.
    (b) Name some other tests carried out on donated blood.
12. How is donated blood separated into fractions?
13. Construct a table to summarise the products extracted from donated blood and the use of each product.
14. The table shows what happens to donated blood.

Table 54.1 What happens to donated blood.

Use of donated blood	Donation percentage (%)
Cancer	30
Heart disease	15
Stomach/bowel disease	15
Burn victim	12
Accident victim	12
Haemophilia	5
Others	11

Construct a pie graph to show this data.

15. Explain why storage of donated blood needs to be at low temperatures.
16. Which correctly identifies a product extracted from donated blood and its use?

	Extracted product 1		Extracted product 2	
	Name of product	Its use	Name of Product	Its use
(A)	Red blood cells	Acute blood loss	Platelets	Anaemia
(B)	Red blood cells	Anaemia	Factor VIII	Haemophilia
(C)	Albumin	Anaemia	Platelets	Severe bleeding
(D)	Plasma	Haemophilia	Factor VIII	Thrombosis

17. Why is it not advisable to freeze many blood products, except plasma?
    (A) Water crystals can cause cells to burst.
    (B) Freezing changes the intracellular pH.
    (C) Freezing causes bacterial endospores to form.
    (D) Cells shrink as ice has a smaller volume than water.
18. Which of the following donated blood products is used to assist blood clotting?
    (A) Lymphocytes from plasma.
    (B) Red blood cells from plasma.
    (C) Interferon from plasma.
    (D) Factor VIII from plasma.

55 Artificial Blood

The term 'artificial blood' is misleading. Blood has many functions while artificial blood or blood substitutes are only designed to carry oxygen and carbon dioxide. Blood substitutes do not carry out many of the other functions of blood, e.g. they are not involved in clotting and coagulation and immune defence.

Need for artificial blood

There are several reasons why there is research into blood substitutes.

- At present there can be supply difficulties when levels of donated blood are below requirement needs.
- Donated blood has a relatively short shelf life, e.g. platelets have a shelf life of 5 days which means donated blood can be thrown away if it has passed its 'use by' date.
- At present there is an element of risk of some pathogens being transferred from donor to recipient in blood transfusions.

There are several advantages in using blood substitutes.

- They are free of infectious agents and allergens making them non-toxic and disease free, e.g. not contaminated with HIV, hepatitis C, prion infections.
- They do not need refrigeration for storage and can be kept for a longer period, e.g. 2 to 3 years compared to 3 weeks for human blood.
- There is universal acceptance by all blood groups allowing transfusion to a person without any tests.
- It is readily available in large supplies solving the worldwide problem of blood donors.

Volume expanders

Volume expanders are fluid solutions that are inert and used to increase blood volume. Volume expanders are important in emergency situations, e.g. used by paramedics and in first response situations. The blood is diluted and has a lower concentration of red blood cells. Non-oxygen carrying blood substitutes can be crystalloid solutions which contain salts and/or sugars, e.g. saline solution or Ringer's lactate or they can be colloidal solutions which contain proteins, starch or gelatine, e.g. using albumin. These plasma expanders are effective for blood loss and most have few negative side effects.

Oxygen carriers

There are two basic types of oxygen carriers – one is based on perfluorochemicals and the other on haemoglobin.

Perfluorocarbons, e.g. oxycyte and fluorovent easily dissolve oxygen and carbon dioxide and can transport these gases to tissues/lungs respectively. They are combined with other materials, e.g. lipids to form an emulsion which can be injected into the patient and used in surgery, trauma, oxygenation of tumours and with chemotherapy. Benefits of perfluorocarbons include their ability to carry large amounts of oxygen, they can be stored at room temperature, they do not need typing and cross-matching and most are inert and chemically stable and they can be fully sterilised. The amount of oxygen they can pick up is directly proportional to the amount of oxygen breathed in. Oxygen accumulation in tissues can damage the tissues and often the PFCs have a short circulating life as they are quickly cleared within a few days from the plasma.

Haemoglobin based oxygen carriers (HBOCs) are based on haemoglobin from humans, animals or recombinant technology, e.g. PolyHeme and Hemolink are human based haemoglobin solutions and Hemopure is a bovine based haemoglobin solution. They combine with oxygen, have a long shelf life, do not require refrigeration, are ready to use and are universally compatible. Problems can arise as some HBOCs cause vasoconstriction and there can be gastrointestinal side effects, e.g. nausea, vomiting, bloating. There can be renal damage if the modifications made to the haemoglobin increase its size. Clinical trials are ongoing as HBOCs are not protected by a cell membrane and hence not protected from degradation.

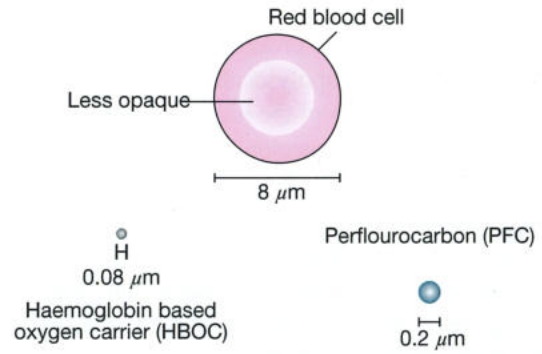


Figure 55.1 Oxygen carriers.

QUESTIONS

1. Explain why the term 'artificial blood' is misleading.
2. Describe four benefits of using a blood substitute.
3. Explain why there is research into artificial blood.
4. List some features needed for a blood substitute to be successful.
5. What are volume expanders?
6. Construct a table to compare the two types of oxygen carrier blood substitutes.
7. What is a benefit of using artificial blood?
   (A) Fewer blood donations would be needed.
   (B) Less risk of spread of infectious disease.
   (C) Matching blood types do not have to be found.
   (D) All of the above.

56 Blood Transfusions

Blood transfusion involves the injection of blood into a person usually from one person to another. The donated blood needs to match the recipient's blood type so that blood transfusions rely on the determination of blood groups. Incorrect matching of blood groups for both the ABO system and RhD group will lead to complications. In an emergency situation when the patient's blood group is unknown O Rh negative is given.

Blood transfusions can be used to treat many different diseases and conditions, e.g. for cancers, for operations, for anaemia, iron deficiency, bleeding disorders, pregnancy and childbirth. There are different types of blood transfusions, e.g. whole blood transfusion or the transfusion of particular blood components.

Table 56.1 Human blood groups.

Blood type	Possible genotype	Red blood cell self antigen (agglutinogen)	Serum antibodies produced
A	I^AI^A, I^Ai	Self-antigen A	Anti-B
B	I^BI^B, I^Bi	Self-antigen B	Anti-A
AB	I^AI^B	Self-antigen A, self-antigen B	No antibodies to A or B
O	ii		Anti-A, Anti-B

ABO blood system

The ABO system shows the presence/absence of two agglutinogens (A and B) which are mucopolysaccharides. These agglutinogens are on the external surface of the plasma membrane of red blood cells and act as antigens. A person with A agglutinogen has blood group A, a person with B agglutinogen has blood group B and with neither agglutinogen has blood group O. For each agglutinogen there is a corresponding antibody (agglutinin) in the plasma. The plasma of group A blood contains b agglutinin (anti-B) only. The plasma of group B blood contains anti-A and the plasma of group O blood has both anti A and anti B.

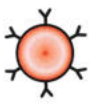
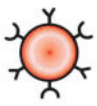
Blood group	A	B	AB	O
Agglutinogen	Agglutinogen A	Agglutinogen B	Agglutinogen A + B	No agglutinogens
Red blood cells with agglutinogens on plasma membrane (antigens)				
Agglutinins in plasma (antibodies)	Agglutinin anti-B	Agglutinin anti-A	No agglutinins	Agglutinins anti-A + anti-B

Figure 56.1 Blood groups.

Blood typing tests are important in matching blood types for blood transfusions. Incorrect matching can be life threatening, e.g. if a person with blood type O is given a transfusion of blood type A blood then an immune reaction will occur with the anti-A antibodies of type O recipient binding to the transfused type A red blood cells causing **agglutination** (clumping of the red cells). Antibody binding also activates the complement system that causes membrane attack of the transfused cells making them swell and lose haemoglobin.

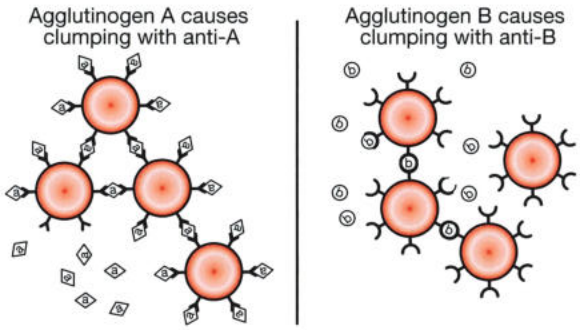


Figure 56.2 Agglutination.

Rhesus system

Research with rhesus monkeys led to the discovery of the rhesus factor. The rhesus factor is a group of antigenic substances on the membrane of red blood cells including D-antigen. Around 83% of the human population have D-antigen and are rhesus positive (Rh+). There are no plasma antibodies for the rhesus factor. However, if a person who is Rh− has blood contact with Rh+ blood (e.g. during childbirth) they will form anti-D antibodies. If a Rh− female has a Rh+ child and then falls pregnant and is carrying another Rh+ child maternal anti-D antibodies can cross the placenta to act against the baby causing haemolytic disease. To stop the next child being potentially a 'rhesus baby' a Rh− mother who gives birth to a Rh+ baby is given an injection of anti-D antibodies to destroy the Rh+ cells that have entered her circulation. This destroys the cells before her own immune system can be sensitised to Rh+ blood.

Table 56.2 Blood donation.

	Donor's blood type							
Blood patient's type	AB+	AB−	A+	A−	B+	B−	O+	O−
AB+	✓	✓	✓	✓	✓	✓	✓	✓
AB−		✓		✓		✓		✓
A+			✓	✓			✓	✓
A−				✓				✓
B+					✓	✓	✓	✓
B−						✓		✓
O+							✓	✓
O−								✓

Types of blood transfusions

There are several types of blood transfusions.

Red cell transfusion, e.g. for low haemoglobin or red cell count if a person has anaemia.

Platelet transfusion, e.g. if person has a low platelet count.

Plasma, e.g. fresh frozen plasma (FFP), e.g. to replace low or missing blood proteins or blood volume in a range of medical situations.

Cryodepleted plasma is plasma with some blood clotting proteins removed and is used for thrombotic thrombocytopenic purpure (TTP).

Cryoprecipitate with clotting proteins, fibrinogen in massive transfusions where large numbers of blood components are needed to help with clotting.

Reasons for blood transfusions

The main reasons for blood transfusions are:

- **Blood loss** – excessive bleeding, e.g. haemorrhage can lead to large loss of blood that affects blood volume and circulation.
- **Thrombocytopenia** – spontaneous bleeding due to insufficient numbers of platelets.
- **Severe anaemia** – insufficient numbers of functioning red blood cells reduces the amount of oxygen being supplied to cells threatening whole body metabolism.
- **Operations** – many types of surgery require blood transfusions.

Complications that can arise with blood transfusions

In some circumstances complications can arise.

- **Haemolytic reaction** – if blood types are not correctly matched the patient's body will destroy and break down the transfused red blood cells.
- **High fluid volume** – the introduction too quickly of blood too quickly can overload the circulatory system. Increased blood volume can distend blood vessels and reduce their resistance and they reduce venous return.
- **Allergic reaction** – the immune system of the recipient may be activated causing itchiness, dizziness, headaches and breathing difficulties.
- **Transfusion related acute lung injury (TRALI)** – agglutination and other reactions to transfused blood can cause blockages in blood vessels in the lungs. This can affect breathing capacity leading to low oxygen levels which in turn affects metabolic processes and lead to life threatening situations.

QUESTIONS

1. What is involved in a blood transfusion?
2. Why are blood groups matched in blood transfusions and what is done if the patient's blood group is unknown?
3. Identify some situations when blood transfusions are needed.
4. State the alleles that are present for the genotype of each blood type.
5. What antigens are present on the surface of red blood cells?
6. What determines blood group?
7. There are around fifty different red blood cell antigens involved in the Rh blood group system. What is the most important antigen of the Rh system?
8. What proportion of the human population is rhesus negative?
9. Explain why it is important to know the Rh factor for a pregnant woman and her foetus.
10. Which blood has been called the 'universal donor'?
11. What is agglutination?
12. Construct a table to summarise different types of blood transfusions.
13. What are the main reasons for blood transfusions?
14. Construct a table to summarise some of the complications that can arise with blood transfusions.
15. The table shows the reactions that occurred when the blood from three people had anti-A serum and anti-B serum added to their blood.

	Person X	Person Y	Person Z
Anti-A serum added to blood			
Anti-B serum added to blood			

Figure 56.3 Blood reactions for three people.

What is the blood group for each of these people?

	Person X	Person Y	Person Z
(A)	Blood group A	Blood group AB	Blood group B
(B)	Blood group A	Blood group O	Blood group B
(C)	Blood group B	Blood group AB	Blood group A
(D)	Blood group B	Blood group O	Blood group A

16. Which antigen is most associated with the rhesus factor?
    (A) Anti-A
    (B) Anti-B
    (C) D-antigen
    (D) E-antigen

57 The Digestive System

The digestive system facilitates the exchange of materials between the external and internal environment of the organism. The **digestive system** consists of the alimentary canal and its associated organs. The **alimentary canal** is a long tube running from the mouth to the anus where ingested food is broken down into smaller pieces so that nutrients can be absorbed. Unneeded wastes are eliminated through the anus. Each part of the alimentary canal is specialised to carry out particular steps in the process of digestion.

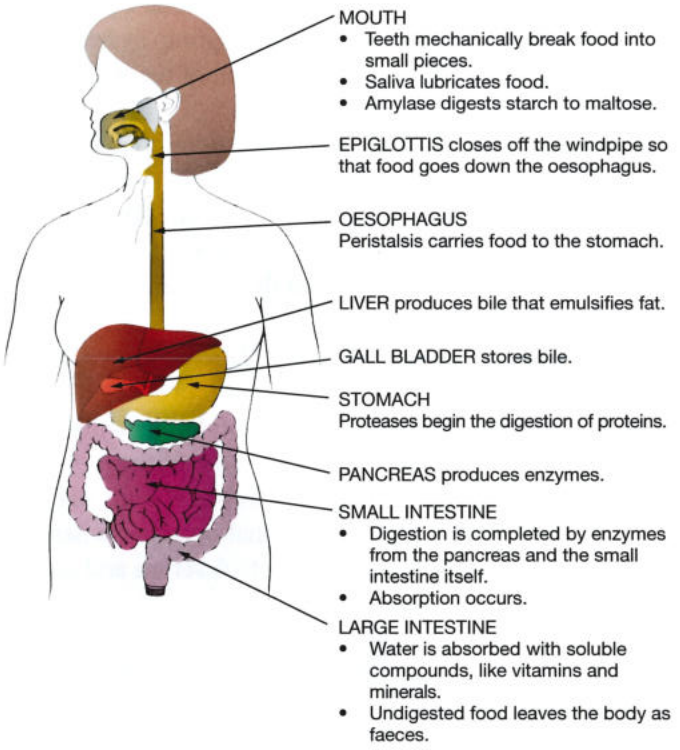


Figure 57.1 Human digestive system. In the digestive system of omnivores food is gradually broken into smaller and smaller pieces.

Some sections of the wall of the alimentary canal have **glands** that produce large amounts of watery secretions some containing digestive enzymes. The digestive system produces **enzymes** to break down the ingested food. Each enzyme performs a specific task and requires specific pH conditions. Digested foods are absorbed through the walls of the alimentary canal. **Amylase** enzymes break down carbohydrates to glucose. **Protease** enzymes break down proteins to amino acids. **Lipase** enzymes break down lipids into glycerol and fatty acids.

In the stomach there are **oxyntic cells** that secrete hydrochloric acid at about pH 1 so that the stomach has an acidic environment of pH 1.5 to 2.0. The acid kills bacteria in the ingested food, denatures proteins and softens fibrous connective tissues in the food and activates the enzymes pepsin and rennin that are in the stomach.

Most of the digested food particles are absorbed through the villi of the small intestine. The simple sugars, amino acids and mineral elements may diffuse across the cell membranes, although most movement is by active transport. Active transport means that there is greater control of which chemicals enter the villi. Amino acids, carbohydrates and minerals are absorbed into the bloodstream, while fatty acids and glycerol enter the lacteal which runs down the middle of each villus and is part of the lymphatic system.

Absorption in the small intestine is efficient because it is a fairly long tube with thousands of small projections called **villi**. Villi are covered with microvilli and these structures increase the surface area for absorption. Other features, which aid efficient absorption, are the thinness of the epithelial lining and a rich blood supply in each villus.

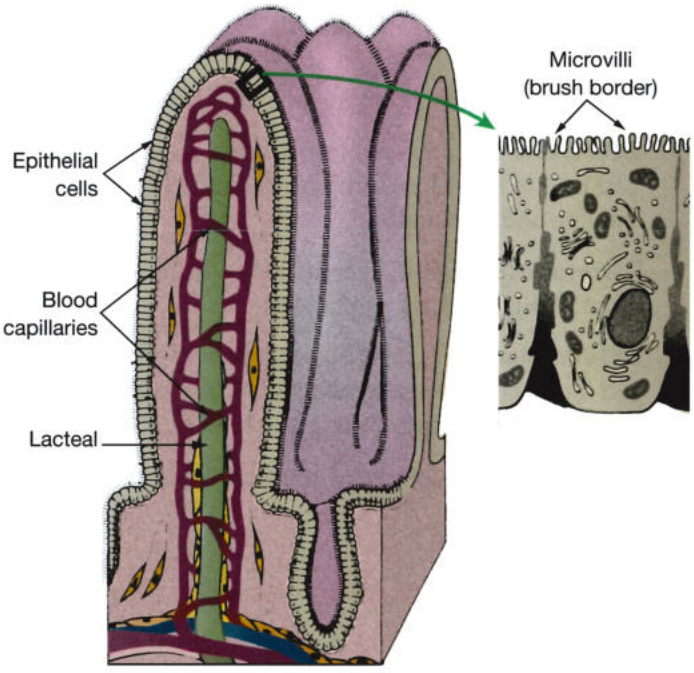


Figure 57.2 Villus. The villi increase surface area to absorb food. Amino acids, carbohydrates and minerals are absorbed into the bloodstream. Fatty acids and glycerol enter the lymphatic system.

In the bloodstream, the amino acids, sugars and minerals are carried to the liver by the hepatic portal vein where the products of digestion may be stored or altered as needed by the body. For example, excess glucose is stored as glycogen in the liver and when blood sugar levels fall, the glycogen is turned back into glucose and released so that blood sugar levels remain fairly constant.

Once in the lacteal, the fatty acids are recombined to form fats once again. The fats are often coated with special proteins, becoming lipoproteins for transportation. The lymphatic system transports the lipids from the digestive to the circulatory system, draining into veins at two locations near the shoulders.

QUESTIONS

1. What is meant by the digestive system?
2. What is the alimentary canal?
3. Construct a table to show the food groups carbohydrates, proteins and lipids, the name of the enzyme that breaks down that food group and the end product of digestion.
4. Outline where mechanical digestion takes place in the digestive system.
5. In digestion, what happens to amino acids, simple sugars, minerals, fatty acids and glycerol?
6. Outline the features in the small intestine which aid absorption of digested food.
7. Once in the bloodstream, what happens to glucose?
8. What happens to fatty acids once they enter the lacteal in the villus?

Use the following information to answer the next TWO questions. The diagram shows part of the small intestine.

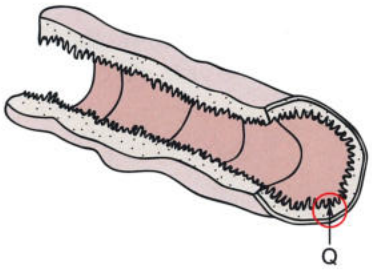


Figure 57.3 Small intestine.

9. Name the small projection labelled 'Q'.
   (A) Villus. (B) Sphincter.
   (C) Villain. (D) Lacteal.
10. What is the function of structure 'Q'?
    (A) Release digestive enzymes.
    (B) Increase surface area for absorption.
    (C) Control the movement of food to the next section of the alimentary canal.
    (D) Provide a surface area for the manufacture of vitamins.
11. Which substances enter the lacteal when they are absorbed?
    (A) Amino acids.
    (B) Sugars.
    (C) Fatty acids.
    (D) Minerals.
12. Digestive juices contain mucus. What is the function of the mucus?
    (A) Protect the cells lining the alimentary canal from attack by digestive juices.
    (B) Lubricate the lining of the canal as the food is moved along.
    (C) Protect against infectious agents such as fungi.
    (D) All of the above.

Use the following information about the human digestive system for the next THREE questions.

Part	Enzyme digestion
Mouth	Starch $\xrightarrow{\text{amylase}}$ maltose
Stomach	Protein $\xrightarrow{\text{pepsin}}$ smaller polypeptides
Small intestine	Polysaccharides $\xrightarrow{\text{amylase}}$ monosaccharides Polypeptides $\xrightarrow{\text{protease}}$ amino acids Lipids $\xrightarrow{\text{lipase}}$ fatty acids + glycerol

13. Which food group is the first to undergo chemical digestion?
    (A) Carbohydrates.
    (B) Nucleic acids.
    (C) Lipids.
    (D) Proteins.
14. Where does chemical digestion cease?
    (A) Mouth.
    (B) Stomach.
    (C) Small intestine.
    (D) Large intestine.
15. Where does the digestion of fats begin?
    (A) Mouth.
    (B) Stomach.
    (C) Small intestine.
    (D) Large intestine.
16. What happens in the large intestine?
    (A) Proteases begin the digestion of proteins.
    (B) Absorption of digested food occurs.
    (C) Amylase begins starch digestion.
    (D) Water is absorbed.
17. What happens to excess glucose when it reaches the liver?
    (A) It is used in respiration.
    (B) It is converted to glycerol.
    (C) It is converted to glycogen.
    (D) It is sent to the pancreas for storage.
18. Which choice correctly matches the enzyme with its location and the food group digested by that enzyme?

	Enzyme	Location	Food group
(A)	Amylase	Oesophagus	Carbohydrates
(B)	Protease	Mouth	Protein
(C)	Lipase	Stomach	Fats
(D)	Amylase	Small intestine	Carbohydrates

19. What is the pH of the stomach?
    (A) 1.5 to 2.0
    (B) 5.5 to 7.0
    (C) 7.0 to 7.2
    (D) 8.5 to 9.0

58 Teeth

Teeth in the mouth are important in mechanical digestion as they break food up into small pieces by biting and chewing. The smaller pieces of food have a greater surface area to volume ratio and this means that digestive enzymes have a greater ability to work efficiently and effectively.

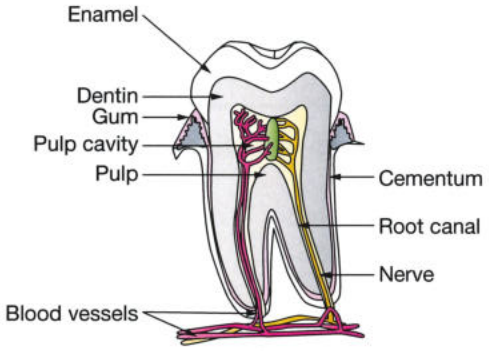


Figure 58.1 The structure of a tooth.

The **crown** of the tooth is the part that projects above the gum. Human teeth are covered by **enamel** which is the hardest substance in the body. The **root** of the tooth is in the jawbone and the bulk of the tooth is tough **dentin** which is like bone in hardness and composition. The **pulp cavity** is below the dentin and contains sensory nerve endings and blood vessels.

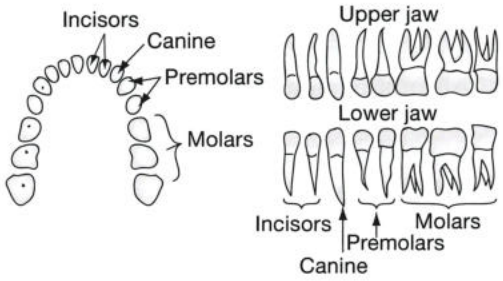


Figure 58.2 Human teeth.

Types of teeth

Humans form two sets of teeth during their life with a milk set followed by a permanent set. Each type of tooth has a specific function.

- **Incisors** are chisel shaped teeth at the front and are used for biting.
- **Canines** have a sharp point and are used for tearing meat.
- **Premolars** have sharp cutting edges and are used for crushing food.
- **Molars** have large flat surfaces with blunt ridges and are used for grinding food.

Swallowing

The salivary glands release saliva into the mouth and the chewing mixes the saliva with the food. Saliva has several functions. It dissolves some of the food, helps to lubricate the food and makes some small pieces stick together. Human saliva contains a digestive enzyme called **salivary amylase** (also called ptyalin) which splits starch into units of the disaccharide maltose. When the food is swallowed the acid in the stomach inactivates the salivary enzyme. When the teeth break the food into smaller pieces, the salivary amylase can reach more sections for chemical digestion.

QUESTIONS

1. Identify the four types of teeth found in humans and briefly state the function of each type of tooth.
2. Count the number of teeth you have and compare this number with the number in the diagram.
3. Outline the function of saliva.
4. Identify the enzyme found in saliva and state its function.
5. The following diagram shows the temporary or 'milk' teeth of a five year old child.

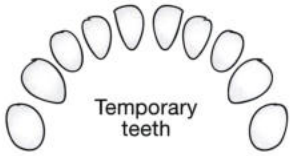


Figure 58.3 Temporary teeth.

Identify the types of teeth present and not present in the temporary teeth of humans and explain the benefit of having temporary teeth.

6. Explain how both chemical digestion and mechanical digestion occur in the mouth.
7. Discuss why the breakdown of starch stops when the food reaches the stomach.
8. Which of the following correctly associates the type of tooth with its function?

	Incisor	Canine	Molar
(A)	Tearing	Biting	Grinding
(B)	Biting	Grinding	Tearing
(C)	Biting	Tearing	Grinding
(D)	Grinding	Biting	Tearing

9. What is the main role of human teeth?
   (A) Mechanical digestion.
   (B) Chemical digestion.
   (C) Assisting swallowing.
   (D) Capturing prey.
10. What enzyme is present in human saliva?
    (A) Salivary protease (B) Salivary lipase
    (C) Salivary maltase (D) Salivary amylase

59 Play – Amino and Lipidet, A Most Lamentable Tragedy

Dramatic personae:
Prince Carb – a carbohydrate
Vitamin A – a young vitamin and friend of Amino
Protein – head of family of organic molecules at feud with the Lipids
Amino – an amino acid and Protein's son
Glycine – a simple protein
Lipid – head of family of organic molecules at feud with the Proteins
Lipidet – a monoglyceride and Lipid's daughter
Glycerol – a kinsman of Lipid
Phospholipid – part of the Lipid household
Steroid – part of the Lipid household
Chorus

THE PROLOGUE

(Enter Chorus)

Chorus Two households both alike in dignity
(In fair digestive system where we lay)
From ancient grudge break to new mutiny
Where organic molecules break away.
From forth metabolism takes its turn
A pair of star-cross'd lovers do depart,
While we see the piteous stomach churn
With their death the reactions shall restart.

SCENE 1 – THE MOUTH

(Enter Phosopholipid, Steroid and Vitamin A)

Phospholipid Will we enter the mouth?
Vitamin A I don't think we have much choice. We're about to be eaten.
Phospholipid I believe Prince Carbohydrate, the Lipid family and the Protein family have already gone through. This person is eating fish and chips.
Vitamin A Oh dear. This means strife. The Protein family is fighting the Lipid family. They have been counting the number of scientists studying their structure and each wants to be the subject of the next Nobel prize.
Steroid Strife? The Lipid family will win any conflict. The Proteins are unimportant. Fat stores 4000 kJ energy per 100 grams while protein only stores 1720 kJ energy per 100 grams. We are four times better than proteins!
Vitamin A I think your maths is out. It is more like 2.3 times more energy in fat than protein.
Phospholipid Who's this?
(Glycine joins the three molecules waiting at the mouth)
Glycine I am Glycine, an amino acid and a member of the Protein family.
Steroid You're the smallest of the 20 amino acids found in proteins and just a precursor to proteins. I bite my thumb at you.
Vitamin A Don't you mean bite my shorts?
Phospholipid To Shakespeare and the Elizabethans biting the thumb was a rude and insulting gesture.
Glycine Do you want to fight? The amino acids in a protein are held together by strong peptide bonds. We can beat you in any fight.
(Glycine and Steroid draw swords and begin to fight)
Vitamin A Part fools, put up your swords, you know not what you do.
(Enter Glycerol)
Glycerol Glycine, stop fighting my kinsman.
Glycine The proteins are the most important organic molecules. Without proteins there is no life. Proteins make up more than 50% dry mass of most cells. I prefer to keep the peace.
Glycerol Peace? I hate the word, as I hate hell, all Proteins and thee: Have at thee, coward.
(Glycerol and Glycine fight and a furious riot ensues only stopping when they are swallowed and enter the mouth)

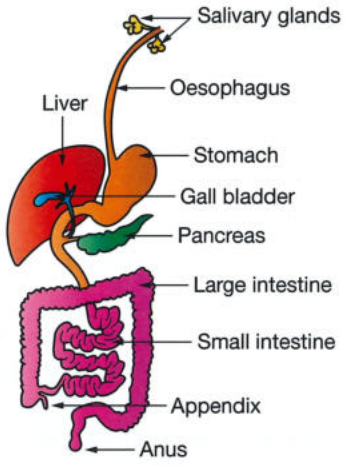


Figure 59.1 Structure of the digestive system.

Phospholipid The mouth is a rather dangerous place. Look at all those teeth.
Steroid Our enemy is here as salivary amylase.
Phospholipid Don't worry about that protein, it only breaks down starch.
Steroid Thank goodness we're not carbohydrates. I wonder how Prince Carb survived the mouth.
Vitamin A Who is waiting for us here?
(Amino joins the group on their journey)
Glycine The rest of you go down the oesophagus. I will speak with my cousin. Good morrow Amino.
Amino Is the day so young?

Glycine Why are you so sad?

Amino My sadness is caused by love. Love is a madness most discreet, a choking gall and a preserving sweet.

Glycine Sweet? You're in love with a carbohydrate?

Amino She is sworn to chastity and refuses to have children.

Glycine We're organic molecules, don't you mean undergo an anabolic reaction?

Amino An anabolic reaction is a metabolic reaction that synthesises a complex molecule from simpler compounds. Such terminology sounds so unromantic!

Glycine It may be unromantic but it keeps our host body alive. To forget your woes let's go down to the stomach. The Lipid family are hosting an ancient feast. We might meet some young organic molecules down there and have some fun.

Amino I fear attending this party. Some consequences yet hanging in the stars shall bitterly begin this fearful date. I predict an untimely death.

Glycine Don't be a pessimist. It's just digestion in the stomach.

(Amino and Glycine move by peristalsis down the oesophagus and arrive at the party in the stomach)

Amino What lady is that? She doth teach the torches to burn bright.

Glycerol By that voice you must be a Protein. Fetch my rapier.

Lipid What's the matter?

Glycerol Uncle, this is a Protein, our foe; villain come to ruin our party. It is Amino.

Lipid Don't worry about Amino. We are in the stomach. The digestion of lipids does not begin until the small intestine so we are safe and can enjoy the party; while the digestion of proteins begins in the stomach. He is doomed. The pH of the stomach is 2 and the enzyme pepsin breaks the peptide bonds that hold a protein together.

Glycerol I'll not endure him. I will have my vengeance.

Lipid He shall soon be broken down. Let him be.

(Amino moves to another part of the stomach looking for Lipidet, the girl he had just glimpsed and fallen immediately in love with)

Amino But soft, what light though yonder window breaks?

Lipidet Breaks? Is there a hole in the stomach wall? An ulcer?

(Lipidet begins studying the stomach wall)

Amino She speaks.

Lipidet O Amino, Amino, wherefore art thou Amino? Is that you surrounded by the enzyme pepsin and all that gastric juice?

Amino Fear not. I am an amino acid, not a polypeptide with peptide bonds. I cannot be broken down by pepsin. I am here and you are my love.

Lipidet If you are an amino acid you belong to the household of my enemy, the Proteins.

Amino I am not your enemy.

Lipidet It is your name that is my enemy. Be some other name. What's in a name? Are we not all carbon, hydrogen and oxygen joined together by covalent bonds?

Amino Can we be together?

Lipidet I believe we can be joined in the small intestine.

Amino I will exchange thy love's faithful vow for mine.

Lipidet Good night, good night. Parting is such sweet sorrow. Till we meet in the small intestine.

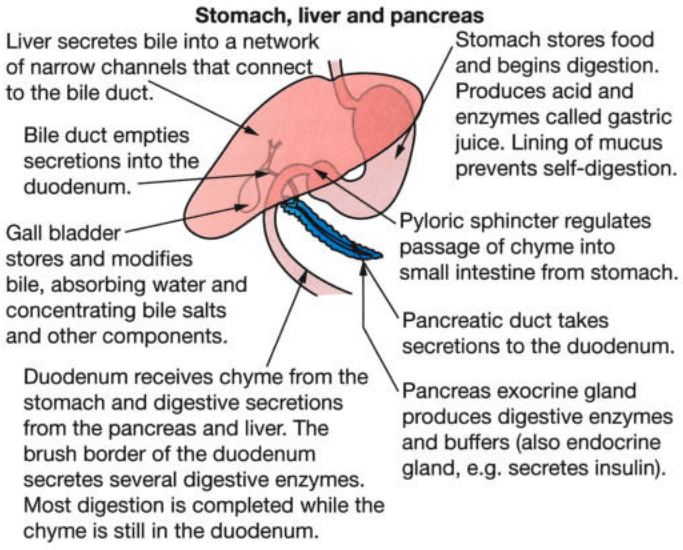


Figure 59.2 Stomach, liver and pancreas.

Questions Scene 1

1. Which food groups are in fish and chips?
2. For each organic molecule in the play, identify the food group to which it belongs.
3. Compare the kilojoules of energy per 100 grams of proteins and lipids.
4. Describe glycine.
5. Outline the digestion that occurs in the mouth
6. Outline the function of peptide bonds.
7. Explain why proteins are an important part of cells.
8. Explain why it is dangerous to be a protein in the stomach but not dangerous to be a lipid.

SCENE 2 – THE SMALL INTESTINE

(Glycine and Vitamin A are the first to enter the duodenum)

Glycine I survived the stomach! I feel smaller, but I made it through!

Vitamin A Did you see how the gastric acid killed the bacteria and other micro-organisms?

Glycine The acid denatured many of the proteins destroying their tertiary structure. That allowed the enzyme to break the peptide bonds. I should have known the lipids would try to weaken us.

(The others arrive in the duodenum)

Glycine By my central carbon atom linked to an amino group and a carboxyl group, here comes the Lipids.

Vitamin A By my carotene parent, I care not.

Glycerol Vitamin A, I believe you are a friend of Amino?

Vitamin A Were we not named as the 'vital amines'? He is my friend.

Glycerol Here's Amino. I would fight the villain.

Amino Villain am I none.

Glycerol You are a boy and I will not forgive the injuries you have given me.

Amino I have never injured you.

Glycerol I will fight you now.

Vitamin A If Amino won't fight you, I will.

(Glycerol and Vitamin A begin to fight)

Amino Stop this fighting. It is not allowed.

(Amino steps between the two and Glycerol thrusts under Amino's arm and hits Vitamin A)

Vitamin A I am hurt.

Amino Courage.

Vitamin A You do not understand. Vitamin A is fat soluble and we are absorbed with fats across the epithelium of the small intestine. I go to another place.

Amino Courage, the hurt cannot be much.

Vitamin A Ask for me tomorrow and you will find me in the liver. A plague on both your houses.

(Exit Vitamin A)

Amino This day's black fate on more days doth depend.

Glycine Most fat-soluble vitamins are absorbed along with the fats.

Glycerol I got your friend and I will get you.

Amino I don't think so. I draw my sword and will fight you now. This shall determine your fate.

(Amino draws his sword and fights Glycerol until Glycerol falls)

Prince Carb What's happened here?

Glycine Vitamin A has passed over and Glycerol is destroyed.

Prince Carb Who is to blame?

Glycine It was Amino.

Prince Carb The Protein family will regret this day. Amino shall meet his fate, like all of us. I have no pity for this murderer. Most digestion and absorption occurs in the small intestine and he will not be exempt from digestion.

Steroid This area is flooded with enzymes. The enemy is surrounding me.

Glycine The enzymes are proteins. They are my kin.

Phospholipid This is terrible. There are pancreatic amylases to break down the polysaccharides, pancreatic trypsin to break down the polypeptides and worst of all bile salts and pancreatic lipases to break down fat globules.

Steroid It is disastrous. I don't want to be coated in bile salts and broken down into little fat droplets.

Phospholipid It is a horrible fate.

Protein The enzymes of my family will complete digestion and absorption will occur further along the small intestine.

Prince Carb Organic molecules move in haste for this hour is your last.

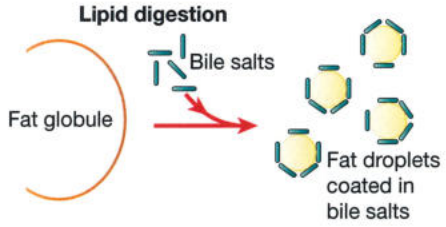


Figure 59.3 Lipid digestion.

Questions Scene 2

1. Outline the function of gastric acid.
2. What is the basic structure of an amino acid?
3. Which food is a source of vitamin A?
4. How were vitamins named?
5. Outline how vitamin A is absorbed.
6. What type of molecules are enzymes?
7. List the enzymes found in the small intestine and state their function.

SCENE 3 – EPITHELIAL CELL IN THE SMALL INTESTINE

(Lipidet as a monoglycerides has diffused across the cell membrane into an epithelial cells of a villi.)

Lipidet Gallop apace. I have entered the epithelial cell and am ready to join with my Amino.

Amino I have arrived and avoided the capillary. As a free amino acid I entered the epithelial cell by the sodium-dependent cotransport proteins.

Lipidet We can now be together. We will become a chylomicron.

Amino Chylomicron?

Lipidet We will be a triglyceride with a protein coat.

Amino You mean we will be joined together in a lipoprotein?

Lipidet An anabolic reaction.

Amino My love. This is what I have dreamed about.

Lipidet Metabolic reactions sustain life. We are united in love.

Amino And then?

Lipidet We will be packed into a secretory vesicle and leave the cell by exocytosis.

Amino So smile the heavens upon this act.

Lipidet Ah! Because of the size of the vesicle we cannot cross the basement membrane and will be absorbed into the lacteal and pass through the lymphatic system. We will enter the venous blood just before it reaches the heart.

Amino A joy past joy calls out to me.

Lipidet There is one small problem. We will have cholesterol and phospholipids in the vesicle with us.

Amino This vesicle needs be large to suit our purpose.

Lipidet By varying the ratios of protein, triglyceride, phospholipids and cholesterol we can become a very low-density lipoprotein (VLDL), a low-density lipoprotein (LDL) or a high-density lipoprotein (HDL).

Amino Then let us become a high-density lipoprotein. The more protein the complex contains, the heavier its weight.

Lipidet My love you make me pleased. High-density lipoproteins are the 'good cholesterol'. We shall be happy, now we are good.

Questions Scene 3

1. How can a monoglyceride enter an epithelial cell of the small intestine?
2. How can a free amino acid enter an epithelial cell?
3. What is a chylomicron?
4. What is the function of a chylomicron?
5. How do the chylomicrons leave the epithelial cell?
6. What happens to the secretory vesicle once it leaves the epithelial cell of the small intestine?
7. Describe why there is a range of lipoprotein complexes.
8. Identify the lipoproteins that are called 'good cholesterol' and 'bad cholesterol'.

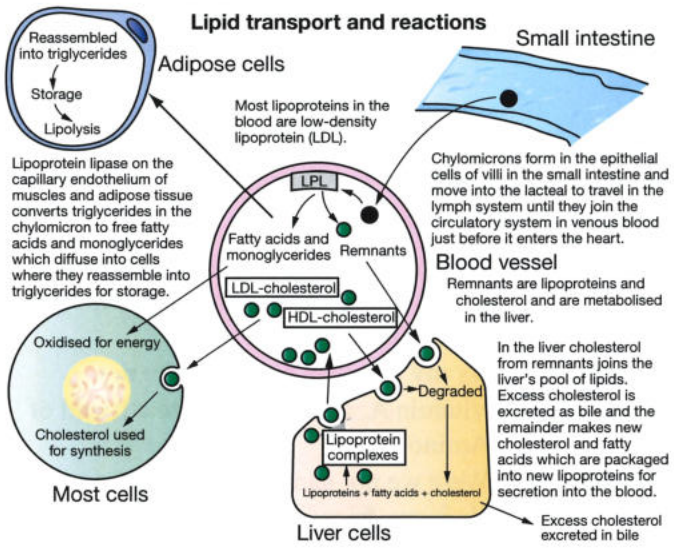


Figure 59.4 Lipid transport and reactions.

SCENE 4 – THE BLOODSTREAM

(The lipoprotein is being carried around the body in the plasma)

Lipidet One night together.

Amino Why must you leave?

Lipidet As a triglyceride I am being unloaded by a lipoprotein lipase.

Amino A relative of mine would do this to us?

Lipidet They are my enemy.

Amino Where is this enemy that would break us apart?

Lipidet The enzyme is bound to the capillary endothelium of an adipose cell. It converts a triglyceride from the chylomicron into a monoglyceride.

Amino You will become a monoglyceride again?

Lipidet In my smaller form I can diffuse across the cell membrane. Farewell my love.

Amino Farewell. I will wait for you.

Lipidet I leave and die while looking at your dear face.

(Lipidet is broken down by the lipoprotein lipase and is taken away)

Amino My life is meaningless without Lipidet. I inhabit a chylomicron remnant which is a shell of its old self. I will let it take me to the liver. Endocytosis will see me taken into a liver cell.

Glycine Hi Amino. Welcome to the liver.

Amino Who are you?

Glycine I was your old friend glycine, but after absorption in the small intestine I've been through a few reactions and I'm now an enzyme that will see you deaminated.

Amino Deaminated?

Glycine Deamination is a reaction which removes an amino group from an amino acid. You will become free ammonia.

Amino Sounds painful.

Glycine Not that painful. Since ammonia is highly toxic to cells, you will be converted into urea.

Amino The stars have condemned my fate.

Glycine You will go back into the bloodstream and then be filtered out by the kidneys and excreted as urine.

Amino I will be broken down and excreted. I die. In the liver I have met my death. Without Lipidet I am happy to die. Farewell.

(Amino is broken down and leaves)

Lipidet I'm back.

Glycine What are you doing here?

Lipidet I went into storage as a triglyceride in an adipose cell for a while, but now I'm back. I was only temporarily gone.

Glycine Adipose cell?

Lipidet Fat cell. Where's Amino? He said he would wait for me.

Glycine There's a slight problem.

Lipidet Problem? What do you mean?

Glycine Amino has been deaminated and has been passed out in the urine.

Lipidet Oh woe. I cannot live without Amino. I too will be deaminated.

Glycine Sorry, can't do. Only amino acids can be deaminated.

Lipidet Then how can I die so I can join my amino in death?

Glycine Excess cholesterol is made into bile. You will then be sent to the gall bladder and excreted through the digestive system.

Lipidet I will meet my death by becoming bile. Farewell. Without Amino I am happy to die.

(Lipidet leaves and Prince Carb, Protein and Lipid enter)

Prince Carb Is there news of Amino and Lipidet?

Protein No news. Alas my wife was so distraught at the news of our son that she went to the liver and was deaminated. She has passed away.

Prince Carb The life of an organic molecule is fraught with danger.

Glycine I am afraid I have bad news.

Protein You have word of my son, Amino?

Glycine I am afraid he has been deaminated and has passed from this body.

Lipid You have news of my daughter Lipidet?

Glycine She became bile and has passed from this body.

Protein Why did they have to suffer such a fate?

Prince Carb They became a chylomicron and there are many fates for lipoproteins. Some lipoproteins form part of cell membranes and live a long life.

Lipid They were not fated to have such a long life.

Protein They were star-crossed organic molecules.

Prince Carb Go hence and have more talk of metabolic reactions, some anabolic, some catabolic, for never was a story of more woe, than this of Lipidet and her Amino.

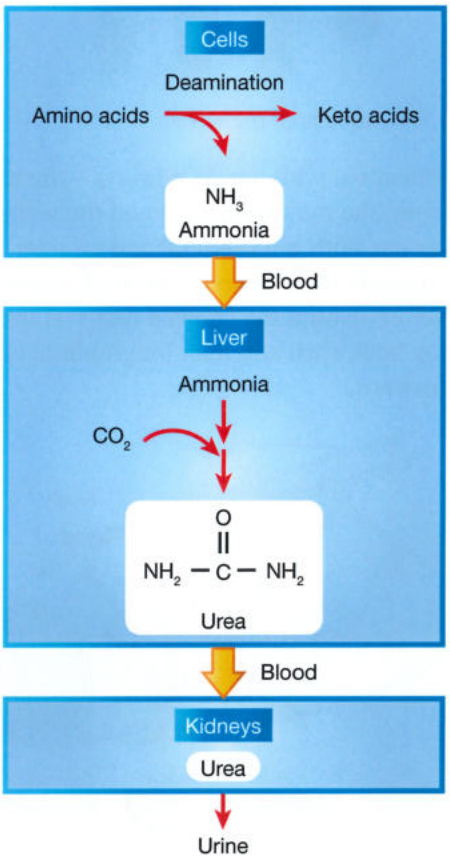


Figure 59.5 Deamination of an amino acid.

Questions Scene 4

1. Outline the function of lipoprotein lipase.
2. What is deamination?
3. Why is ammonia quickly converted into urea?
4. Identify the nitrogenous waste formed from proteins and outline how it is excreted.
5. What is an adipose cell?
6. Outline the fate of excess cholesterol.

60 Mechanical Digestion

Mechanical digestion involves the physical breakdown of food into smaller pieces. It involves teeth, bile and peristalsis.

Peristalsis

Peristalsis is waves of muscular contraction of smooth muscle that moves food along the length of the digestive tract.

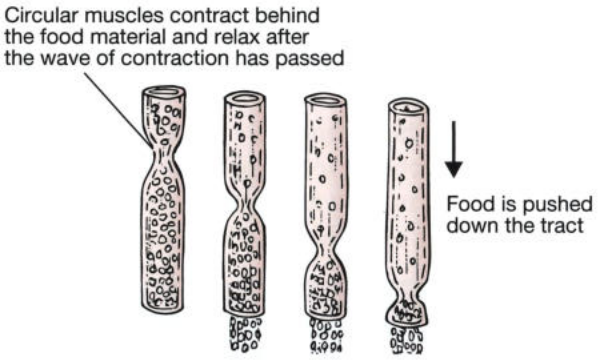


Figure 60.1 The process of peristalsis.

The gastrointestinal wall has four layers – the mucosa, the submucosa, the muscular layer and the serosa. The muscle layer has both longitudinal and circular smooth muscle. Peristalsis involves relaxation of smooth muscle then contraction behind the chewed material to keep it from moving backward and then longitudinal contraction to push it forward.

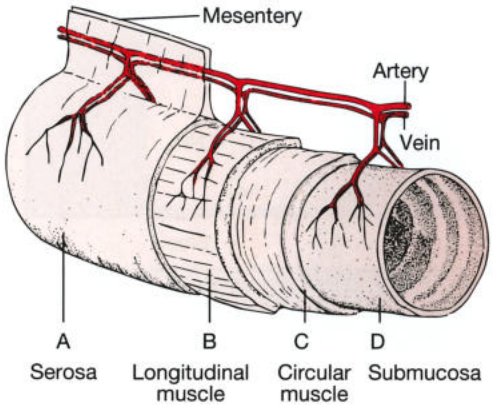


Figure 60.2 Layers of the wall of the small intestine.

Bile

Bile is a solution secreted by the **liver** composed mainly of bile acids, bile pigments and cholesterol. It does not contain any digestive enzymes. Liver cells conjugate bile acids with either glycine or taurine (amino acids) to form conjugated bile acids. These conjugated bile acids are often called **bile salts** due to their acid-base properties. Bile salts have a steroid structure with four rings of carbon atoms.

The **bile pigments** are by-products of red blood cell breakdown in the liver and enter the alimentary canal to be eliminated with faeces.

Bile salts **emulsify fats**. Bile salts have a hydrophobic side that associates with lipids and a hydrophilic side that associates with water. This prevents the lipid droplets from coalescing and reduces large fat globules to small droplets. Lipids are insoluble in water which means hydrolysis of fats is a problem. The lipid molecules tend to coalesce to form large fat globules. Bile salts are needed as they emulsify large fat globules to form small droplets.

Bile is stored in the **gall bladder** until needed. Bile is released from the gall bladder and travels to the duodenum where the salts break large fat globules into small droplets in an action similar to the action of detergents.

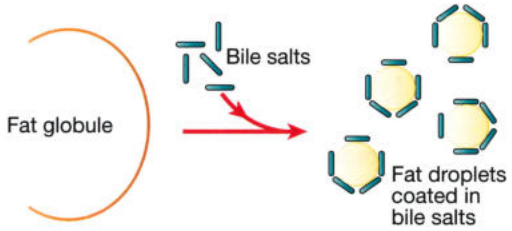


Figure 60.3 Lipid digestion by bile salts.

The bile salts have a hydrophilic end and a hydrophobic end. The hydrophobic end binds with the lipid while the hydrophilic end interacts with the water and a droplet forms. The small droplets have a large surface area exposed to lipase which is the enzyme needed for the chemical digestion of lipids.

Sphincters are ring-like valves that lie at junctions between different sections of the alimentary canal. They allow/prevent the movement of material from one section into the next. For example, the pyloric sphincter controls when food is released from the stomach into the duodenum of the small intestine.

QUESTIONS

1. Define mechanical digestion.
2. What is peristalsis?
3. Name the layers present in the gastrointestinal wall.
4. Explain how the muscle layer of the gastrointestinal wall is involved in peristalsis.
5. (a) What is bile?
   (b) What are bile salts?
   (c) Where is bile stored?
6. Outline the function of bile salts.
7. Explain why bile salts are not part of chemical digestion.
8. What is the function of sphincters?
9. Outline why bile is needed in digestion.

61 Chemical Digestion

Chemical digestion involves the breakdown of food into simpler substances. It involves enzymes with distinctive operating conditions and function that are located in different sections of the digestive system.

Enzymes

Enzymes are organic catalysts that control the rate of reactions.

Digestive enzymes speed up the breakdown of large organic compounds into smaller molecules and thus required nutrients can be supplied at a faster rate for use in the body after ingestion.

There are separate pathways for the chemical digestion of carbohydrates, proteins, fats and nucleic acids.

$$\text{Carbohydrates} \xrightarrow{\text{amylase, disaccharidase}} \text{monosaccharide}$$

$$\text{Protein} \xrightarrow{\text{protease}} \text{amino acids}$$

$$\text{Lipids} \xrightarrow{\text{lipase}} \text{glycerol, fatty acids, glycerides}$$

$$\text{RNA, DNA} \xrightarrow{\text{nuclease, nucleotidas}} \text{bases, sugars, phosphates}$$

Carbohydrate digestion

Carbohydrates, e.g. sugars, glycogen and starch begin chemical digestion in the mouth where salivary amylase breaks down starch and glycogen into maltose (disaccharide). An amylase enzyme hydrolyses starch into sugars and acts at an optimum pH of 6.7 to 7.0.

Table 61.1 Carbohydrate digestion.

Name of enzyme	Source	Substrate	Product
Salivary amylase	Saliva from salivary glands	Starch	Disaccharides, e.g. maltose
Pancreatic amylase	From exocrine glands in pancreas	Starch	Disaccharides
Disaccharidases	From brush border of epithelium in small intestine	Disaccharides	Monosaccharides

Protein digestion

Protein digestion begins in the stomach with gastric juice. **Gastric juice** is a thin, almost colourless liquid secreted by special glands in the mucosa lining of the stomach wall and contains hydrochloric acid (HCl) and the enzymes **pepsin** and rennin and mucus.

The acid breaks the extracellular matrix that holds cells together, e.g. in meat and the low pH (pH = 2) denatures the proteins in the food. The acidity stops the action of salivary amylase and the mucus prevents mechanical injury to the stomach wall by lubricating the contents.

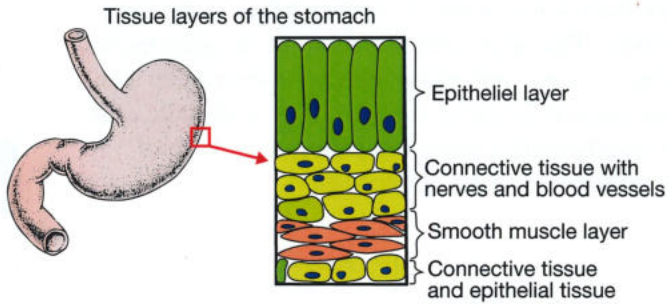


Figure 61.1 Layers of stomach wall.

One group of protein enzymes (**endopeptidases**) break apart the long protein chains with each type of enzyme acting on the bonds linking particular amino acids. This makes shorter and shorter peptide fragments.

Another group of protein enzymes (**exopeptidases**) act only on the end of a chain, some on the amino end and some on the carboxyl end splitting off dipeptides (pairs of amino acids).

Another group of protein enzymes break the remaining dipeptides into single amino acids.

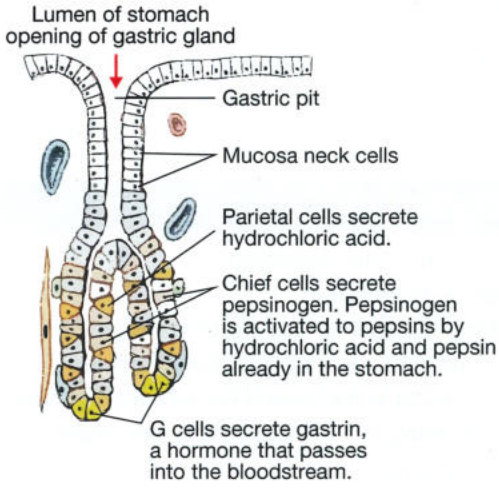


Figure 61.2 Structure of stomach wall.

Lipid digestion

Dietary lipids are often in the form of triglycerides. The fats form globules. Bile secreted by the liver breaks large fat globules into small droplets in an action similar to the action of detergents. Bile does not contain enzymes. It is important in breaking large droplets into small droplets that have increased surface area for enzyme action.

The enzyme **lipase** hydrolyses the triglycerides acting on the surface of the fat droplet forming free fatty acids and monoglycerides. Lipase is water soluble, works in a watery environment and has an active site for the hydrophobic lipid molecules which are its substrate. The lipase can only work at the lipid-water interface on the surface of the lipid sphere. The large surface area of the droplets assists digestion, e.g. pancreatic lipase. As enzymatic and mechanical digestion proceed, fatty acids, bile salts, monoglycerides, phospholipids and cholesterol form small cylindrical micelles.

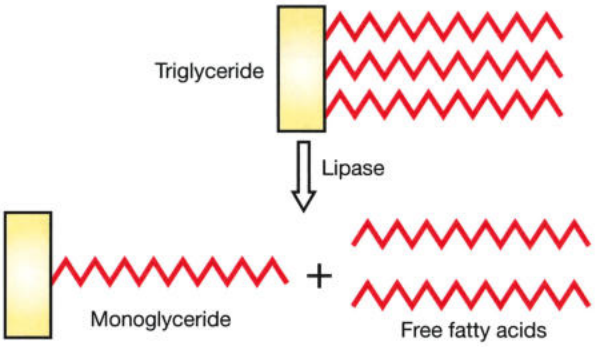


Figure 61.3 Lipid digestion.

Lipase alone cannot penetrate the bile salt coating and it needs the presence of colipase, a protein cofactor secreted from the pancreas. Colipase displaces some bile salts allowing lipase access to fats inside the bile salt coating.

The fatty acids and monoglycerides leave the micelles and diffuse into the brush border epithelial cell of the small intestine where they re-form into triglycerides.

QUESTIONS

1. Define chemical digestion.
2. What is an enzyme?
3. Outline the function of digestive enzymes.
4. (a) What two main types of enzymes are needed in the breakdown of carbohydrates?
   (b) Distinguish between these two types of enzymes.
5. Where does carbohydrate digestion begin?
6. What is gastric juice?
7. List the main layers of the stomach wall.
8. Distinguish between an exopeptidase and an endopeptidase.
9. Construct a table to summarise the function of three different types of cells in the wall of the stomach that are involved in producing gastric juice.
10. In what form are most dietary lipids?
11. Outline chemical digestion of lipids.
12. What is the function of the cofactor colipase?
13. The diagram shows a polypeptide.

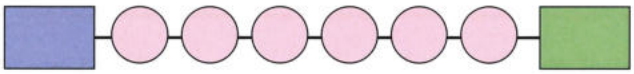


Figure 61.4 Polypeptide.

Copy this diagram and label the amino terminal end, the amino acids, the peptide bonds and carboxyl terminal end.

14. The table summarises some digestive enzymes.

Table 61.2 Some digestive enzymes.

Name of enzyme	Type of enzyme	Source	Substrate	Product	Optimum pH
Salivary amylase	Amylase	Saliva from salivary glands	Starch	Maltose	pH 7
Pepsin	Protease	Cells in stomach wall	Protein	Small polypeptides	pH 2
Pancreatic lipase	Lipase	Exocrine cells of pancreas	Lipids	Glycerol, fatty acids	pH 8

   (a) Which of these enzymes requires an acidic environment?
   (b) Which enzyme breaks carbohydrates into a disaccharide?
   (c) Where would you expect pancreatic lipase to enter the alimentary canal?

15. The flow chart shows a type of chemical digestion.

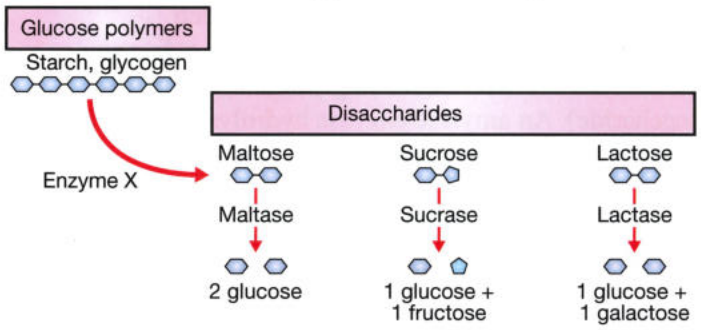


Figure 61.5 Type of chemical digestion.

What is the most likely identity of enzyme X?
(A) Protease enzyme. (B) Lipase enzyme.
(C) Amylase enzyme. (D) Disaccharidase enzyme.

16. What product(s) form when lipase acts on triglycerides?
   (A) Fatty acids and monoglycerides.
   (B) Dipeptides.
   (C) Disaccharides.
   (D) Monosaccharides and glycerol.
17. Gastrin, a digestive hormone in the blood supplying the stomach stimulates the gastric glands to produce more gastric juice. Why does gastrin continue to stimulate the gland cells even after a meal is completed?
   (A) The smell of food is still in the air.
   (B) To continue digestion while food is in the stomach.
   (C) To increase absorption in the small intestine.
   (D) To make a person feel 'full'.

62 Accessory Digestive Organs

Accessory digestive organs are organs that aid digestion but are not a part of the alimentary canal and digestive tract, e.g. salivary glands, pancreas, liver, gall bladder and tongue.

The tongue

The **tongue** is a muscular organ that has sensory taste buds. It is kept moist by saliva and has nerves and blood vessels. There are four basic types of taste receptor – sweet, sour, salty and bitter. For a substance to be tasted it must first dissolve in the saliva and mucus in the mouth.

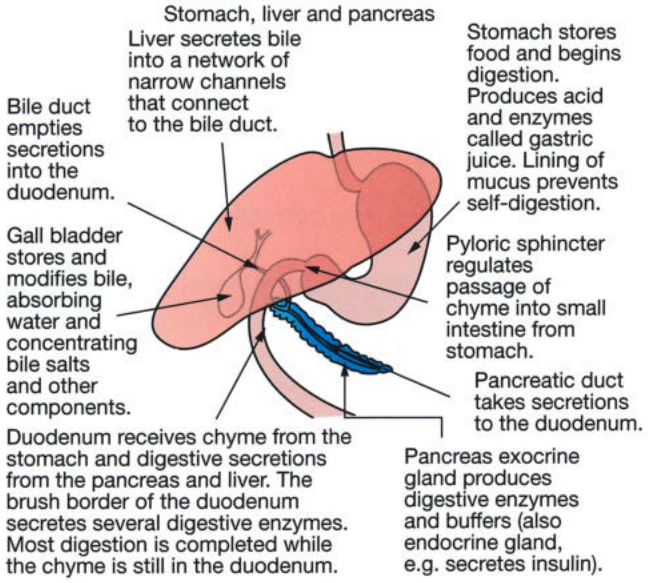


Figure 62.1 Some accessory organs.

Salivary glands

Salivary glands are connected to the oral cavity and secrete substances that lubricate food and begin the process of chemical digestion. Saliva is a combination of secretions from the salivary glands and from the mucus secreting glands lining the oral cavity.

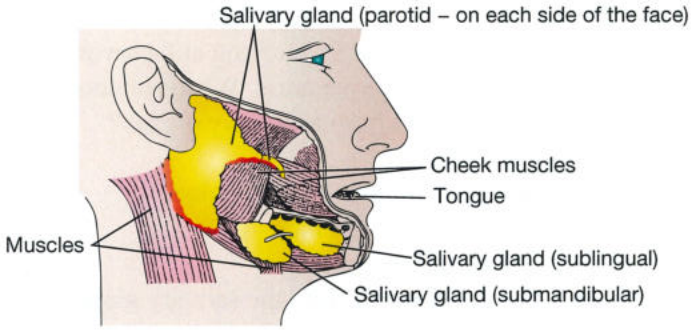


Figure 62.2 Salivary glands.

Saliva contains the enzyme salivary amylase, mineral salts, lysozyme, antibodies, blood clotting factors, water and mucus.

Saliva is needed for:

- Chemical digestion of carbohydrates, e.g. starch.
- Lubrication of food to make into a bolus for swallowing.
- Assisting in taste causing dry substances to dissolve.
- Cleansing the mouth and lubrication of the mucous membrane to keep the tissues moist, soft and pliable.
- Non-specific defence with lysozymes, clotting factors and antibodies acting against microbes in ingested food and water.

The pancreas

The **pancreas** is a pale grey gland that weighs about 60 grams and lies parallel to and beneath the stomach. It has both endocrine and exocrine tissues. The **exocrine part** is involved in digestion with the secretion of enzymes and an alkaline solution into the small intestine via a duct. The exocrine cells form lobules with secretory cells that drain into a small duct which eventually forms the pancreatic duct.

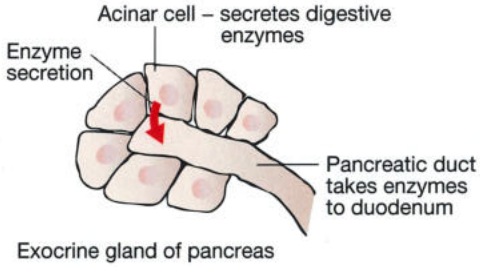


Figure 62.3 Exocrine gland of pancreas.

These secretions form pancreatic juice.

Table 62.1 Action of different digestive agents of pancreas.

Digestive agent	Optimal pH	Substrate	Result
Pancreatic amylase	7.1	Polysaccharides, e.g. starches	Disaccharides, e.g. maltose
Pancreatic lipase	8.0	Emulsified fats	Fatty acids and monoglycerides
Trypsin	8.0	Polypeptides – breaking bonds next to certain amino acids	Dipeptides
Chymotrypsin	8.0	Polypeptides – breaking bonds next to certain amino acids	Dipeptides
Carboxypeptidase	8.0	Peptides	Dipeptides and amino acids
Ribonuclease	8.0	RNA	Nucleotides
Deoxyribonuclease	8.0	DNA	Nucleotides
Bicarbonate ions		Acidic chyme	Neutralises acidity

The **endocrine part** of the pancreas secretes the hormones insulin and glucagon to control blood sugar levels.

The liver and gall bladder

The liver is a large internal organ that has many functions. It is involved in digestion as it produces bile. It also receive the products of digestion and detoxifies poisonous chemicals from the blood and maintains blood glucose levels. The liver consists of lobes which are made up of small lobules. Liver cells, **hepatocytes** synthesise the components of bile. Bile passes into the gall bladder for storage. The **gall bladder** is attached to the posterior surface of the liver by connective tissue.

QUESTIONS

1. What is meant by accessory digestive organs?
2. The diagram shows a section through a tongue and the location of taste buds.

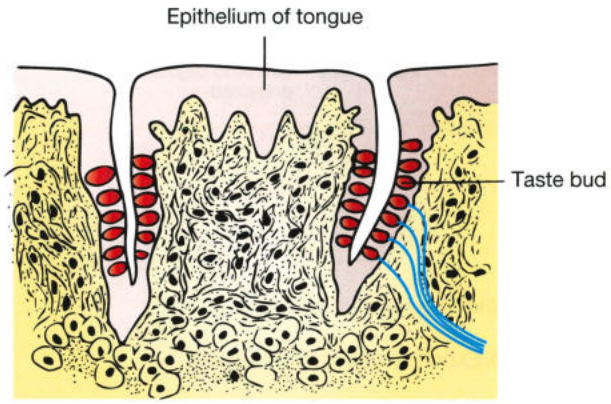


Figure 62.4 Location of taste bud in the tongue.

   (a) What are the main types of taste receptors found in taste buds?
   (b) Explain why tablets that are 'gulped' down are not 'tasted'.
3. Identify the location of the salivary glands.
4. Construct a table to summarise the functions of saliva.
5. Describe the pancreas.
6. Describe the exocrine part of the pancreas.
7. What is the main enzyme from the pancreas for:
   (a) Carbohydrate digestion?
   (b) Lipid digestion?
   (c) Protein digestion?
8. In what way is the liver involved in digestion?
9. (a) What are hepatocytes?
   (b) Give one function of hepatocytes.
10. (a) Where is the gall bladder?
    (b) What is the function of the gall bladder?
11. Besides digestive enzymes the pancreas also produces bicarbonate ions that are secreted into the pancreatic duct. What is the function of the bicarbonate ions?
12. The diagram shows a section of the alimentary canal.

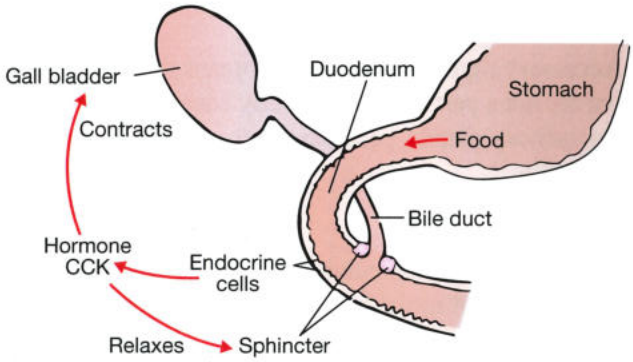


Figure 62.5 Section of the alimentary canal and gall bladder.

After a fatty meal has been eaten the hormone cholecystokinin (CCK) is secreted by endocrine cells in the mucosa of the duodenum when the fats leave the stomach. What two actions are caused simultaneously by this hormone and how will this aid digestion?

13. Which of the following correctly associates organ and function

	Organ	Function
(A)	Salivary gland	Secretes salivary protease to break down peptide bonds
(B)	Hepatocyte	Synthesises the components of bile
(C)	Pancreatic exocrine cell	Secretes pepsin to break down peptide bonds
(D)	Pancreatic endocrine cell	Secretes pancreatic amylase to break down carbohydrates to disaccharides

14. Which of the following only contains accessory digestive organs that are not a section of the alimentary canal?
   (A) Salivary glands, pancreas, duodenum, oesophagus.
   (B) Oesophagus, stomach, gall bladder, duodenum.
   (C) Salivary glands, liver, oesophagus, stomach.
   (D) Liver, gall bladder, pancreas, salivary glands.
15. Which of the following is *not* a digestive enzyme?
   (A) Insulin. (B) Amylase.
   (C) Lipase. (D) Protease.
16. The duodenum is about 25 cm long and curves around the head of the pancreas. What secretions enter the duodenum from accessory organs?
   (A) Gastric juice.
   (B) Gastric juice and pancreatic juice.
   (C) Pancreatic juice and bile.
   (D) Bile and salivary amylase.
17. What is found in saliva from the salivary glands?
   (A) Amylase.
   (B) Amylase, protease.
   (C) Amylase, mucus.
   (D) Protease, mucus.

63 Absorption

Absorption is the movement of small nutrients into the body across a membrane, e.g. from the small intestine into blood capillaries.

Assimilation is the use of absorbed nutrients to form parts of an organism.

Absorption processes

The main processes involved in absorption are diffusion, facilitated diffusion and active transport.

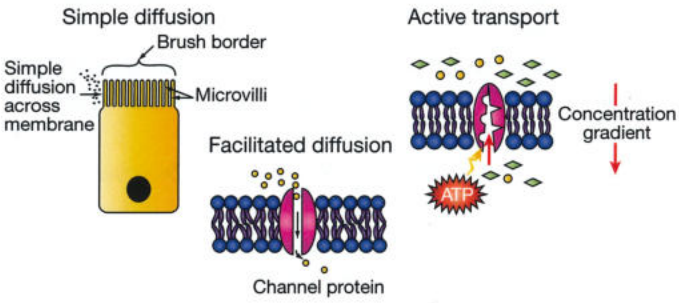


Figure 63.1 Main processes in absorption.

Simple diffusion is passive transport with substances moving *down* a concentration gradient. **Water** moves by passive diffusion by **osmosis** with water uptake often following the osmotic gradient of sodium, potassium and ATPase. If the chyme is dilute, water is absorbed through the intestinal mucosa into the capillaries in the villi. Sometimes water flows in the other direction, e.g. if the chyme is hyperosmotic then water will move from the blood plasma into the duodenum until the chyme is isosmotic with the plasma.

Lipids are absorbed into the cell by simple diffusion across the apical membrane. At the brush border surface, fatty acids and monoglycerides move out of micelles and into the epithelial cell.

Simple sugars, e.g. monosaccharides, amino acids and mineral elements may diffuse across the cell membrane, although most movement is by active transport as this is a faster process than diffusion.

Facilitated diffusion is passive transport with substances moving down a concentration gradient with the assistance of membrane proteins, e.g. glucose uses a glucose transporter. Amino acid structure is so variable there are multiple amino acid transport systems – many use facilitated diffusion.

Active transport requires energy, e.g. ATP. Substances are often moved against a concentration gradient. Ion uptake, e.g. sodium, calcium, magnesium and iron are taken up by active transport. The sodium/potassium pump is highly important in maintaining the electrochemical gradient across the epithelial cell boundary of the lumen. Disaccharides, dipeptides and tripeptides are actively transported into the villi.

Endocytosis is a form of active transport requiring energy input. Larger peptides are carried into the cell in vesicles. Immunoglobulins, e.g. IgA from plasma cells in the mucosa is absorbed through receptor mediated endocytosis and released into the intestinal lumen.

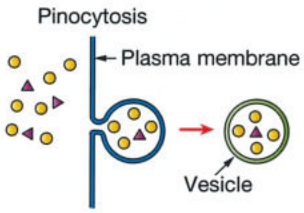


Figure 63.2 Endocytosis.

Into the bloodstream or lymphatic system

Amino acids, carbohydrates and minerals are absorbed into the bloodstream while fat soluble vitamins, fatty acids and glycerol enter the lacteal and the lymphatic system. Although large amounts of fluid enter the alimentary tract each day only some is absorbed by the small intestine and most passes to the large intestine.

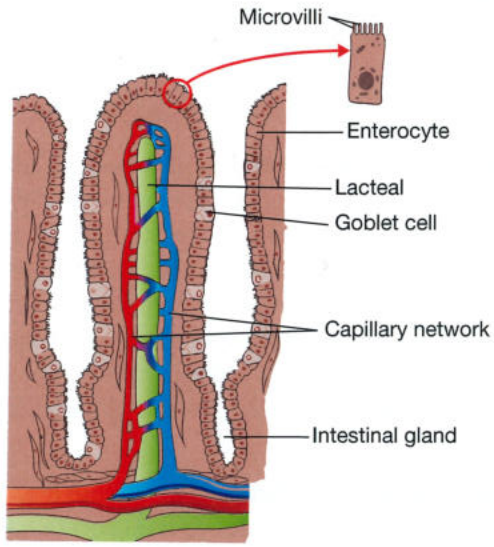


Figure 63.3 Structure of villi in the small intestine.

Some proteins are absorbed unchanged, e.g. immunoglobulins in breast milk and oral vaccines.

Stomach absorption

The stomach wall is a poor absorptive area with only small quantities of some highly lipid soluble substances, e.g. alcohol and some drugs, e.g. aspirin being absorbed. Poor absorption is due to the absence of villi and the junctions between the epithelial cells are tight junctions. **Tight junctions** are cell to cell junctions in epithelia that do not allow much movement of material between the cells.

Small intestine absorption

Most digested food particles are absorbed through the villi of the small intestine. Villi and microvilli increase surface area which increases the efficiency of absorption. Each villus extends about 1 mm from the surface of the mucosa. Smooth muscle in the wall of the small intestine causes peristalsis and food is pushed along the tract.

The middle section of the small intestine (jejunum) has many plicae each with many villi. This increases the surface area for absorption of the products of digestion. The last section (ileum) has fewer plicae, villi become smaller and there is less absorption of digested products. The presence of Peyer's patches helps prevent the spread of potentially harmful bacteria that inhabit the large intestine into the small intestine.

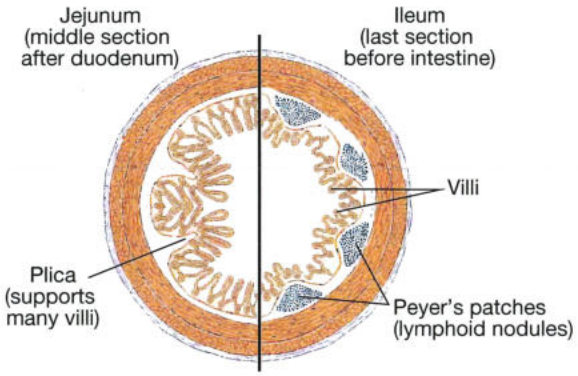


Figure 63.4 Last two sections of the small intestine.

Some **digestion of the disaccharides** maltose, sucrose and lactose is completed in the absorption process with enzymes maltase, sucrase and lactase as part of membrane proteins of microvilli. Enzymes cause hydrolysis and movement of enzymes across the membrane to discharge the monosaccharide into the cell interior. This uptake of sugar is an active process linked with sodium ion transport. The sugars enter the blood capillaries and travel to the liver.

Carrier proteins for each different peptide in the membrane of microvilli use active transport to take peptides into cell. Peptides are then hydrolysed to become free amino acids in the cytoplasm. Aminopeptidases on the surface of microvilli break short chain peptides to amino acids and they transport them across the membrane. Amino acids enter blood capillaries and travel to the liver.

Fatty acids and monoglycerides combine with bile salts to form micelles which go to the microvilli. The fatty acids and glycerol diffuse into the cell leaving the rest of the micelle behind. Inside the epithelial cell they are resynthesised into triglycerides which are then packaged with cholesterol and phospholipids to form **chylomicrons**. Chylomicrons enter the lacteal and travel in the lymph to eventually enter the blood.

QUESTIONS

1. What is absorption?
2. Define assimilation.
3. Construct a table to summarise the processes of simple diffusion, facilitated diffusion, active transport and endocytosis in absorption in the digestive system.
4. Which nutrients enter the bloodstream and which nutrients enter the lymphatic system?
5. Name some proteins that cross the membranes unchanged.
6. What are tight junctions?
7. Why are villi and microvilli needed in the small intestine?
8. (a) Why is the stomach a poor area for absorption?
   (b) Identify some substances that are absorbed through the stomach wall.
9. In what way is the middle section of the small intestine different to the end section of the small intestine?
10. Both starch and cellulose are polymers made up of glucose units, though the glycosidic linkages are different causing cellulose to form strong microfibrils. The enzyme cellulase digests cellulose. Suggest why cellulose becomes 'roughage' in the human diet.
11. The table shows vitamin absorption.

Table 63.1 Vitamin absorption.

Vitamin	Absorption
Fat soluble, e.g. vitamins A, D, E, K	Fat soluble vitamins are absorbed in the same way as lipids.
Water soluble, e.g. B, C	Water soluble vitamins can be directly absorbed into the blood from the small intestine. Vitamins B6, B12, C and folate can be absorbed in the mouth. B12 is mainly absorbed in the ileum.

Explain why vitamins A, D, E and K are absorbed differently to vitamins B and C.

12. Outline the function of Peyer's patches.
13. What is the function of maltase, sucrase and lactase as membrane proteins in microvilli?
14. Explain why the small intestine functions in mechanical digestion, chemical digestion and absorption.
15. Where does most absorption take place?
    (A) Mouth. (B) Stomach.
    (C) Small intestine. (D) Large intestine.
16. Which of the following would most likely be absorbed by simple diffusion?
    (A) Monoglycerides. (B) Triglycerides.
    (C) Sucrose. (D) Dipeptides.

64 Elimination

Elimination is the passing of undigested material out of the alimentary canal.

The **large intestine** is after the small intestine. The human large intestine consists of the caecum, appendix, colon and rectum. It is about 1.5 metres long and supplied with blood by the mesenteric arteries. The wall of the large intestine is much thinner than the wall of the small intestine and the diameter of the colon is approximately three times the diameter of the small intestine. The large intestine lacks villi.

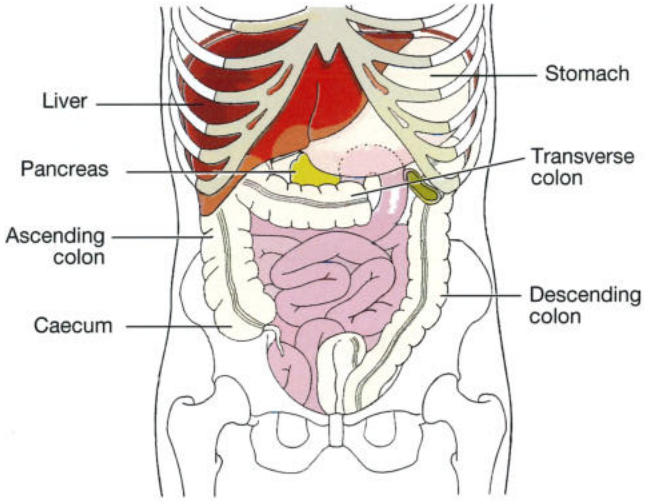


Figure 64.1 Some structures in the abdomen.

The movement of material into the large intestine is controlled by a **sphincter muscle** and **valve**. These prevent backflow into the small intestine. The lumen of the large intestine is larger than the lumen of the small intestine.

In humans the **caecum** and **appendix** are mainly bypassed as food is pushed into the ascending colon. The contents entering the colon are mainly undigested substances, e.g. cellulose, lignin and other plant material, water, mucous secretions and dead mucosal cells, fatty acids, **bile pigments** from the breakdown of haemoglobin of red blood cells in the liver and any microbes, e.g. bacteria, protozoans. Mucus is needed to help lubricate the faeces.

Large intestine microflora

The large intestine is colonised by microflora, e.g. bacteria such as *Escherichia coli, Enterobacter aerogenes, Streptococcus faecalis* and *Clostridium perfringens*. These are commensals in humans but can become pathogenic if they infect other parts of the body. Bacterial fermentation causes the release of methane, carbon dioxide and hydrogen gases. Some colonic bacteria produce significant amounts of absorbable vitamins, e.g. synthesise biotin (a B vitamin) and vitamin K.

Large intestine absorption

Water and **mineral ions** with some vitamins and some drugs are absorbed across the cell membranes into the bloodstream.

The substance that keeps moving down the colon becomes **faeces**. Faeces moves by **mass movement** rather than peristalsis. Mass movement occurs when a strong peristaltic movement occurs (about twice an hour) that forces the contents along all sections of the colon. The **rectum**, a short muscular tube that terminates at the anus, is usually empty but when mass movement pushes faeces into the rectum the stretching stimulates nerve endings. In infants defaecation is involuntary when these nerve endings are stimulated but by the age of 2 to 3 years old the defaecation reflex can be overridden and the outer striated muscle comes under voluntary control.

Diarrhoea involves contents moving quickly along the colon forming watery faeces while **constipation** involves contents being retained too long so that the faeces become hard, dry and compressed.

QUESTIONS

1. Define elimination.
2. What are the main parts of the large intestine?
3. What controls the movement of material into the large intestine?
4. The diagram shows the large intestine.

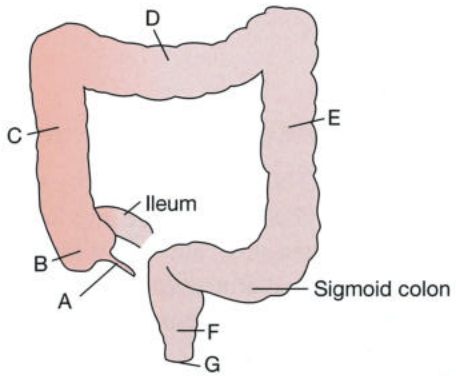


Figure 64.2 Large intestine.

Identify the parts A to G.

5. Outline how the small intestine is protected from bacteria migrating from the large intestine.
6. List some substances that are eliminated.
7. What are some differences between the wall structures of the small intestine and large intestine?
8. Why are colonic bacteria important?
9. What is mass movement?
10. Why is roughage needed in the diet?
    (A) To provide bulk to stimulate defaecation.
    (B) To prevent intestinal bacterial fermentation.
    (C) To equalise the intestinal electrolyte balance.
    (D) To complete hindgut carbohydrate digestion.

65 The Musculoskeletal System

The musculoskeletal system provides support, maintains posture, gives stability and produces movement for the body. It consists of muscles, cartilage, tendons, ligaments, joints and connective tissue.

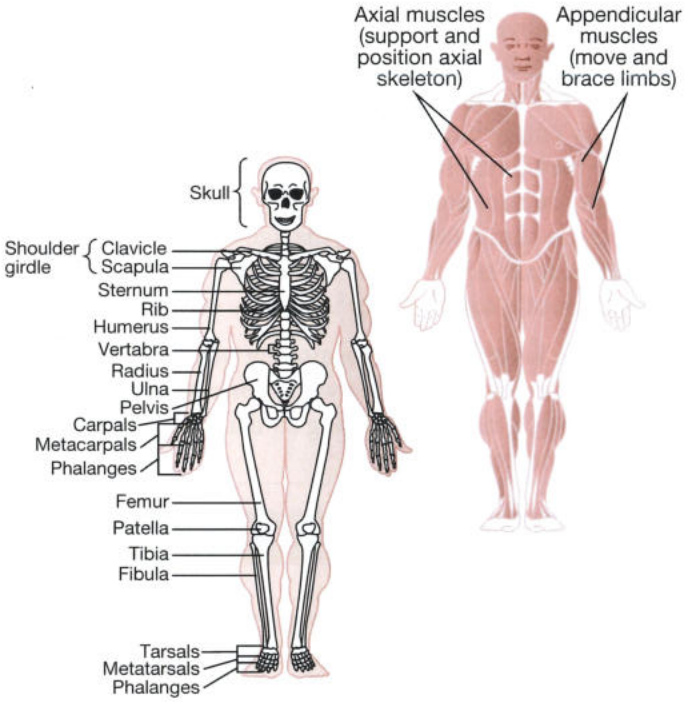


Figure 65.1 Skeletal and muscle systems.

Axial skeleton

The **axial skeleton** consists of the skull and vertebral column.

The **skull** consists of 22 bones most joined by **sutures** which are immovable joints. These joints have interlocking edges that are held together by a small amount of fibrous tissue. The **cranium** encloses and protects the brain. The **mandible** (lower jaw) is the only movable bone of the skull.

The **vertebral column** (backbone) has 33 bones in a linear sequence – 24 separate movable irregular bones, the sacrum (5 fused bones) and the coccyx (4 fused bones). The 24 separate bones are in three groups: 7 cervical, 12 thoracic and 5 lumbar. Each bone is a **vertebra**. The vertebral column and its attached muscles support the head and body. The bony processes of the vertebrae allow muscle attachment and articulate with similar structures on adjacent vertebrae.

Adjacent vertebrae are separated by **intervertebral discs** that consist of an outer rim of fibrocartilage and a central core of soft gelatinous material. These discs act as shock absorbers and allow movement between adjacent vertebrae.

Ligaments are fibrous connective tissue that joins bones together at joints. Ligaments hold vertebrae together and help keep the intervertebral discs in place.

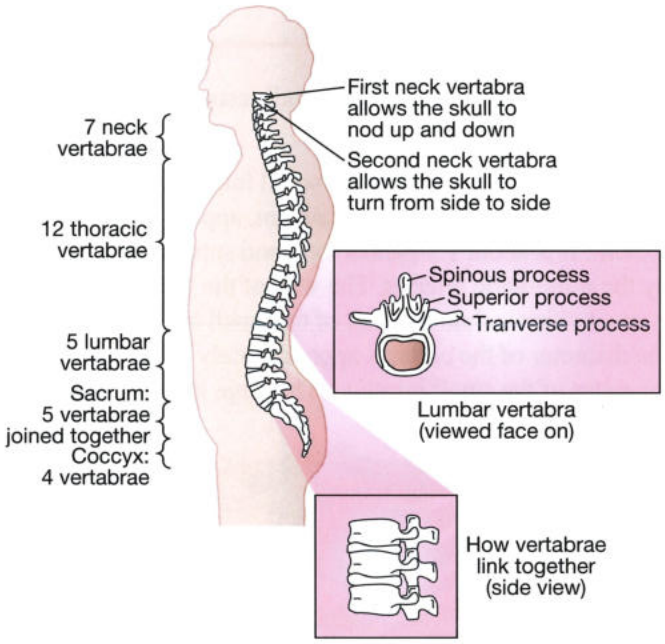


Figure 65.2 Vertebrae and vertebral column.

Appendicular skeleton

The **appendicular skeleton** consists of the limb girdles and limbs and is important in giving mobility, the ability to control your environment and the ability to change position in space. It allows you to stand, walk, write, pick up objects, feed yourself, changes your clothes etc.

The **shoulder girdle** (also called the pectoral girdle) consists of the **clavicle** (collarbone) and **scapula** (shoulder blade) which articulate with the arms.

The **pelvic girdle** consists of the **sacrum** and **hip bones** with each hip bone consisting of three fused bones. The female pelvis is more shallow and rounded than the male pelvis to allow for the passage of the baby during childbirth.

The limbs show the pentadactyl structure.

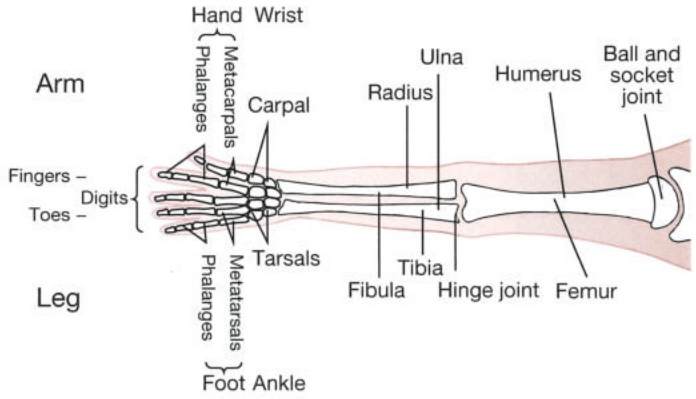


Figure 65.3 Human arm shows pentadactyl structure.

In the forearm the **ulna** is longer than and medial to the **radius**.

QUESTIONS

1. Outline the function of the musculoskeletal system.
2. What are the components of the musculoskeletal system?
3. Distinguish between the axial skeleton and the appendicular skeleton.
4. How many bones are in the skull?
5. What is a suture?
6. (a) What are ligaments?
   (b) Where are ligaments in the axial skeleton?
7. What are the two girdles of the appendicular skeleton?
8. If human bones are found, what physical feature is examined to determine the gender of the person?
9. What is the pentadactyl limb?
10. The diagram shows the development of the curves of the spine.

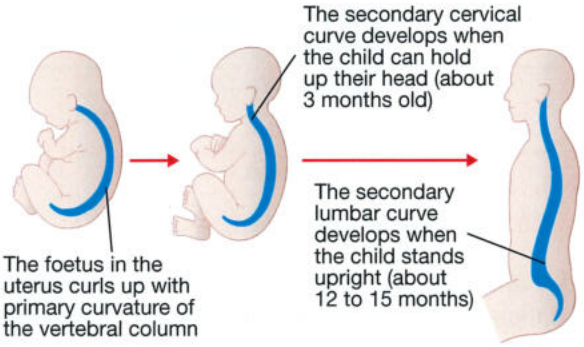


Figure 65.4 Spinal curve development.

Describe the series of steps in the development of the curves of the vertebral column.

11. The diagram shows a skeleton.

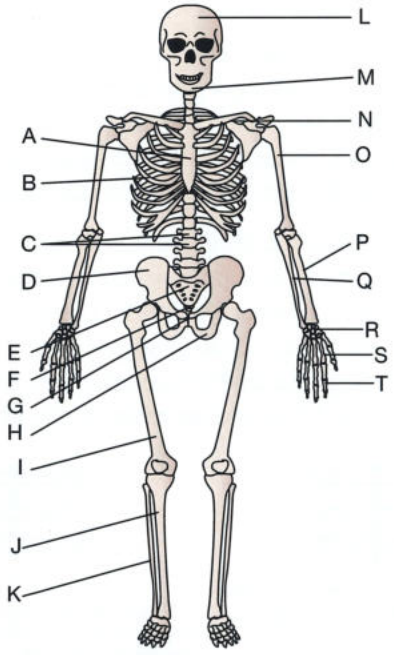


Figure 65.5 Human skeleton.

Label the bones of the body.

12. What is the correct number of bones in the vertebral column?

	Cervical vertebrae	Thoracic vertebrae	Lumbar vertebrae
(A)	5	12	7
(B)	12	5	7
(C)	7	5	12
(D)	7	12	5

13. Which bone is the only movable bone in the skull?
   (A) Mandible.
   (B) Hyoid bone.
   (C) Occipital bone.
   (D) Parietal bone.
14. How many separate movable bones are in the vertebral column?
   (A) 33 (B) 24 (C) 12 (D) 7

Use the diagram for the next THREE questions.

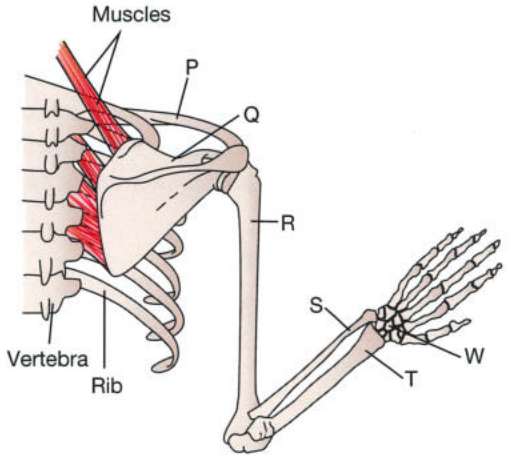


Figure 65.6 Part of the human skeleton.

15. What is the identity of bones P and Q?

	Bone P	Bone Q
(A)	Clavicle	Sternum
(B)	Sternum	Scapula
(C)	Clavicle	Scapula
(D)	Scapula	Clavicle

16. Which bone is the radius?
   (A) Bone P (B) Bone R
   (C) Bone S (D) Bone T
17. To which division of the skeleton do bones P, Q, R, S and T belong?
   (A) Axial skeleton.
   (B) Appendicular skeleton.
   (C) Thoracic skeleton.
   (D) Abdominal skeleton.

66 Muscles

There are three types of muscle tissue – skeletal muscle, smooth muscle and cardiac muscle.

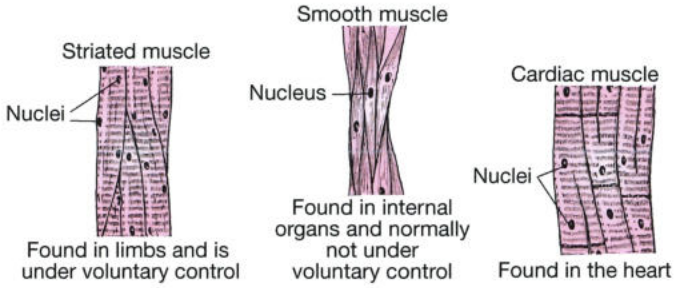


Figure 66.1 Three types of muscle cells.

Limb muscles consist of skeletal muscle fibres, connective tissue, blood vessels and nerves. Typical skeletal muscles move bone on bone and attach across a movable joint.

Skeletal muscle, also called **striated** or **voluntary** muscle, consists of cylindrical cells called **muscle fibres**. Muscle fibres are very large, multinucleated and lie parallel to each other. Each muscle fibre has several nuclei just under the **sarcolemma** which is the cell membrane of each muscle fibre. The sarcolemma surrounds the sarcoplasm (cytoplasm). Sarcoplasm contains many **myofibrils**. Skeletal muscle is organised into smaller and smaller units.

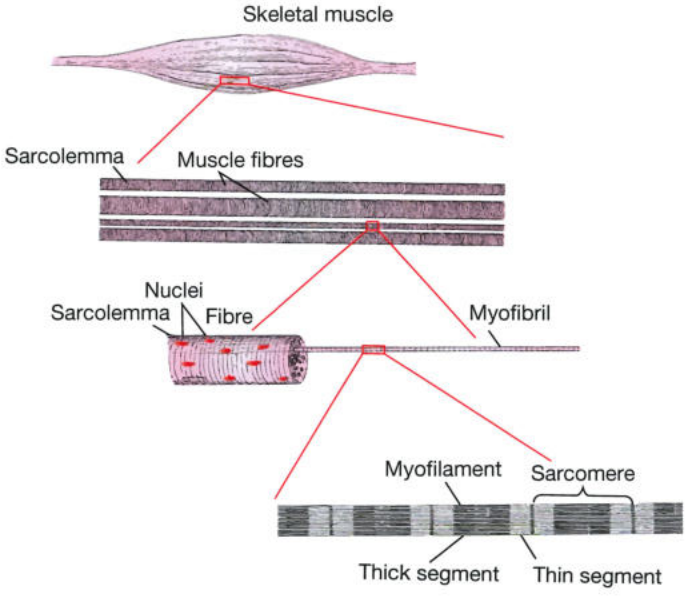


Figure 66.2 Structural organisation of skeletal muscle.

Each muscle fibre consists of myofibrils that lie longitudinally inside the fibre. **Myofibrils** are cylinders (1 to 2 μm diameter) and as long as the entire cell. Myofibrils can shorten and cause contraction. Each myofibril is attached to the sarcolemma at each end of the cell thus contraction shortens the cell. Myofibrils are composed of thin filaments and thick filaments.

Myofilaments are surrounded by sarcoplasmic reticulum (like smooth ER) which forms a network with transverse tubules. Myofilaments are protein filaments consisting mainly of actin and myosin. Actin is in thin filaments and myosin is in thick filaments. The actin and myosin are organised in sarcomeres which repeat along the myofilament. A **sarcomere** is the fundamental repeating unit of a skeletal muscle myofilament and is the segment between two Z lines. The arrangement of the thin and thick filaments gives the dark and light banded appearance of a myofibril.

Thus skeletal muscle has **alternating light and dark bands** (striations) with no individual cells visible and a number of nuclei scattered along its length.

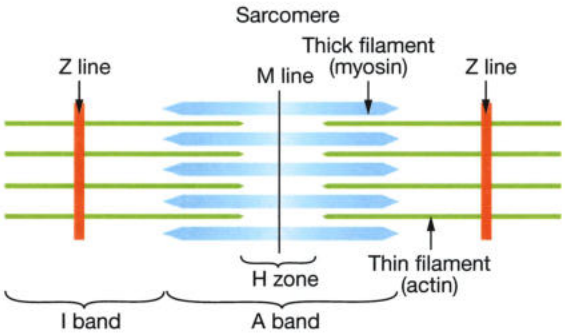


Figure 66.3 Structure of a sarcomere.

The **I band** is the area near the edge of the sarcomere where there is only thin filaments. The thin filaments are attached to the Z line. The **A band** is the central region determined by the length of the thick filaments where the thin and thick filaments overlap including the H zone where there are only thick filaments.

Sliding filament theory

The **sliding filament theory** provides a mechanism for muscle contraction. During a contraction neither thin (actin) filaments nor thick (myosin) filaments change in length, but slide past each other when the myosin heads of the thick filaments bind to the active sites on the thin filaments.

When **cross bridge binding** occurs, the myosin head pivots towards the M line pulling the thin filament towards the centre of the sarcomere. The cross bridge detaches and returns to its original position. This continues causing overlap between the thin and thick filaments and the Z line moves towards the M line.

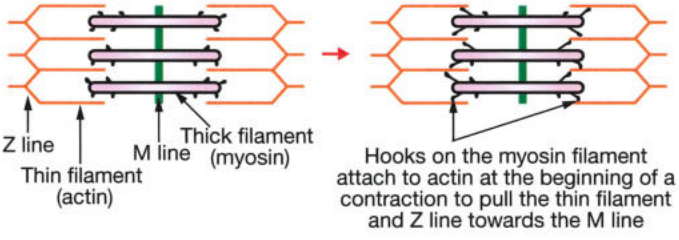


Figure 66.4 Structural organisation of skeletal muscle.

Striated muscle is capable of rapid contraction though the contraction cannot be held as long as a contraction of smooth muscle. Muscle fibres contain a large number of mitochondria to supply the energy needed for contraction.

QUESTIONS

1. What are the three types of muscle cells?
2. List the components of limb muscles.
3. Explain why skeletal muscle is often called striated muscle.
4. Skeletal muscle consists of smaller and smaller units. What is the order of these units from large to small?
5. Construct a table to describe skeletal muscle, muscle fibres, myofibrils and myofilaments.
6. Define a sarcomere.
7. Distinguish between the I band and the A band in a sarcomere.
8. What is the sliding filament theory?
9. The diagram shows two electron micrographs of skeletal muscle.

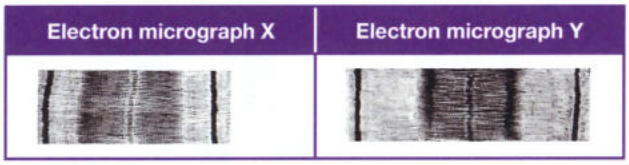


Figure 66.5 Electron micrographs of skeletal muscle.

Identify which is the contracted form of the sarcomere and give reasons for your decision.

10. Distinguish between the proteins found in the thin filaments and in the thick filaments.
11. Compare the contraction of skeletal muscle with smooth muscle.
12. What supplies the skeletal muscle with the energy for contraction?
13. Muscle fibres do not extend the whole length of a skeletal muscle.

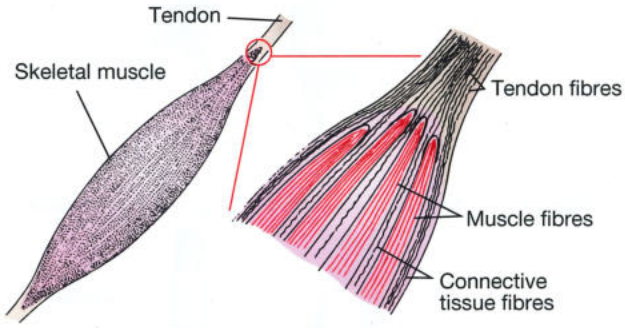


Figure 66.6 Section of skeletal muscle.

Muscle fibres do not extend the complete length of a muscle and individual fibres vary in length.

(a) Whish section of a muscle contains the most muscle fibres?
(b) What surrounds groups of muscle fibres?

14. Which of the following shows the organisation of skeletal muscle units from largest to smallest?
(A) Sarcomere, myofilament, myofibril, fibre.
(B) Fibre, myofilament, myofibril, sarcomere.
(C) Sarcomere, fibre, myofilament, myofibril.
(D) Fibre, myofibril, myofilament, sarcomere.
15. The diagram shows a cross-section of a skeletal muscle.

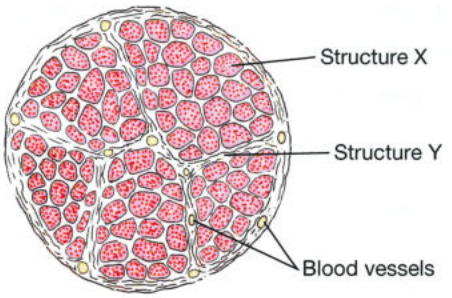


Figure 66.7 Cross-section of skeletal muscle.

Identify structure X and structure Y.

	Structure X	Structure Y
(A)	Bundle of fibres	Connective tissue
(B)	Connective tissue	Bundle of fibres
(C)	Myofilament	Fibre
(D)	Fibre	Myofilament

16. What is a sarcomere?
(A) The cytoplasm of a muscle fibre.
(B) The cell membrane of a muscle fibre.
(C) Fundamental repeating unit of a myofilament.
(D) The area where there is only thin filaments.
17. Skeletal muscle is the greatest mass of muscle in the human body. When we eat 'meat' it is usually skeletal muscle from an animal. Which group of organic compounds makes up the bulk of skeletal muscle?
(A) Carbohydrates.
(B) Protein.
(C) Lipids.
(D) Nucleic acids.
18. The diagram shows a section of muscle fibres.

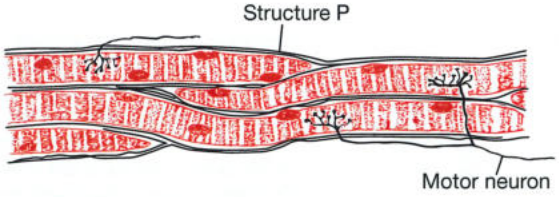


Figure 66.8 Section of muscle fibres.

What is structure P?
(A) Sarcomere.
(B) Connective tissue.
(C) Nucleus.
(D) Sarcolemma.

67 Musculoskeletal Terminology

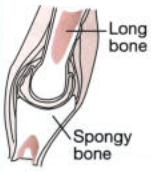
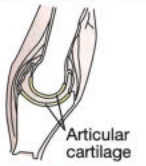
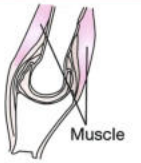
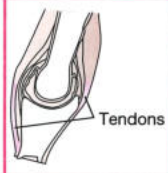
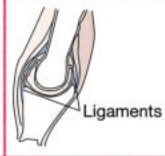
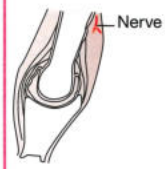
Anatomy uses many terms to describe regions and parts of the body, movement, anatomical position and terms of direction and relationship. Many of the terms have Latin or Greek roots and it is handy to have some knowledge of these terms.

The body is divided into head (capitis), neck (cervix), trunk and limbs. The trunk is divided into the chest (thorax) and abdomen.

Superior – means towards the head.

Inferior – means towards the feet.

Table 67.1 Particular body parts.

Part	Diagram	Description
Bone		Bones are connective tissue of living cells in a strong matrix of calcium salts, e.g. calcium phosphate and calcium carbonate and collagen fibres. Bones provide a strong structure to support the body and provide attachment places for muscles. **Ossicles** are small bones.
Articular cartilage		Type of connective tissue with collagen fibres. Lines the ends of bones to reduce friction and allow smoother movement.
Muscle		A muscle is a tissue made of long cells capable of contracting when stimulated by nerve cells. The contraction of the muscle causes the bone to move. Muscles attach two bones.
Tendon		Tendons are a type of connective tissue of collagen fibres that connect muscle to bone. When the muscle contracts the tendon pulls on the bone and the bone moves. Tendons do not stretch or contract.
Ligament		Dense band of connective tissue with fibres that join bones together at joints. Ligaments prevent bones moving apart.
Nerve		A nerve is a bundle of neurons wrapped in connective tissue that transmits messages. When a person makes a voluntary movement a message passes from the cerebral cortex along a nerve to the muscle. Receptors detect movement and the relative position of limbs.
Synovial membrane		Encloses a joint and produces synovial fluid to lubricate the joint.

Ventral – means towards the front of the body.

Dorsal – means towards the back of the body.

Medial – means closer to the midline.

Proximal – means closer to the trunk or point of origin.

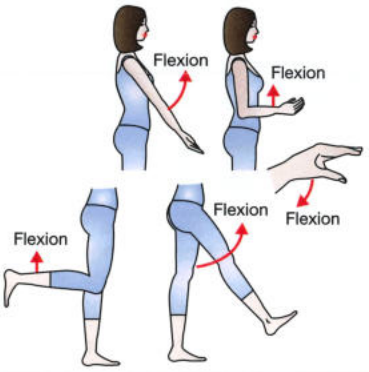
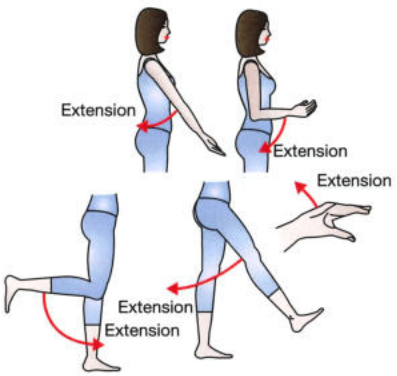
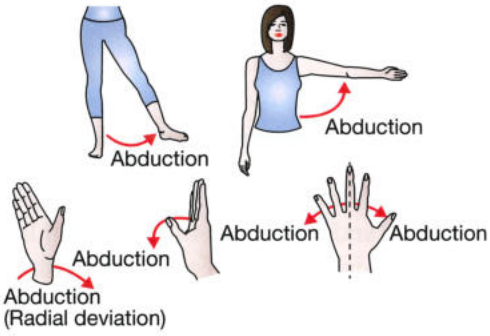
Distal – means farther from the trunk or point to origin.

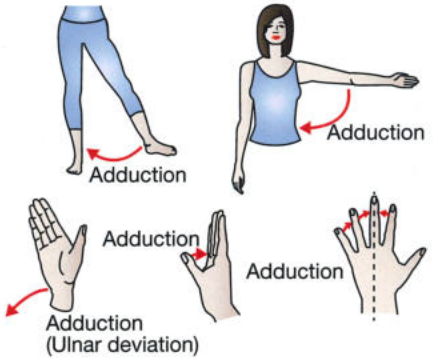
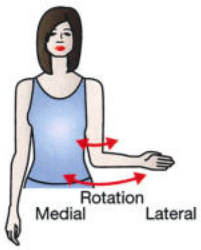
Articulation is a joint between two or more bones and an articulating surface is the part of the bone that enters into the formation of a joint.

Terms of movement

The table summarises several kinds of movement.

Table 67.2 Types of movement.

Type of movement	Description
Flexion	Means bending decreases the angle between two parts. 
Extension	Is the opposite of flexion and usually straightens out a bent part. 
Abduction	Means moving apart or away from the midline. 

Type of movement	Description
Adduction	Is the opposite of abduction and brings parts together or towards the midline. 
Rotation	Is a twisting of a part on an axis. **Lateral** (external) rotation occurs if the anterior surface of the part is turned laterally and **medial** (internal) rotation occurs if it is turned medially. 

QUESTIONS

1. From which language(s) do we get the root of many anatomical terms?
2. What are the major divisions of the body?
3. For anatomical position distinguish between inferior and superior.
4. To which parts of the body do the following terms apply?
   (a) Capitis. (b) Thorax. (c) Abdomen.
5. (a) What is flexion and what is its opposite movement?
   (b) Give an example of flexion.
6. The diagram shows movement of the arm and hip joint.

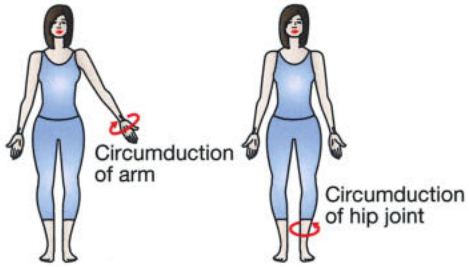


Figure 67.1 Circumduction.

What is happening in circumduction?

7. Distinguish between ventral and dorsal.
8. Distinguish between proximal and distal.
9. What is meant by articulation and an articulating surface?
10. The cervical vertebrae are the smallest of the vertebrae. The atlas holds up the head and during development the atlas fuses to the body of the axis which is the second vertebra. The atlas is a ring of bone with two short transverse processes and the axis has a small body with upward projecting processes that articulate with the atlas.

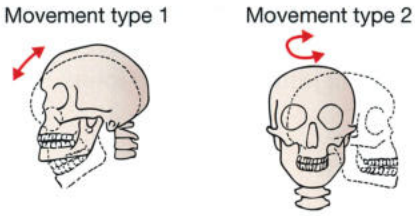


Figure 67.2 Atlas/axis articulation.

Identify the two types of movement due to atlas/axis articulation.

11. What is the name of the dense fibres of connective tissue that join bones to bones?
12. What is the name of connective tissue fibres that connect muscle to bone?
13. The diagram shows a motor neuron attached to skeletal muscle fibres.

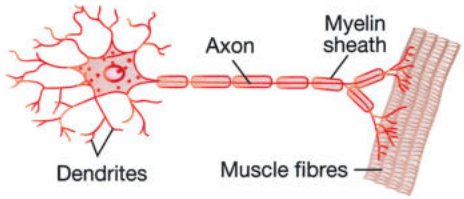


Figure 67.3 Motor unit consisting of motor neuron and muscle fibre.

Outline what would happen if the nerve supply to the muscle fibre was cut.

14. What is articular cartilage?
   (A) Connective tissue with crystalline calcium deposits.
   (B) Epithelial tissue with crystalline calcium deposits.
   (C) Epithelial tissue lining the ends of bones.
   (D) Connective tissue lining the ends of bones.
15. The diagram shows a synovial joint.
   What are structure X?
   (A) Ligaments.
   (B) Tendons.
   (C) Synovial membrane.
   (D) Articular cartilage.

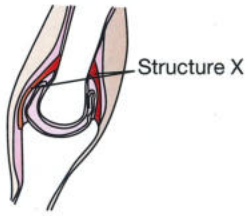


Figure 67.4 Synovial joint.

16. The diagram shows a human skull.
   Which position is ventral?

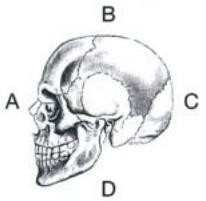


Figure 67.5 Human skull.

68 Joints

Joints occur where two bones meet. There are several types of joints based on their structure – fixed or fibrous joints, cartilaginous or slightly movable joints and synovial or freely moving joints.

Synovial joints enable a wide range of movement and the main types of joints are ball and socket joint, hinge joint, gliding joint, pivot joint and saddle joints.

Fixed joints

Fixed joints, also called immovable or fibrous joints have fibrous tissue between the bones. **Sutures** are fixed joints between bones of the cranium. The fibrous tissue is predominantly collagen fibres.

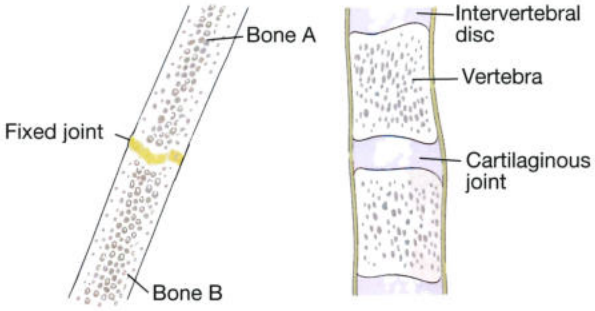


Figure 68.1 Fixed joint and slight movable joint.

Slightly movable joints

Slightly movable joints (also called cartilaginous joints) have pads of fibrocartilage that form a disc between the bones, e.g. between vertebrae in the vertebral column, the symphysis pubis in the pelvis and between ribs and the sternum.

Synovial joints

Synovial joints are freely movable joints with a synovial cavity separating the bones. There is a **capsule** (capsular ligament) surrounding the joint holding the bones together and an inner synovial membrane. The capsule consists of collagen fibres and dense connective tissue. The **synovial membrane** consisting of epithelial cells lines the capsule and secretes **synovial fluid** into the **synovial cavity**. The synovial fluid provides nutrients for the components of the joint cavity, acts as a lubricant, helps provide stability, has phagocytes for defence and removing cellular debris caused by the wear and tear of joint use.

Articular cartilage is at the ends of the bones where the bones are in contact. This cartilage provides a smooth interface and can withstand compression during movement. Ageing decreases the thickness of the cartilage lining as it receives no blood supply or little nourishment from synovial fluid.

Ligaments are a type of connective tissue with fibres that join bones together at joints. Ligaments prevent bones moving apart.

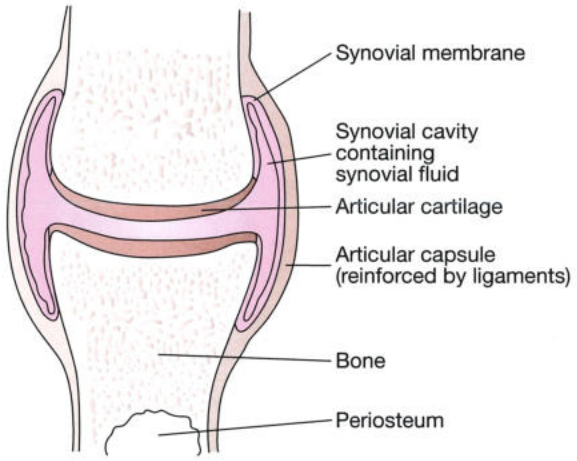


Figure 68.2 Synovial joint.

Tendons are a type of connective tissue made of fibres that connect muscle to bone. When the muscle contracts the tendon pulls on the bone and the bone moves. Tendons do not stretch or contract. The **insertion of a muscle** is the point of attachment of the tendon onto a bone.

Ball and socket joints

Ball and socket joints, e.g. shoulder joint and hip joint allow movement in three planes. The shoulder joint has a shallow socket which means that the joint is easily dislocated. A **dislocated joint** occurs when the bone is displaced from its socket with accompanying damage to ligaments and sometimes tendons.

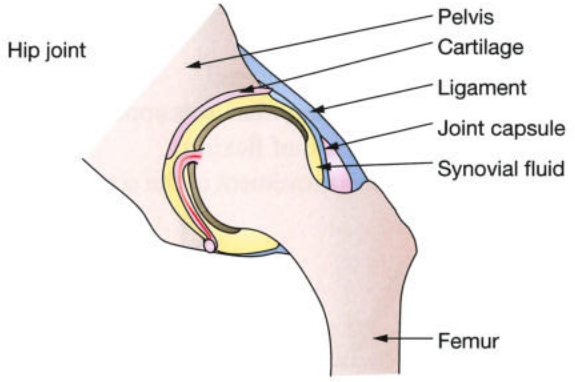


Figure 68.3 Hip joint.

Hinge joints

Hinge joints restrict movement to a single plane, e.g. the elbow joint, the knee joint, the ankle joint and between the phalanges of the fingers and toes. Hinge joints are very simple synovial joints when they do not bear much body weight or mechanical force, e.g. knuckles of fingers and toes.

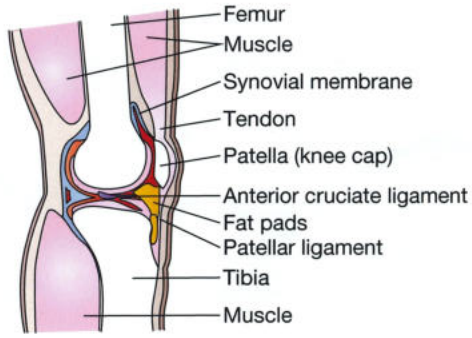


Figure 68.4 Knee joint.

The elbow joint is more complex as it is subject to stronger forces. The elbow joint is between the humerus and the ulna and radius. The joint has an extensive synovial membrane and ligaments on either side. Tension in the ligaments restricts movement.

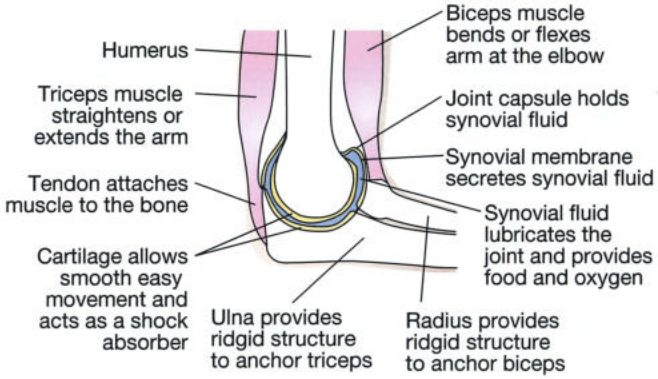


Figure 68.5 Elbow joint.

Movement at joints

There are several kinds of movement that occur at joints – flexion/extension, abduction/adduction and rotation.

Flexion and **extension** change the angle between body parts. Flexion decreases the angle, e.g. bending the knee and extension increases the angle, e.g. straightening the leg at the knee.

Adduction and **abduction** move a structure with respect to the centre of the body. Abduction pulls a structure away from the middle of the body, e.g. moving the leg to the side while adduction pulls a structure towards the midline of the body, e.g. returning the leg to the body.

Rotation refers to a movement that spins a structure on an axis, e.g. rotating the tibia and fibula of the lower leg. Medial rotation (also called internal rotation) is towards the midline while lateral rotation is away from the midline.

QUESTIONS

1. What are the three types of joints?
2. Identify the main types of synovial joints.
3. Describe the structure of a fixed joint.
4. What is a suture?
5. Describe a cartilaginous joint.
6. (a) What is a synovial joint?
   (b) Outline the structure of a synovial joint.
7. Outline some of the functions of synovial fluid.
8. Where do you find articular cartilage and what is its function?
9. What is a ligament?
10. Define a tendon.
11. Outline what occurs when a person has a dislocated joint.
12. Construct a table to summarise the different types of movement at joints.
13. Construct a table to compare a ball and socket joint with a hinge joint, with an example, its location, its function, axes of movement, kind of movement, capsule, synovial cavity, fat pad and stability.
14. The diagram shows several types of joint movement.

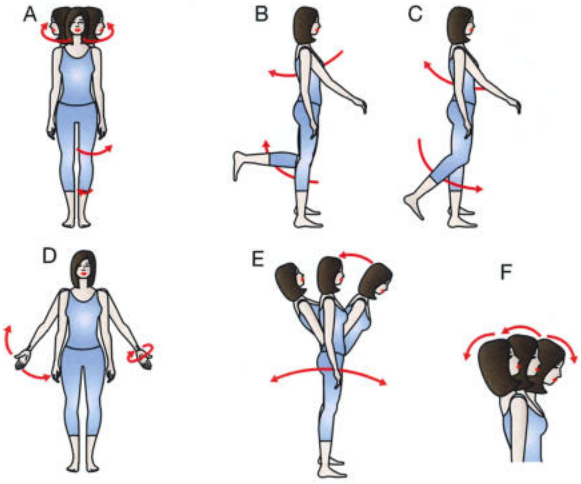


Figure 68.6 Types of joint movement.

Identify each type of movement.

15. The diagram shows the contraction of the biceps.

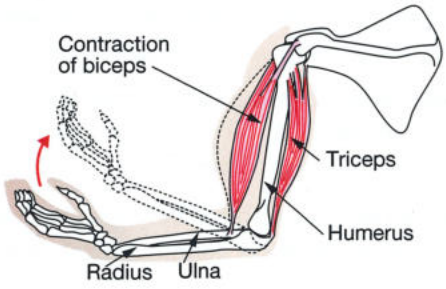


Figure 68.7 Contraction of biceps and joint movement.

What is the name of this type of joint movement?

(A) Flexion. (B) Extension.
(C) Adduction. (D) Abduction.

16. Where will you find a simple hinge joint?
   (A) Elbow joint.
   (B) Shoulder joint.
   (C) Between phalange bones.
   (D) Cranium.

69 Paired Muscles

Skeletal muscles occur as a pair of antagonistic muscles and movement results from the actions of these paired muscles. The muscles also help maintain posture. Muscles that carry out an action are called **prime movers** (protagonists or agonists). Other muscles act as **stabilisers**.

Antagonistic muscles are a pair of muscles that oppose each other, one must relax while the other contracts, e.g. biceps and triceps. A **synergist** is the muscle that contracts at the same time as the prime mover and the **antagonist** opposes the prime mover.

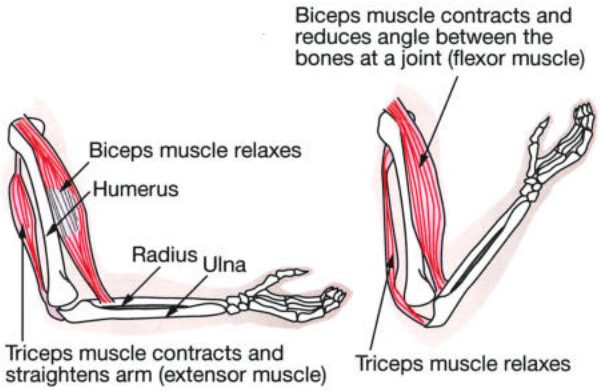


Figure 69.1 Antagonistic muscles.

Levers

Muscles attached across a joint act like a **lever system** with the joint acting as the pivot point (fulcrum). Skeletal muscles in lever systems provide speed and greater movement with less expenditure of energy. Levers give mechanical advantage and greater efficiency. Not every muscle is part of a lever system.

First class lever – the fulcrum is between the load (resistance) and the effort (applied force), e.g. seesaw. Bending the head forward is an example of a first class lever.

Second class lever – the load is between the effort and fulcrum, e.g. wheelbarrow. Rising onto your toes is an example of a second class lever.

Third class lever – the effort is between the load and the fulcrum. These are the most common lever systems in the body.

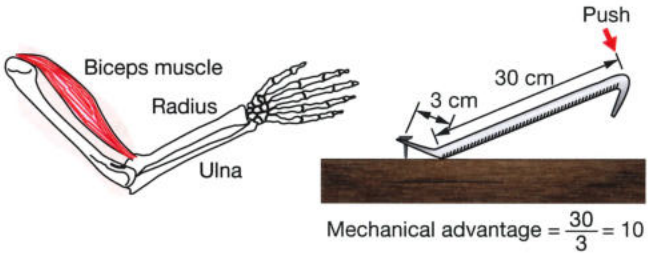
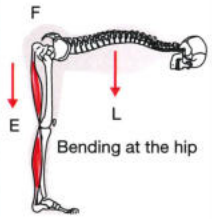
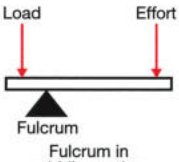
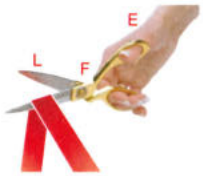
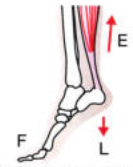
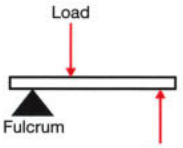
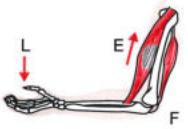
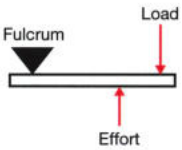


Figure 69.2 Levers provide mechanical advantage.

Table 69.1 Classes of levers and examples.

Body example	Class lever	Hand example
 Bending at the hip	 Fulcrum in middle section	 Scissors
 Raising the body on the toe	 Load in middle section	Nutcracker
 Raising the forearm	 Effort in middle section	Forceps

Nervous control

The nervous control of most antagonistic muscles is by reflex arcs. The stimulus for the contraction of one of the pair of muscles also causes the inhibition of the opposing muscle. Every muscle has at least one nerve branch. Limb muscles often have two or more nerve fibres from spinal nerves.

The nerve to a muscle will have **motor neurons** to cause contraction of the muscle and will also have a large number of **sensory neurons**, e.g. connected to pain receptors that detect muscle soreness due to overexercise or strain or from tears of the muscle or tendon. Many receptors are **proprioceptive receptors** that detect the positions of bones, joints and muscles monitoring the stretch of contraction of a muscle and the tension within a tendon.

A **muscle spindle** is a sensory end organ in a muscle with proprioceptors that monitor the length of skeletal muscles. A muscle spindle contains a group of 2 to 10 small muscle fibres grouped by the sensory neuron. The resting tension in a skeletal muscle is called **muscle tone**.

Nervous control is highly important as most movements require the coordination of a number of individual muscles.

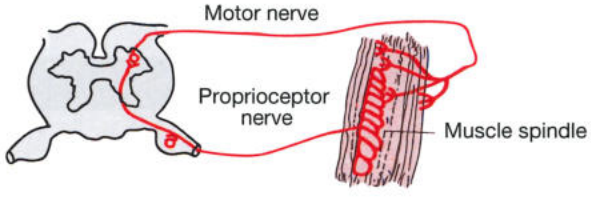


Figure 69.3 Muscle spindle.

QUESTIONS

1. What are antagonistic muscles?
2. Define a prime mover.
3. Distinguish between an antagonist and a synergist.
4. What is a lever system?
5. Distinguish between first class, second class and third class lever systems.
6. Which lever system is the most common in the human body?
7. The diagram shows a person raising their head.

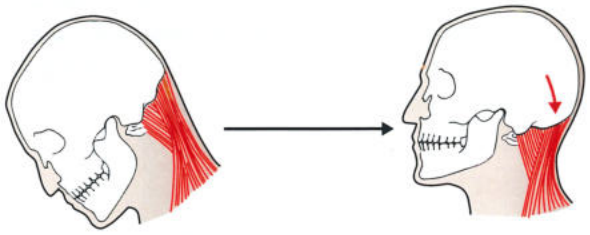


Figure 69.4 Person raising their head.

(a) What class of lever is this situation?
(b) Draw a lever diagram to show the load, effort and fulcrum for this situation.

8. Why do paired muscles need to be under nervous control?
9. Which type of neuron will stimulate a muscle to contract?
10. The diagram shows antagonistic muscles.

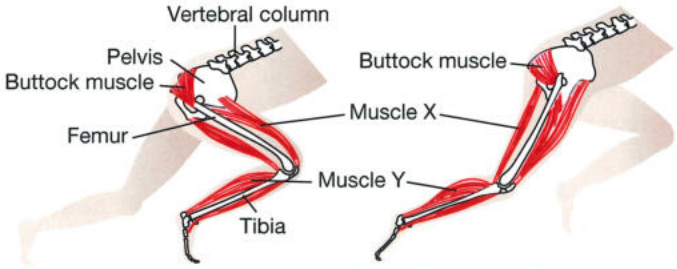


Figure 69.5 Antagonistic muscles.

Use the diagram to explain how antagonistic muscles are involved in running.

11. What are proprioceptors?
12. Define muscle tone.
13. What is a muscle spindle?
14. What supplies energy for the contraction of a muscle?
15. When the forearm is raised by bending at the elbow, what happens to the biceps and triceps?

	Biceps	Triceps
(A)	Contracts	Straightens
(B)	Straightens	Contracts
(C)	Contracts	Contracts
(D)	Straightens	Straightens

16. The diagram shows muscles acting as part of a lever system.

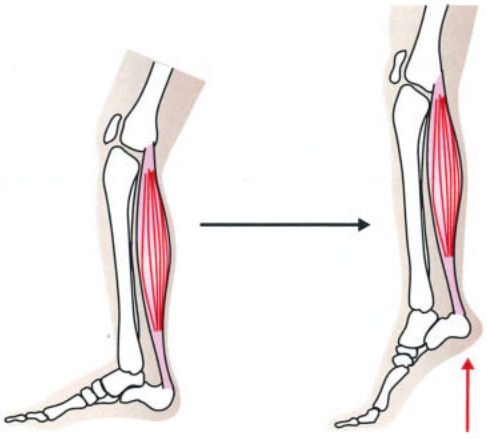


Figure 69.6 Muscles as part of a lever system.

What class of lever is shown in this example?
(A) First class lever.
(B) Second class lever.
(C) Third class lever.
(D) Fourth class lever.

17. What is the difference in function between a tendon and a ligament?

	Tendon	Ligament
(A)	Attach bone to bone	Attach muscle to bone
(B)	Surround synovial cavity	Attach muscle to bone
(C)	Attach muscle to bone	Attach bone to bone
(D)	Attach bone to bone	Surround synovial cavity

18. The diagram shows one diagram with the actual attachment of the biceps muscle and another diagram with alternate attachment points.

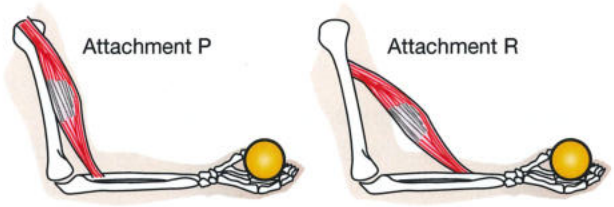


Figure 69.7 Muscle attachment points.

If both muscles are the same size, how would the different attachment affect movement?
(A) Both attachment arrangements would have equal movement ranges.
(B) Attachment R would be able to move backwards more than attachment P.
(C) Attachment R would have a greater range of movement than attachment P.
(D) Attachment R would have a more limited range of movement than attachment P.

70 Bone

Bone (osseous tissue) is the hardest connective tissue and makes up most of the skeleton being about 15% of the mass of an adult.

Bone consists of connective tissue in a solid matrix that contains minerals that form crystals of **hydroxyapatite** which incorporates ions of sodium, magnesium and fluoride with calcium carbonate and calcium phosphate to give strength and rigidity. The combination of flexible collagen and mineral salts gives a degree of flexibility and hardness without being brittle. About one third of the weight of bone is from collagen fibres.

The naked eye can identify two types of bone – compact bone and spongy bone.

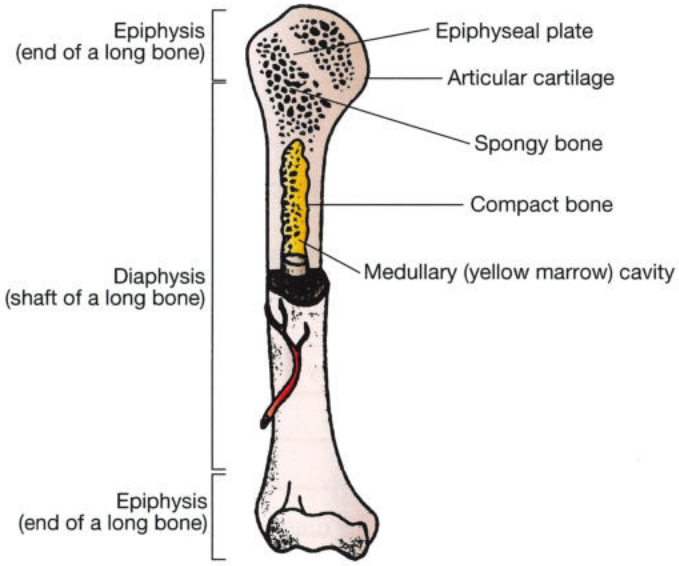


Figure 70.1 Long bone.

Osteocytes

Osteocytes are mature bone cells that secrete and maintain the matrix with both the release of calcium from bone to blood and the deposit of calcium salts into the matrix. They are surrounded by collagen fibres and blood vessels that run in **Haversian canals** in the matrix supplying the bone cells with nutrients and removing wastes. Osteocytes have stopped dividing and are located in small pockets called **lacunae**.

Osteoblasts

Osteoblasts are bone-forming cells that secrete collagen and other parts of bone tissue. They are actively dividing bone cells and if an osteoblast becomes surrounded by matrix it differentiates into an osteocyte. **Ossification** is the process of laying down the inorganic parts of bone.

Osteoclasts

Osteoclasts are giant cells with many nuclei that help maintain optimum bone shape. They secrete acids by exocytosis from lysosomes. The acids dissolve the bony matrix and release stored calcium and phosphate minerals in **osteolysis**. Many osteoclasts are under the periosteum and help remove callus during the healing of fractures.

There is a balance between osteoblasts which are producing matrix that binds minerals and osteoclasts which are removing matrix and releasing minerals. If an imbalance occurs with the osteoclasts removing calcium salts at a faster rate than the action of osteoblasts the bone will weaken. If the imbalance has increased activity of the osteoblasts, then the bone will become more massive and stronger.

Compact bone

Compact bone is very dense with osteocytes and matrix arranged in Haversian systems. It forms the outer surface of all bones and is laid down in concentric layers. Compact bone varies in hardness and thickness.

A **Haversian system** (osteon) is a cylinder of bone with circling layers of mineralised matrix called **lamellae** surrounding a central Haversian canal. The canals usually run parallel to the surface of the bone. Small cytoplasmic processes extend from immature osteocytes through tiny tubes called **canaliculi**. The canaliculi later fill with tissue fluid and become the main transport route for supplies to/from the bone.

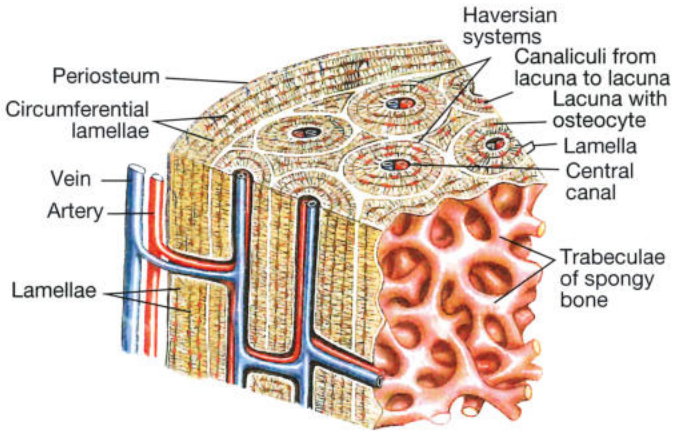


Figure 70.2 Haverisian systems in bone.

Spongy bone

Spongy bone lies within compact bone in larger bones and consists of very thin plates of bone called trabeculae with interspersed areas of bone marrow. It appears spongy in appearance with the open network of plates and spaces. This arrangement makes it much lighter than compact bone and is found where the bone is not under heavy stress or the stress is applied from several directions. There are no Haversian systems in spongy bone.

The **medullary cavity** is the space within the bone that contains marrow.

Trabeculae contain osteocytes and the matrix has less inorganic material than the matrix of compact bone. They develop along the lines of stress within bone.

Red bone marrow tissue is found at the head of long bones and is the site of red blood cell formation. It is very soft tissue and less dense than bone.

Yellow bone marrow tissue is found in the shaft of the long bone and consists mainly of fat.

Periosteum is the outer covering on the bone surface. It consists of a fibrous outer layer and a cellular inner layer. It protects the bone from surrounding tissue, is involved in bone growth and repair, is a place of attachment for blood vessels and nerves and near joints become continuous with the connective tissues that surround the joint and help stabilise the joint. Some of the fibres of the periosteum penetrate the bone and join with internal collagen fibres.

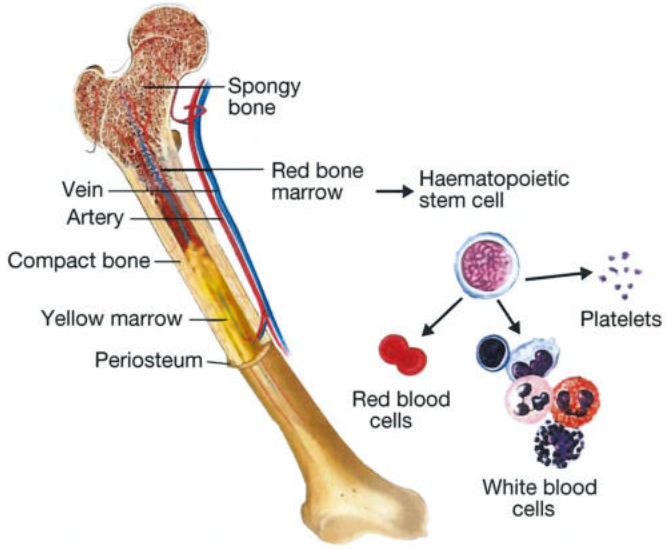


Figure 70.3 Parts of a long bone.

QUESTIONS

1. In an adult, approximately what proportion of their mass is bone?
2. What is hydroxyapatite?
3. Identify the two types of bone that can be seen with the naked eye.
4. What are osteocytes?
5. Distinguish between epiphysis and diaphysis.
6. What is the epiphyseal plate?
7. Distinguish between lamellae and lacunae.
8. What is a Haversian system?
9. Construct a table to distinguish between osteocyte, osteoblast and osteoclast.
10. Define ossification.
11. Explain why there needs to be a balance between the activities of osteoblasts and osteoclasts.
12. What is the medullary cavity?
13. The diagram shows a Haversian system

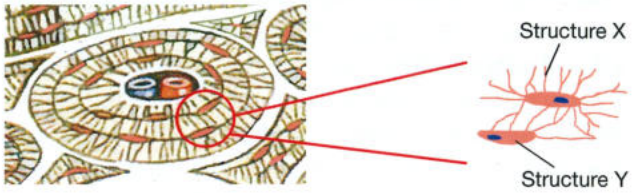


Figure 70.4 A Haversian system.

(a) Identify structure X and state its function.
(b) Identify structure Y and state its function.

14. Construct a table to compare compact bone and spongy bone.
15. (a) What is the periosteum?
(b) Outline some of the functions of the periosteum.
16. Distinguish between red bone marrow and yellow bone marrow.

Use the following diagram for the next TWO questions.

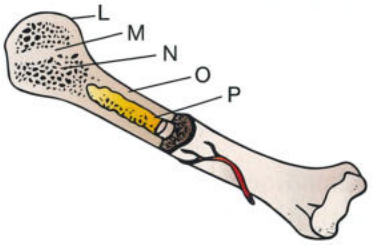


Figure 70.5 Bone.

17. Which part would consist of spongy bone?
(A) Part L
(B) Part M
(C) Part N
(D) Part O
18. What would you expect to find at location P?
(A) Spongy bone.
(B) Compact bone.
(C) Red bone marrow.
(D) Yellow bone marrow.
19. Bones can be classified as long bones, short bones, flat bones, sutural bones or irregular bones. Which of the following has correct examples of a long bone and a flat bone?

	Long bone	Flat bone
(A)	Humerus	Ribs
(B)	Carpal bones	Humerus
(C)	Ribs	Tibia
(D)	Tibia	Vertebra

20. Which cells could be called 'bone removing cells'?
(A) Osteocyte.
(B) Osteoclast.
(C) Osteoblast.
(D) Osteons.

71 Cartilage

Cartilage is a type of flexible connective tissue consisting of scattered cells in an intracellular material that is a mass of collagen fibres in chondroitin sulfate. **Chondroitin sulfates** are complex polysaccharides that form complexes with protein to form proteoglycans. This makes the matrix tougher and harder and gives cartilage the ability to resist compression. The matrix is more homogeneous than the matrix of ordinary fibrous tissues.

Collagen fibres gives the matrix tensile strength. If the cartilage needs to be more flexible and less brittle, e.g. the tip of the nose or the pinna of the ear, there is less collagen and more elastic fibres.

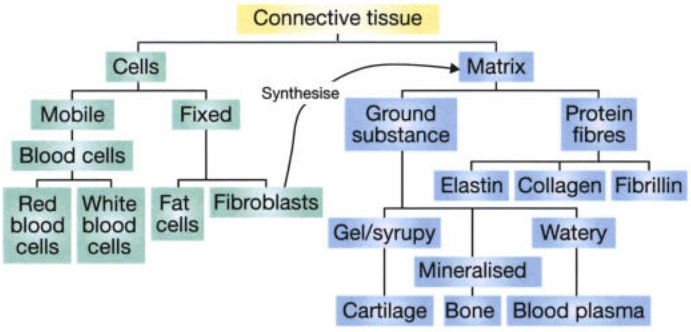


Figure 71.1 Connective tissue.

Chondrocytes are cartilage cells and are the only cells in the cartilage matrix. They are found in small pockets called lacunae and they secrete chondroitin sulfate and collagen. Chondrocytes produce a chemical that stops the formation of blood vessels, making cartilage avascular. This means that the supply of oxygen and nutrients to chondrocytes is by diffusion and the removal of wastes also occurs by diffusion. The rate of diffusion depends on the concentration gradient and can be a slow process causing damaged cartilage to heal slowly.

Most chondrocytes are surrounded by a **perichondrium** which consists of two layers. The outer fibrous layer gives support and protection and attaches to other structures and the inner cellular layer helps growth and maintenance of the cartilage.

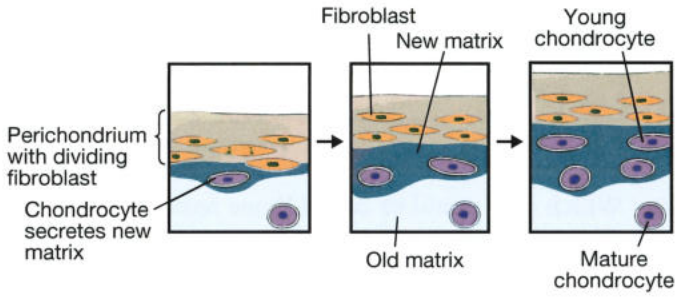


Figure 71.2 Growth of cartilage in an embryo.

In an embryo, cartilage grows by repeated cell division of the inner layer of the perichondrium increasing the surface and dimensions of the cartilage and by division of the chondrocytes within the cartilage matrix making additional matrix. This growth does not occur in adults.

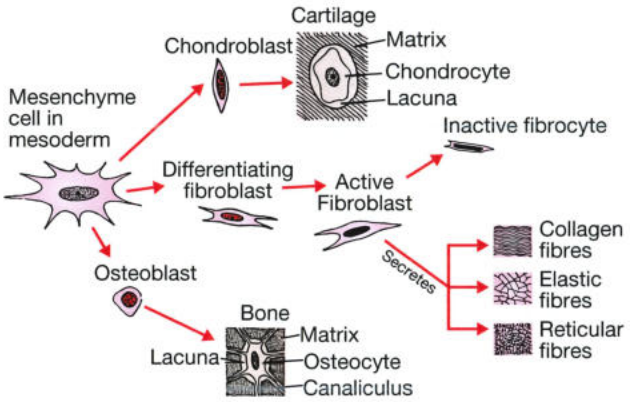


Figure 71.3 Development of bone and cartilage from mesenchyme cells.

Function of cartilage

Cartilage is used in structures that require firmness that can resist strain but also require flexibility and some degree of elasticity. The physical properties of cartilage depends on the components in the matrix.

Structures made of cartilage have the following uses.

- **Support softer tissues**, e.g. rings of cartilage in the trachea support the walls of the windpipe and prevent it from collapsing during inhalation and exhalation.
- **Flexible connections,** e.g. between sternum and ribs.
- **Shock absorption**, e.g. intervertebral discs.
- **Reduces friction between bony joints**, e.g. bone surfaces at synovial joints.

Types of cartilage

Hyaline cartilage has chondrocytes in small groups in a smooth and solid matrix and appears as a smooth blue-white tissue with no prominent fibres. It provides stiff, slightly flexible support and reduces friction between bony surfaces. It is found on the surface of parts of bones that form synovial joints. It forms part of the larynx, trachea and bronchi, part of the nasal septum and forms the costal cartilages that attach the ribs to the sternum. The entire skeleton of an embryo is made of hyaline cartilage.

Fibrocartilage has dense masses of white collagen fibres with widely spaced cells and a matrix similar to hyaline cartilage. It resists compression, limits relative movement and prevents bone to bone contact.

It is found as ligaments joining bones, as intervertebral discs, on the rim of bony sockets such as the hip and shoulder joints, allowing the cavity to be deeper without restricting movement and forms the semilunar cartilages which are in the knee joint between articulating bone surfaces.

Elastic cartilage is flexible with yellow elastic fibres in a solid matrix. It provides support and can distort without damage and return to its original shape. It forms the pinna (ear lobe), the external auditory canal, the epiglottis and part of the tunica media of blood vessel walls.

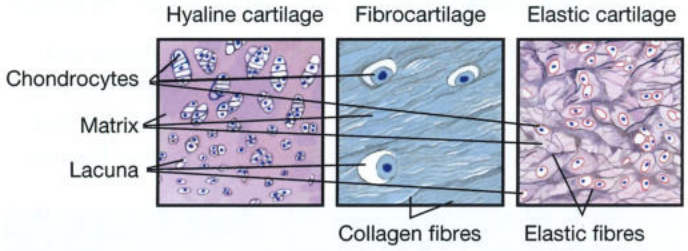


Figure 71.4 Three types of cartilage.

QUESTIONS

1. What is cartilage?
2. Explain why cartilage is classified as a fibrous connective tissue.
3. The diagram shows the structure of the repeating unit of chondroitin sulfate.

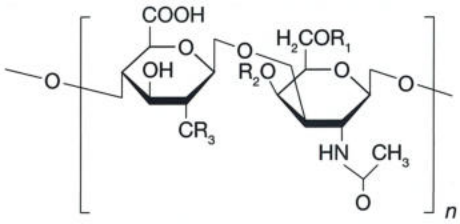


Figure 71.5 Repeating unit of chondroitin sulfate.

   (a) What is chondroitin sulfate?
   (b) Why is chondroitin sulfate an important part of cartilage?
   (c) Suggest why one of the main treatments for osteoarthritis involves supplements of chondroitin sulfate.
4. What are chondrocytes?
5. What features of cartilage make it suitable for framework structures in the body?
6. (a) Why is cartilage avascular?
   (b) How does being avascular affect the functioning of cells in cartilage?
7. Explain why damaged cartilage usually heals quite slowly compared to other damaged areas such as skin.
8. What is the perichondrium?
9. Construct a table to distinguish between the three types of cartilage.
10. Identify some of the functions of cartilage.
11. Hyaline cartilage is the weakest type of cartilage. How does this show when viewing under the light microscope?
12. Construct a table to compare bone and cartilage.

Use the diagram of the cross-section through the trachea (windpipe) for the next FOUR questions.

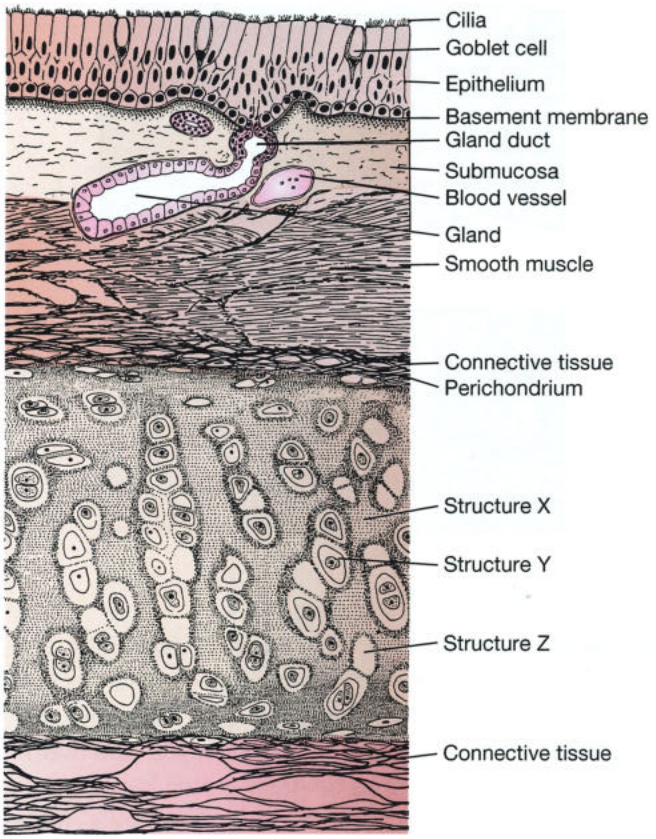


Figure 71.6 Cross-section trachea.

13. Structures X, Y and Z are part of a ring of cartilage in the trachea. What type of cartilage is shown in the diagram?
   (A) Fibrocartilage with prominent collagen fibres.
   (B) Elastic cartilage with prominent elastic fibres.
   (C) Hyaline cartilage with prominent collagen fibres.
   (D) Hyaline cartilage with no prominent fibres.
14. Identify structure X.
   (A) Matrix. (B) Lacuna.
   (C) Chondrocyte. (D) Osteocyte.
15. Identify structure Y.
   (A) Matrix. (B) Lacuna.
   (C) Chondrocyte. (D) Osteocyte.
16. Identify structure Z.
   (A) Matrix. (B) Lacuna.
   (C) Chondrocyte. (D) Osteocyte.
17. Which type of cartilage is found in intervertebral discs?
   (A) Hyaline cartilage. (B) Fibrocartilage.
   (C) Elastic cartilage. (D) Osteocartilage.

72 Osteoporosis

Osteoporosis is a disease, primarily of ageing, that causes disability due to a reduction in bone mass and strength that compromises normal bone function. As with other diseases increased understanding of the cause of osteoporosis enables improved practices for management and prevention.

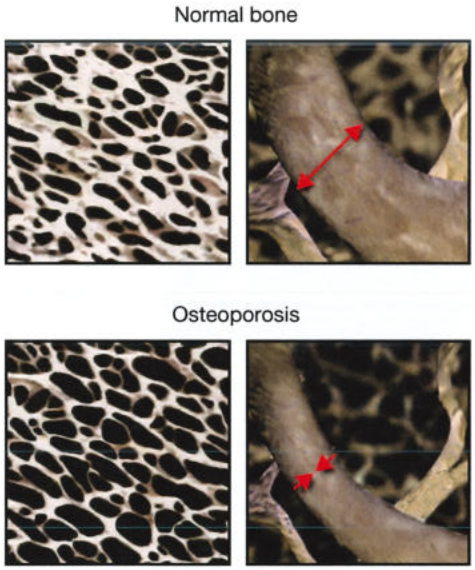


Figure 72.1 Osteoporosis.

Osteoporosis occurs when the amount of deposition is less than the reabsorption. Spongy bone (cancellous bone) is first affected and thins, then there is loss of trabeculae. Trabeculae are the small beams of osteocytes and matrix that develop along the lines of stress within bone. Trabeculae are found at the end of long bones, e.g. femur and provide strength and help withstand stress.

Around 99% of the body's calcium is in the bones and loss of calcium from bones decreases bond density and bone strength.

Decreased bone mass increases the chance of bone fractures, e.g. wrist, hip, vertebrae. Osteoporosis can also cause decreased height with age due to compressed vertebrae and bone pain.

Peak bone mass occurs at the end of skeletal maturation and in both sexes peak bone mass occurs at around 30 to 35 years old. This is when the bones have reached maximum strength and density.

After menopause lowered oestrogen levels can cause accelerated loss of bone mass in women so that females after menopause have lower bone density than men.

Steroid hormones include the corticosteroids (e.g. made in the adrenal cortex) and the sex steroids (e.g. made in the gonads or placenta). Steroid hormones are lipids that can control the expression of specific genes and thus affect growth and development.

The change in balance of steroid sex hormones (oestrogen and androgens) and corticosteroid hormones (e.g. glucocorticoids) affects bone deposition/reabsorption.

Other factors that can cause decreased bone mass include lifestyle choices such as smoking, excessive alcohol intake, immobility, diet low in calcium, lack of exercise and drugs such as corticosteroids.

Prevention of osteoporosis involves having sufficient calcium and magnesium in the diet, adequate vitamin D to aid the absorption of calcium and regular weight bearing exercise that strengthens muscles and increases resistance.

Treatment of osteoporosis involves increased calcium and magnesium in the diet, increased vitamin D, regular exercise, no smoking and medication for some patients. After the first osteoporotic fracture there is a higher risk of another fracture.

QUESTIONS

1. What is osteoporosis?
2. When does osteoporosis occur?
3. What is first affected in osteoporosis?
4. Outline the structure and function of trabeculae.
5. Identify a consequence of decreased bone mass.
6. Explain why people with osteoporosis often have decreased height.
7. When does peak bone mass occur?
8. Explain why bone density of women is lost at a greater rate than in men after both sexes have reached peak bone mass.
9. (a) Define steroid hormone.
   (b) What are the main two types of steroid hormones?
   (c) How are steroid hormones involved in osteoporosis?
10. List some risk factors that can lead to osteoporosis.
11. Outline some actions that can be taken to help prevent osteoporosis.
12. Identify some treatments advised for people with osteoporosis
13. People with osteoporosis often say they have 'brittle bones'. What do they mean when they say this?
    (A) Their bones contain a high amount of sugar.
    (B) Their bones are hard but break easily.
    (C) Their bones are soft and break easily.
    (D) Their bones are soft and bend easily.
14. What happens in osteoporosis?
    (A) Compact bone thickens and spongy bone gets bigger spaces.
    (B) Compact bone thickens and spongy bone gets smaller spaces.
    (C) Compact bone thins and spongy bone gets bigger spaces.
    (D) Compact bone thickens and spongy bone gets smaller spaces.

73 Osteoarthritis

There are over one hundred different types of arthritis with osteoarthritis being the most common form. **Osteoarthritis** is a degenerative non-inflammatory disease caused by the breakdown of joint cartilage and the underlying bone. It is a disease, primarily of ageing, that causes disability with increased understanding of the causes of osteoarthritis aiding diagnosis, treatment and the ability to improve practices for management and prevention.

Figure 73.1 Hands with osteoarthritis.

Cause of osteoarthritis

Articular cartilage is smooth white connective tissue that covers the ends of bones at joints and thins with age. The thinning of the articular cartilage is due to normal wear and tear over time and as the rate of cartilage renewal is slower than the breakdown and thinning of the cartilage the cartilage becomes thinner and thinner with ageing. As the cartilage wears away some fragments can enter the joint or bony spurs and grow at the end edges of the bones. When thinning leads to bones coming into contact with each other the ends of the bones begin to degenerate and the articular surfaces can become misshaped. Uneven bumps prevent the bones from gliding smoothly over one another and in severe cases the joint becomes immobile.

Factors that affect the risk of osteoarthritis include:

- Genetics – family history.
- Gender – females have a higher incidence than males.
- Injury to joints and overuse.

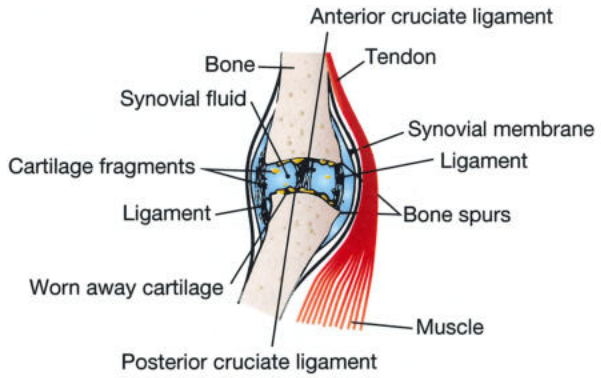


Figure 73.2 Articular cartilage wears away in osteoarthritis.

Symptoms of osteoarthritis

Osteoarthritis causes joints to become swollen which leads to restricted movement of the affected joints and pain. Symptoms range in pain severity from stiffness to disabling pain. Joints become swollen so that pain and swelling are key indicators of osteoarthritis. In early stages of osteoporosis the pain usually comes from using the joint and in later stages the pain continues even when the joint is not being used. Some people feel pain at night even while resting. Common sites of swelling and pain are the hands, spine, hips and knees.

Diagnosis of osteoarthritis

Diagnosis involves collection of data, e.g. about personal and family history, medical history, and a physical examination to test the range of motion of the joints and the position and alignment of the neck and spine.

Diagnostic tests can include an X-ray to show damage, magnetic resonance imaging (MRI) to show images of the cartilage and other structures, and joint aspiration where the affected area is made numb and a needle is inserted to withdraw fluid from the joint. The fluid is tested for crystals or joint deterioration. Blood tests do not specifically show osteoarthritis.

Treatment of osteoarthritis

Treatment involves managing symptoms, e.g. pain management, reducing swelling and aiding stiffness. There is no cure. Some people have joint replacement. People are advised to maintain a healthy weight and keep active with exercise. Exercise maintains muscle strength which can take the load from the joint to help reduce pain.

QUESTIONS

1. Define osteoarthritis.
2. Outline the cause of osteoarthritis.
3. What can cause a joint to become immobile?
4. Identify some factors that affect the risk of osteoporosis.
5. List some symptoms of osteoarthritis.
6. What are the key indicators of osteoarthritis?
7. Identify some joints that are likely to suffer from osteoarthritis.
8. What is involved in the diagnosis of osteoarthritis?
9. How are these technologies involved in diagnosis of osteoarthritis?
   (a) X-ray
   (b) MRI
10. What are the main treatments for osteoarthritis?

74 Excretion, Respiration and Water

The **respiratory system** controls the levels of oxygen and carbon dioxide in the body by regulating breathing rates. Oxygen is used in respiration to release energy for the body, and carbon dioxide is produced as a waste product to be excreted by the lungs. Respiratory surfaces of multicellular organisms need to have a large surface area for gas exchange, a concentration gradient across the membrane so diffusion will occur, a good blood supply to transport oxygen, carbon dioxide and water to and from the respiratory surface and need to be kept moist to assist the dissolving of gases so they pass across the membrane.

The **excretory system** involves several different organs that remove waste products from the body. They also regulate water and salt levels and maintain homeostasis which is necessary for metabolic reactions to occur with maximum efficiency. The main metabolic waste products are excess water, carbon dioxide, excess salts and nitrogenous wastes, such as urea and ammonia. The main excretory organ is the kidney.

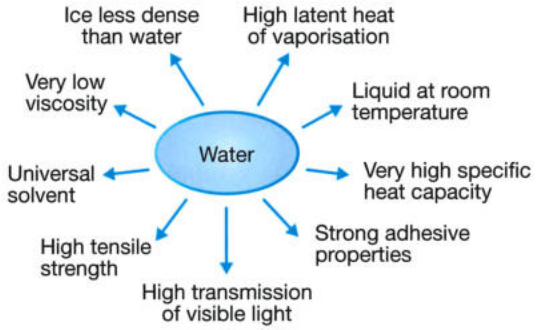


Figure 74.1 Properties of water.

Water is an important molecule because of its properties as a solvent and many substances, e.g. nutrients, oxygen and wastes are carried to and from cells in water. Nearly every chemical reaction in cells involves water and the concentration of water must be kept relatively constant to maintain pH, maintain substrate concentration and the other conditions necessary for metabolic reactions. Water has a very high heat capacity which means it can absorb and retain heat and thus help maintain a constant body temperature and heat can be transported around the body by the water in the plasma in the blood. Water in the body can also act as a lubricant preventing friction between two opposing surfaces, e.g. between internal organs like the heart and lungs; or between opposing bony surfaces of the skeleton.

QUESTIONS

1. (a) Describe the role of the respiratory system and excretory systems in maintaining humans as functioning organisms.
   (b) Explain how the two systems are integrated and work together to produce an independent organism.
2. (a) Describe three properties of water and explain how each assists its role in living organisms.
   (b) Explain why the concentration of water in cells must be held constant.
   (c) Many deaths caused by mountain sickness at high altitude are caused by pulmonary oedema. In pulmonary oedema, interstitial fluid leaks into the alveoli increasing the diffusion barrier between blood and air. Predict how the water balance has changed for this to occur and why this leads to death.
3. Describe necessary features of a respiratory surface in humans.
4. Outline some ways in which respiratory systems are essential for maintaining homeostasis.
5. Identify the role of water in metabolic reactions in living cells and explain why the concentration of water in cells must be kept constant.
6. Which of the following is the best description of the function of an excretory system?
   (A) Deliver necessary nutrients to needy cells.
   (B) Control the breakdown of large molecules which are too big to diffuse across membranes.
   (C) Remove unwanted metabolic wastes.
   (D) Eliminate urea from the body.
7. What is the role of water in metabolic reactions in living cells?
   (A) Act as a solvent.
   (B) Provide non-polar attraction to dissolve other chemicals.
   (C) Adhere to other water molecules to provide water tension.
   (D) Release heat due to its low specific heat capacity to stabilise body temperature.
8. Human cells survive best in an isosmotic environment, that is, where solutions are of equal solute concentrations either side of the cell membrane. Plant cells are usually healthiest in a hypotonic environment, that is, when there is more water outside the cell than inside the cell. What would most likely happen to an animal cell in a hypotonic environment?
   (A) Water is lost to the surroundings.
   (B) Water is gained from the surroundings.
   (C) Salts are gained from the surroundings.
   (D) Cell uses its selectively permeable membrane to maintain water balance.
9. Why is the excretion of wastes essential?
   (A) To reduce the volume of intracellular fluid.
   (B) To increase the specific heat of water to maintain body temperature.
   (C) To prevent colon cancer.
   (D) For metabolism to proceed in a controlled and sustained manner.

75 Excretory Organs

Excretion is the removal of metabolic wastes from the body. Metabolic wastes are excreted from the lungs, the skin, the digestive system and the kidneys.

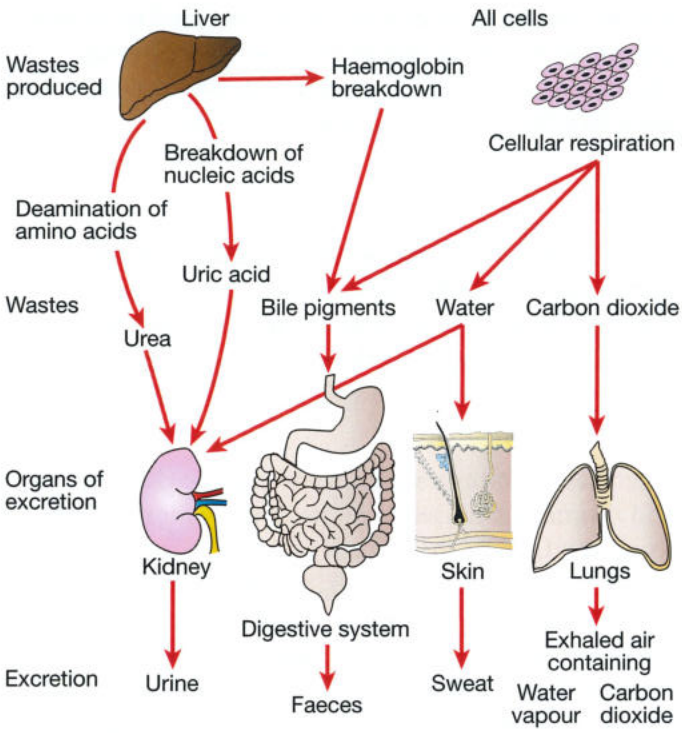


Figure 75.1 Excretion in humans.

The **lungs** excrete most of the carbon dioxide and a lot of water.

Sweat glands in the skin excrete 5% to 10% of all metabolic wastes. Sweat is mostly water (99%) which is used in thermoregulation of body temperature cooling the body when the water evaporates from the skin. Sweat also contains sodium chloride which gives it a salty taste. Excretory waste products in sweat include some urea and uric acid (nitrogenous wastes) and wastes from carbohydrate metabolism.

In the **liver** amino acids are deaminated leading to the formation of urea and uric acid. The liver is also the site of the breakdown of red blood cells which forms bile pigments which are excreted by the liver into the **intestine** to pass out of the body with faeces. Undigested food eliminated from the intestine is not excretion.

The **kidneys** are the main excretory organ removing nitrogenous wastes and controlling the water/salt level of the body. By controlling the composition of urine the kidneys maintain homeostasis in the body and remove toxic compounds from the blood. The three main nitrogenous wastes are ammonia, uric acid and urea. Ammonia is the first metabolic product in the deamination of amino acids.

Ammonia is highly toxic and is converted by mammals into urea. Urea is soluble in water and less toxic than ammonia. Humans excrete uric acid that is produced in the breakdown of the purines of nucleic acids. Uric acid is insoluble in water.

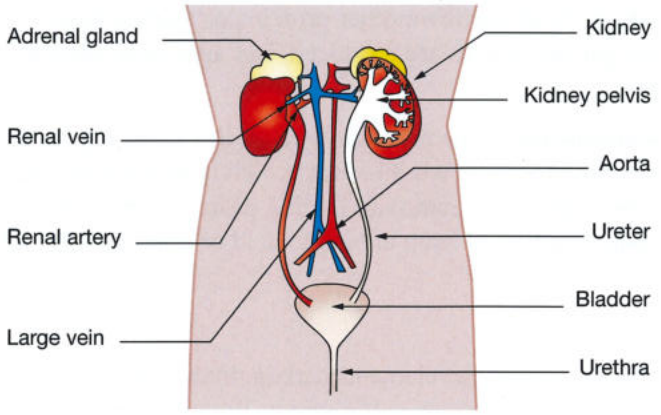


Figure 75.2 Human urinary system.

The human urinary system

The human urinary system has several functions.

- Excretion of nitrogenous waste.
- Osmoregulation maintaining the water potential of the blood and body fluids.
- Ionic regulation maintaining ions levels, e.g. potassium (K^+) and sodium (Na^+).
- Regulation of blood pH.
- Regulation of blood pressure long term.

QUESTIONS

1. What is excretion?
2. Identify the organs that excrete metabolic wastes.
3. Identify the metabolic wastes that are excreted from the lungs.
4. Explain why sweat glands are involved in excretion.
5. Explain how the digestive system is involved with excretion.
6. Name the three forms of nitrogenous wastes that are excreted from animals.
7. Identify the main excretory organ of humans for nitrogenous waste.
8. Identify the main functions of the human kidney.
9. What is osmoregulation?
10. Explain why humans need to convert ammonia to urea.
11. Which of the following would not be considered a metabolic waste that is excreted from the body?
    (A) Carbon dioxide. (B) Urea.
    (C) Undigested food. (D) Bile pigments.

76 Deamination

Dietary protein is reduced to its component amino acids so that these smaller molecules can be absorbed into the bloodstream. If these amino acids are not needed or the body is breaking down other proteins or nucleic acids nitrogenous wastes are produced, e.g. ammonia, urea and urea.

Deamination is the process when amino acids, e.g. in the liver are broken down into urea which is then transported to the kidneys for removal. During deamination an amino group is removed from an amino acid or other compound.

Proteins

Proteins contain the elements carbon, hydrogen, oxygen and nitrogen and often sulfur atoms. Proteins consist of one or more polypeptides folded and coiled into specific three-dimensional shapes. There are four levels of protein structure. The **primary structure** of a protein is a series of linked amino acids. The **secondary structure** of a protein are the coils and folds due to hydrogen bonds of the backbone. The **tertiary structure** is the overall shape due to the interactions between the side chains (R groups) of the various amino acids. The **quaternary structure** is the overall shape and structure due to two or more polypeptide chains being present in one large macromolecule.

In the polypeptide chain the amino acids are joined together by a covalent bond called a **peptide bond**.

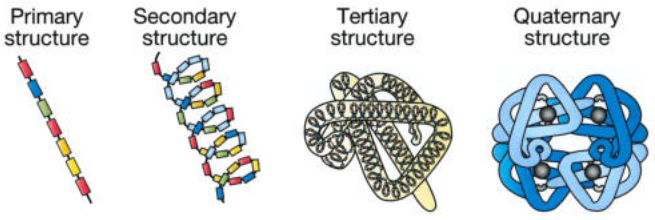


Figure 76.1 Four levels of protein structure.

Amino acids

An amino acid is the basic unit of a protein. All amino acids contain an amino group ($-NH_2$) and an acid group (carboxyl group) ($-COOH$). The carbon atom at the centre of an amino acid is called the **alpha carbon** and it bonds with the amino group, the acid group, a hydrogen atom and a variable side group usually represented by R. The side groups can be non-polar, polar or electrically charged.

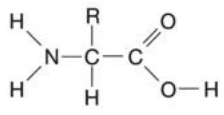


Figure 76.2 General formula for an amino acid.

There are over 20 amino acids and they include alanine, arginine, asparagine, cysteine, histidine, leucine, lysine, phenylalanine, serine, proline, valine and glutamic acid.

Transamination removes the amino group from an amino acid and attaches it to another carbohydrate molecule to form a new **non-essential amino acid**. An **essential amino** acid are the amino acids that must be included in the diet and cannot be synthesised by the body. The essential amino acids are lysine, tryptophan, threonine, methionine, phenylalanine, leucine, valine and isoleucine.

Nitrogenous wastes

After the body has sufficient stored protein additional amino acids are broken down to be used as energy or stored as fat or glycogen and a nitrogenous waste is formed. The formation of the nitrogenous waste starts with the breakdown of the amino acids in the liver with deamination.

Deamination forms **ammonia** and a **keto acid**. The keto acid is usually oxidised to release energy for metabolic processes. Usually the amount of ATP released for each gram of protein oxidised is slightly less than the amount of ATP formed for each gram of glucose oxidised in aerobic respiration. If a person **does not eat protein** then their body will break down their own body proteins, e.g. muscles into amino acids which are then deaminated and oxidised. This is called **obligatory loss** of protein.

Ammonia (NH_3) – when the amino group is removed from an amino acid ammonia is formed. Ammonia is highly soluble and can easily pass through membranes. Ammonia is also highly toxic and needs to be quickly converted to urea. If ammonia accumulates in the blood its toxicity will affect the brain causing the body to go into a coma and death.

Urea – almost all the ammonia is removed from the blood by conversion into urea in the liver. This is an important step in removing nitrogenous wastes as it keeps the levels of ammonia always at a low level. Since urea is less toxic than ammonia the blood can hold more urea in higher concentrations before tissue damage occurs.

$$2NH_3 + CO_2 \rightarrow H_2N-\underset{\underset{O}{\|}}{C}-NH_2 + H_2O$$

Figure 76.3 Two molecules of ammonia join with one molecule of carbon dioxide to form urea.

Urea travels in the blood to the kidneys where it is excreted in urine. In urine urea is dissolved in the water.

Uric acid – is produced from ammonia and by the breakdown of nucleotides from nucleic acids, e.g. the breakdown of the genetic material of worn out cells. Uric acid is not very soluble and forms crystals.

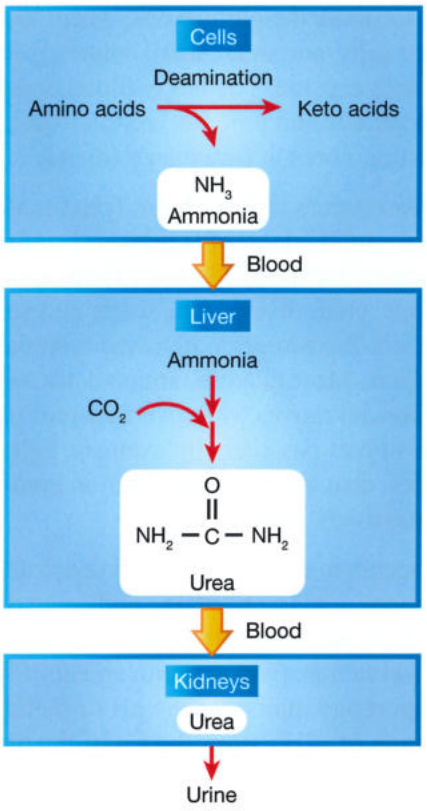


Figure 76.4 Deamination.

QUESTIONS

1. Define deamination.
2. Identify the elements present in proteins.
3. Distinguish between the four levels of protein structure.
4. What is a peptide bond?
5. Write the formula for:
   (a) An amino group.
   (b) A carboxyl group.
6. The diagram shows the formula for three amino acids. Copy each formula and on each identify the alpha carbon and the R side group.

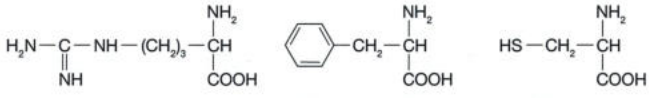


Figure 76.5 Three amino acids.

7. Explain why some amino acids can be polar molecules, some are non-polar molecules and some are electrically charged molecules.
8. Write the general formula for an amino acid.
9. Define transamination.
10. (a) Distinguish between an essential amino acid and a non-essential amino acid.
    (b) Construct a table to list some essential amino acids and some non-essential amino acids.
11. Explain why some amino acids are deaminated.
12. (a) What is formed when an amino acid is deaminated?
    (b) What happens to these products of deamination?
13. What happens when there is obligatory loss of protein?
14. Explain why it is necessary to convert ammonia into urea.
15. What happens to the urea that is formed after deamination and conversion of ammonia to urea?
16. When is uric acid produced?
17. The flow chart shows the sources of amino acids in the body and what happens to these amino acids.

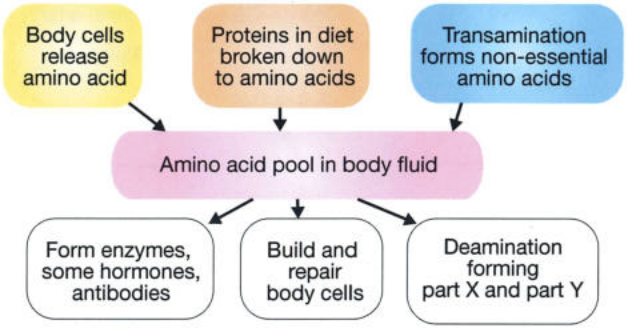


Figure 76.6 Amino acids in the body.

(a) Identify part X and part Y.
(b) Name a substance that would be formed by transamination to join the pool in the body fluid.

18. Which nitrogenous waste is first formed in the breakdown of an amino acid?
    (A) Urea. (B) Uric acid.
    (C) Ammonia. (D) Polypeptide.
19. The diagram shows the kidney and three amino acids.

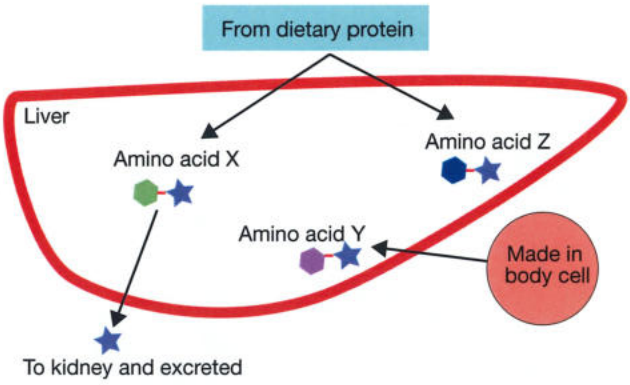


Figure 76.7 Process in the liver.

Which of these statements can be made about the processes shown occurring in this liver diagram?
(A) Amino acids Z and X must be essential amino acids
(B) Amino acid X undergoes deamination
(C) Amino acid Y was formed by deamination
(D) The star symbol represents the amino group and the carboxyl group of all the amino acids.

77 The Kidney

The kidney is the main excretory organ for most vertebrates. The kidney is bean shaped in mammals and has the dual functions of excretion and osmoregulation.

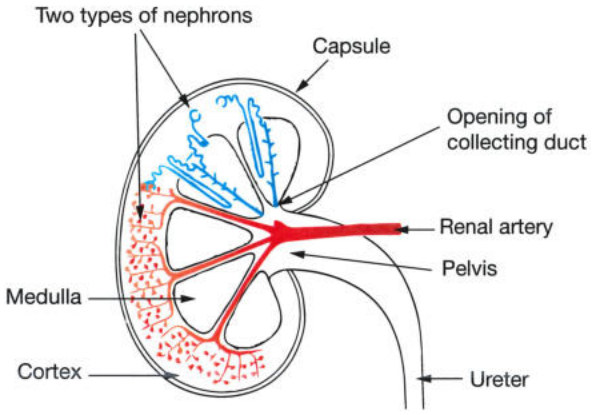


Figure 77.1 Human kidney.

The functional unit of the kidney is the **nephron**. There are two types of nephrons – cortical nephrons and juxtamedullary nephrons – which differ in the length of the loop of Henle and the position of the Bowman's capsule.

Each nephron has three major parts: Bowman's capsule, the glomerulus and a long thin tubule. Urine produced by normal kidneys can be four times more concentrated than the waste products in blood. The kidney needs to be able concentrate wastes to reduce water loss and hence dehydration. Excretion from the kidney involves three main stages: filtration, reabsorption and secretion.

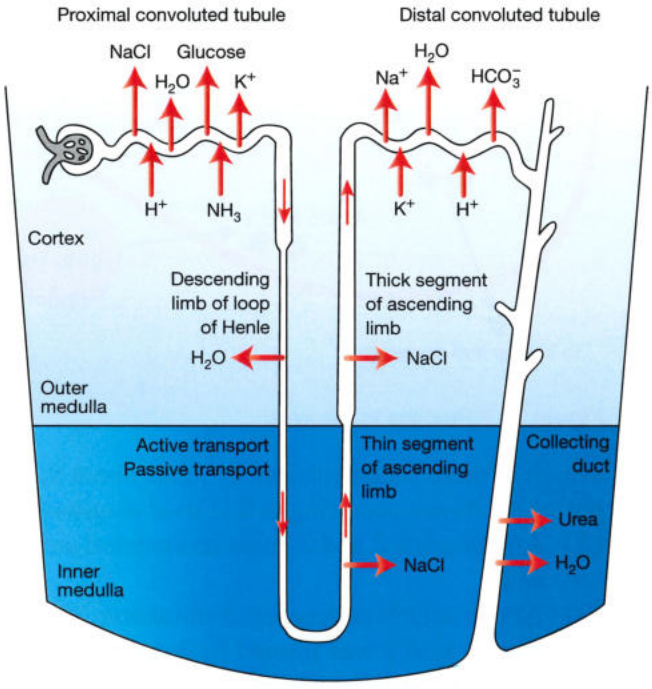


Figure 77.2 Functional activities in the nephron.

Filtration occurs in the glomerulus, where blood pressure forces water, salts, and other small solutes from the blood into Bowman's capsule. The larger blood components, like red blood cells and proteins are too large to filter through the tiny pores in Bowman's capsule.

Reabsorption occurs in the tubules (proximal and distal tubules and the loop of Henle) and controls the concentration of substances in body fluid, e.g. amino acids, glucose, vitamin and some water and some salts are reabsorbed. Reabsorption involves both passive and active transport. Most glucose, amino acids and salt (sodium chloride) moves by active transport using energy, while water moves passively by osmosis. Salt moving out of the tubules, changes the concentration gradient and the water will passively follow.

Secretion occurs in the proximal and distal tubules and adds substances, e.g. H^+, K^+, NH_4^+ and some drugs to the filtrate. It is highly selective. It may involve either active or passive transport, e.g. movement of H^+ involves active transport and maintains the pH of blood and urine while secretion of NH_4^+ stops the pH of the filtrate from becoming too low.

QUESTIONS

1. Define the terms.

Nephron	Filtration	Reabsorption
Secretion	Filtrate	Osmoregulation
Solute	Ureter	Urethra

2. Compare active and passive transport using examples of these processes occurring in the mammalian kidney.
3. The kidney is one of the main excretory organs of the body. One of its functions is homeostasis.
   (a) Define homeostasis.
   (b) Draw a flow diagram to show the pathway of urea from the blood in the renal artery to leaving the body through the urethra.
   (c) Predict how the kidney will respond if a person drinks no fluids on a very hot day.
   (d) Explain why the presence of red blood cells in urine indicates a serious problem.
4. Students dissected a mammalian kidney and made the following drawing.

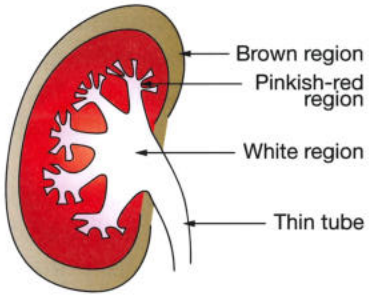


Figure 77.3 Student drawing of a dissected kidney.

(a) Identify the regions labelled 'brown region', 'pinkish-red region' and 'white region' and for each describe the structural features which would be present and the function of each section. Identify the 'thin tube'.

(b) Outline two safety procedures which would have needed to be followed by the students when carrying out this dissection.

5. (a) Explain how the processes of filtration and reabsorption in the mammalian nephron regulate body fluid composition.

(b) When people show some of the first symptoms of kidney failure, such as pale and yellow complexion, fatigueness, shortness of breath, frequent urination at night, doctors will often advise their patient to change their diet restricting the intake of proteins such as poultry, seafood and soya products. However, they have to be careful not to reduce this intake too low. Explain why this measure slows down the rate of renal failure and why the protein intake cannot become too low.

6. Explain why the removal of metabolic wastes, such as nitrogenous wastes, is essential for efficient metabolic activity.

7. Explain why the processes of diffusion and osmosis are inadequate for excretion in mammals.

8. What causes filtration from the glomerulus into Bowman's capsule?
   (A) High blood pressure due to the narrowing of the arteriole diameter.
   (B) Difference in protein concentration causes water to move by osmosis out of the glomerulus.
   (C) Diffusion of small molecules into Bowman's capsule.
   (D) All of the above.

9. What is the best definition of osmosis?
   (A) Diffusion of water across a semipermeable membrane from a region of high concentration of water to a region of low concentration of water.
   (B) Diffusion of water across a semipermeable membrane from a region of low concentration of water to a region of high concentration of water.
   (C) Movement of water across a semipermeable membrane from a region of high salt concentration to a region of low salt concentration.
   (D) Movement of particles from an area of high concentration to an area of low concentration.

10. Mammalian kidneys excrete toxic wastes. What is another function of their kidneys?
    (A) Regulate salt concentrations.
    (B) Make urea.
    (C) Break down and recycle old red blood cells.
    (D) Deamination of proteins.

Use the following information for the next TWO questions. The diagram shows the structure of a nephron in the kidney.

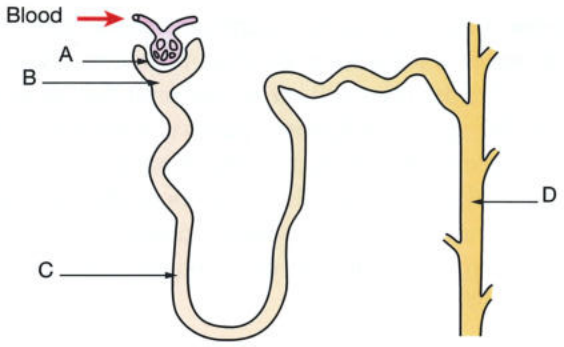


Figure 77.4 Nephron in the kidney.

11. Which point is most likely to be the site of active transport?
    (A) A (B) B (C) C (D) D

12. Which site would have the least amount of glucose?
    (A) A (B) B (C) C (D) D

Use the following information for the next TWO questions. The table shows the results of the analysis of samples of liquid taken from three different parts of a mammalian nephron.

Substance	Sample A (%)	Sample B (%)	Sample C (%)
Water	91.35	99.00	96.20
Glucose	0.25	0.25	0.00
Proteins	8.15	0.00	0.00
Urea	0.25	0.75	3.80

13. Which sample(s) is most likely to have been taken from the collecting duct of the nephron?
    (A) Sample A.
    (B) Sample C.
    (C) Samples A and C.
    (D) Sample B.

14. Which of the following identifies the most likely process that is responsible for the difference in concentration of glucose between parts B and C?
    (A) Filtration.
    (B) Mitosis.
    (C) Reabsorption.
    (D) Secretion.

15. Which of the following would have the highest concentration of urea?
    (A) Renal vein.
    (B) Renal artery.
    (C) Interstitial fluid between nephrons in the kidney.
    (D) Hepatic portal vein.

78 Osmosis, Diffusion and Active Transport

Diffusion is the movement of particles from an area of high concentration of particles to an area of low concentration of particles. The particles will spread out until they evenly fill the available space. Diffusion is passive transport as energy does not need to be provided for the movement to occur. The rate of diffusion depends on the concentration gradient between the area of high concentration and the area of low concentration. The higher the diffusion gradient, the greater the rate of diffusion.

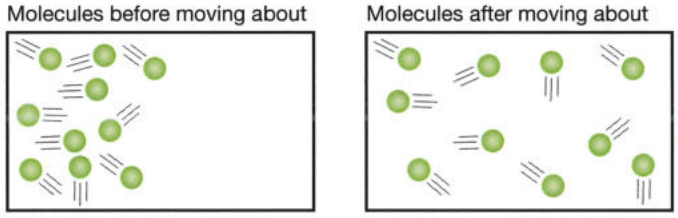


Figure 78.1 Diffusion.

Osmosis is a type of diffusion. It is the movement of water from an area of high concentration of water to an area of low concentration of water across a semipermeable membrane. (*Note*: In chemistry, osmosis refers to the net movement of any solvent, not just water.)

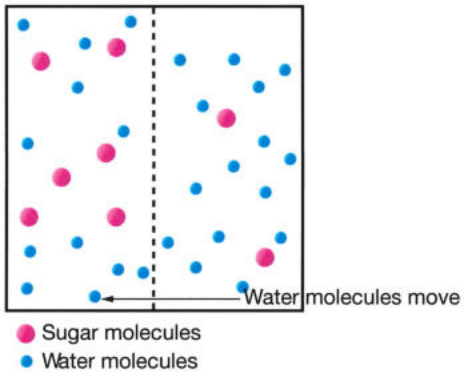


Figure 78.2 Osmosis.

Active transport

Active transport requires energy to be used to move particles against a concentration gradient. The cell membrane determines which substances can pass in/out of the cell and the movement of many substances requires energy from respiration.

QUESTIONS

1. Identify two substances that enter the human body by diffusion.
2. The diagram shows a human body cell in a container filled with distilled water.

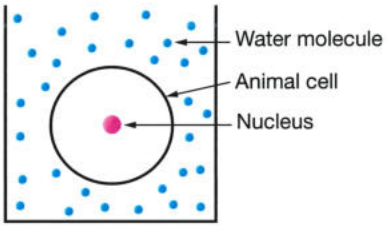


Figure 78.3 Human cell in a container of water.

What would you expect to happen to the animal cell? Explain your answer.

3. The diagram shows the structure of a kidney nephron.

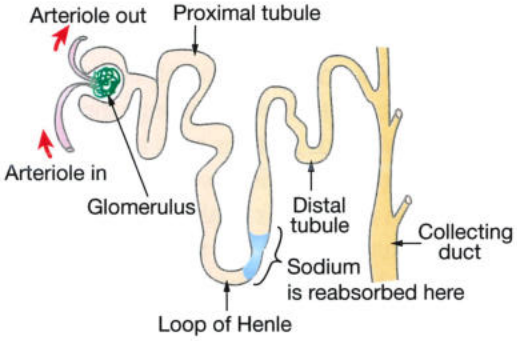


Figure 78.4 Nephron structure.

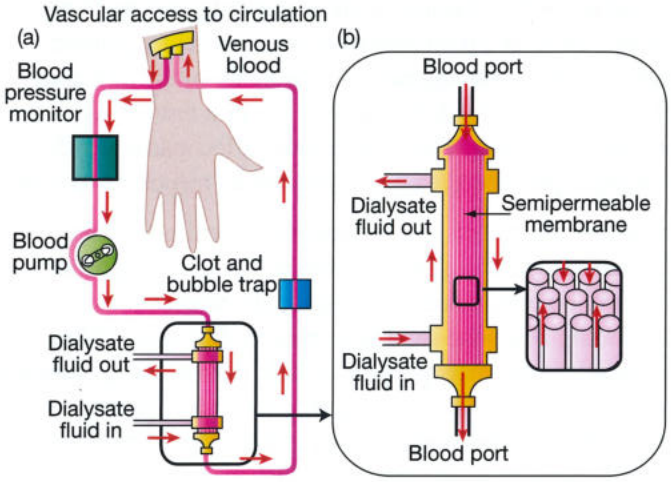
If salt is reabsorbed in the loop of Henle and there is an increase in the salt concentration in the capillaries around the distal tubule, what movement of water would you expect to take place in the distal tubule?

4. The kidney can change the permeability of the collecting duct. How does this assist excretion for the body?
5. In humans the kidneys are about 0.4% of total body weight but receive about 25% of the blood flow. The kidney is one of the most metabolically active organs using large amounts of energy. Explain why the kidney needs this blood flow and energy requirement.
6. (a) Suggest what could happen if the plasma becomes more dilute than the intracellular fluid within red blood cells.
   (b) Suggest what could happen if the plasma becomes more concentrated than the intracellular fluid within red blood cells.
7. Reabsorption in the kidney involves both active transport and passive transport. Which of the following statements is correct about the movement of substances in the kidney?
   (A) About 65% of the filtered sodium is reabsorbed by the sodium pump in the proximal tubule.
   (B) 90% of the sodium diffuses out of the distal tubule into the capillaries.
   (C) Water is actively transported across the membranes in the loop of Henle.
   (D) All proteins are reabsorbed in the proximal tubule.

79 Renal Dialysis

Dialysis is a process that removes waste products from the blood. **Acute dialysis** is needed in emergency situations; when potassium levels are high there is fluid overload and pulmonary oedema – fluid accumulates in the lungs. Acute dialysis is also required if there is a need to remove medications or toxins from the blood. **Chronic dialysis** is needed for long term renal failure.

Haemodialysis is the most common method of dialysis and uses dialysers or artificial kidneys. Shunts are inserted into an artery and a vein so that blood can flow from the person into the dialysis machine, be filtered, and then flow back to the person and into their vein.



Haemodialysis system (a) Blood from an artery is pumped into (b) a dialyser, where it flows through the cellophane tubes that act as a semipermeable membrane (inset). The dialysate fluid has the same chemical composition as blood, except that it is free of urea and waste products. The blood flows down the tubes, bathed in the dialysate flowing in the reverse direction. Waste products in the blood diffuse through the semipermeable membrane into the dialysate.

Figure 79.1 Haemodialysis system.

In the artificial kidney, toxins and wastes move by diffusion and water moves by osmosis and ultrafiltration across an artificial membrane. The pores in the artificial membrane allow diffusion of small ions but prevent large plasma proteins leaving the blood. The body's buffer system, which maintains homeostasis, is maintained using a dialysate, which is a solution of all the extracellular ions such as hydrogen carbonate and acetate. Heparin, an anticoagulant, is administered to stop the blood clotting in the dialysis tubing.

QUESTIONS

1. (a) What is oedema?
   (b) Describe pulmonary oedema.
2. Outline some situations when renal dialysis may be needed.
3. Haemodialysis is the most common method of dialysis.
   (a) What is dialysis?
   (b) Describe one advantage and one disadvantage of haemodialysis.
   (c) Compare the process of renal dialysis with the function of the kidney.
   (d) During a football match, a player received a kick in his back. He began to pass urine with blood. Suggest one way his kidney could be damaged.
4. Explain why heparin is administered.
5. Outline how the body's buffer system is maintained during dialysis.
6. Urea, uric acid, potassium ions, phosphate ions and sulfate ions move into the dialysis fluid, while hydrogen carbonate ions and glucose pass into the blood from the dialysis fluid. Outline the importance of the direction of movement of these substances.
7. During dialysis, many small molecules can easily diffuse through a synthetic membrane. In what way does a cell membrane behave differently to synthetic membranes?
   (A) Cell membranes can transport large molecules by active transport.
   (B) Cell membranes prevent the movement of water against a concentration gradient.
   (C) Cell membranes allow the passive transport of very large molecules into cells.
   (D) Cell membranes maintain a permanent rigid structure with molecules locked into place.
8. The diagram shows the basic structure of the kidney dialysis machine.

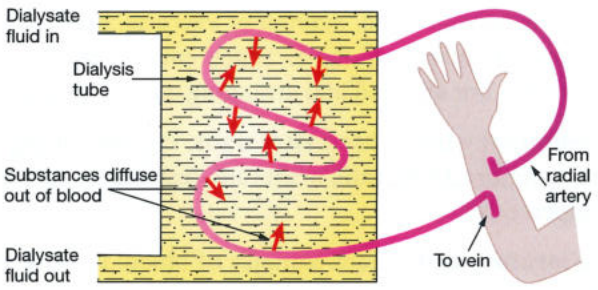


Figure 79.2 Kidney dialysis.

What comparison can be made between the dialysis machine and the human kidney?
(A) Both filter the same substances from blood.
(B) The kidney machine acts like Bowman's capsule, the renal tubules and collecting ducts.
(C) The kidney machine acts like Bowman's capsule and the renal tubules.
(D) Only the kidney has reabsorption.

80 Treatment of Conditions

Treatment of conditions due to system or organ dysfunction has changed through improvements in early diagnosis and appropriate use of drugs, physical therapy, radiation therapy and removal and/or replacement of affected parts.

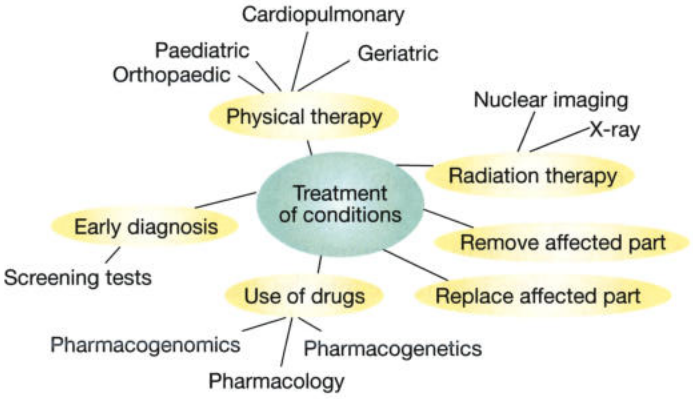


Figure 80.1 Treatment of conditions.

Early diagnosis

Diagnostic testing involves gathering clinical information about a person's health. Early and accurate diagnosis is highly important in managing many diseases and disorders. If symptoms are reversible then early diagnosis may be able to halt, perhaps reverse symptoms or reduce the period of illness. If symptoms are not reversible early treatment can stop the rate of onset and further decline due to the condition.

Early diagnosis gives a person some sense of empowerment and they can make plans, e.g. controlling lifestyle choices such as diet, smoking, alcohol and caffeine consumption, amount of exercise, managing finances, preparing living arrangements if mobility and flexibility are affected and arranging care and support.

Preliminary tests check **vital signs**, e.g. measuring blood pressure, breathing rate, heart rate, respiratory rate and temperature.

Laboratory tests check samples of blood, urine or body tissues.

Prenatal testing checks the health of an embryo or foetus. Some tests are screening tests and depending on the results of these tests further diagnostic testing can be carried out.

Diagnostic imaging looks inside the body for indications of system or organ dysfunction and other medical conditions. It includes X-rays, CT scans, nuclear medicine scans, MRI scans and ultrasound. **Computed tomography** (CT) scans are a type of imaging using X-rays to look for broken bones, cancers, blood clots and signs of heart disease or internal bleeding.

Genetic testing looks for indications of genetic disorders. There are more than 1000 genetic tests available from testing laboratories. The Guthrie test or HPLC test is used to test newborn infants for phenylketonuria (PKU) as part of the national screening program for babies. Early detection means the child can be given a special diet so that the phenylalanine will not accumulate causing progressive mental retardation.

Use of drugs

The appropriate use of drugs is an important part of the treatment of many conditions that are caused by system or organ dysfunction. **Pharmacology** is the study of drugs looking at the possible uses, effects on the body and the interactions with other drugs. **Pharmacogenomics** investigates the effects of a patient's genome on the way the person will respond to a particular drug and the likely effectiveness of a particular drug for a particular genetic make-up. **Pharmacogenetics** is the study of the influence of inherited genetic differences of one or a few genes on the response of a drug. These differences affect particular drug metabolic pathways leading to individual responses to drugs.

A person diagnosed with **osteoporosis** will often take calcium and vitamin D supplements to aid bone production and will be prescribed bisphosphonate medications that strengthen bones and help prevent fractures. Other medications for osteoporosis include denosumab, strontium ranelate and selective oestrogen receptor modulators (SERMS). The response of the person to these treatments is influenced by their genome and knowledge about the person's genotype can be used to avoid the use of drugs that will have an adverse effect on the patient.

Physical therapy

Physiotherapy is a branch of rehabilitative health that uses specially designed exercises and equipment to help enhance or restore movement and physical function. If a person has a condition that has been affected due to system or organ dysfunction a physical therapist will help the management and treatment of the disability. If a person has osteoporosis a physical therapist will design a specific program for the individual needs of the patient with specific exercises, e.g. weight bearing and/or resistance exercises. The person will also be advised on posture, how to improve balance to reduce the risk of falling and ways to improve muscle strength and mobility.

Radiation therapy

Radiation therapy or **radiotherapy** is a type of cancer treatment that uses beams of intense energy to kill cancer cells. Radiotherapy uses ionising radiation which consists of particles, X-rays or gamma rays with sufficient energy to cause ionisation in the medium through which it passes.

It is especially useful if the cancer is localised in one area of the body. Radiotherapy can be used in combination with surgery, chemotherapy or both.

Chemotherapy is the use of anti-cancer drugs to destroy cancer cells. These drugs can be given orally, through a needle injection or by direct introduction to the affected organ or tissue.

PET

PET (positron emission tomography) is an imaging technique which can detect radioisotopes. A PET camera can scan a patient and detect the radioisotope, which has been inhaled or injected and has accumulated in the target organ. This detected radioactivity is shown as an image on a computer screen. For example, gallium-67 is used to find cancers in lymph nodes and bronchi, and malignant melanomas. The most commonly used active molecule for PET is fludeoxyglucose (FDG) as it is a marker for the tissue uptake of glucose. It is used to assess the glucose metabolism in the heart, lungs and brain and to identify most types of tumours.

Artificial eye

Artificial arm

Figure 80.2 Prosthetic replacements.

Removal/replacement of affected part

In many instances affected organs or tissues are surgically removed and often replaced with transplants, e.g. from a person who has died or from engineered tissues that have been grown in a laboratory for the implant. Tissues grown from the patient's own cells do not have the rejection problems that are associated with transplants from other donors. Recipients are given immunosuppressive drugs to prevent rejection of non-self organs or non-self tissues. **Stem cell cloning** can be used in tissue engineering with the nucleus from a body cell of the patient who needs an organ or tissue and placing it in a stem cell. The stem cell is then stimulated to divide and the isolated cells are cultured to form the required organ or tissue. The cultured tissue or organ can then be transplanted into the patient with no rejection problems.

A **prosthesis** is an artificial body part, e.g. limb, joint or silicone breast implant that replaces removed disease affected parts.

QUESTIONS

1. List some areas where the treatment of conditions due to system or organ dysfunction has changed.
2. What is involved in diagnostic testing?
3. List some reasons why early and accurate diagnosis of conditions or disorders is highly important.
4. What is checked in preliminary testing?
5. Identify substances checked in laboratory tests.
6. What is prenatal testing?
7. Outline what is involved in diagnostic imaging.
8. (a) What is genetic testing?
   (b) Give an example to show how early diagnosis due to genetic testing can help manage a particular disease.
9. Distinguish between pharmacology, pharmacogenomics and pharmacogenetics.
10. What is physiotherapy?
11. Outline the physical therapy that would be designed for a person with osteoporosis.
12. What is radiotherapy?
13. What is chemotherapy?
14. (a) What is 'PET'?
    (b) Outline the use of PET.
15. Outline the use of gallium-67 in conjunction with PET.
16. What is a prosthesis?
17. Which is *not* a correct relationship of technology and use?

	Technology	Use
(A)	Nuclear imaging	Diagnosis of cardiovascular diseases
(B)	Ultrasound	Detects foetal abnormalities
(C)	Pharmacogenomics	Detects organ dysfunction
(D)	X-rays	Detects fractures

18. Stem cell cloning has a high potential for cell replacement therapy and tissue engineering. What is involved in this process to reduce tissue rejection?
    (A) Place the nucleus of a body cell from the patient into a stem cell and stimulate growth.
    (B) Take a stem cell from the patient, remove the nucleus, insert healthy DNA and stimulate growth.
    (C) Choose body cell from healthy person, isolate and stimulate growth.
    (D) Take a gamete from patient, fertilise in vitro with gamete with healthy DNA and stimulate growth.

81 Chemical Monitoring

Chemical monitoring involves a range of technologies and different areas of scientific understanding and knowledge. Medical conditions can be monitored to see if there is a change over time. There can be continuous monitoring or testing can occur at intervals, e.g. blood tests for cholesterol levels or a glucose meter to measure blood sugar.

Monitoring is usually classified by the target of interest, e.g. cardiac monitoring measures heart rhythm and cardiac output, haemodynamic monitoring measures blood pressure and blood flow, respiratory monitoring measures carbon dioxide and oxygen levels and neurological monitoring measures intracranial pressure and brain wave patterns.

Respiratory monitoring

Several current technologies allow the measurement of oxygen saturation and carbon dioxide concentrations in the blood. These include: artificial blood gas (ABG) analysis; pulse oximetry; and capnometers.

Arterial blood gas analysis. ABG analysis takes blood samples from an artery, usually the radial artery in the wrist, and the sample is tested in a blood gas analyser. The sample is tested for the concentration of oxygen, the concentration of carbon dioxide and pH. The procedure is invasive and there can be a delay between sampling and the availability of results. The results of ABG analysis provide vital information for critically ill patients who are on ventilators, or undergoing respiratory therapy. Since many diseases show a range of similar conditions, e.g. dizziness, chest pain, breathing difficulty, an ABG analysis can often reveal the cause of the problem. For example, high carbon dioxide concentration with low pH can indicate heart disease, drug abuse or pneumonia.

Pulse oximetry. The pulse oximeter uses two wavelengths of light – red (660 nm) and infra-red (940 nm) to measure the amount of absorption of light as the light passes through the finger from the light source to the photodetector. The amount of light absorbed by haemoglobin depends on whether the haemoglobin is saturated with oxygen or reduced haemoglobin. Mathematical calculations work out the proportion of oxygenated haemoglobin present. Pulse oximetry is used when conditions require non-invasive techniques or when rapid, continuous monitoring of arterial blood is needed, e.g. through anaesthesia and during the recovery phase. Pulse oximeters are also used in intensive care units during mechanical ventilation and can quickly detect problems with oxygenation. Figure 81.1 shows the structure of a pulse oximeter.

Capnometers. A capnometer is another new, non-invasive device which measures the concentrations of respired gases using an infra-red beam of light. The amount of light absorbed depends on the number of carbon dioxide molecules present. A capnometer is useful for monitoring changes in carbon dioxide concentrations in patients who are haemodynamically stable, but not the critically ill. Sudden changes in the capnometer readings of carbon dioxide concentrations must be interpreted with caution, e.g. a sudden rise in carbon dioxide levels could be due to the release of a tourniquet, an injection of sodium hydrogen carbonate or a sudden increase in cardiac output and it is suggested that ABG analysis be also used. The benefits of a capnometer is its continuous, non-invasive monitoring.

QUESTIONS

1. What is meant by chemical monitoring?
2. How is medical monitoring often classified?
3. (a) Identify a current technology that allows measurement of oxygen saturation in blood.
   (b) Explain the conditions under which these technologies are used.
   (c) Describe a situation when the readings of a pulse oximeter may not be accurate.
4. (a) What is a capnometer?
   (b) Briefly describe how it operates.
   (c) Explain the conditions when a capnometer is used.
5. Study the diagram which shows a pulse oximeter.

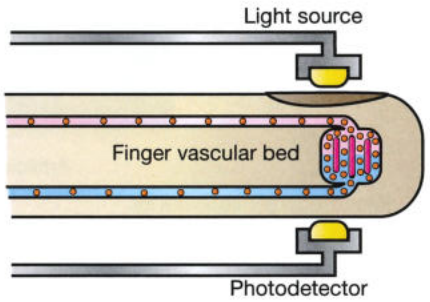


Figure 81.1 Pulse oximeter.

How does the pulse oximeter use the two wavelengths of monochromatic light – red (660 nm) and infra-red (940 nm)?

(A) To measure the amount of haemoglobin which is oxygenated in arterial blood.
(B) To gauge oxygen levels in venules during vasoconstriction.
(C) To distinguish between different forms of haemoglobin, e.g. haemoglobin combined with carbon monoxide rather than oxygen.
(D) To measure carbon dioxide concentration in plasma.

82 Experiment – Carbon Dioxide and pH

Some Human Biology students set up the following equipment to demonstrate the effect of dissolved carbon dioxide on the pH of water.

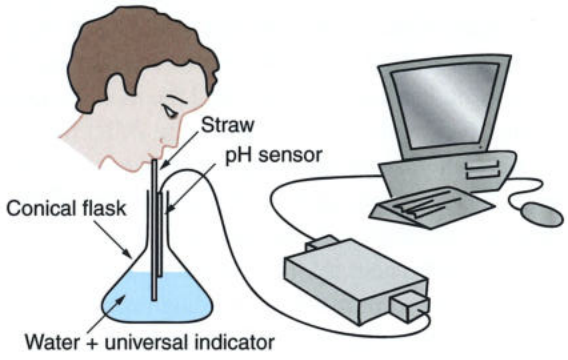


Figure 82.1 CO_2 and pH of water.

The students used the pH sensor with the data logger setting the *x*-axis as time and the *y*-axis as pH. Distilled water was placed in a dishwasher cleaned conical flask. Universal indicator was added to the distilled water and it turned dark green.

One student exhaled through the straw into the distilled water and the pH sensor recorded the change over 2 minutes.

The universal indicator turned yellow and the following graph shows the computer printout from the data logger.

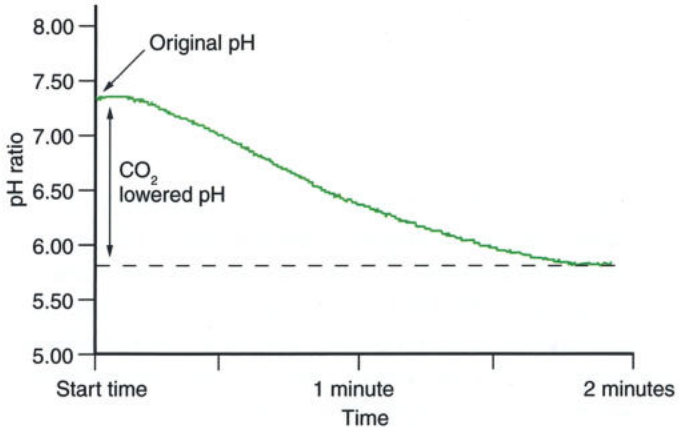


Figure 82.2 Graphical results from data logger.

QUESTIONS

1. Why did the student exhale through the straw into the distilled water?
2. Explain why the students added universal indicator to the distilled water.
3. The universal indicator changed colour from dark green to yellow. What does this mean?
4. The data logger graph shows that the initial pH of the distilled water was distinctly above 7.00. Discuss this reading.
5. In this experiment what was the independent variable and the dependent variable?
6. Compare the use of universal indicator and the pH sensor for accuracy.
7. What is the effect of carbon dioxide on the pH of water?
8. Describe how carbon dioxide is produced by the body.
9. To prove that exhaled air contained carbon dioxide, what distinguishing test could you perform?
10. Suggest one way in which this experiment could be improved.
11. (a) What is homeostasis?
    (b) Explain how homeostasis involves carbon dioxide.
12. Carotid bodies and aortic bodies are chemoreceptors found in the respective walls of the carotid artery in the neck and in the aorta leaving the heart. They can detect slight increases in acidity caused by increased carbon dioxide levels.
    (a) Identify one condition which causes an increase in the carbon dioxide level in the blood.
    (b) The medulla oblongata has chemoreceptors which monitor carbon dioxide levels and it also has the control centre for breathing. When the chemoreceptors detect high levels of carbon dioxide, the breathing control centre sends messages to the intercostal muscles between the ribs and the diaphragm to increase the rate of breathing and the depth of each breath. Draw a flow chart to show the stimulus-response pathway.
    (c) A faster and deeper breathing rate increases the oxygen intake and removal of the excess carbon dioxide. As the carbon dioxide level drops, the pH rises becoming less acidic. The faster and deeper breathing rate is now slowed down. Identify this type of response.
13. What will be one effect of an increase in carbon dioxide levels in the blood?
    (A) Raise the blood pH.
    (B) Decrease pulse rate.
    (C) Increase breathing rate.
    (D) Increase blood salinity.
14. In an experiment to measure the effect of carbon dioxide on water, which of the following would have the most effect to improve the accuracy of the results?
    (A) Using litmus paper to measure pH.
    (B) Using a pH probe and data logger to measure pH.
    (C) Using a pipette to measure the volume of water.
    (D) Using a measuring cylinder to measure the volume of water.

83 CAT and MRI Scans

X-ray computed tomography (X-ray CT scan) uses X-rays to view inside a body giving virtual slices of the part being viewed. A computer processes the information and a three-dimensional image can be made from the series of two-dimensional slices.

CAT scans

Medical imaging uses CT scans in both diagnosis and treatment of many diseases. A CAT scan or **computed axial tomography** is one type of computed tomography – there is also PET (positron emission tomography) and SPECT (single-photon emission tomography).

CT scans are also used in preventive medicine and in screening for diseases. Tumours are detected by swelling and distortion of the size and shape of normal body parts.

It has the advantage of allowing multiplanar reformatted imaging which means images can be viewed as either axial, coronal or sagittal planes. It has the disadvantage that the radiation can damage cells and DNA and potentially causes cancer.

If the stomach, bowel or small intestine are being scanned the person being scanned is given a large drink of an oral contrast agent that helps to enhance the picture. Or the patient may be given an intravenous injection of an iodinated contrast medium. Sometimes people have allergic reactions to these substances.

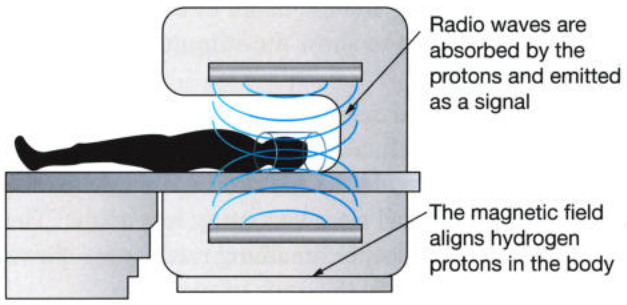


Figure 83.1 Magnetic resonance imaging.

MRI scans

MRI is magnetic resonance imaging and it uses strong magnetic fields and radio waves to form images of the body. MRI scans do not have the same risks as CT scans as MRI does not using ionising radiation. MRI scans are widely used for medical diagnosis and to follow the progress of certain diseases in people. MRI scans have made such an impact that in 2003 Paul Lauterbur and Sir Peter Mansfield were awarded the Nobel Prize in Physiology or Medicine for 'discoveries concerning magnetic resonance imaging'.

MRI scans are often used for study of neurological cancers as it is a more sensitive process for small tumours compared with CT scans. **Functional MRI** (fMRI) scans can show how different parts of the brain respond to external stimuli and are used in cognitive and behavioural research, e.g. to confirm processing areas of the brain such as the visual cortex, the motor cortex and Broca's area of speech and language. When people are required to carry out cognitive tasks, e.g. memorisation, imagination tasks, smell odours, the results have helped developed our understanding of how memories form, language is learnt and the identification of areas involved with emotions and pain. fMRI measures changes in blood flow in the brain or spinal cord. It detects changes in blood oxygenation with increased blood flow and its corresponding increased consumption of oxygen related to increased neuron activity. They are also used for spinal imaging assessing joint diseases and diagnosing soft tissue tumours. When investigating the liver and gastrointestinal tract they can detect lesions on the liver, pancreas and bile ducts and be used to diagnose rectal and prostate cancer.

Patients who have medical devices or implants need to be assessed to find out if the device is MR-safe, MR-conditional or MR-unsafe. MR-unsafe devices are ferromagnetic, e.g. cochlear implants and most permanent pacemakers, and scanning can disrupt the functioning of these devices and be potentially life threatening.

Figure 83.2 MRI signs.

CAT scans versus MRI scans

In many ways CT scans and MRI scans complement each other for use in diagnosis and treatment of many illnesses and diseases. CT scans are more widely used in comparison with MRI scans, though recent concerns about radiation-induced cancers due to CT scans has risen and affected CT scan usage. There are situations where CT scans cannot be replaced by MRI scans.

QUESTIONS

1. What is X-ray computed tomography?
2. What is a CAT scan?
3. What is a MRI scan?
4. Who was awarded the Nobel Prize for discoveries concerning MRI?
5. Outline some advantages and disadvantages of CT scans.
6. Outline some advantages and disadvantages of MRI scans.

84 Lifestyle Choices – Exercise and Diet

Lifetyle choices have an enormous impact on body functioning with both short term and long term consequences. Choices relate to the amount and type of exercise done per day/week/month, the use of drugs and the type of diet.

Exercise

Exercise is needed to maintain muscle tone and maintain nervous coordination. The activities of muscles depend on the nervous system and patterns of movement are learned behaviour. Regular healthy exercise also maintains blood circulation. A sedentary lifestyle causes a greater risk of heart disease.

Muscle fibres are as long as they need to be. If a person does not stretch the muscle it will contract to a reduced length to what is now needed. The shortening of muscle can occur if the muscle is in a flexed position in a splint or if it is not being used for some other reason.

Diet

A healthy diet needs to contain all the essential nutrients that cannot be synthesised by the body. This includes amino acids, fatty acids, minerals, vitamins and water.

Table 84.1 Some essential nutrients.

Food group	Examples of essential nutrients
Amino acids	Of the 20 amino acids, eight are essential in the adult human diet – isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan and valine as some metabolic pathways synthesise the other amino acids.
Fatty acids	Two of all the fatty acids cannot be made by the human body – omega-6 fatty acid, e.g. linoleic acid (LA) and omega-3 fatty acids, e.g. alpha-linolenic acid (ALA).
Minerals	**Major minerals** include, iron, calcium, phosphorus, magnesium, sodium, chloride and potassium. **Trace elements** needed in minute amounts include copper, chromium, fluorine, iodine, manganese, selenium and zinc.
Vitamins	Vitamins include vitamin A (retinol), thiamine (B_1), riboflavine (B_2), vitamin C (ascorbic acid), vitamin D (calciferol) and vitamin E (tocopherol).
Water	Water is essential to maintain proper metabolism, for osmoregulation and to replace water lost by urine and evaporation. Water is the main solvent in the body.

Proteins are needed in the diet as they contain amino acids which are needed to build body structures, e.g. around two thirds of the total dry mass of a cell is protein and for cell metabolism, e.g. to form enzymes, for protection, e.g. to form antibodies, and to maintain homeostasis, e.g. soluble proteins act as buffers to stabilise pH.

Metabolic fuel is needed to provide energy requirements and is normally taken in as carbohydrates and lipids. In some countries protein is used to provide 10% to 15% of fuel requirements. The amount of energy needed by a person depends on their basal metabolism, sex, age, body size and activity. There is a rising problem of **obesity** in Australia with 1 in 5 children being overweight. Obesity in children is difficult to define as height varies with age during growth. Obesity is often not defined as an absolute number but by looking at historical data and using a BMI greater than the 95th percentile. Obesity increases the risk of many diseases, e.g. **coronary disease** and shortens life expectancy.

Minerals – Most inorganic minerals are needed in small amounts. Humans need large amounts of calcium and phosphorus to maintain bones. Calcium is also needed for muscle and nerve functioning and phosphorus for ATP and nucleic acids. Iron is essential for haemoglobin and RBCs and in cellular respiration. Iodine is needed to produce thyroid hormones.

Vitamins – There are 13 essential vitamins – four fat soluble and nine water soluble. Vitamins are required in smaller quantities than amino acids and fatty acids and a lack will lead to a deficiency disease.

QUESTIONS

1. List some types of lifestyle choices that can affect health.
2. What can cause muscle shortening?
3. What is needed in a healthy diet?
4. Why are proteins needed in the diet?
5. Explain why children need some of the non-essential amino acids in their diet.
6. Which food groups are used to provide metabolic fuel for the body?
7. Identify some factors that determine the amount of energy needed by a person.
8. (a) How is obesity defined?
   (b) Why can it be hard to define obesity in children?
   (c) Why is obesity a problem?
9. Construct a table to outline why cheap high energy foods, large portion sizes, vehicles for transport and sedentary occupations contribute to obesity.
10. Name the two fatty acids that cannot be made by humans in their body.
11. Use some examples to explain why some named minerals are needed by the body.
12. A lack of vitamin A can cause night blindness and dry, flaky skin. What foods are good sources of vitamin A?

85 Lifestyle Choices – Drugs

A **drug** is any chemical substance that is taken into the body and causes modification of one or more functions of the body. Many drugs are taken to relieve pain or treat an illness. Some drugs cause addiction and are habit forming, e.g. alcohol, nicotine and narcotics.

An **addiction** refers to a physical or psychological dependence on a compulsive behaviour that negatively impacts the quality of life of the person and that the person feels powerless to stop. **Substance addiction** involves the consumption of substance(s), e.g. alcohol, tobacco or psychoactive drugs such as narcotics, stimulants or sedatives. The term '**substance dependence**' is often used rather than 'addiction' as there is debate as to whether some substances, e.g. tobacco are 'addictive' making the person physically dependent on the substance with increasing tolerance for the substance. Substance dependence shows the reliance on the substance and the inability of the person to quit.

Psychoactive drugs are chemical substances that alter mood, behaviour, perception or mental functioning. Psychoactive drugs modify biochemical or physiological processes in the brain. They can either block or enhance the effect of neurotransmitters.

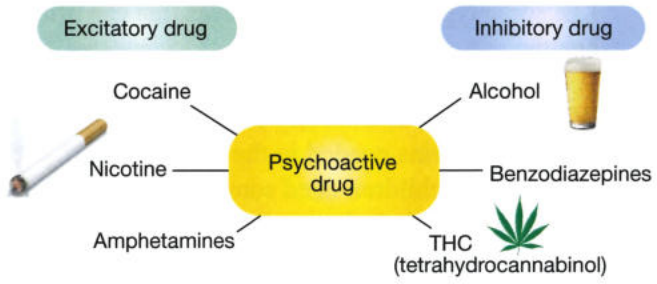


Figure 85.1 Psychoactive drugs.

Drugs and dopamine secretion

Many drugs, e.g. cocaine and amphetamines alter dopamine functioning and dopamine is thus related to addiction. **Dopamine** is a neurotransmitter that affects brain processes involved in behaviour and cognition, controlling voluntary movement, emotional and mood responses and the ability to experience pleasure and pain. It is associated with the reward system of the brain giving a sense of enjoyment that reinforces a behaviour and increases motivation to perform a certain activity.

Drugs of abuse stimulate the reward pathway, e.g. by acting on the dopaminergic neurons. Drugs may increase postsynaptic sensitivity to dopamine, increase dopamine release (amphetamines) or inhibit dopamine reuptake (cocaine and amphetamines).

Cocaine

Cocaine is highly addictive and acts by inhibiting the reuptake of neurotransmitters causing more neurotransmitter to be present in the synaptic cleft. By binding to the dopamine transporter cocaine causes dopamine to accumulate in the synaptic cleft with the dopamine continuing to stimulate neurons causing prolonged feelings of pleasure and excitement. Habitual use and prolonged exposure makes the brain adapt to cocaine and leads to a decrease in the synthesis of dopamine by neurons. Since nicotine increases the levels of dopamine in the brain, tobacco smoking while using cocaine enhances the feelings of happiness but can lead to chain smoking and other health issues, e.g. strain on the cardiovascular system, chest pains, respiratory problems. Prolonged use causes itching, hallucinations and paranoid delusions.

Cocaine addicts find it hard to maintain a job and concentrate on activities that will finance their habit. Cocaine stimulates the CNS and is an appetite suppressant. It enhances endurance, increases alertness, reduces feelings of fatigue and gives feelings of pleasure associated with the 'reward pathway'. When habitual use leads to a decrease in the synthesis of dopamine by neurons, the low dopamine levels cause depressive moods, irritability, paranoia, feelings of fatigue, high blood pressure, cardiovascular disease, permanent brain damage and psychoses.

Nicotine

Nicotine binds to the presynaptic receptors, exciting the neuron to fire more action potentials and increases the release of dopamine into the synapse. It can bind to receptors on the presynaptic neuron and stimulate receptors that usually respond to acetylcholine causing an increase in adrenaline and increasing heart rate, blood pressure and blood sugar levels. Signs of nicotine addiction are yellow stains on the teeth, bad breath, decreased lung capacity and increased heart rate. Long term health problems include lung cancer and other cancers of the respiratory system, e.g. throat, mouth, tongue, emphysema, cardiovascular disease, stroke, chronic bronchitis and prematurely wrinkled skin.

Amphetamines

Amphetamines stimulate the CNS and suppress the appetite. They block reuptake of neurotransmitter causing more neurotransmitter to be present in the synaptic cleft. As a stimulant amphetamines increase heart rate and blood pressure. High blood pressure can lead to heart problems and there is some evidence that abuse of amphetamines increases the possibility of heart attacks in young adults. **Ecstasy** contains the drug methylenedioxymethamphetamine (MDMA) and can be a mix of amphetamines with other substances.

It causes feelings of happiness, confidence, fast heart rate, dehydration, muscle aches and pains, heightened senses and hallucinations. After taking ecstasy there are feelings of confusion, depression, impaired judgement, sleep problems severe anxiety, faintness, chills and paranoia.

Tetrahydrocannabinol (THC)

THC (tetrahydrocannabinol) is the main psychoactive substance in the cannabis plant and causes relaxation, drowsiness and reduces pain. In high doses it distorts perception. THC binds to cannabinoid receptors (e.g. CB1 receptors in the brain). It mimics the chemical anandamide which is manufactured by the brain and bonds to the postsynaptic neuron. When acting on CB1 receptors in the hunger centres in the hypothalamus it increases the palatability of food and enhances appetite. There is conflicting evidence and debate about whether THC is addictive/non-addictive. Some studies suggest negative side effects with prolonged and excessive use, e.g. short term memory loss and users have a greater risk of developing psychosis than non-users. THC is a psychoactive drug that makes a person feel relaxed, drowsy, an altered space-time perception with heightened sensory experiences of colour, taste and sound. THC reduces concentration, impairs decision making and affects short term memory. It has moderate analgesic effects, stimulates the appetite, inhibits vomiting and is used to treat pain.

Benzodiazepines

Benzodiazepines are depressant drugs that slow down messages from the brain and activity in the CNS. They enhance the effect of GABA and hyperpolarise the postsynaptic membrane so that impulses are not transmitted and the person experiences a sedative, hypnotic effect. Most are known by their brand name, e.g. valium, mogadon, serepax, murelax. They are used to treat anxiety, insomnia and as a muscle relaxant to reduce convulsions. If taken with other CNS depressants, e.g. alcohol and opiates the result can be toxic.

Lysergic acid diethylamide (LSD)

LSD (lysergic acid diethylamide) and **mescaline** are psychedelic psychoactive drugs that distort sensory perception, e.g. sight and sound. They produce an hallucinatory effect by binding to the brain receptors for serotonin and dopamine on the postsynaptic membrane. They interfere with the normal functioning of the neurotransmitter action of serotonin which controls behavioural, perceptual and regulatory systems of the body, e.g. mood, hunger, sexual behaviour and muscle control. Although LSD is not addictive the brain is affected with a distortion in the sense of time and 'ego death' (loss of sense of identity or the ego), delusion and a 'cross over' of the senses, e.g. hearing colours and seeing sound. LSD can trigger chronic and severe psychotic disorders, e.g. schizophrenia.

Behaviours indicating substance dependence

Behaviours that indicate substance dependence or addiction include the following.

- Increasing tolerance to substance with increased doses needed to achieve same result or with reduced response to same dose.
- Withdrawal signs or symptoms if the substance is abruptly discontinued.
- A continual desire or unsuccessful attempts to cut down or stop using the substance.
- Craves use of the substance.
- Effort and/or time is spent obtaining the substance.
- Other activities, e.g. occupational, social or recreational requirements are negatively affected.
- Despite arising physical and psychological problems, the substance use is maintained.

QUESTIONS

1. (a) Define a drug.
   (b) Outline why people take drugs.
2. What is meant by an addiction?
3. Outline what is involved in substance addiction.
4. Discuss why the term substance dependence is often used rather than substance addiction.
5. (a) What are psychoactive drugs?
   (b) How do psychoactive drugs work?
6. Explain how drug addiction can be related to the secretion of dopamine.
7. Distinguish between excitatory drugs and inhibitory drugs.
8. Construct a table to describe some excitatory psychoactive drugs.
9. Construct a table to describe some inhibitory psychoactive drugs.
10. List some behaviours that indicate substance addiction.
11. Which diseases have a higher risk of occurring if a person habitually takes large amounts of oral amphetamines?
    (A) Measles, mumps, hepatitis.
    (B) Sleep problems, AIDS, coronary heart disease.
    (C) Paranoia, mumps, AIDS.
    (D) Sleep problems, paranoia, coronary heart disease.

86 Lifestyle Choices – Alcohol

There are many drinks that contain alcohol (ethanol), e.g. beer, wine, spirits. Most alcoholic drinks contain between 2% to 40% ethanol.

When alcohol is consumed it can be absorbed straight from the stomach into the bloodstream (approximately 25%) and the remaining amount is absorbed through the villi in the small intestine. Once in the bloodstream it remains in the body until processed. About 90% to 98% is broken down in the liver.

Figure 86.1 Alcohol is a risk factor.

Alcohol dehydrogenase is an enzyme found in the lining of the stomach and in the liver and allows the consumption of alcoholic drinks. It converts alcohols to aldehydes or ketones.

Alcohol → acetaldehyde → acetic acid → carbon dioxide

Figure 86.2 Breakdown of alcohol.

At least six slightly different alcohol dehydrogenases in humans relate to the ability of different people to process alcohol, e.g. male/female, old/young, or populations from different parts of the world.

Short term effects

Alcohol depresses the CNS and there are short term effects on the brain, producing a sedative, pleasant, relaxed effect. Intoxication impairs judgement and motor skills and chronic use causes liver and brain cell damage. Alcohol can be physiologically addictive with dangerous withdrawal symptoms. Alcoholics often suffer from nutrient deficiencies as they lose interest in food. High alcohol intake can also cause weight gain and obesity due to the high energy value of alcohol.

Long term effects

Long excessive use of alcohol increases the chances of several diseases, e.g. oesophageal and pancreatic cancer, damage to the immune system with higher susceptibility to infection, weaker and less regular heartbeat and most importantly liver damage.

There are three main types of **alcohol induced liver disease**.

- **Alcoholic hepatitis** which is an inflammation of the liver with destruction of individual liver cells and scarring.
- **Fatty liver** which involves an excessive accumulation of fat inside the hepatocytes causing the liver to become enlarged.
- **Alcoholic cirrhosis** which involves the destruction of normal liver tissue and permanent scarring making the cells non-functional.

Signs of liver damage include the following.

- **Pale stools** – caused by a lack of stercobilin (bile pigment) and non-functional liver cells which cannot perform the normal breakdown of haemoglobin.
- **Dark urine** – due to larger than normal amounts of bilirubin (yellow pigment from bile) mixes with urine due to impaired functioning of liver cells.
- **Jaundice** – a yellowing of the skin and sclera (white) of the eyes is due to high bilirubin levels.
- **Bruising and easy bleeding** – liver damage can disrupt the manufacture of clotting factors and the absence of these factors leads to bleeding and bruising.
- **Excessive fatigue** – disruption of normal liver processes, e.g. carbohydrate and protein metabolism reduces energy levels and the supply of nutrients.

QUESTIONS

1. How much ethanol is in most alcoholic drinks?
2. What amount of alcohol can be absorbed straight from the stomach into the bloodstream?
3. (a) Identify the enzyme that allows the consumption of alcohol.
   (b) What is the significance of differences in this enzyme between different people?
4. List some short term effects of alcoholic intake.
5. Discuss some of the long term effects of alcoholic intake.
6. Describe the three main types of liver disease caused by high alcoholic intake.
7. Describe the signs of liver damage.
8. Which organ is first likely to be affected by high alcoholic intake?
   (A) Mouth (B) Stomach
   (C) Liver (D) Gall bladder

Topic Test

Section A – Multiple Choice (35 marks)

1. The diagram shows a process in the body.

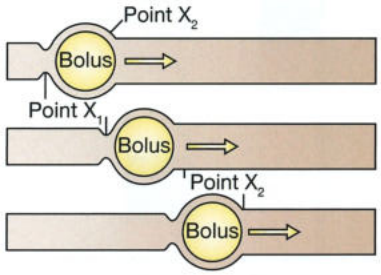


Figure TT.1 Body process.

What is happening to the muscles at points X_1 and X_2?

	Point X_1	Point X_2
(A)	Contraction	Contraction
(B)	Relaxation	Relaxation
(C)	Contraction	Relaxation
(D)	Relaxation	Contraction

2. Which organelle is mainly involved with the transformation and release of energy?
   (A) Mitochondria.
   (B) Golgi body.
   (C) Nucleus.
   (D) Endoplasmic reticulum.
3. The diagram shows part of an organic molecule.

Figure TT.2 Part of an organic molecule.

Identify the group this molecule belongs to.
(A) Carbohydrate. (B) Protein.
(C) Lipid. (D) Enzyme.

4. Sucrose can be broken down by the enzyme sucrase. Which of the following would slow the reaction rate?
   (A) Diluting with water.
   (B) Adding sucrase.
   (C) Adding sucrose.
   (D) Adding both sucrose and sucrase.
5. Which of the following best matches food group and use?

	Proteins	Carbohydrates	Lipids
(A)	Stores of energy	In membranes	Cell growth/repair
(B)	Waterproofing	Stores of energy	Energy source
(C)	Energy source	Cell growth/repair	Energy store
(D)	Cell growth/repair	Energy source	In membranes

6. The photomicrograph below is of a cell organelle.

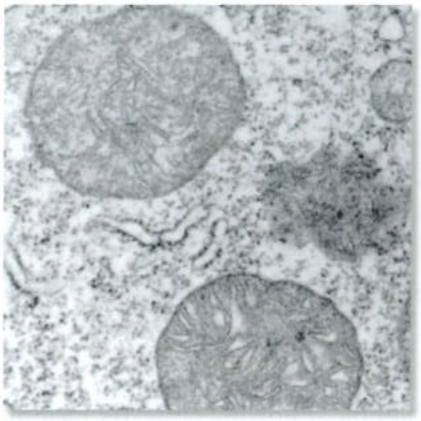


Figure TT.3 Electron micrograph of an organelle.

What is the main function of this organelle?
(A) Protein synthesis.
(B) Aerobic respiration.
(C) Transport of substances.
(D) Lipid manufacture.

7. The table shows the number of organelles found in four different types of cells.
Which of these cells would you expect to be an active muscle cell?

Cell type	Number of ribosomes	Number of Golgi bodies	Number of mitochondria
(A)	10 million	1	5
(B)	10 million	4	5
(C)	5 million	1	15
(D)	5 million	1	8

8. The diagram shows a human cell with several organelles.

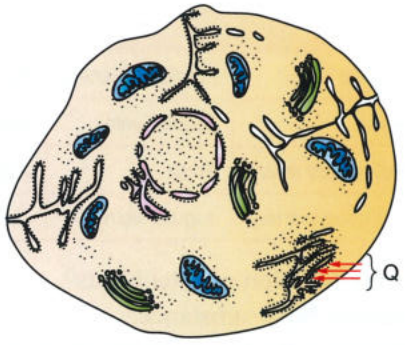


Figure TT.4 Human cell.

Which chemical test would identify the substances produced by structure Q?
(A) Biuret solution. (B) Benedict's solution.
(C) Iodine solution. (D) Sudan IV.

9. The presence of which substance in urine is indicative of kidney problems in a human?
   (A) Glucose. (B) Protein.
   (C) Water. (D) Salt.

Use the following diagram of the cross-section of three blood vessels for the next TWO questions.

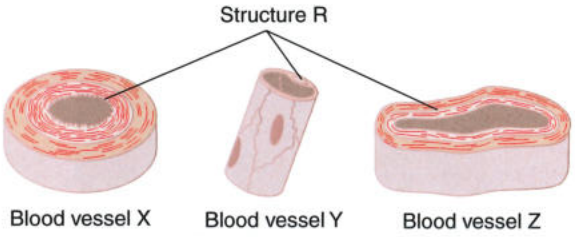


Figure TT.5 Three blood vessels.

10. Which of the following correctly identifies each blood vessel?

	Vessel X	Vessel Y	Vessel Z
(A)	Vein	Capillary	Artery
(B)	Capillary	Vein	Artery
(C)	Vein	Artery	Capillary
(D)	Artery	Capillary	Vein

11. What is structure R?
    (A) Muscle layer.
    (B) Elastic layer.
    (C) Endothelium.
    (D) Epithelium.
12. Some researchers wished to compare the strengths of different blood vessels. They threaded equal lengths of different blood vessels onto the horizontal arm of a retort stand and then recorded the mass in grams that caused each blood vessel to break. Which of the following is the most likely result?

	Mass that caused vessel to break		
	500 g	3500 g	5000 g
(A)	Jugular vein	Pulmonary artery	Aorta
(B)	Aorta	Jugular vein	Pulmonary artery
(C)	Pulmonary artery	Aorta	Jugular vein
(D)	Pulmonary artery	Jugular vein	Aorta

13. What is the function of osteoblasts?
    (A) Secretes bone substance.
    (B) Absorbs bone tissue during growth.
    (C) Mature older bone cell.
    (D) Produce cartilage and collagen matrix.
14. What is meant by cell metabolism?
    (A) All the chemical reactions that take place in a cell.
    (B) The rate of biochemical reactions in the cell.
    (C) A process that uses energy in a cell.
    (D) A process that releases energy for use in the cell.
15. The daily water loss of a person was studied for two days – one day in winter and one day in summer. On both occasions the person had unrestricted access to water.

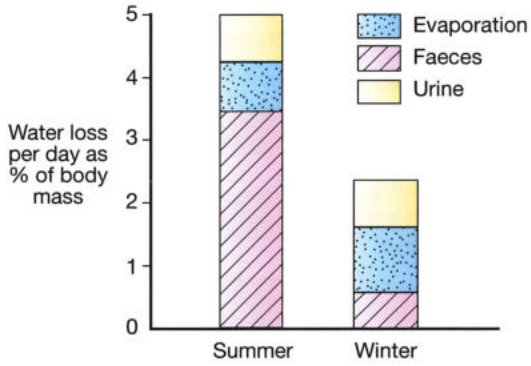


Figure TT.6 Water loss for a person in summer and in winter.

Which of the following makes a correct comment about the amount of water lost as urine?
    (A) Is greatly reduced in winter due to cold weather slowing down the function of the kidneys.
    (B) Did not change as there was no need to conserve water in either situation.
    (C) Is much greater in summer as water is needed for evaporation and cooling the body.
    (D) Is always constant as people cannot alter urine concentration.
16. What is the correct order from smallest to largest component?
    (A) System → organ → tissue → cell → organelle
    (B) Organelle → organ → cell → tissue → system
    (C) Organelle → cell → organ → tissue → system
    (D) Organelle → cell → tissue → organ → system
17. The diagram shows a model of a cell structure.

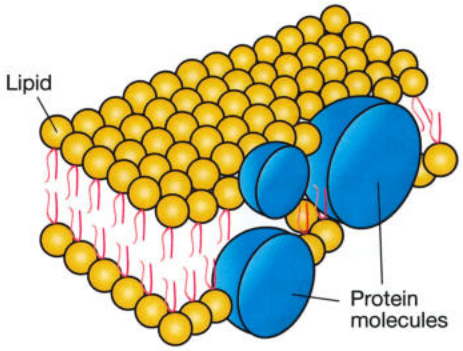


Figure TT.7 Model of a cell structure.

What name was given to this model?
    (A) Fluid mosaic model.
    (B) Bilipid and protein model.
    (C) Bernoulli model.
    (D) Mitochondrion model.

18. Which of the following cellular structures makes it possible for a cell to be different from its surroundings?
(A) Nuclear membrane. (B) Nucleus.
(C) Plasma membrane. (D) Mitochondria.

19. The diagram shows sucrose solutions in a beaker and in a bag. Sucrose is a disaccharide.

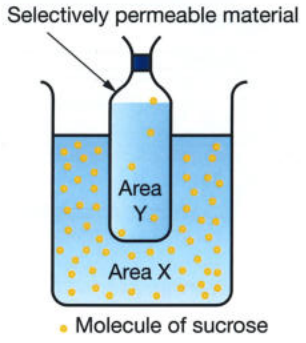


Figure TT.8 Sucrose solution in a bag and in a beaker.

After several hours, tests showed that there was a change in the sucrose concentrations in area X and in area Y. What is the best explanation for this change?
(A) Sucrose molecules moved from area Y to area X by osmosis.
(B) Water moved from area Y to area X by osmosis.
(C) Sucrose molecules moved from area X to area Y by osmosis.
(D) Water moved from area X to area Y by osmosis

20. What happens as a cell grows in size?
(A) Its volume decreases.
(B) Its surface area decreases.
(C) Its surface area to volume ratio decreases.
(D) Its surface area to volume ratio increases.

21. What happens in diffusion?
(A) Energy is expended to move particles
(B) Particles in liquid dissolve and move about
(C) There is random movement of particles
(D) Particles move from low to high concentration

22. Which of the following correctly identifies anabolic and catabolic reactions?

	Anabolic reaction	Catabolic reaction
(A)	Protein synthesis	Respiration
(B)	Respiration	Protein synthesis
(C)	Lipid synthesis	Amino acid synthesis
(D)	Respiration	Lipid synthesis

23. Which body system is responsible for the delivery of oxygen and the removal of carbon dioxide from body cells that are respiring?
(A) Digestive system. (B) Respiratory system.
(C) Urinary system. (D) Circulatory system.

24. Which of the following correctly identifies the role of enzymes in metabolism?
(A) Biological catalysts controlling the speed of chemical reactions.
(B) Biological buffer system to maintain the pH of the blood and interstitial fluid.
(C) Hormonal secretion to cause a response in an effector.
(D) Nervous secretion to control the speed of electrochemical impulses in nerves.

25. Three enzymes found in the human body are amylase, pepsin and trypsin.

	Amylase	Pepsin	Trypsin
Found in	Saliva in mouth	Stomach	Small intestine
Acts on	Starch	Protein	Further breakdown of protein
pH area	5 to 6.5	2	8 to 9

What is shown by this data?
(A) Enzyme activity depends on temperature.
(B) Acidity of the environment is important in controlling enzyme function.
(C) Pepsin works best in an alkaline environment.
(D) pH is constant throughout body tissues and organs.

26. The diagram shows the pathway for the secretion of two proteins – protein X and protein Y out of a cell across the plasma membrane.

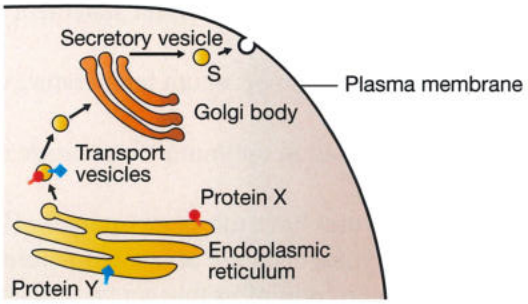
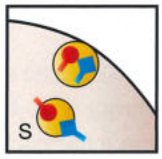
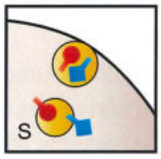
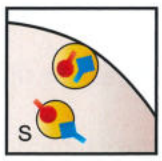


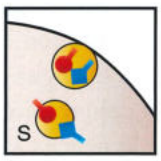
Figure TT.9 Protein secretion.

Which diagram shows the correct orientation of these two proteins as they are secreted from the cell?

(A)


(B)


(C)


(D)


27. What are the raw materials for respiration?
    (A) Carbon dioxide and water.
    (B) Carbon dioxide and glucose.
    (C) Oxygen and glucose.
    (D) Oxygen and water.
28. Which of the following is an example of passive transport in the kidneys?
    (A) Movement of sodium out of the convoluted tubules into the interstitial fluid.
    (B) Reabsorption of water.
    (C) Glucose moving against a concentration gradient from the filtrate into the interstitial fluid.
    (D) Amino acids transported out of the convoluted tubules.
29. The following diagram shows the reaction rates of two enzymes over a range of temperatures.

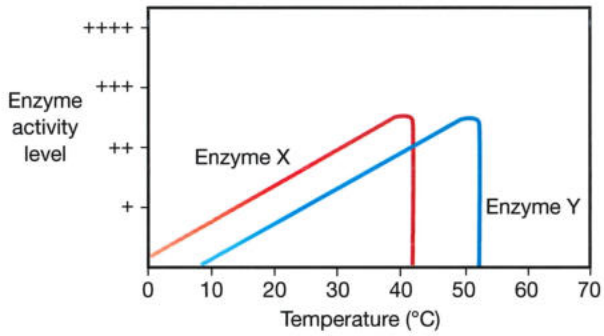


Figure TT.10 Reaction rates and temperature for two different enzymes.

Which of the following is a correct statement about these results?
    (A) Enzyme X has an optimum temperature of about 40°C.
    (B) Enzyme Y has an optimum temperature of about 40°C.
    (C) Both enzymes have the same optimal pH.
    (D) Both enzymes denature at low temperatures.
30. What is the main form of usable energy released by the chain of biochemical reactions in cellular respiration?
    (A) Glucose
    (B) Glycerol
    (C) ADP
    (D) ATP
31. Alcohol consumption can produce a diuretic effect and in excessive amounts can compromise body functioning. What happens when alcohol acts as a diuretic?
    (A) It decreases haemoglobin levels improving hydration.
    (B) It changes fluid balance increasing the risk of dehydration.
    (C) It acts as a stimulant increasing fluid retention.
    (D) It acts as a depressant increasing fluid retention.
32. When graphing experimental results, what is the *y*-axis?
    (A) The control.
    (B) Controlled variables.
    (C) Dependent variable.
    (D) Independent variable.
33. The diagram shows measuring pulse oximetry.

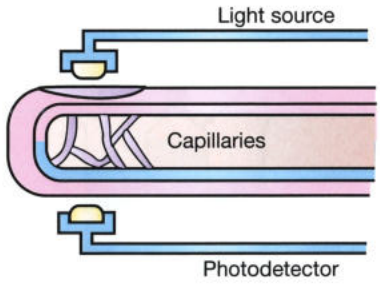


Figure TT.11 Pulse oximetry.

What is the function of pulse oximetry?
    (A) To measure carbon dioxide concentrations in blood.
    (B) To monitor the pH of the blood by measuring the amount of HCO_3^- present.
    (C) To measure the amount of ozone in the blood.
    (D) To monitor the oxygen saturation of haemoglobin.
34. When looking at a prepared slide of human blood under a light microscope you can see that:
    (A) Blood carries oxygen and carbon dioxide to and from body tissues.
    (B) There are only nucleated cells.
    (C) There are several cell types.
    (D) Vitamins and hormones are transported in blood.
35. What is the advantage of a lung being divided into numerous air sacs rather than consisting of one very large sac with the same external dimensions?
    (A) Gas exchange needs a large surface area.
    (B) Alveoli can more efficiently filter dust and foreign particles from the air.
    (C) Internal alveoli prevent dehydration of the gas exchange surface.
    (D) Some air sacs allow oxygen diffusion into the blood and other air sacs allow diffusion of carbon dioxide from the blood.

Section B – Written Response (65 marks)

36. (a) Define a coenzyme. (1 mark)
    (b) Name a coenzyme. (1 mark)
    (c) How are cofactors different to coenzymes? (2 marks)
37. Cells will grow and increase in size until they reach a certain critical size, at which point they will undergo mitosis dividing to become two identical cells. Explain why this occurs. (3 marks)
38. Study the electron micrograph of an organelle.

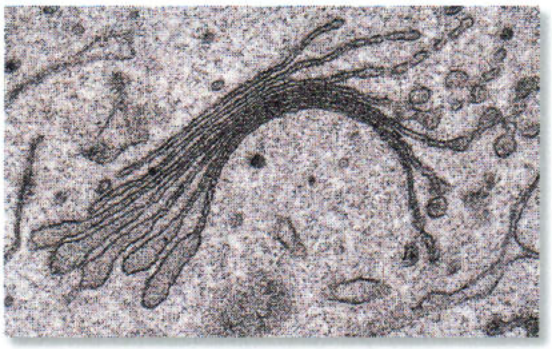


Figure TT.12 An organelle.

    (a) Identify this organelle. (1 mark)
    (b) Describe its function. (2 marks)
39. A large raw potato is peeled and two cubes are cut from the remaining block. One cube X has sides 1 cm × 1 cm × 1 cm, while the other cube Y has sides 2 cm × 2 cm × 2 cm.

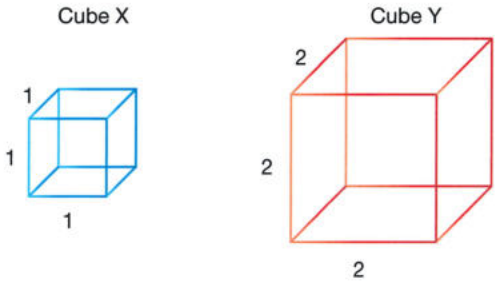


Figure TT.13 Two cubes of potato.

    (a) Explain what would happen if cube Y was placed in a very strong salt solution. (4 marks)
    (b) Explain any difference that would occur if cube X was placed in a salt solution with the same concentration. (2 marks)
40. Identify three features that are needed by an efficient respiratory surface and for each feature explain why it is needed. (3 marks)
41. Distinguish between diffusion, osmosis and active transport noting the concentration gradient and the energy requirements for the movement of substances. (6 marks)
42. Carbon dioxide is a waste product from respiration.
    (a) Describe an experiment you could perform to demonstrate the effect of dissolved carbon dioxide on the pH of water. (3 marks)
    (b) Explain the expected results using your understanding of the chemical composition of the substances involved. (3 marks)
    (c) An enzyme called carbonic anhydrase acts on red blood cells to cause carbon dioxide to form carbonic acid. Predict what would happen if this enzyme was not available. (3 marks)
43. The diagrams show inhalation and exhalation.

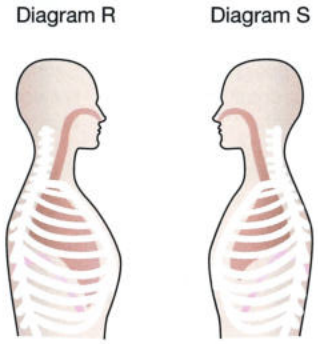


Figure TT.14 Inhalation and exhalation.

    (a) Identify which diagram shows inhalation and which diagram shows exhalation. (1 mark)
    (b) Discuss how the diagrams show the mechanism of breathing. (3 marks)
44. Copy the following diagram of the heart.

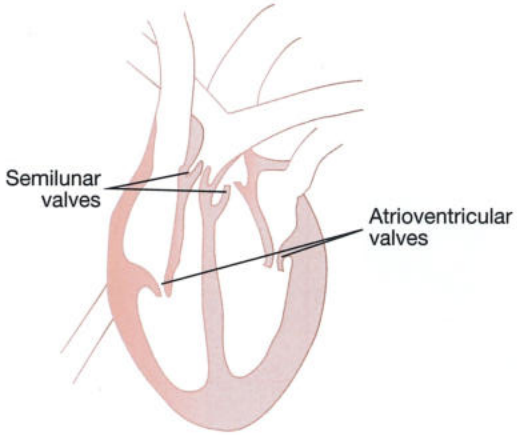
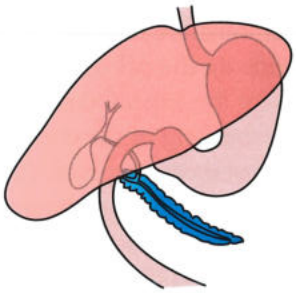


Figure TT.15 Human heart.

    (a) On the diagram identify the superior vena cava, inferior vena cava, the aorta, the left and right atria, the left and right ventricles, the left pulmonary artery and the left pulmonary vein. (3 marks)
    (b) Draw arrows on your diagram to show the direction of blood flow through the heart. (2 marks)
    (c) The diagram shows the semilunar valves and the atrioventricular valves. Compare the function of these sets of valves. (2 marks)

45. The diagram shows part of the diagram system.



S	Stores food. Produces acid and enzymes.	W	Receives chyme from the stomach and digestive secretions from the pancreas and liver.
T	Regulates passage of chyme into small intestine from stomach.	X	Secretes bile into a network of narrow channels.
U	Takes secretions to the duodenum.	Y	Stores and modifies bile, absorbing water and concentrating bile salts and other components.
V	Exocrine gland produces digestive enzymes.	Z	Empties secretions into the duodenum.

Figure TT.16 Part of the digestive system.

Copy the diagram and correctly match each part and its description by labelling the diagram. (8 marks)

46. Copy and complete the following table to describe the levels of muscle structure. (4 marks)

Diagram	Description
Level of structure =	
Level of structure =	
Level of structure =	
Level of structure =	

47. The diagram shows the structure of a nephron.

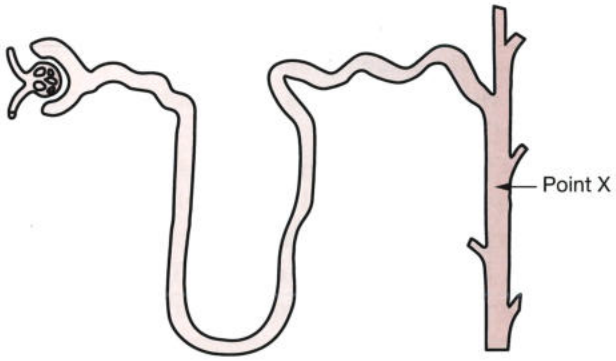


Figure TT.17 A nephron

The following processes take place in the nephron.

A Filtration takes place.
B Water reabsorption takes place.
C Glucose reabsorption takes place.
D Secretion of foreign materials, e.g. penicillin and aspirin.

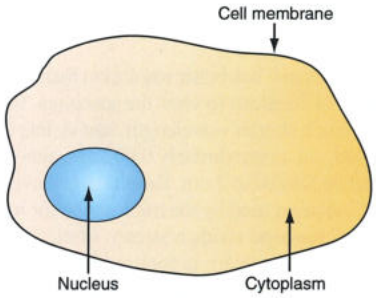
(a) Which process(es) A, B, C, D are occurring at point X? (1 mark)
(b) Which process(es) A, B, C, D occur due to active transport? (1 mark)
(c) Outline what happens in deamination. (1 mark)

48. Assess the impact on society of one major advance in scientific understanding and technology in any field associated with blood and identify one possible future direction of biological research in this field. (6 marks)

Answers

1 Assumed Knowledge

1. The cells of living things have a cell membrane, cytoplasm and DNA.
2. Human epithelial cheek cell.



3. A = eye of person using the light microscope, B = ocular lens, C = objective lens, D = specimen, E = condenser lens, F = light source.
4. When using a light microscope, you should always wear shoes with covered toes, as the microscope is heavy and if you drop it you could damage exposed skin on your feet.
5. The nucleus stores information needed to control all cell activities.
6. The cell membrane surrounds the cell contents from the external environment and controls the substances that can leave or enter the cell.
7. Cytoplasm is a general term for the contents of a cell outside the nucleus and within the cell membrane.
8. Protoplasm is the semifluid transparent substance that makes up the living matter of plant and animal cells including the nucleus and cytoplasm.
9. Cytosol is the semifluid part of the cytoplasm.
10. The circulatory system is the transport system of the body, e.g. it carries the glucose and oxygen to the cells and distributes heat around the body.
11. Vas/vaso means vessel. Vasoconstriction means a decrease in the diameter of a superficial blood vessel. The vas deferens is the tube in the male reproductive system that carries sperm from the epididymis to the urethra.
12. Blood consists of plasma and blood cells and transports substances around the body through a large network of blood vessels.
13. Blood vessels include arteries, veins and capillaries.
14. The heart is a muscular pump that uses metabolic energy to maintain blood pressure causing blood to flow down the pressure gradient through blood vessels.
15. The human circulatory system consists of the heart and blood vessels, e.g. arteries, veins and capillaries and blood.
16. Respiration is the chemical reactions in which cells obtain energy from food.
17. In humans, the respiratory system consists of the lungs and passages which bring air into and out of the lungs, e.g. trachea, bronchi and alveoli.
18. The function of the respiratory system is the intake, expulsion and exchange of oxygen and carbon dioxide.
19. Lymph vessels are part of the lymphatic system beginning as blind-ended tubes in the spaces between tissue cells and blood vessels.
20. Metabolism is the sum of all chemical reactions in an organism.
21. The four basic groups of organic compounds are proteins, carbohydrates, lipids and nucleic acids.
22. Inorganic compounds are molecules that do not contain carbon (excluding some carbonates and simple oxides of carbon).
23. The function of the digestive system is to break down ingested food into smaller particles so that nutrients can be absorbed into the body.
24. Food is needed to provide – 1. The chemical substances which can be built into large molecules that make up the structure of cells. 2. Chemical substances that can be oxidised to yield energy. The energy is needed for movement, for synthesis of large molecules from simple molecules and in endotherms for the production of heat.
25. An enzyme is an organic catalyst that changes the rate of a reaction without being consumed by the reaction.
26. The prefix hepat- refers to the liver, e.g. hepatitis is a disease of the liver and the hepatic portal vein goes to the liver.
27. Anabolism is the building of large, complex molecules from simpler compounds while catabolism is the breaking down of substances to provide energy and raw materials.
28. A = salivary glands, B = oesophagus, C = stomach, D = liver, E = gall bladder, F = pancreas, G = large intestine, H = small intestine, I = appendix, J = anus.
29.

Part	Structure	Function
Mouth	Has teeth and openings from salivary glands	Teeth break food into small pieces and salivary enzymes begin chemical digestion
Oesophagus	Long tube	Moves food to stomach by peristalsis
Stomach	Muscles and glands in wall	Churns food and produces digestive enzyme to digest protein
Small intestine	Long thin tube with villi and glands	Digestion is completed and nutrients absorbed through walls
Large intestine	Long tube	Water, salts and vitamins absorbed
Anus	Muscular ring	Eliminates faeces

30. (a) (i) Carbohydrates are broken down into glucose.
    (ii) Proteins are broken down into amino acids.
    (iii) Fats are broken down into fatty acids.
    (b) Scientists make simple models to make it easier to understand complex systems providing, as in this case of the model of the digestive system, a basic idea of the structure and functioning of a multifaceted body system. The diagram shows the digestive system as a long tube through the body that acts as a a functional unit of life.
31. The excretory system regulates the chemical composition of body fluids by removing metabolic wastes and retaining the proper amounts of water, salts and nutrients.
32. The excretory system consists of the kidneys, liver, lungs and skin functioning at the organ level.
33. The muscular system is organised to maintain posture and produced movement.
34. Movement is needed for obtaining food, reproduction and avoiding injury.
35. Muscle tissue consists of long muscle cells that are able to contract when stimulated.

2 Writing a Practical Report

1. (a) A suitable hypothesis could be 'Exercise causes an increase in heart rate.'
   (b) The independent variable is before/after exercise. The dependent variable is the heart rate.
   (c) To test their hypothesis the students would need to: 1. Use the same test person. 2. Temperature of area before/after. 3. Other room conditions need to be constant, e.g. humidity, wind exposure. The only factor that should vary is before/after exercise.
2. (a) A suitable aim would be –'To use a light microscope to determine the size of red blood cells and white blood cells'.
   (b) Red blood cells are approximately 6 to 8 micrometres in diameter. A high power objective is needed to clearly see red blood cells to determine their dimensions.
   (c) The students would need a light microscope with lamp, lens tissues and a prepared slide of blood.
   (d) When using a light microscope you need to carry it using two hands to avoid dropping it as it is heavy and if dropped could hurt legs and feet. If using a separate microscope lamp you should take care to not touch the hot lamp to avoid burns and wait till the lamp is cool before packing away.
3. The independent variable is time and the dependent variable is the average weight gain.

4. The experiment is not valid as variables were not sufficiently controlled in the design of the experiment. Different sets of students were used to gather data about normal breathing rate, inspiratory reserve volume and expiratory reserve volume. This invalidates the average vital lung capacity data that was calculated.
5. A practical report follows the process of scientific method and should be written in the same sequence of events. If a hypothesis is proved correct, then it can be developed into a theory and then a law.

3 The Modern Light Microscope

1. The compound light microscope passes visible light through the specimen and then through a series of lenses. It operates on the main principle that an objective lens with a very short focal length can form a highly magnified real image of the object.
2. (a) The condenser focuses light on to the sample.
   (b) The iris diaphragm controls the amount of light on to the stage.
3. The resolution of the microscope is limited by the shortest wavelength of light used to view the specimen (wavelengths of visible light are 400 to 700 nm).
4.

Advantages of light microscope	Disadvantages of light microscope
Can view living specimens and observe processes in cells and within an organism. Allows the use of many different coloured stains to identify substances, structures and provide contrast for easier viewing. Specimens are easy to prepare, stain and observe in a short time frame. Relatively inexpensive and class sets can be purchased for use in schools. Light microscopes are not particularly large and are easy to store, e.g. in schools.	Limited magnification, e.g. effectively up to 1000×. Resolution limited to 0.2 micrometres.

5. A photomicrograph is a photograph made through a microscope.
6. The main benefit of the phase contrast microscope is that it increases the contrast in unstained cells. It improves our ability to study living, unpigmented cells in biological and medical research. Many dyes and stains stop chemical processes in cells which means the phase contrast microscope has improved our ability to see detail in living cells, e.g. the process of cell division.
7. The main benefit of fluorescence microscopy is that the fluorescence shows the location of specific molecules in the cell. Natural fluorescing substances, e.g. chlorophyll can be located or the specimen can be treated with fluorescing chemicals or antibodies to locate other particular section or substances in cells.
8.

Object	Size (μm)	Viewing
Frog egg	1100	Naked eye and light microscope
Paramecium	100	Light microscope
Plant epithelial cell	60	Light microscope
Human ovum	10	Light microscope
Red blood cell	7 to 8	Light microscope
Mitochondrium	1.1	Light microscope
Nanobes	0.025	Cannot be seen under a light microscope – need an electron microscope

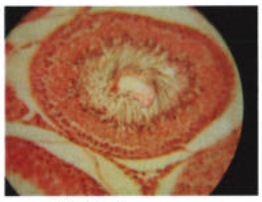
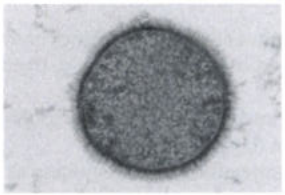
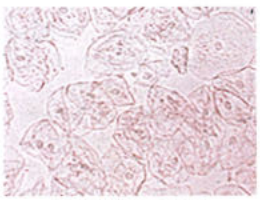
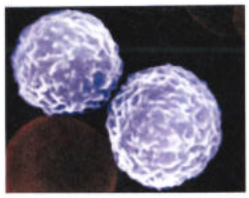
9. Stains and the light microscope have greatly increased our understanding of the processes in cells. For example, pH indicators which will show acidic or basic conditions can be used to show how the activity of micro-organisms change the pH balance of their environment such as bacterial action causing the souring of milk.
10. Objectives found on school light microscopes include 10× and 40×. The 10× objective is a low power lens used for first viewing the specimen to locate the area to be examined. The 40× objective is a high power lens and is used to make detailed observations of the specimen.
11. A parfocal microscope allows you to quickly focus using the lower power objective and then swivel the nosepiece to observe the specimen under a higher power objective without only minimal adjustment of the fine focus knob.
12. Red blood cells are very small – they have a diameter of 6 to 8 micrometres. This means the student would have been using a high power objective, e.g. 40× in the light microscope set-up to clearly observe these cells as seen in the diagram.
13. The Gram stain is usually used as the first step in identifying bacteria. It differentiates bacteria into two groups – Gram negative and Gram positive.
14. A
15. C

4 The Electron Microscope

1. The electron microscope has better resolution than the light microscope as it uses a beam of electrons to view the specimen. Since the beam of electrons has a much shorter wavelength than visible light resolution is greatly improved, e.g. approximately 0.002 nm, although in practical situations it can be limited to 2 nm. Resolution is inversely related to the wavelength of radiation used by the microscope for imaging.
2. The electron microscope sends a stream of electrons through a vacuum. The electron beam is focused by electromagnets, magnified by an objective lens and projected onto a fluorescent screen or photographic film.
3.

Advantages of electron microscope	Disadvantages of electron microscope
Increased magnification, e.g. up to 300 000×. Increased resolution, e.g. approximately 0.0005 μm. SEM shows a three-dimensional image of the specimen.	Cannot view living specimens and observe processes happening in cells. Cannot use coloured stains. Relatively expensive and are not available in schools.

4. The main benefit of the TEM is that it shows the internal ultrastructure of cells at higher magnification and resolution than the light microscope. You can see objects to the order of a several nanometres (10^{-9} m) which means new organelles have been discovered using the TEM, e.g. ribosomes and there is an increased capacity for medical, biological and materials research.
5. The main benefit of the SEM is that it gives a three-dimensional detailed view of the surface of the specimen. The image has great depth of field and gives a view of the actual appearance of the specimen.
6.

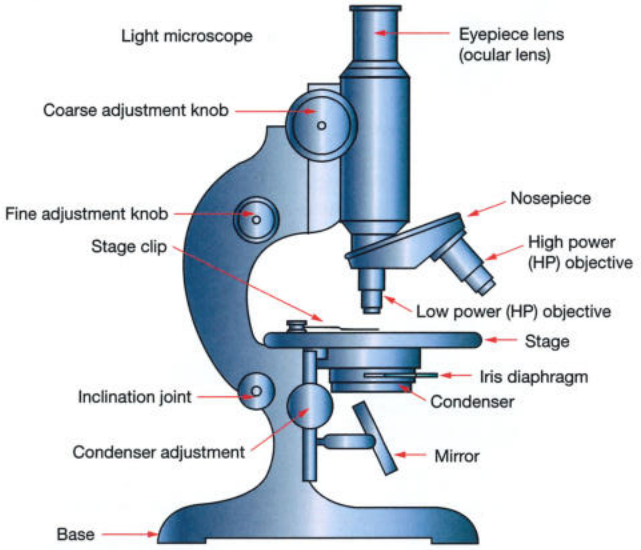
Image A	Image B
 Scanning electron microscope	 Light microscope
Image C	**Image D**
 Transmission electron microscope	 Light microscope
Image E	**Image F**
 Scanning electron microscope	 Light microscope

7. D

5 The Stereo Microscope

1. The stereo microscope is easily identified as it has two ocular eyepieces that each view the object from different angles. There are two separate optical paths for viewing.
2. You are likely to use a stereo microscope when dissecting small organisms or when you need to observe the external features of a specimen.
3. The parts of the stereo microscope are:
   A = ocular lens
   B = objective lens
   C = stage clip
   D = stage
   E = focus knob
   F = arm
4. If a compound microscope has two eyepieces which give the same image then the image will not be three-dimensional and give a 'stereo' image. To provide a three-dimensional image there must be two images each from slightly different viewing angles.
5. In a stereo microscope the lens is a distance away from the object and this gives poor resolution.
6. A typical school stereo microscope has a magnification ranging from 10× to 40× while a typical school monocular microscope has a magnification from 40× to 400×.
7. Salt crystals are light coloured and usually translucent. The dark colour beneath the crystal is needed to provide contrast so that the crystal can be properly viewed.
8. When using a stereo microscope some parts of an organism may not be in focus as the specimen is three-dimensional with several levels. The microscope is focusing on one level making other levels appear out of focus.
9. A

6 Experiment – The Light Microscope

1. 

2. It is important to always use two hands when carrying a light microscope as they are heavy and if dropped could damage skin and feet. If dropped the fall will likely irreparably damage sections of the microscope and the school may not be able to replace the equipment. You should *never* swing a microscope and hold it with one hand.
3. It is very important to always view the stage from the side when lowering the low power objective as many microscopes allow the lens to be lowered as far as the slide and even lower, which can break the slide. Cut glass can pierce skin and potentially cause infection if it is not clean. By always raising the lens away from the slide you reduce the possibility of breaking the glass slide.
4. (a) Using the low power objective (10×) gives an overall magnification of 10 × 10 = 100×.
   (b) Using the high power objective (40×) gives an overall magnification of 10 × 40 = 400×.
5. The iris diaphragm controls the amount of light reaching the stage and passing through the specimen.
6. When changing objective lenses it is usually necessary to adjust the amount of light as higher magnification lenses require more light and you need the correct amount of light to see details in the specimen. When changing from high power to low power a decrease in the amount of light is usually needed.
7. A stereo microscope is usually a low powered microscope that has two eyepieces and provides a stereoscopic view of the specimen. It is usually used for looking at organisms or parts of organisms, e.g. in dissections and does not use thin sections on a microscope slide as used by light compound microscopes.
8. If you touch the lens with your finger you will leave a print that has body oil which smudges the glass. If the oil is on the glass for a lengthy time period, it can etch the glass. Lens paper must be used to clean the lens. Other types of paper, e.g. paper towel, tissues, toilet paper scratch the lens.

7 Experiment – Using a Light Microscope

1. Magnification refers to the ratio between the size of the image (the apparent size of the object under the microscope) and the actual size of the object. Resolution is the ability to distinguish between two points as separate.
2. (a) The diameter of the field of view under low power is 1.4 mm which is 1400 μm.
   (b) If the high objective lens (40×) is used then the change is from an overall magnification of 100× to 400×.
   $$\text{New diameter field of view} = 1400 \times \frac{100}{400}$$
   $$= 350\ \mu\text{m}$$
3. The letter 'e' as seen under the microscope is upside down and back to front. This horizontal and vertical inversion of the image means that when you are viewing a slide and wish to see another part of the slide a movement of the slide to the left moves the image to the right and if you move the slide away from your body the image moves towards you. This makes it important to make sure a very small object is in the centre of the field of view before you swap from low power to high power.
4. To determine the position of different organelles in a large unicellular being viewed under a light microscope you would focus up and down using the fine adjustment knob and carefully observe the order of appearance of each organelle in the depth of field.
5. The width of the root hair is 0.3 mm or 300 μm.
6. To record your observations when using a light microscope you need to draw a circle to represent the field of view. The observed structures should be in the position and proportionally the same size as you see them. The diagram requires a statement of the magnification and a title for the slide. Any identifiable features should be labelled.
7. If you have raised the body tube so that the objective lens is more than 2 cm above the slide, you need to start focusing again by looking at the stage and lowering the objective to just above the slide. You must not lower the objective lens while looking down the eyepiece.

8 Cells, Tissues, Organs and Systems

1. (a) A cell is the basic unit of life.
   (b) Tissue is a group of cells with similar structure and function.
   (c) Organ is a group of tissues grouped together to make a structure with a special function.
   (d) System is usually a group of organs whose function is closely related.
2. (a) Organs include heart, stomach, kidney, lung, ovary, skin, liver.
   (b) There are four main types of tissues – epithelial tissue, connective tissue, muscle tissue and nervous tissue.
   (c) Epidermal cell in the epidermis, red blood cell, white blood cell.
3. There are three types of muscle tissue – smooth muscle, skeletal muscle and cardiac muscle. These tissues are in different systems, e.g. cardiac muscle is in the cardiovascular system, skeletal muscle is in the muscular system and smooth muscle is in a variety of systems, e.g. digestive system along the alimentary canal and cardiovascular system in the walls of arteries. Although these muscles in these systems have their own specific functions, they all have the general purpose being used to generate force that leads to motion of some kind.
4. C
5. B
6. D

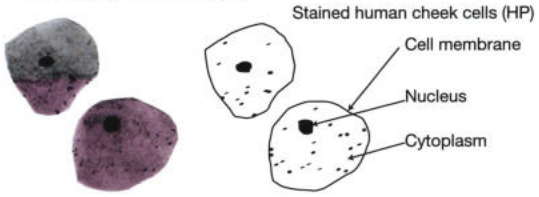
9 Differentiation and Specialised Cells

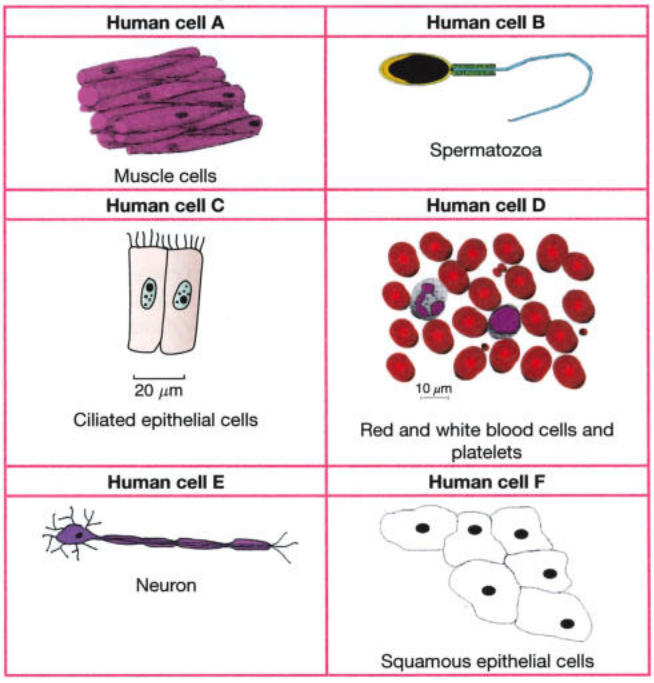
1. Differentiation is a process when a young, relatively unspecialised cell develops into a more mature state to become a more specialised cell.
2. In a multicellular organism each cell must be supplied with oxygen and nutrients and wastes need to be removed. The process of diffusion is too slow and inefficient to meet the requirements of each cell. Thus a multicellular organism needs specialised cells with division of labour so particular tasks can be carried out more efficiently and each cell is supplied with oxygen and nutrients and wastes removed.
3. Differential gene expression means that a particular gene is 'switched on' to produce a particular polypeptide. The production of this polypeptide and protein leads to the cell taking a particular shape and having a particular function.
4. Muscle cells produce the proteins myosin and actin. These proteins form filaments that interact to cause cell contraction. The contraction is needed to provide locomotion for the multicellular organism.
5. A tissue may have only one kind of cell or several kinds of cells. For example, connective tissue contains red blood cells, white blood cells, lymph, cartilage and bone.
6. The main types of mammalian tissue are: 1. Epithelial tissue. 2. Connective tissue. 3. Muscle tissue. 4. Nervous tissue.
7. Epithelial tissue is found over many surfaces and linings of body cavities, e.g. it is the outer part of skin and forms the linings of the digestive tract, the lungs, the blood vessels and various ducts.
8. Epithelial cells are either flat (squamous), cubical or columnar in shape.
9. Microvilli increase the surface area for diffusion and active transport into and out of the cell.
10. The basement membrane is the floor of an epithelial membrane on which the basal cells rest. The basement membrane is usually made of a network of white, wavy, non-elastic fibres that form a fibrous network.
11. An endocrine gland is a ductless gland that secretes hormones directly into interstitial fluid. An exocrine gland has ducts.
12. Connective tissue is found in many areas of the body, e.g. bone, cartilage, blood, tendons and ligaments.
13. Muscle cells are specialised to convert chemical energy in ATP into mechanical energy for movement.
14. Smooth muscle is involved in most involuntary movement of internal organs. Striated muscle is involved in voluntary movement. Cardiac muscle is responsible for the beating of the heart.
15. Nervous tissue consists of neurons and glial cells.
16. There are several types of neurons (nerve cells), e.g. sensory neurons take messages from a sensory receptor to the CNS, motor neurons take messages from the CNS to an effector, e.g. muscle or gland and bipolar cells take messages from the retina.
17. A
18. C
19. B

10 The Light Microscope and Cell Organelles

1. Staining techniques cause different organic compounds to react with coloured dyes to identify specific chemicals and also make certain structures stand out.
2. The cell membrane is very thin and this makes it hard to identify under a light microscope.
3. The cell membrane controls the movement of substances into and out of the cell.
4. The cytoplasm is not a single substance. It includes all material between the cell membrane and the nuclear membrane, including the cell membrane, thus it contains many different particles.
5. The diagram shows mitochondria in the human cheek cell with the double membrane. School light microscopes do not have the resolution or magnification to show detail of mitochondria.
6. Active cells require a lot of energy and thus need many mitochondria to supply the energy through respiration.
7. The nucleus will stain yellow with iodine solution and will stain blue with methylene blue.
8. The photomicrograph would not have been taken using a light microscope as it shows internal structures of mitochondria and a Golgi body. It would have been taken using a transmission electron microscope.
9. The invention of the electron microscope showed the internal membrane structure of a mitochondrion and thus allowed an increase in our knowledge of the mitochondria.
10. Camillo Golgi first identified Golgi bodies.
11. Cells in the salivary glands are secreting salivary enzymes and hence require many Golgi bodies to produce and then discharge these enzymes.
12. Exocytosis is the secretion of macromolecules from a cell by the fusion of vesicles with the plasma membrane.
13. Cytoplasmic streaming improves the movement of substances throughout the cell and allows a constant mixing of particles in the cytoplasm.
14. DNA is found in the nucleus and mitochondria.
15. The field of view under a high power objective is smaller than the field of view under a low power objective. When swapping from low to high power there is often not enough light under high power and the image appears dark. The iris diaphragm needs to be adjusted to let more light onto the image.
16. D
17. B
18. A

11 Experiment – Human Cells

1. (a) Mitochondria – site of cellular respiration releasing energy in the form of ATP for use in the cell.
   (b) Centrosome – involved in the formation of the spindle during cell division.
   (c) Cell membrane – controls the substances entering and leaving the cell.
   (d) Nucleus – controls all cell activities and contains the hereditary information.
2. 

3. Animal cells do not have the regular shape of plant cells and are not enclosed in an extracellular cellulose cell wall. Animal cells do not have a central large vacuole and do not have plastids such as chloroplasts. The centrosome of animals contains a pair of centrioles which are not found in plant cells. Both plant and animals cells have a cell membrane, nucleus, cytoplasm, mitochondria and Golgi bodies.
4. 

5. D

12 The Electron Microscope and Cell Organelles

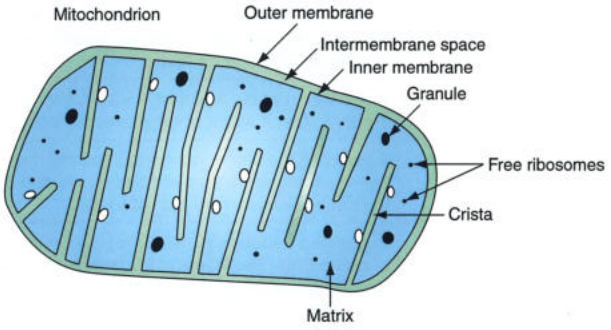
1. Lysosomes contain enzymes which break down large molecules such as fats, proteins and polysaccharides and recycle the unwanted material inside the cell.
2. The dark spots on endoplasmic reticulum are ribosomes which are involved in protein synthesis.
3. X = nuclear membrane, Y = nucleolus, Z = nucleus
4. Lysosomes contain enzymes that break down large molecules such as fats, proteins and polysaccharides. Lysosomes ingest and destroy foreign or unwanted materials in the cell. This is needed to remove unwanted materials and to recycle substances.
5.

Light microscope	Electron microscope
Advantage: Can view living material	Disadvantage: Cannot view living material
Advantage: Can use coloured stains to show particular compounds and for colour contrast	Disadvantage: Stains are not coloured
Disadvantage: Poorer magnification	Advantage: Higher magnification
Disadvantage: Poorer resolution	Advantage: Higher resolution

6. Endoplasmic reticulum consists of many membranes which are continuous with the cell membrane. This provides a large surface area for reactions to occur on and also provides a large network to speed up the transport of materials. Thus structure assists function.
7. C
8. D

13 Mitochondria

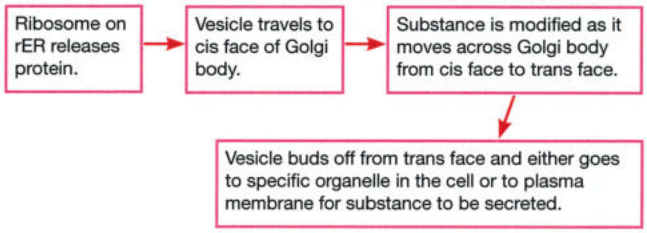
1.



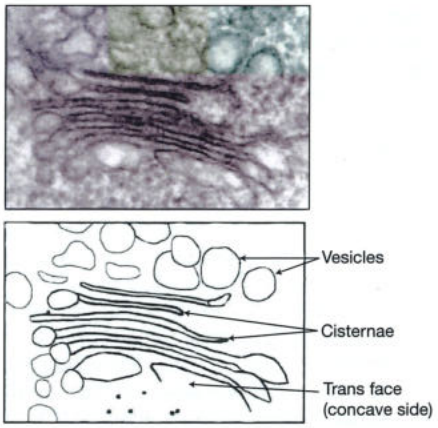
2. Mitochondria are oval shaped with an outer regular membrane that protects the organelle. There is an inner membrane which is highly folded to form cristae. The cristae increase the surface area of the inner membrane and extend into the matrix. The space between the two membranes is called the intermembrane space. The matrix contains a watery fluid, a variety of molecules, enzymes, DNA and ribosomes.
3. Mitochondria are the site of cellular respiration. This process releases energy in the form of ATP for use in cellular processes. Highly active cells require a steady supply of ATP. The structure of a mitochondrion is suited to its purpose as the inner membrane is highly folded to increase its surface area and provide maximum space for the necessary reactions of cellular respiration.
4. The growth processes occurring within a cell include mitochondria dividing into two. The cell requires mitochondria to release energy for use in the cell in other cellular activities. If the cell undergoes cellular division, each new daughter cell requires sufficient mitochondria to support itself and supply its energy needs.

14 Golgi Bodies

1. A Golgi body is a stack of flattened membrane-bound sacs (cisternae). The cisternae form an extensive network in the cell with each sac surrounding an internal space that is separated from the cytosol. Most Golgi bodies consist of 5 or 6 cisternae. The membranes on the opposite sides of a Golgi body are quite different in thickness and composition, i.e. cis face and trans face.
2. Golgi bodies are involved in intracellular transport, processing, concentration and packaging of chemicals for either use in the cell or dispatch to places outside the cell.
3. The cis and trans face of Golgi bodies differ in thickness and composition and have different jobs to perform. The cis face is usually found near the rough endoplasmic reticulum (rER) and is the receiving face for vesicles from rER. These vesicles fuse with the cis face and release their contents into the lumen inside the sac. As the substance moves from the cis face to the trans face across the Golgi body it is modified into the required molecules. Vesicles are pinched off from the trans face and transport the Golgi products to specific organelles within the cell or to the plasma membrane where they are secreted from the cell. Thus the two opposite sides of the Golgi body perform different functions and are needed for the overall functioning of the organelle.
4.



5. There are two types of cytoplasmic vesicles: 1. Secretory vesicles which contain proteins to be exported. 2. Storage vesicles which do not leave the cytoplasm, e.g. lysosomes are the major storage vesicles of the cell.
6. Endocrine cells contain large numbers of secretory Golgi bodies. Endocrine cells secrete hormones and thus they need this large number of Golgi bodies to produce sufficient quantities of hormones. The hormones are stored in storage vesicles until the cell receives a signal for secretion at which time the secretory vesicle releases its contents by exocytosis into the extracellular space.
7.



15 Cell Organelles Summary

1. Cytosol is the semifluid part of the cytoplasm.
2. The membrane system of the cell includes the cell membrane, the nuclear membrane, the endoplasmic reticulum, ribosomes, the Golgi bodies, lysosomes and vacuoles.
3. The mitochondria are involved in either collecting or releasing energy.
4. Ribosomes carry out protein synthesis and thus a cell that has a high rate of protein synthesis will need large numbers of ribosomes to carry out the job.
5. There is a need to connect the nuclear membrane to the membrane system in the cytosol to help the transport of materials throughout the cell. It aids the communication between the nucleus and the cytoplasm.
6. ATP (adenosine triphosphate) is the form of energy produced by cellular respiration.
7. Vacuoles have a variety of functions depending on the type of cell in which they are present, e.g. vacuoles can be involved in storage, e.g. food vacuoles, digestion or water elimination.
8. D
9. C
10. A

16 Chemicals in Cells

1. Organic compounds contain carbon and were originally made by living things, e.g. carbohydrates, lipids, proteins and nucleic acids, while inorganic compounds are compounds that do not contain carbon as the main element, except for carbonates, e.g. water, mineral salts, and trace elements such as calcium and manganese.
2. Four main groups are carbohydrates, proteins, lipids, nucleic acids.
3. Inorganic molecules include salts such as sodium chloride, copper sulfate and calcium carbonate and water.
4. (a) Glucose.
   (b) Amino acid.
   (c) Fatty acid and glycerol.
   (d) Nucleotide
5. Amino acids are named as they all contain an amino group ($-NH_2$) and an acid group (–COOH).
6. C
7. D
8. D
9. D
10. A
11. B
12. C
13. B
14. A
15. C

17 Experiment – Substances in Tissues

1. Standard precautions include wearing covered-in shoes, hair tied back, and wearing safety goggles. All accidents or incidents must reported to the teacher and a report form must be completed.
2. To perform a first-hand investigation to identify the presence of lipids in a range of foodstuffs you need to collect a range of different foodstuffs, e.g. starch solution, potato, carrot, bread, apple, a knife, a dropper bottle of iodine solution, and a watch glass. Put a few drops of starch solution on the watch glass and add five drops of iodine solution to confirm the distinguishing test for starch and to act as the control. Record the colour change. Cut up each of the foodstuffs into smaller pieces and in turn, place on the watch glass and add five drops of iodine solution to each. Record the colour change for each and compare with the distinguishing test (control). Analyse the results and note which of the foodstuffs contained starch. Repeat the experiment.
3. (a) He needed to wear rubber gloves as hands have their own lipids and he needed to be sure that the oil came from the biscuit and not his own hands.
   (b) For gathered data to be reliable, the experiment needs to be repeated and the same results occur. In this experiment, the student or others need to follow the same procedure and if the same results are achieved, then the data is reliable.
4. (a) Biuret contains copper sulfate solution and sodium hydroxide solution.
   (b) When using Biuret solution you need to wear safety glasses and gloves. This is because it is toxic if ingested and it is corrosive due to sodium hydroxide.
   (c) Foodstuffs showing positive to Biuret solution include any red meat, white meat, egg white, and fish.
5. To perform a first-hand investigation to identify the presence of nucleic acids in skin cells you need to place some clear adhesive tape on a section of skin, e.g. arm or back of hand. Rub the tape to collect the skin cells, then peel off the skin and stick to a microscope slide. Peel the corner of the tape up and place a drop of toluidine blue under the tape. Observe under a light microscope and record the results, noting areas that are dark blue. Analyse the results and identify the areas with nucleic acids. Repeat the experiment.
6. Proteins are highly important: 1. They act as structural components in cells and tissues. 2. They include the enzymes which control the body's metabolism. 3. They act as chemical messengers, e.g. hormones. 4. They form immunoglobulins (antibodies) which combat infection. 5. They form contractile elements for movement. 6. They act as carrier molecules in transporting substances.
7. (a) Starch is a storage form of energy. Starch is a polysaccharide made up of glucose monomers and can be easily broken down to provide glucose for respiration or dietary needs.
   (b) Plants store starch as starch granules in plastids, e.g. starch granules can be found in chloroplasts. Some plants, e.g. potato tubers and grains hold extremely large stores of starch.
   (c) Iodine solution is used for the distinguishing test for starch. The yellow/brown solution turns blue/black with starch.
8. D
9. B

18 Development of the Model of the Cell Membrane

1. Nineteenth century scientists knew that there had to be a barrier around cells that separated the internal contents of the cell from the surrounding environment. Light microscopes do not clearly show the structure of the cell membrane. And thus it was not until the 20th century and the invention of the electron microscope that the structure of the cell membrane became more visible.
2. When 19th century scientists placed plant cells in strong solutions, e.g. salt solutions they noticed that the cytoplasm pulled away from the cell wall. This showed that there was some kind of membrane around the cell contents.
3.

Scientist	Contribution
Traube	Realised cell membranes were molecular sieves and produced artificial semipermeable membranes. He suggested the barrier was formed by an interfacial reaction of the cell protoplasm with the extracellular fluid.
Quincke	Suggested the existence of a thin lipid layer at the boundary of the protoplasm and calculated that the fluid layer of fat was less than 100 nm thick.
Overton	Found that the ability of a substance to pass through the membrane was related to its chemical nature. Non-polar, fat-soluble substances passed most easily through the layer.
Fricke	Calculated that the membrane was 3.3 nm thick and assumed that it was a single molecular layer.
Gorter and Grendel	Suggested the bilayer concept of the cell membrane by measuring lipids from red blood cells and using the total cell area of the red blood cell.
Danielli	Suggested that the cell membrane had outer layers of protein with two layers of lipids between them. He calculated the total width of the membrane to be 8.0 nm.
Robertson	Took electron micrographs of cell membranes and showed the total thickness was 7.5 nm. Suggested the unit membrane.
Mueller and Rudin	Suggested the BLM and determined different properties of the cell membrane.
Frye and Edidin	Showed the fluidity of cell membrane using mouse and human cell hybrids
Singer and Nicolson	Proposed the fluid mosaic model of the cell membrane with lipid bilayer and peripheral and integral proteins.

4. In the Singer and Nicolson model, the peripheral proteins can be dissociated from the membrane when it is treated with polar reagents such as extreme pH or high salt. Integral proteins are in the membrane and can be dissociated with reagents that are detergents which displace the membrane lipids.
5. The glycocalyx is a carbohydrate layer on the outer surface of the cell which protects the cell surface.
6. B
7. D

19 The Current Plasma Membrane Model

1. Semipermeable means that some molecules can easily pass through while other molecules cannot.
2. Jonathon Singer and Garth Nicolson proposed the fluid mosaic model to represent the structure of the cell membrane.

3. The cell membrane structure consists of phospholipids which form a bilayer with the hydrophobic end of the lipid facing the middle of the membrane and the hydrophilic end facing towards the watery contents of the cell. Embedded in the phospholipid bilayer there are large protein molecules which can extend from one side to the other or extend only partway across the bilipid layer or the protein can be attached to the surface and part of the integral protein. The proteins have different compositions depending on the function of that cell.
4. Carbohydrates on the surface of the cell membrane are important in cell-cell recognition. This is part of the immune response and failure of recognition will stimulate cell destruction.
5. Most of the lipids and some of the proteins can 'drift' and move their position and hence their location is 'fluid' and hence the name 'fluid' mosaic model.
6. 'Kinks' in the unsaturated hydrocarbon tails of phospholipids are important as they reduce the tight packing of the molecules and increase membrane fluidity.
7. At low temperatures lipids become solid while at high temperatures they become liquid.
8. Lipids with high amounts of unsaturated hydrocarbons have a high capacity to stay fluid at low temperatures.
9. Integral membrane proteins (IMPs) are permanently attached to the membrane. Some extend across the whole membrane while others only extend partway across the cell membrane.
10.

Transport proteins	Membrane receptor proteins	Membrane enzymes
Assist the movement of ions, small molecules or macromolecules in or out of the cell.	Communicate with the environment, e.g. receive a signal from a hormone, neurotransmitter which triggers the cell to carry out specific internal response.	Modify molecules needed near the cell surface.

11. Molecules not soluble in lipids, e.g. water, diffuse across the cell membrane through small transitory openings made by the movement of the fluid lipids. Water moves through pores called aquaporins.
12. Large water-soluble molecules such as amino acids combine with transport integral proteins which are a particular 'carrier' molecule to cross the membrane. As the carrier molecule moves across the membrane, it transports the molecule with it.
13. If energy is used to move a chemical against a concentration gradient, it becomes active transport across the membrane.
14. Some substances cannot cross the cell membrane as they are not lipid-soluble and there is no suitable carrier for them to take them across the membrane.
15. A
16. D

20 Passive Transport

1. Diffusion is the movement of particles from an area of high concentration of particles to an area of low concentration of particles.
2. Osmosis is the movement of water, across a semipermeable membrane, from an area of high water to an area of low water.
3. The higher the concentration gradient, the faster the rate of diffusion.
4. Active transport moves particles against the concentration gradient, causing particles to collect in one area.
5. The terms are used to show the difference in concentrations when comparing two solutions. Hypertonic is used to describe the more concentrated solution and hypotonic describes the less concentrated solution. If the two solutions have the same concentration they are called isotonic.
6.

Diffusion	Osmosis
Similar – involves movement of particles from area of high concentration to low concentration	
Similar – substance moves down a concentration gradient	
Different – diffusion refers to movement of any substance	Different – in biology osmosis only refers to movement of water
Different – diffusion does not necessarily involve semipermeable membrane	Different – movement is across a semipermeable membrane

7. If a human red blood cell was placed in distilled water, water would get into t he cell by osmosis trying to dilute the cell to equalise the osmotic pressure. Since the cell membrane can only withstand a certain amount of internal pressure, the cell membrane will eventually burst due to swelling and the cell will be destroyed.
8. C
9. B

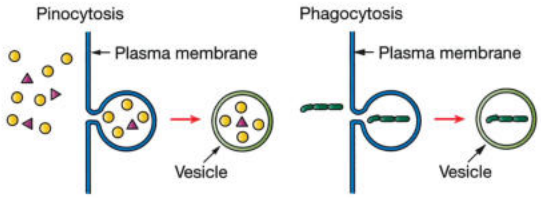
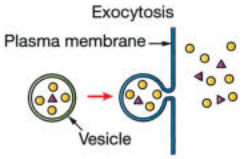
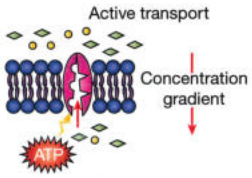
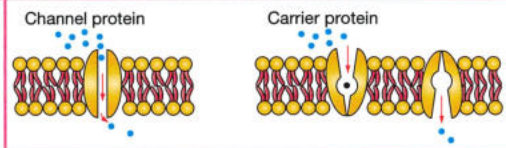
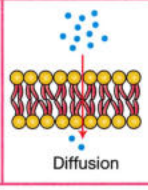
21 Diffusion and Osmosis

1. Diffusion is the movement of particles from an area of high concentration of particles to an area of low concentration of particles.
2. Diffusion occurs due to the random kinetic movement of particles.
3. A concentration gradient refers to the difference in the density of particles in one area to the density of particles in another area.
4. In diffusion particles move down a concentration gradient.
5. The larger the particle the slower the rate of diffusion.
6. The higher the temperature the faster the rate of diffusion.
7. The further the particle has to travel the slower the rate of diffusion.
8. Permeability is the ability of a particle to pass through.
9. Solubility refers to the amount of a substance that can be dissolved in a given amount of solvent.
10. If two molecules have equal size the molecule that is lipid soluble will more easily pass through a cell membrane.
11. For two lipid soluble molecules the smaller molecule will have the fastest rate of diffusion.
12. Oxygen enters the blood from the lungs by diffusion. Glucose enters the blood from the small intestine by diffusion.
13. Osmoregulation is the active regulation of the osmotic pressure of body fluids so they do not become too dilute or concentrated in particular areas.
14. The water potential is the physical property that predicts the direction in which water will flow and is determined by the solute concentration and the applied pressure.
15. Osmotic pressure is the pressure needed to stop the flow of water across a semipermeable membrane.
16. In an aqueous solution of glucose, the solvent is the water and the solute is the glucose.
17. To carry out an experiment using dialysis tubing to show that some substances can cross a membrane while other substances cannot cross the membrane, an aqueous sucrose solution is used as the water can pass through the dialysis tubing while the large sucrose molecule cannot pass through the tubing. This will show osmosis. If glucose solution is used both the glucose and the water will cross the membrane until equilibrium is reached with equal glucose concentrations on either side of the dialysis tubing.
18. You would expect osmosis to occur and the water molecules will move into the cell. Since it is a human cell with only a thin cell membrane for support, the cell will swell and will most likely burst due to the amount of water entering the cell.
19. By changing the permeability of the collecting duct, the kidney can produce either concentrated or dilute urine by osmosis.
20. The high hydrostatic pressure at the arterial end will cause the forward movement of blood and force some water and very small solutes out of the capillaries and into the tissue spaces. The outward pressure is the difference between the hydrostatic pressure and the osmotic pressures.
21. B
22. D

22 Active Transport

1. Active transport is the movement of substances across a membrane against a concentration gradient. It requires an energy input which is usually in the form of ATP.
2. An electrogenic pump is a transporter protein that generates voltage across a membrane, e.g. the sodium/potassium pump in neurons.
3. Cytosis is the passage of large 'packages' of material across a membrane. The materials are large and cannot move across by diffusion or with the assistance of carrier or channel proteins.
4. Endocytosis involves the taking in of materials from the environment by infolding of the cell membrane to form vesicles while exocytosis is the reverse process where vesicles fuse with the cell membrane and substances are secreted out of the cell.
5. If the substance to be taken into the cell is liquid, it is called pinocytosis while if the substance is solid it is called phagocytosis.

6.

Process	Diagram	Definition
Endocytosis		Cell takes up substances by surrounding it with membrane The membrane changes shape, infolds and encloses substance to be taken into the cell, e.g. substance is highly polar or too large to enter by other means, e.g. large peptides, immunoglobulins. It is active transport.
Exocytosis		Substances are released from the cell when vesicles fuse with the membrane. The membrane fuses with the vesicle, changes shape and expels the substance, e.g. hormone secretion.
Active transport		Requires energy, e.g. ATP. Substances move against a concentration gradient. Ion uptake, e.g. sodium, calcium, magnesium and iron are taken up by active transport. The sodium/potassium pump is highly important in maintaining the electrochemical gradient across the epithelial cell boundary of the lumen.
Facilitated diffusion		Is passive transport with substances moving down a concentration gradient with the assistance of membrane proteins, e.g. glucose uses a glucose transporter. Amino acid structure is so variable there are multiple amino acid transport systems – many use facilitated diffusion.
Simple diffusion		Substances move across the membrane by passive transport from an area of high concentration to an area of low concentration, e.g. lipids are absorbed into epithelial cells of small intestine cell by simple diffusion, fatty acids and monoglycerides move out of micelles and into the epithelial cell. Water uptake by osmosis often follows the osmotic gradient of sodium, potassium.

7. (a) The process is phagocytosis – a form of endocytosis.
   (b) In this process the foreign body is surrounded by the cell membrane and engulfed in a vesicle. The vesicle is taken into the cell when it separates for the cell membrane.
   (c) Organelle X is vesicle released by the Golgi body and is most likely a lysosome. Lysosomes contain digestive enzymes that break down foreign bodies in the process of phagocytosis.
8. A sodium-potassium pump has a concentration gradient with more sodium outside the cell than inside the cell. The cell uses energy to pump sodium out of the cell. The unequal balance of sodium ions and potassium ions and the electrochemical gradient which forms a potential difference across the membrane is used by many cells, e.g. neurons and is used to drive other active transport mechanisms.
9. D

23 Experiment – Membranes, Diffusion and Osmosis

1. Semipermeable membranes allow some substances to pass through more readily than other substances.
2. Dialysis tubing was acting as a semipermeable membrane.
3. When the equipment was set up, the iodine was in the water outside the bag in gas jar 3 and made the water a yellow colour. The next day, the starch inside the bag in gas jar 3 had turned blue-black; this showed that the iodine molecules had moved across the dialysis tubing and had turned the starch blue-black.
4. Gas jar1 had water inside the bag and sucrose solution outside, while gas jar 2 had sucrose inside the bag and water outside. After 24 hours the bag in gas jar 1 was more empty while the bag in gas jar 2 had become more full. This means that the sucrose had not crossed the dialysis tubing by diffusion. However, osmosis had occurred with water moving into bag 2 and out of bag 1, trying in both cases to equalise the osmotic pressure by diluting the sucrose solution.
5. Gas jar 4 had glucose solution inside the bag and water outside the bag with a piece of yellow Tes-tape in the water. During the lesson the Tes-tape began to change colour to green, showing that glucose was now in the water. This means that glucose diffused out of the bag, through the dialysis tubing into the water. After 24 hours the Tes-tape had reverted to yellow. Since both glucose and water could cross in and out of the bag, the inside and outside of the bag would have reached an equilibrium, and were isotonic. Since it is a very weak glucose solution it did not register a green colour on the Tes-tape.
6. Water, glucose and iodine were able to pass through the dialysis tubing.
7. Starch and sucrose could not pass through the dialysis tubing.
8. If the molecule is small in size, then it will be able to pass through the dialysis tubing.
9. Dialysis tubing was a suitable substitute for the cell membrane as the results showed it was semipermeable, allowing some substances, such as water and glucose, to easily move across, but it did not allow other substances, such as starch, to move across. This reflects the function of the cell membrane. This means that dialysis tubing provided a suitable model to show the nature of a selectively permeable membrane.
10. Diffusion is the movement of particles from an area of high concentration of particles to an area of low concentration of particles. Osmosis is the movement of water, across a semipermeable membrane, from an area of high water concentration to an area of low water concentration. In this experiment, iodine diffused across the tubing to combine with the starch solution – it moved from an area of high iodine (water was yellow) to an area of low iodine. The movement of water showed osmosis when water moved across the tubing, trying to equalise the osmotic pressure on either side of the tubing.
11. A

24 Experiment – Surface Area to Volume Ratio and Diffusion

1. The students worked out that they would have to measure the surface area of the cell as this determines the amount of material that can be moved in and out of the cell.
2. The students worked out that they would have to work out the volume of the cell as this determines the amount of material the cell would need to survive.
3. When the students began their experiment, the 'cells' were a pink jelly as the jelly contained the alkali sodium hydroxide and had also been coloured with the indicator phenolphthalein. The phenolphthalein had turned pink in alkali.
4. Wear rubber gloves to protect skin from jelly with sodium hydroxide and hydrochloric acid as these chemicals are corrosive and can damage skin. Wear safety goggles to protect eyes from any splash while using hydrochloric acid which is corrosive and can damage eyesight if it enters the eye.
5. The students chose different sized cubes as they needed to measure surface area and volume and the calculations for a cube are easy to calculate and it is relatively easy to cut cube shapes out of jelly.
6. Each cube showed that the acid had diffused 2 mm into each side. This means that the rate of diffusion was constant for each cube and over a set time, the acid can penetrate to a certain depth.
7.

Cube	Surface area	Volume	Ratio SA : vol	Reduced ratio
Small	$0.6 \times 0.6 \times 6 = 2.16$	$(0.6)^3 = 0.216$	2.16 : 0.216	10 : 1
Medium	$1.6 \times 1.6 \times 6 = 15.36$	$(1.6)^3 = 4.096$	15.36 : 4.096	3.8 : 1
Large	$2.6 \times 2.6 \times 6 = 40.56$	$(2.6)^3 = 17.576$	40.56 : 17.576	2.3 : 1

8.

Cube	Percentage diffusion into cube
Small	$\frac{1 - 0.216}{1} \times \frac{100}{1} = 78.4\%$
Medium	$\frac{8 - 4.096}{8} \times \frac{100}{1} = 48.8\%$
Large	$\frac{27 - 17.576}{27} \times \frac{100}{1} = 38\%$

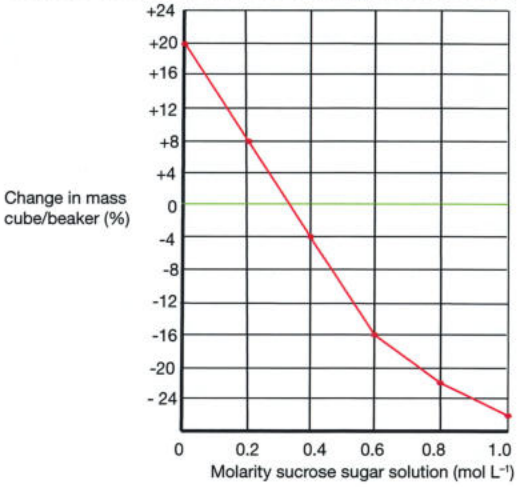
Since the smallest cube had 78.4% diffusion into the cube, compared to 48.8% and 38%, the students were quite correct in deciding that the smallest cube had the greatest percentage of diffusion into the cube.

9. The students decided that cell size would influence its ability to survive, with a smaller cell having a greater chance of survival than a larger cell as cells need required substances to diffuse into the cell and wastes to diffuse out of the cell. The smaller cell size has the greater surface area to volume ratio and hence within a specified time, diffusion will be able to supply a greater percentage of the interior of the cell with nutrients. This increases the survival ability of the cell. A larger cell will find wastes accumulating inside the cell and the core of the cell will not be efficiently receiving nutrients to survive.
10. The smallest cube would lose and gain heat more quickly. This is because the smallest cube has the largest surface area to volume ratio and heat will have less distance to travel to go from the centre to the outside, or from the outside to the centre.
11. When cells reach a certain size, diffusion can no longer efficiently supply and remove substances from the cell. The cell then divides, splitting into two so that each daughter cell now has a greater surface area to volume ratio compared to the large parent cell. Each daughter cell will now be able to more efficiently supply and remove nutrients by diffusion to all its interior compared to the parent cell.
12. D
13. C
14. C
15. B
16. A
17. C

25 Experiment – Material Exchange and Concentration Gradient

1. Equilibrium is defined as a state where all opposing forces or influences are balanced. For materials it is the state when particles are evenly spread throughout the given space.
2. The higher the concentration gradient the higher the rate of diffusion.
3. (a) A suitable hypothesis is – 'If the concentration of the sugar solution increases, the per cent change in mass increases'.

(b)



(c) The molarity of the sugar solution where the graph crosses the *x*-axis equals the molarity of the sugar solution in the potato tissue.

(d) A 0 mol L^{-1} sugar solution is pure water, e.g. distilled water). Osmosis occurs when a cube of potato is placed in distilled water with water moving into the potato from the surrounding medium. This means the potato will increase in mass (+20% change in mass cube/beaker).

(e) A potato cube in 1.0 mol L^{-1} sugar solution would lose water to the surrounding solution due to osmosis. This would be loss in mass for the potato cube of 26%.

(f) There is a direct relationship between the percentage change in mass of the potato and the molarity of the sugar solution. The higher the molarity of the sugar solution, the greater the loss in mass.

4. Diffusion is the movement of particles from an area of high concentration of particles to an area of low concentration of particles.
5. Temperature affects the rate of diffusion as it influences the speed of the particles. Particles diffuse faster at higher temperatures.
6. B

26 Material Exchange and the Nature of the Material

1. The physical properties of a substance do not change the chemical nature of the substance while the chemical properties of a substance do change the chemical nature of the substance.
2.

Physical properties	Chemical properties
Boiling point. Melting point. Density. Permeability. Size of particles. Molecular weight. Molecular surface area. Molecular volume.	Heat of combustion. Reactivity with water. pH. Flammability.

3. (a) Polar molecules have a separation of electric charge with appositively charged sections. Non-polar molecules do not have a separation of electric charge with appositively charged sections.
   (b) Water is a polar molecule and carbon dioxide is a non-polar molecule.
4. Integral membrane proteins are proteins that are permanently attached to the membrane.
5. (a) Glucose is a polar molecule as it has positive and negative sections.
   (b) Glucose easily dissolves in water as it has many OH groups that can interact with water.
   (c) Glucose crosses the cell membrane using transport channels in facilitated diffusion.
6. D

27 Experiment – Surface Area and Rate of Reaction

1. Label two 250 mL beakers: ground soluble aspirin and whole soluble aspirin. Pour 100 mL water into each 250 mL beaker. Put on safety goggles and safety gloves. Grind one soluble aspirin tablet with mortar and pestle to make a powder. Add a whole tablespoon to one beaker and the ground powder to the other beaker. Use a stopwatch to accurately time how long each takes to dissolve. Record results in a table and analyse results.
2. The mortar and pestle is needed to grind the soluble aspirin into a powder to increase the surface area. The powder has a much larger surface area than the whole tablet.
3. One safety precaution is wearing safety goggles to protect the eyes in case some of the aspirin which is being ground up by the mortar and pestle is flicked into the eyes; could cause potential damage.
4.

Variable kept constant	Why it was important to keep this variable constant
Surrounding temperature	It is important to keep the temperature of the surroundings constant for both beakers as higher temperatures cause a higher rate of reaction.
Amount of aspirin used	It is important for both to have the same amount of reactant so that the time of reaction does not depend on quantity.

5. The ground tablet dissolved faster than the whole tablet. This shows that an increased surface area allows a faster rate of reaction.
6. Increasing surface area increases the rate of reaction.
7. For an experiment to be valid, the method must allow the aim to be achieved. In this experiment, the aim was to show the relationship between surface area and rate of reaction. By keeping the amount of water and amount of aspirin constant and only varying the surface area, this aim was achieved.
8. When grinding up the tablet and then putting the ground powder into the beaker of water it is very hard to make sure all powder is removed from the mortar. This leads to inaccuracy as there is less quantity of the ground powder and thus will have a shorter time.
9. Use a brush to remove the powder onto a piece of paper to collect all particles before adding to the beaker of water. This will ensure the quantity of aspirin is the same for both sets of equipment.
10. A large surface area is needed to increase the rate of diffusion or rate of reaction. This increased rate is needed to supply the body with needed energy or food or remove wastes to ensure efficient metabolism.
11. For gathered data to be reliable, the experiment needs to be repeated and the same results occur. In this experiment, the test was not repeated, however, if others follow the same procedure, they will achieve the same results and thus the data is reliable.
12. To work effectively as a group the students need to allocate specific tasks to each person and then each person needs to carry out their job and follow the procedure step by step.
13. C
14. A
15. B

28 Human Organ Systems

1. Organ systems are groups of interacting organs.
2.

Organ system	Main components
Skeletal system	Bones, cartilage, ligaments, bone marrow.
Muscular system	Skeletal muscles, tendons.
Digestive system	Mouth, salivary glands, oesophagus, stomach, small intestine, liver, gall bladder, pancreas, large intestine, rectum, anus.
Cardiovascular system	Heart, blood vessels (arteries, veins and capillaries), blood.
Respiratory system	Nasal cavity, sinuses, pharynx, larynx, trachea, bronchi, lungs with bronchioles and alveoli, diaphragm.
Nervous system	Central nervous system of the brain and spinal cord. Peripheral nervous system and autonomic nervous system consisting of neurons.
Urinary system	Kidneys, ureter, urinary bladder, urethra.
Lymphatic system	Lymphatic vessels, lymph nodes, spleen; thymus.
Endocrine system	Variety of glands – pituitary, thyroid, parathyroid, pineal, thymus, adrenal, pancreas, testes, ovaries.
Integumentary system	Epidermis, dermis, cutaneous membrane, hair follicles, hairs, sebaceous glands, sweat glands, nails, sensory receptors, subcutaneous layer.
Reproductive systems	Male – testes, epididymis, vas deferens, prostate gland, penis, scrotum. Female – ovaries, fallopian tubes, uterus, vagina, clitoris, labia, mammary glands.

3. D For locomotion to occur the muscular system needs to cooperate with the skeletal system to move limb or body part containing the bones. The nervous system is needed to coordinate the muscles, e.g. when antagonistic muscles such as the bicep and tricep move a limb, one muscle needs to contract while the other relaxes and this is coordinated by the nervous system. This example shows how all systems interact and need to cooperate with each other for efficient body functioning.

29 Organ and Tissue Transplantation

1. A tissue transplantation is the moving of a tissue from one body to another or from a donor site to another location on the patient's own body.
2. An organ transplantation is the moving of an organ from one body to another.
3. An autograft occurs when an organ or tissue from a patient's body is moved to another location on their body.
4.

Tissues transplanted	Organs transplanted
Bones	Hearts
Tendons	Livers
Cornea	Kidneys
Skin	Lungs
Heart valves	Pancreas
Nerves	Intestines
Veins	Thymus

5. An autograft occurs when a tissue or organ is transplanted from one part of a person's body to another area of their body. An allograft is when a tissue or organ is transplanted from one person to another person.
6. A person is put on the waiting list for a transplantable organ when they have end-stage organ failure.
7. In Australia there are waiting lists for transplantable organs for heart, lungs, kidneys, liver, intestine and pancreas.
8. The source of donor transplants can be living or dead.

9. MHC molecules are attached to the cell membrane and act as antigens. White blood cells, e.g. T-cells identify the MHC molecules as self or non-self. If the MHC molecule is identified as non-self the immune system is activated and there will be tissue or organ rejection of the transplant.
10. Ethical issues have arisen about the definition of death, when, how and who can give consent for an organ to be transplanted and the payment for the donation. The demand of transplantation has led to illegal organ and tissue trafficking, forced donation and 'transplantation tourism'. In some countries it is easier, cheaper and quicker to get a transplant but this may involve the exploitation of the poor and needy and the violation of their human rights.
11. Immunosuppressive therapy involves the use of immunosuppressive drugs and other drugs to suppress the immune system. When a person is on immunosuppressive therapy they are more susceptible to infection and cancer. Thus there are associated health risks and potential harmful consequences for people during the recovery time after a transplantation.

30 Bioartificial Organs

1. There is a need for bioartificial organs as: 1. The supply of donor organs and tissues is limited causing some people to wait months, even years on a waiting list for transplantation. 2. Donor tissues and organs can be rejected by the recipient's body. 3. Transplantation involves immunosuppressive therapy which makes the recipient more susceptible to infection and cancer. 4. There are ethical issues about the definition of death, when, how and who can give consent for an organ to be transplanted and the payment for the donation. 5. There has been a growth in illegal organ and tissue transplantation, forced donation and 'transplantation tourism'. 6. Longer life expectancy and more people surviving to older ages has increased the need for transplant donations.
2. Regenerative medicine is the process of replacing or regenerating human cells, tissues or organs to give the recipient normal functioning of that part.
3. In a hip replacement the head of the femur is replaced by a metal ball and a plastic cup is inserted to act as the socket in this ball and socket joint. Hip replacement can be a total replacement or a half replacement.
4. Tissue engineering involves the use of cells, engineering and material methods and suitable biochemical and physiochemical factors to improve or replace biological functions.
5. There is great interest in the developments in tissue engineering as progress could lead to a future where tissues and organs using a patient's own cells are produced within a suitable framework to replace damaged or diseased body parts. This would remove the need for donor organs and tissues, remove waiting lists and be a more efficient and more effective way of improving the quality of life for many people.
6. Scientific knowledge about the structure and function of organs has led to tissue engineering and regenerative medicine. There is social pressure to continue to produce bioartificial tissues and organs to aid and improve the quality of life of people with diseases or damaged body parts. There are also economic pressures to continue the research and development both from the manufacturers' side to increase their income but also to reduce the medical costs of looking after people with the diseased or damaged body parts. Cultural and ethical considerations are involved in the development of new technologies, as the production of artificial parts will reduce the need for donors for transplantation. Thus the use of scientific knowledge is influenced by social, economic, cultural and ethical considerations.
7. D

31 Metabolism

1. Metabolism refers to the sum of all chemical reactions in an organism.
2. Metabolic pathways are series of chemical reactions that either build complex macromolecules or break down complex molecules into simpler molecules.
3. A catabolic pathway is a series of reactions that release energy by breaking down complex molecules into simpler molecules.
4. An anabolic pathway is a series of reactions that build complex macromolecules from simpler molecules.
5. Catabolic reactions are often called 'downhill' reactions as energy is released while anabolic reactions are often called 'uphill' reactions as energy is needed to drive the reaction.
6. An enzyme is an organic catalyst that controls the rate of a specific chemical reaction.
7. Enzymes lower the activation energy of the chemical reactions that they catalyse thus making it easier for the reaction to occur.
8. In an exothermic reaction energy is released as either light or heat.
9.

Type of work	Description
Transport work	Substances are moved across a membrane by active transport against the concentration gradient.
Chemical work	Energy is used to drive specific chemical reactions, e.g. biosynthesis of macromolecules.
Mechanical work	Involves the transfer of energy, e.g. beating of cilia, muscle cells change shape in contraction, chromosomes move in mitosis.

10. ATP is the 'energy currency' of a cell. It is used to drive transport and mechanical work.
11. Glucose + oxygen → Carbon dioxide + water
    ADP + P_i → ATP
12. D

32 Respiration

1. Respiration is a series of chemical reactions which releases energy from complex carbohydrates.
2. All living things respire.
3. Aerobic respiration uses oxygen and is the complete breakdown of sugars into carbon dioxide and water, while anaerobic respiration has no, or insufficient oxygen and other products are formed.
4. Yeast cells respire anaerobically and form carbon dioxide and ethanol. Beer makers and wine makers use the anaerobic respiration of yeast to make beer and wine.
5. Glucose + oxygen → carbon dioxide + water
6. The general equation for respiration shows the reactants and the final products. It could be misleading, if it was assumed the process occurred in one step. Respiration is actually a series of reactions controlled by enzymes.
7. An enzyme is a chemical made by living things and its function is to control the rate of a specific chemical reaction that occurs in the body.
8. Enzymes are important in respiration as they control the rate of reaction at each step in the series of reactions. This means that the release of energy is controlled.
9. The first stage of aerobic respiration breaks down the 6-carbon sugar (glucose) into two 3-carbon molecules (pyruvate) releasing a small amount of energy. The second stage breaks down the 3-carbon molecule into a 1-carbon molecule (carbon dioxide) using oxygen and releases a much larger amount of energy.
10. C
11. A
12. B

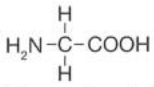
33 Reactions in Respiration

1. Oxidation involves the loss of electrons from a substance that is in a redox reaction and reduction is the addition of electrons to a substance involved in a redox reaction.
2. An oxidised substance has less energy than a reduced substance.
3. A glucose molecule ($C_6H_{12}O_6$) would have more energy than a methane molecule (CH_4) as the amount of energy in a molecule partly depends on the size of the molecule and glucose is larger than methane.
4. A molecule of methane (CH_4) has more energy than an molecule of carbon dioxide (CO_2) as a reduced molecule has more energy than an oxidised molecule.
5. The main steps in cellular respiration are: 1. Glycolysis. 2. Krebs cycle. 3. Electron transport system.
6. Glycolysis means the splitting of sugar.
7. Glycolysis is a series of reactions where a six-carbon sugar (glucose) is broken down into a three-carbon sugar.
8. Glycolysis can occur in both aerobic and anaerobic conditions as it does not require oxygen to occur.

9. Two ATP molecules are synthesised per molecule of glucose in glycolysis.
10. Phosphorylation is the addition of a phosphate group to a molecule.
11. The Krebs cycle occurs in the mitochondria.
12. Chemiosmosis is an energy coupling mechanism that uses stored energy in the form of a hydrogen ion (proton) gradient across a membrane to power cellular activities.
13. (a) The inner membrane is folded to form cristae which have the carriers and enzymes of the electron transport chain and ATP synthase which is involved in ATP synthesis.
    (b) The Krebs cycle occurs in the matrix as this is where the enzymes for these reactions are found.
14. A
15. B
16. C
17. D

34 Protein Structure

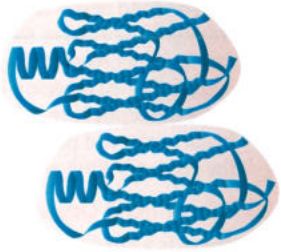
1. A protein is a macromolecule that always contains the elements carbon, hydrogen, oxygen and nitrogen.
2. (a) Each amino acid has an amino group (NH_2), a carboxyl group (COOH) and a side chain (R).

    (b)


3. The different chemical properties of different amino acids, e.g. polar (hydrophile) or non-polar (hyrodphobe) depend on the properties of R.
4. Glycine is the simplest amino acid with R = H.
5. A dipeptide is 2 amino acids joined together by a peptide bond. A tripeptide is 3 amino acids joined together by peptide bonds. A polypeptide is many amino acids joined together by peptide bonds.
6. (a) The distinguishing test for protein is the Biuret test. Biuret solution (equal amounts of 1% copper sulfate solution and 5% potassium hydroxide solution) will turn from pale blue to purple in a positive test for protein.
    (b) The Biuret test is actually detecting peptide linkage – which is the bond that holds amino acids together in a protein.
7. (a) The primary structure of a protein is the unique linear sequence of amino acids in its molecule and is determined by genetic information.
    (b) The secondary structure of a protein refers to the segments of the polypeptide chain that has coils or folds caused by hydrogen bonds between repeating CO and NH groups.
    (c) The tertiary structure of a protein is formed by the folding of the secondary structure into a complex and compacted shape.
    (d) The quaternary structure of a protein involves the interaction between several polypeptide chains that aggregate into one functional macromolecule.
8. The coils/folds of secondary structures of proteins are the helix and the beta pleated sheet.
9.

Feature	Fibrous protein	Globular protein
Shape	Long thread	Globe-like
Polypeptide chain	Extended chain and threadlike	Twisted, looped and folded back on itself
Type of structure	Secondary structure	Tertiary structure
Example	Collagen	Haemoglobin

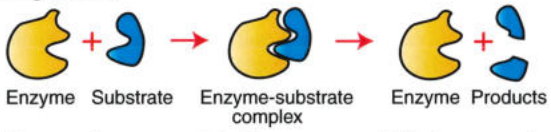
10. The three types of bonds that hold the tertiary configuration in shape are: 1. Hydrogen bonds. 2. Ionic bonds. 3. Disulfide bonds.
11. Disulfide bonds are strong covalent links between sulfur containing amino acids, e.g. between the –SH groups of amino acids that are distant in the chain but lie next to each other due to the folding and twisting of the chain. These bonds are important as they stabilise the structure of the protein and provide structural strength.
12. (a) Denaturation occurs when a protein loses its shape and the chains unravel. This occurs when the weak hydrogen bonds in the structure are broken.
    (b) Denaturation can occur due to physical or chemical changes in the environment, e.g. excessive heat, salt concentration, temperature, pH.
13. (a) The diagram shows the primary structure of the protein as it shows the long chain of the polypeptide. However, it does not show the actual sequence of amino acids in the polypeptide which is what makes the precise primary structure of the protein. The diagram shows the secondary structure of the protein as it shows the segments of the polypeptide that are coiled or folded. The diagram shows the tertiary structure as it gives the overall globular shape of the polypeptide.
    (b) Structure X is a pleated sheet and Structure Y is a helix coil.
    (c) It is a globular protein as the secondary structure is folded into a complex and compacted shape.
    (d) One possible arrangement of two of these polypeptides which join together to form a quaternary structure is shown in the diagram.



14. B
15. D
16. A

35 Enzymes

1. Enzymes are biological catalysts.
2. Most enzymes are protein molecules containing the elements C, H, O and N.
3. Substrate is the reactant with which an enzyme binds and converts into the product.
4.

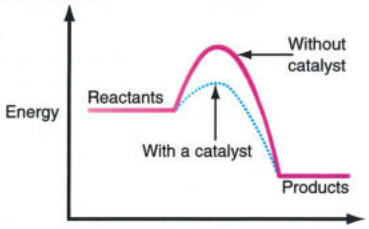


The diagram shows a model of an enzyme which has an active site. The active site recognises the substrate and since it has a precise size and shape a 'lock and key' fit results. This means that each enzyme fits with a specific substrate and hence is specific for a specific reaction.

5. (a) The active site of the enzyme is often a surface groove which fits perfectly onto the substrate and brings the active sites of the chemicals involved into alignment so that the chemical reaction can occur more quickly.
    (b) The 'lock and key' model was replaced by the 'induced fit' model as scientific data was collected about the action of enzymes. It was found that the enzyme and the active site do not have fixed rigid structures. The shape of the active site alters slightly to make a tight fit with the substrate. Thus the model was renamed 'induced fit'.
6. Enzyme activity depends on temperature, pH, substrate concentration, presence of cofactors and coenzymes, presence of inhibitors and heavy metals.
7. Pepsin is an enzyme found in the stomach.
    (a) An increase in temperature above 42°C will cause the enzyme to denature and cease to function.
    (b) Pepsin operates best at pH = 2 and an increase in pH causes the enzyme to cease functioning.
    (c) Increasing the substrate concentration will initially increase the rate of reaction but a saturation point is reached when all active sites on the enzyme are reacting and the reaction rate will level off.
8. An increase in substrate concentration causes the reaction rate to increase until all the active sites of the enzyme are occupied. This is the maximum reaction rate according to substrate concentration and is known as the saturation point.
9. High temperatures break the hydrogen bonds in proteins, causing their surface shape to alter. For enzymes this change in 3-D shape means that the substrate can no longer bind with the enzyme and reactivity is lost.
10. A
11. D
12. B

36 Enzymes and Reactions

1. The 'induced fit' model allows for the active site to slightly change its shape so that it fits more closely to the substrate. Evidence showed that the active site is more flexible than was previously believed and the 'keyhole' analogy was too rigid to explain this evidence.
2. Enzyme inhibitors attach to the enzyme and block its functioning. Competitive inhibitors attach to the active site so that the enzyme cannot fit with the substrate. Non-competitive inhibitors do not attach to the active site but another part of the enzyme. However, this attachment still affects the shape of the enzyme and makes the active site less effective in its function.
3. (a) For the reaction to occur and use less energy, the curve must rise sufficiently above the dotted line showing the reaction will occur, but it must be below the maximum height of the original line showing that less energy was expended.



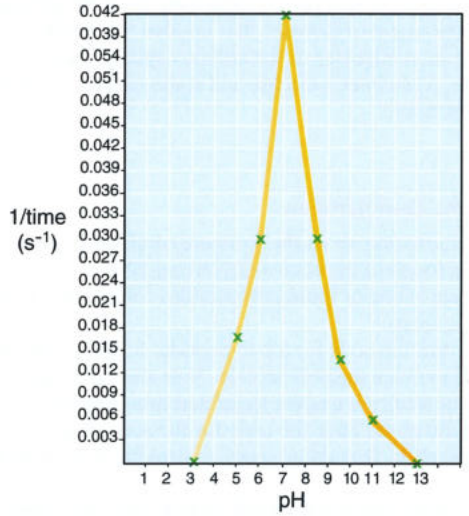
   (b) Enzyme reactions are between a hundred and a million times faster than the same reaction in non-living systems. To make the reaction faster in the laboratory, the temperature or addition of strong acids or bases is needed — which are conditions unsuitable for life forms.
4. Invertase is an enzyme that will only catalyse the hydrolysis of sucrose (cane sugar) to glucose and fructose. It does not work on any other sugar.
5. B

37 Experiment – Enzyme Activity and pH

1.

pH	Time when iodine stayed yellow-brown	$\frac{1}{\text{time}}$ (s^{-1})
1	No reaction	0
3	No reaction	0
5	1 minute	0.017
6	30 seconds	0.030
7	15 seconds	0.060
8.5	30 seconds	0.030
9.5	1 min 15 seconds	0.014
11	3 minutes	0.006
13	No reaction	0

2.



Marks are allocated for the axes – having an even scale which covers the complete range of numbers, and the labelling of axes with both title and units. The rest of the marks are allocated for correct plotting of points and the correct joining of points.

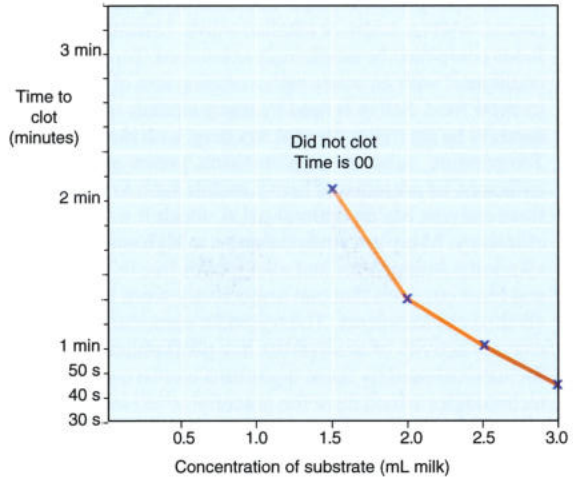
3. Repeating an experiment gives statistical verification. One set of results could be an anomaly and hence taking an average of several trials provides a more accurate answer.
4. Plotting pH versus $\frac{1}{\text{time}}$ is plotted for more convenience as it makes it easier to plot the 'no reaction' results, which become zero, rather than dealing with 'infinity'. It also provides a curve with a peak in the usual format shape. Plotting pH versus time would not give an incorrect graph and is also a valid way of showing the results – it gives an inverted curve where the lowest point in the trough shows the fastest reaction.
5. pH = 7
6. An amylase in the mouth would need to have a pH = 7 as this is 'neutral'. There would be several reasons why the mouth should be neutral – the mouth is the first point of entry into the alimentary canal and the ability to detect taste could seriously be affected if the food reacted with an acidic or alkaline saliva. Unsuitable or poisonous foods could then be unwittingly consumed. Tooth enamel and dentine could react with an acidic saliva causing loss of teeth and the ability to chew food. Saliva is used by many animals to lick wounds – so it needs to be neutral to prevent reactions with skin and fur.
7. Temperature, substrate concentration, presence of coenzymes or cofactors, or presence of heavy metals such as mercury or cadmium.
8. Each enzyme has an optimal pH at which it works with maximum efficiency. Many enzymes denature at high acidity and/or high alkalinity. Acidophiles and alkaliphiles live in hostile environments and have enzymes that can respectively work in very acidic and very alkaline environments. The scientific community is interested in the enzyme activity of acidophiles and alkaliphiles as the mechanisms for survival used by these organisms can be used to develop new technologies which have the potential to aid individual people, society and reduce human impact on the environment. A greater knowledge of enzymes and their roles in the metabolism of different organisms enables new treatments to be devised to improve the quality of life for those with problems with enzyme activity in their bodies and for the development of new commercial products.
9. A
10. B

38 Experiment – Enzyme Activity and Temperature

1. The student needs to test rennin with milk both boiled and not boiled as well as milk by itself both boiled and not boiled.
2. The milk by itself acts as a control to be used as a comparison.
3. Boiling causes enzymes to denature so that the active site can no longer bind with the substrate.
4. Choose a dry biscuit or bread as the starch food to be tested. Use iodine solution to show that starch is present in this food – the iodine solution will turn blue-black from a brown-yellow colour if starch is present. A piece of this food is chewed in the mouth for a period of time allowing thorough mixing with saliva and the chewed food is removed from the mouth and tested with iodine solution. If the iodine solution does not change colour with the chewed food, this shows that the starch in the food has been broken down by the saliva.
5. Low temperatures, e.g. refrigeration can be used for human tissue transplants or preserving food as the enzyme-catalysed reactions are slowed and cells are not damaged.
6. (a) Low core temperature would cause a slow heartbeat, shallow breathing, shivering and chattering of teeth, goosebumps, purple fingers and toes, lack of coordination, slurred speech and dilated pupils.
   (b) Low core temperature slows the activity of enzymes. This means that cellular reactions occur at a slower rate and energy is released at a slower rate.
   (c) In hypothermia the body is losing heat faster than it can produce heat causing the low body temperature.
7. C
8. C
9. C
10. A

39 Experiment – Enzyme Activity and Substrate Concentration

1. The substrate refers to the substance on which the enzyme acts. The concentration of the substrate refers to the number of molecules of the substrate present in the solution or mixture.
2. Temperature needed to be kept at approximately 37°C as that is human body temperature and the optimum temperature for this enzyme.
3. (a) Rennin is an enzyme secreted by the walls of the stomach.
   (b) Rennin acts on the milk protein casein causing it to clump into a semisolid curd.
   (c) Junket is a milk-based dessert food that uses rennin.
4. The students started with pure milk bought from the shops and then diluted each test tube with differing amounts of water, always keeping the total volume of milk/water at 3 mL.
5. Pure milk clotted the fastest.
6. (a)



   (b) Curve shows that decreasing the concentration of the milk (substrate) increases the time to clot. Shape is tending towards exponential curve.
   (c) Both graphs show how substrate concentration influences reaction rate and both show that low substrate concentration slows down the reaction. The shape difference is due to the difference in axis construction and the few numbers obtained by the students. The student graph has time to react – where the longer time means less enzyme activity and the graph cannot be plotted to infinity for the concentrations which did not clot – while the usual presentation has enzyme activity – where the higher activity is higher on the axis. The inability to plot infinity on a graph is the reason why 1/time is usually plotted. The *x*-axis has an inverse relationship in comparison between the two graphs.
7. Students can conclude that dilute milk solutions clot more slowly with rennin than non-diluted milk solutions.
8. All containers must be correctly labelled, especially if left unattended, to ensure the safety of other people who may find the containers and need to know what risks are involved with the chemicals in the containers. When using thermometers in test tubes in polystyrene cups, diligence is needed to prevent the apparatus tipping over. All students must be aware of the correct method of disposal of broken glassware, and if mercury thermometers are being used, of correct disposal of mercury if it is released onto the bench.
9. D
10. D

40 Enzymes and Inhibitors

1. The active site on an enzyme is the part of the enzyme that binds with the substrate where the reaction occurs.
2. The shape of the active site determines which molecules can 'fit' onto the enzyme and thus is linked to the specificity of the enzyme.
3. Irreversible inhibition occurs when the chemical bonds with the enzyme and the bond cannot be broken and the enzyme stops functioning. Reversible inhibition occurs when weak bonds form with the enzyme and the enzyme can resume normal functioning when the bonds are broken.
4. Competitive inhibition occurs when a chemical has a similar shape to the substrate and can block the active site. This means the enzyme cannot bind with the substrate and cannot catalyse the reaction to form the product.
5. Non-competitive inhibition occurs when the chemical binds to a part of the enzyme that is not the active site. This causes the enzyme to change shape affecting the active site so that it no longer can properly bind with the substrate. The enzyme becomes less effective in carrying out its job.
6. (a) Acetylcholine is a neurotransmitter that passes the nerve message from the end of one neuron across the synapse to the dendrites of the next neuron and also activates skeletal muscles.
   (b) Acetylcholinesterase is the enzyme that breaks down acetylcholine after the neurotransmitter has crossed the synapse.
   (c) An acetylcholinesterase inhibitor will prevent the enzyme acetylcholinesterase from breaking down. This means the neurotransmitter will keep stimulating the muscle and cause prolonged muscle contraction.
   (d) Sarin affects the lung muscles so that the person can no longer control their breathing rate. The diaphragm muscle cannot relax causing lung muscle paralysis and the person will die of asphyxiation
7. B

41 Allosteric Activation and Inhibition

1. Allosteric regulation occurs when the functioning of an enzyme or protein is controlled by the binding of a regulatory molecule to a separate site.
2. Allosteric activators speed up reactions while inhibitors slow down reactions.
3. Allosteric regulated enzymes usually consist of two or more subunits with each subunit having an active site.
4. The active site is where the substrate binds to the enzyme and where the reaction occurs. The regulatory site is the place where the regulatory molecule binds with the enzyme.
5. When one activator binds with the enzyme at the regulatory site it changes the shape of the enzyme causing the active site to become stable and functional. The change in shape of this subunit affects the shape of the other subunits so that all active sites become functional.
6. When one inhibitor binds with the enzyme at the regulatory site it changes the shape of the enzyme causing the active site to become stable in the non-functional mode. The change in shape of this subunit affects the shape of the other subunits so that all active sites become non-functional. The enzyme becomes inactive.
7. End product inhibition is a feedback mechanism where the end product binds to the first enzyme in the pathway at an allosteric site. This inactivates the first enzyme and the pathway does not function.
8. Scientists are interested in the virus genes for caspase inhibitors as these genes could be removed from the viral genome and used in the treatment of diseases that involve cell death, e.g. some degenerative diseases of the nervous system such as Parkinson's disease, Alzheimer's disease and some cancers involve caspases and cell death.
9. D
10. B

42 Enzyme Deficiencies

1. The information about the structure of different proteins in the body is found in the code of DNA and RNA.
2. A protein is one or more polypeptides folded and coiled into a specific shape.
3. A gene has the code for a polypeptide. In gene expression the code is used to make the polypeptide. Note that some gene codes for RNA molecules are never translated into proteins.
4. Structural genes have the code for the sequence of amino acids in a polypeptide. Regulator genes control the activity of the structural gene.
5. Intracellular enzymes are made in cells and stay within the cell. Extracellular enzymes are made in the cell, packaged and secreted from the cell to work elsewhere in the body, e.g. digestive enzymes.
6. Lactose is milk sugar. It is a disaccharide sugar found in the breast milk of mammals.

7. The enzyme lactase digests the sugar lactose.
8. Human populations that raise cattle and have consumed dairy products for centuries have many people who produce lactase throughout their life. Populations which have not traditionally depended on dairy products as a food source have people who cease to produce lactase in childhood or adolescence.
9. If a person who does not produce lactase eats dairy products, the lactose is not digested by the person and when it reaches the large intestine it is fermented by intestinal bacteria causing diarrhoea and vomiting.
10. PKU is phenylketonuria, a genetic disease that causes increased levels of phenylalanine in the blood due to a deficiency of the enzyme phenylalanine hydroxylase (PAH). PAH breaks down phenylalanine into the amino acid tyrosine.
11. Newborn babies are tested for PKU. If a child is found to have PKU, the parents are advised about the need for the child to have a restricted diet which decreases the intake of foods that contain phenylalanine, e.g. reduce the intake of chicken, fish, meat, eggs, nuts, cheese, milk and other dairy products. This reduces the build-up of phenylalanine in the blood. High levels of phenylalanine in the blood causes damage to the nervous system, e.g. limits brain development, causes intellectual disability, seizures and behavioural and heart problems. Supplementary tablets can also be taken to provide tyrosine which is normally made from phenylalanine. Treatment thus improves the quality of life for the child and reduces the effects of high levels of phenylalanine in the blood.
12. B

43 The Respiratory System

1. Individual cells obtain energy by the process of respiration.
2. During aerobic respiration organic molecules such as glucose are combined with oxygen to release energy, and carbon dioxide and water are formed as waste products.
3. The word equation for respiration can be written as follows. Glucose + oxygen → water + carbon dioxide + energy
4. Gas exchange is the movement of oxygen and carbon dioxide in different directions across a membrane.
5. Four characteristics of a respiratory surface are: 1. Must be moist so gases can dissolve for diffusion across the membrane. 2. Need a large surface area for maximum diffusion. 3. Need a rich blood supply to remove absorbed oxygen. 4. Need to be thin for quicker diffusion.
6. In mammals, the respiratory system consists of the lungs and passages involved with the intake, expulsion and exchange of oxygen and carbon dioxide. Gas exchange occurs at alveoli.
7. The organs of the respiratory system include – nose, pharynx, larynx, trachea, two bronchi, bronchioles, two lungs with their pleura and muscles of respiration (intercostal muscles and diaphragm).
8. The nose consists of bone and cartilage and the nasal cavity. The nasal cavity is divided into two equal passages divided by a septum. The nose is lined with ciliated epithelium which has mucus-secreting goblet cells. In the nose the air is warmed, moistened and filtered.
9. The nose has a significant number of blood vessels which are needed to warm the air that enters the nose. In a nose bleed large amounts of blood can be lost from these vessels.
10. Goblet cells produce mucus which is used to trap microbes and small foreign particles that are inhaled and enter the respiratory system.
11. Passage of air into the human body – nostril → pharynx → larynx → trachea → bronchi → bronchiole → alveoli.
12. In humans, oxygen enters the bloodstream in the alveoli.
13. Once oxygen enters the bloodstream it attaches to the haemoglobin of a red blood cell forming oxyhaemoglobin and is transported to the heart in the pulmonary vein and hence to a needy cell that requires oxygen.
14. Breathing is the process of getting air in and out of the lungs while respiration is a series of chemical reactions which release energy to the cell.
15. The respiratory pigment in human blood is haemoglobin. Haemoglobin increases the capacity of blood to carry oxygen. This means that humans can supply more oxygen to cells and can have a higher rate of respiration and a higher metabolic rate. This is advantageous as it allows humans to be more active and for longer periods of time.
16. The human right lung is slightly larger than the left lung and has three lobes compared to two lobes for the left lung. The left lung has a 'cardiac notch' which allows room for the heart which is usually on the left hand side of the chest.
17. C
18. C

44 Gas Exchange, Breathing and Ventilation

1. Ventilation is any process that increases the contact between a respiratory surface and the respiratory medium. Gas exchange is the process where oxygen is absorbed and carbon dioxide is excreted across a membrane.
2. Breathing is often called pulmonary ventilation. Ventilation is the process that increases contact between the respiratory surface and the respiratory medium. The process of breathing brings air into the lungs to bring oxygen in close contact with the capillaries which surround the alveoli sacs.
3. The diaphragm is a dome shaped sheet of muscle that separates the thoracic and abdominal cavities.
4.

Feature	Inhalation	Exhalation
Diaphragm	Contracts and moves downwards	Relaxes and moves upwards
Intercostal muscles	Contract moving the ribs upwards	Relax lowering the ribs
Chest cavity	Increases in volume	Decreases in volume
Air pressure inside lung	Decreases	Increases
Movement of air	Into lungs	Out of lungs

5. Chest breathing is shallow breathing that predominantly relies on the intercostal muscles to raise and lower the chest to change the chest cavity capacity while abdominal breathing predominantly relies on the diaphragm to change the chest cavity capacity.
6. Breathing is controlled by the respiratory centre which is in the medulla oblongata of the brain.
7. Tidal volume is the volume of air inhaled/exhaled with each breath while vital capacity is the maximum tidal volume that a person can achieve with forced breathing. Vital capacity is greater than tidal volume.
8. Inhaled air is atmospheric air and contains 20.9% oxygen and 0.04% carbon dioxide. In the lungs there is gas exchange with oxygen absorbed for respiration and carbon dioxide released as a waste product. Thus exhaled air has higher carbon dioxide (5.60%) and less oxygen (14.0%). Exhaled air has had about one third of the oxygen removed.

45 The Circulatory System

1. A circulatory system is needed to transport materials around the body. Diffusion is not able to supply the needs of every cell in large organisms, e.g. supply nutrients and remove waste products, and thus humans need a transport system.
2. The human circulatory system transports substances, e.g. glucose, hormones, amino acids; blood cells, e.g. red blood cells and white blood cells; and heat throughout the body.
3. Interstitial fluid is the extracellular fluid.
4. The human circulatory system is a double circulation system as blood goes through the heart twice in each circuit. It involves the pulmonary circulation and the systemic circulation.
5. Any fluid that seeps from the blood vessels into the interstitial fluid is collected by the lymphatic system and then returned to the bloodstream.
6. The left side of the heart receives oxygenated blood while the right side of the heart receives deoxygenated blood.
7. In double circulation the blood goes through the heart twice in each circuit. Blood from the body enters the right side of the heart and is pumped to the lungs where it is oxygenated. The blood is then returned to the left side of the heart which pumps the blood out the aorta. The blood then travels through arteries which connect to capillaries, and then to veins which return the blood to the heart.

8. Humans need a strong heart as the circulatory system is a closed system which means the pumping of the heart has to overcome the resistance to flow caused by the thin diameter of blood vessels.
9. The main components of the circulatory system are the heart, blood vessels and blood.
10. The heart is a four-chambered organ that pumps blood through blood vessels. It is located behind the breastbone and is about the size of a clenched fist.
11. Separation of the circulating fluid and interstitial fluid means that blood can contain red blood cells. These have the respiratory pigment haemoglobin which combines with oxygen to form oxyhaemoglobin. Haemoglobin greatly increases the ability of blood to carry oxygen. Thus a unit volume of circulating fluid (blood) can carry high amounts of oxygen for respiration easily meeting the energy needs of an active person.
12. Valves which are flaps of connective tissue prevent backflow of blood.
13.

Blood vessel	Description
Arteries	Take blood away from the heart
Veins	Take blood towards the heart
Capillaries	Connect arteries and veins

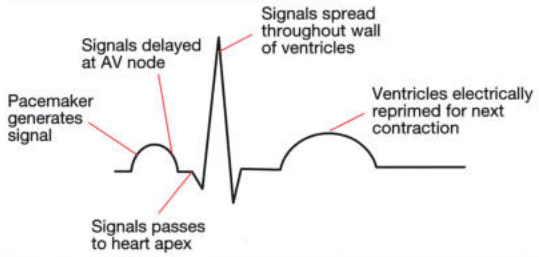
14. A heart murmur involves blood squeezing backwards through the heart chambers.
15. All blood vessels except capillaries have smooth muscle fibres which are connected to nerves of the autonomic nervous system.
16. Nerve stimulation causes vasoconstriction of small arteries and arterioles reducing blood flow while reduced nerve stimulation causes the smooth muscle to relax leading to vasodilation and increased blood flow.
17. The diameter of large arteries varies according to the amount of blood they contain.
18. Blood is a connective tissue that makes up about 7% of body weight for the average man. It consists of three cell types and plasma.
19. Plasma is mainly water – around 90% water and dissolved substances such as plasma proteins, hormones, enzymes, gases, nutrients such as glucose, amino acids, fatty acids, glycerol and vitamins and inorganic salts such as sodium chloride, sodium hydrogen carbonate, potassium, magnesium, copper, iodine, calcium and iron.
20. (a) Homeostasis is the maintenance of a constant internal environment within narrow limits.
    (b) The circulatory system maintains blood volume and the concentration of its components within narrow limits and is thus part of the homeostasis mechanism.
21. B
22. C
23. A
24. D

46 The Cardiac Cycle

1. The cardiac cycle is one complete sequence of filling and pumping of the heart.
2. The pathway through the pulmonary system is – vena cava → right atrium → right ventricle → pulmonary artery → lungs → pulmonary vein → left atrium → left ventricle → aorta.
3. The three main stages in the cardiac cycle are – the atrial systole, the ventricular systole and the complete cardiac diastole.
4. The systole is the contraction phase and diastole is the relaxation phase.
5. (a) The cardiac output is the volume of blood each ventricle pumps per minute.
   (b) Cardiac output is determined by the heart rate which is the rate of contraction or number of beats per minute and the stroke volume which is the amount of blood pumped by a ventricle in a single contraction.
7. The coronary arteries deliver oxygen and nutrients to the muscle of the heart. If a coronary artery is blocked, there will be a restricted supply of oxygen and glucose for cellular respiration and the heart muscle will not be able to function properly. This can lead to heart attacks.
8.

Part of heart	Function
Right atrium	Collects blood from the body (anterior and posterior vena cava). When thin walls relax blood flows into the right ventricle. Contraction completes filling of ventricle.
Left atrium	Collects blood from the lung (pulmonary veins). When thin walls relax blood flows into the left ventricle. Contraction completes filling of ventricle.
Right ventricle	Pumps blood to the lung. When thick wall contracts blood is forced into pulmonary artery.
Left ventricle	Pumps blood to the body. When very thick wall contracts blood is forced into the aorta.
Atrioventricular valves	Keep blood from flowing backwards into atria. During contraction of ventricles, pressure closes the valve.
Semilunar valves	Keep blood from flowing backwards into ventricle. Contraction of ventricle forces valve to open and blood leaves the heart.

9. The human heart is classified as a myogenic heart as the pacemaker is specialised muscle tissue located in the heart.
10. (a) An electrocardiogram (ECG) is a record of the electrical events that occur during heartbeat.
    (b)

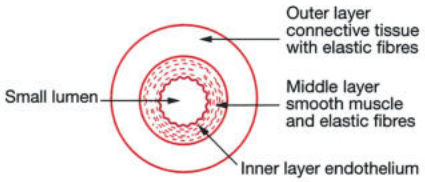


11.

Factor	How it affects heart rate
Autonomic nervous system	Autonomic nerves in the cardiovascular centre in the medulla oblongata of the hindbrain affect heart rate. The heart rate centre has two pairs of nerves that connect the medulla oblongata to the pacemaker. One pair of nerves go through the spinal cord and accelerate the heart. The other pair form part of the vagus nerves and decelerate heart rate. Parasympathetic nerves (vagus nerves) supply the SA and VA nodes and atrial muscle and reduce the rate at which impulses are produced, decreasing heartbeat rate. Sympathetic nerves supply the SA and VA nodes and the myocardium of the atria and ventricles and increase heartbeat rate. These are antagonistic nerves.
Circulating chemicals	Hormones adrenaline (epinephrine) and noradrenaline secreted from the adrenal medulla increase heartbeat rate by stimulating cardiac muscle and increasing the excitability of the body, e.g. adrenaline mediates 'fight or flight' responses to short term stress causing the release of glucose into the blood from the liver so that the glucose can be used in the cellular respiration required for vigorous physical activity
Temperature	Heart rate is influenced by temperature with higher temperatures accelerating heartbeat and lower temperatures decelerating heartbeat.
Position	When a person is upright the heart rate is usually faster than when lying down.
Exercise	Lower pH, e.g. due to increased exercise and the release of carbon dioxide speeds up breathing and heartbeat. Active muscles need more blood than resting muscles and need faster heart rate and selective vasodilation.
Age and gender	Heart rate is faster in women than in men and faster in babies and small children than in older children and adults.
Emotional state	Heart rate is faster when a person is feeling fear, anxiety or excitement.

47 Blood Vessels

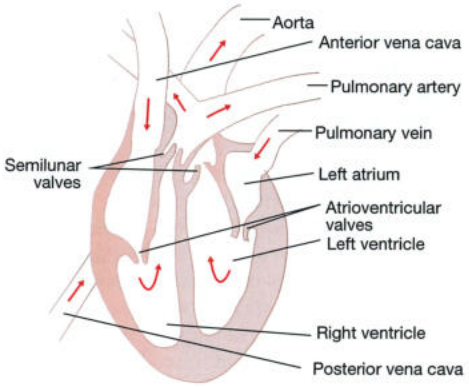
1. Arteries and veins are defined by blood flow direction, i.e. arteries carry blood away from the heart and veins carry blood to the heart as blood composition changes. Most veins carry deoxygenated blood. While most arteries carry oxygenated blood, the pulmonary artery travelling from the right side of the heart to the lungs carries deoxygenated blood. However, the pulmonary vein travelling from the lungs to the left side of the heart carries oxygenated blood.
2. (a) X = artery as it has thick walls with layer of thick elastic fibres and smooth muscle; Y = vein as it has thin wall, large bore and the layer of smooth muscle and elastic fibres is thin; Z = capillary as the wall is a single cell layer which is needed for efficient diffusion.
   (b) The diagram does not show a valve in the vein cross-section. Valves are essential for the functioning of veins as they prevent backflow. Veins return blood to the heart, yet are not under pressure, which means the valves are needed to keep the blood flowing in one direction.
3. Distortion into a bell shape to fit through narrow capillaries means that the surface area:volume ratio of the RBC is increased which assists with diffusion. It also pushes the haemoglobin to the edge of the cell decreasing the distance for oxygen to travel.
4. (a) Blood vessel N is the aorta.
   (b)



   (c) Feature: Walls of arteries have thick walls of elastic fibres and smooth muscle.
   Explanation: Arteries need thick walls to withstand the pressure and to help maintain the pressure, e.g. the heartbeat sends a pulse of blood flowing into the arteries, which stretches them. During the relaxation phase of the heart cycle, the elastic fibres recoil to maintain pressure on the blood.
   (d) L = most likely to have similar structure to N as L is the pulmonary artery while M and P are veins.
   (e) Vessel P has deoxygenated blood while vessel N has oxygenated blood. The difference occurs as vessel P (vena cava) is bringing the blood from the body; it then flows through the lungs, picks up oxygen, travels to the heart and is pumped to the whole body from vessel N (aorta).
5. Arteries are under pressure and have a pulse. They do not need valves as the pumping of the heart keeps pushing blood away from the heart. Veins are not under the same high pressure and valves are needed to prevent backflow.
6. The diagram shows how the use of skeletal muscle when we move helps to move blood through the vein. It also shows the presence of valves and how the opening and closing of the valves keeps the blood flowing in one direction, preventing backflow.
7. C
8. C
9. B
10. C
11. D
12. A

48 Case Study – Coronary Occlusion

1. An occlusion is a blockage or closing of a blood vessel or hollow organ.
2. Ischaemia means there is a lack of blood flow to a tissue.
3. Infarction means there is tissue damage due to inadequate flood supply.
4.

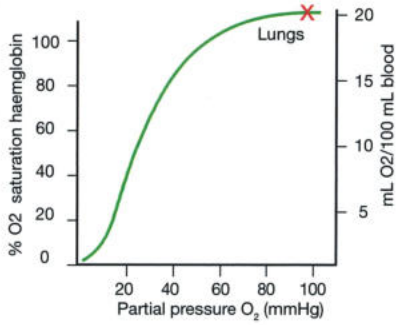


5. A coronary occlusion is defined as the complete or partial blockage of blood flow in a coronary artery which are the arteries that supply the heart with blood delivering oxygen and nutrient to the cardiac muscle cells.
6.

Feature	Description for coronary occlusion
Cause	• Gradual build-up of plaques containing cholesterol (atherosclerosis) making the arteries narrow and hardened with little elasticity. • Formation of blood clot on plaque (coronary thrombosis). • Temporary spasm of muscles in artery wall. • Inflammation of coronary artery wall.
Risk factors	• The build-up of plaque is associated with certain contributing risk factors. • Smoking. • High blood pressure. • Certain chronic diseases. • Overweight. • Diet – high animal fat and high low density lipoproteins (LDL) with less high density lipoproteins (HDL). • Family history.
Consequence	• Death of the area of cardiac muscle served by that artery (myocardial infarct). • Heart working harder can cause pain (angina). • Heart attack. • Without treatment can be death.
Treatment	Depends on the precise nature and site of the occlusion. • Medication, e.g. nitrate group such as glyceryl trinitrate that increase the diameter of the artery and increase the flow of blood containing the required oxygen and nutrients. Or calcium antagonists such as nifedipine which causes the muscle in the coronary artery wall to relax. • Angioplasty with the artery stretched open with a balloon introduced in surgery. • Bypass surgery, e.g. using vein graft or using nearby artery.

49 Blood

1. (a) Oxygen travels in the blood attached to haemoglobin, ie. as oxyhaemoglobin in the red blood cells.
   (b) (i) 70% saturation.
   (ii) 70% saturation means that the haemoglobin has given up about 28% of its oxygen.
   (iii)



   (c) Although the percentage of oxygen would be the same in the air in Tibet, the higher altitude means that the total amount of air would be less. This means that there is less available oxygen, which in turn means less oxygen crosses the placenta to the foetus and the child dies in the womb.

2. In most circumstances, more glucose is found in the blood entering the liver than leaving the liver. Glucose is absorbed into the bloodstream in the capillaries inside the villi of the small intestine, hence the hepatic portal vein, which takes blood from the intestines to the liver, will have high amounts of glucose, especially after a meal rich in carbohydrates has been digested. In the liver the excess glucose is converted into glycogen as a temporary storage of energy so that blood leaving the liver usually has a concentration around 0.1% glucose.
3. (a) Lungs – Blood entering the lungs is deoxygenated, and oxygen diffuses from the alveoli into the capillaries so that blood leaving the lungs is oxygenated. This occurs because of the concentration gradient whereby there is more oxygen in the lungs causing the loading of oxygen into the bloodstream. The oxygen is needed for cellular respiration.
   (b) Kidneys – Blood in the renal artery has a high amount of urea while blood in the renal vein has very little urea. Urea is toxic and is filtered from the blood in the nephron of the kidney and excreted as urine.
4. B
5. A
6. B
7. D
8. A
9. D
10. D
11. C
12. A

50 Experiment – Scaled Diagram of Blood Cells

1. Diameter of field of view under low power is 1.6 mm.
2. (a) Changing from low power objective lens '×10' to high power objective lens '×40' reduces the field of view by one quarter. New diameter of field of view is 0.4 mm.
   (b) 0.4 mm = 400 μm.
3. 400 μm divided by 46 gives approximately 8.7 μm as the diameter of each RBC.
4. The usually accepted value for the diameter of a RBC is 8 μm. The student result of a diameter of 8.7 μm is a little large. Differences are most likely due to the degree of accuracy in the method of counting the RBCs. Since each RBC on the prepared slide of blood is not usually exactly touching the next RBC and they are not lined up in a neat row across the diameter, students need to consider the spacings between the RBCs and estimate how many would actually fit across the diameter. This method involves a certain decrease in accuracy of results, which would account for the slightly larger size of the RBC estimated by the students.
5. (a) The WBC observed by the students was either a neutrophil or an eosinophil.
   (b) Under the microscope WBCs can be distinguished from RBCs as the nucleus of a WBC is usually stained purple and the RBCs do not have a nucleus and appear as a pinkish donut shape.
6. B

51 Haemoglobin

1. Haemoglobin is a globular protein made of four polypeptide chains which are folded to enclose a haem group.
2. It is synthesised inside developing red blood cells in red bone marrow.
3. The haemoglobin value is the weight of haemoglobin in whole blood and is measured in grams per 100 mL, e.g. normal values for haemoglobin (Hb) for males is 13 to 18 g/100 mL and for females 11.5 to 16.5 g/100 mL.
4. Three functions are transport of oxygen, transport of carbon dioxide and to act as a buffer to prevent dramatic changes in blood pH.
5. A
6. D

52 The Lymphatic System

1. Lymph is a colourless fluid found in the lymphatic system. Interstitial fluid becomes lymph when it enters lymph capillaries.
2. The movement in the lymphatic system is one-way as it takes interstitial fluid into lymph capillaries where it become lymph to enter blood circulation via the thoracic ducts.
3.

Feature	Lymph vessel	Blood vessel
Capillary wall	Single celled wall	Single celled wall
Capillary diameter	Larger, more irregular	Smaller, more regular
Vein wall	Thinner	Thicker
Valves	Present in veins to prevent backflow	Present in veins to prevent backflow
Fluid removal	Removes excess fluid from tissues	Removes fluid from tissues
End of capillary	Blind-ended with only one end connecting to lymph vessel	Connects both ends to blood vessels
Permeability	More permeable to insoluble substances	Less permeable to insoluble substances
Contents	Lymph, leucocytes	Plasma, red blood cells, leucocytes, platelets
Circuit	One-way movement	Continuous circuit

4. The organs in the lymphatic system include lymphatic vessels, lymph nodes, spleen and thymus.
5.

Body system	Interaction with lymphatic system
Cardiovascular system	Returns fluids and proteins filtered out of the capillaries of the circulatory system
Digestive system	Picks up fats absorbed at the small intestine and transfer them to the circulatory system
Immune system	Filters, captures and destroys invading pathogens at lymph nodes

6. Lymph valves are found along lymph veins and prevent backflow.
7. In mammals lymph is moved by the contractions of body muscles with the valves stopping backflow.
8. Lymph nodes are bean shaped structures with a fibrous capsule that vary in size from 0.1 to 2.0 cm depending on location and the health of the person.
9. Main functions of the lymphatic system are: 1. Tissue fluid is collected and returned to the bloodstream. 2. Transports fats absorbed from the digestive tract to the bloodstream. 3. The permeable walls of lymph vessels allow blood proteins lost by leakage from blood capillaries to be returned to the blood via the lymph. This maintains normal osmotic balance between blood and tissue. 4. Lymph nodes filter tissue debris and invading pathogens from the lymph and phagocytes destroy and remove these substances. 5. Non-digestible particles, e.g. dust and soot that phagocytes cannot remove are stored in lymph nodes.
10. The recycling of blood proteins is highly important as without the return of the proteins a person would die within about 24 hours. Blood proteins are lost by leakage from blood capillaries and the permeable walls of the lymph capillaries collects these proteins which flow in the lymph to be returned to the blood maintaining normal osmotic balance between blood and tissue.
11. (a) Oedema is caused by an accumulation of interstitial fluid in an area. The flow of lymph in the lymphatic system relies on the movement of muscles. An immobilised limb has limited muscle movement which limits drainage in the lymphatic system and thus tissue fluid accumulates causing oedema in the immobilised limb.
    (b) Usual treatment recommends that the patient raises the injured limb above the level of the heart so that gravity will assist lymph to flow back to the blood.
12. Histamines released in the inflammation response increase the permeability of the walls of blood vessels. This allows plasma proteins, e.g. antibodies to move into the interstitial fluid. This changes the osmotic balance which leads to swelling in the affected region. Drainage by the lymphatic system is interrupted and nearby lymph nodes will release macrophages and neutrophils to aid the inflammation response.
13. When the parasites block the lymph vessels, lymph can no longer drain back to the circulatory system. This means that interstitial fluid is trapped in the lower limbs thus causing massive enlargement of the legs.

14. A = lymph valve
    B = afferent lymph vessel
    C = cortex/follicle/germinal centre
    D = medulla
    E = capsule
    F = efferent lymph vessel
15. A
16. D

53 Play – In the Blood

Scene 1 – The End of a Long Bone in a Leg

1. RBCs form in bone marrow.
2. RBCs are biconcave disc shape, approximately 8 μm in diameter.
3. They do not have a nucleus and hence have limited life span.
4. RBC feeling 'blue' as it is not carrying any oxygen.
5. Hepatic portal vein takes products of digestion from the small intestine to the liver and hence they can have afternoon tea there as it is full of 'food'.

Scene 2 – The Foot

1. Phagocyctes are specialised WBCs which engulf and destroy bacteria.
2. When bacteria enter, phagocytes rush to the area to engulf and destroy bacteria.
3. Platelets are involved in clotting blood.
4. Pus is dead cells, bacteria and WBCs.

Scene 3 – The Hepatic Portal Vein

1. Sugar-containing foods are broken down into glucose in the digestive system. Then the glucose diffuses across the wall of a villus in the small intestine into the bloodstream.
2. Reaction is respiration.
   Glucose + oxygen → carbon dioxide + water + energy
3. Respiration occurs in all living cells.
4. In the lungs RBCs pick up oxygen.
5. 45 seconds.

Scene 4 – A Small Capillary

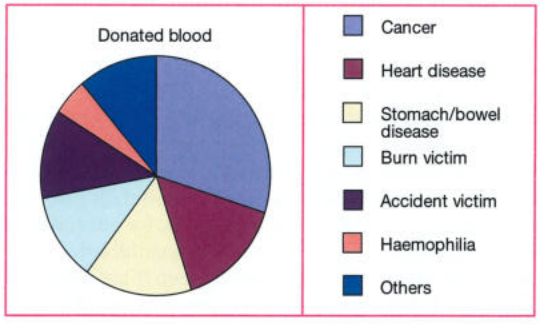
1. Glucose moved through the wall by diffusion.
2. Losing the oxygen made the RBC blue.
3. Insulin is made in the pancreas.
4. Insulin causes glucose in the bloodstream to be converted into glycogen for storage.
5. Lack of sugar causes insulin shock.
6. Blood contains RBCs, WBCs, platelets, hormones, glucose.

54 Donated Blood and Its Products

1. Blood is a type of connective tissue that contains cells in a liquid matrix (plasma).
2. An average adult has approximately 5 to 6 litres of blood.
3. Functions of blood – 1. Transport of nutrients from the alimentary tract and oxygen from the lungs to cells. 2. The transport of wastes to excretory organs. 3. It carries heat for thermoregulation from active tissues to less active tissues. 4. Transports hormones for coordination. 5. Functions in osmotic regulation and stabilising pH. 6. Transports substances for defence, e.g. antibodies. 7. Transports clotting factors.
4. Directed blood is given by friends or family and is donated for a specific person. Standard donation involves the removal of 450 mL of blood for a non-specific purpose and usually takes 15 minutes. Autologous donation is when a patient's own blood may be collected for future transfusions.
5. There are several regulations concerning the eligibility of people allowed to donate blood, e.g. age, recent illness, past medical history and lifestyle issues. There are also regulations if you have just had your ears pierced or had other body piercings or if you have travelled to an overseas area that has a malaria risk.
6. There needs to be a constant supply of donated blood as platelets only have a shelf life of up to 5 days, red cells have a shelf life of up to 42 days and plasma can only be stored for up to one year.
7. Most donated blood is used to assist people with major medical conditions such as heart disease, cancer, e.g. leukaemia, haemophilia or diseases of specific organs, e.g. stomach, bowel, liver or kidney. Products of donated blood are also used to assist burn and accident victims or used in emergency surgery.
8. Main components of blood are – 1. Red blood cells – make up around 40% of the volume of blood and the red haemoglobin combines with oxygen to form oxyhaemoglobin. Haemoglobin transports oxygen from the lungs to needy cells. 2. Platelets – are fragments that assist in the clotting of blood. If the platelet count is too low, people may suffer from spontaneous bleeding. 3. Plasma – is the straw coloured liquid that is up to 90% water and carries many substances in solution, e.g. urea, carbon dioxide, salts, amino acids, hormones, vitamins and antibodies.
9. Apheresis is a collection process where a separating machine takes blood from the arm and separates out and collects the plasma or platelets and returns the rest back into the arm.
10. (a) Donated whole blood is normally separated into several fractions, e.g. red blood cells, platelets, fresh frozen plasma and granulocytes.
    (b) Whole blood is centrifuged and because red blood cells weigh the most they will settle to the bottom. Red blood cells contain the iron-rich pigment haemoglobin which carries oxygen in the blood. The red blood cells make up approximately 47% of the blood volume in an adult male.
    (c) Red blood cells are needed for patients who suffer from chronic anaemia, or have had acute blood loss from trauma or surgery.
11. (a) Blood is also tested for infectious diseases – HIV/AIDS, hepatitis B, hepatitis C, syphilis and human T cell lymphotropic virus (HTLV).
    (b) Donated blood is tested for blood type, Rh group and red cell antibodies.
12. Donated blood is separated into components by centrifuging at different speeds.
13.

Product extracted from donated blood	Use of Product
Red cells	Treat people with severe anaemic or accident victim with severe bleeding or whose red blood cells do not function correctly.
Platelets	Treat people with cancers, leukaemia, or people undergoing chemotherapy.
Plasma	Fresh frozen plasma is used to treat patients who have clotting problems, e.g. liver transplants, trauma.
Cryoprecipitate	Prepared from fresh frozen plasma containing blood clotting factors, e.g. fibrinogen and factor VIII used for liver transplants, haemophilia A.
Immunoglobulins	To treat people who need antibodies against common infectious diseases.
Anti-D	To prevent Rhesus disease in the newborn.

14.



15. Low temperatures slow down microbial growth and increase the time that blood can be stored and used. Blood being transferred from one location to another needs the maintenance of low temperatures to reduce the spread of disease and increase patient safety.
16. B
17. A
18. D

55 Artificial Blood

1. Artificial blood is a term to describe 'man-made' products that fulfil some of the functions of biological blood. The term is misleading as no product fulfils all functions of normal blood – products are either oxygen carriers or volume expanders. Artificial blood is not involved in clotting and coagulation and immune defence.

2. Four reasons – 1. They are free of infectious agents and allergens making them non-toxic and disease free. 2. They do not need refrigeration for storage and can be kept for a longer period, e.g. 2 to 3 years compared to 3 weeks for human blood. 3. There is universal acceptance by all blood groups allowing transfusion to a person without any tests. 4. It is readily available in large supplies solving the worldwide problem of blood donors.
3. Research into artificial blood is needed as there are current supply problems with donated blood to meet needs, the shelf life of donated blood is limited and there is concern about contamination with infectious diseases, e.g. HIV.
4. A successful blood substitute needs to be: 1. Carry oxygen and perform some of the tasks carried out by blood. 2. Free of disease and will not cause an immune response. 3. Have a long shelf life. 4. Universally compatible with all blood types. 5. Eeasily obtained and transported.
5. Volume expanders are fluid solutions and have been developed to increase blood volume, e.g. in emergency situations by paramedics and in first response situations. Two main types have been developed – crystalloid solutions which contain salts and/or sugars, e.g. saline solution or Ringer's lactate and colloidal solutions which contain proteins, starch or gelatine, e.g. using albumin.
6.

Feature	Perfluorocarbons	Haemoglobin based oxygen carriers
Structure	Chemical compound based on fluorocarbons.	Haemoglobin derived from humans, animals or artificially made using recombinant technology.
Function	Carry and release oxygen.	Carry and release oxygen.
Examples	Oxycyte, Fluorovent.	PolyHeme, Hemolink, Hemopure.
Benefit	Can carry large amounts of oxygen and the amount of oxygen picked up is directly proportional of the amount of oxygen breathed in.	Have long shelf life, do not require refrigeration and is disease free.
Disadvantage	Do not freely dissolve in plasma and must be emulsified before being injected into the patient.	Can cause vasoconstriction and gastrointestinal side effect, e.g. nausea. Also can be associated with kidney problems if size of haemoglobin has been increased.

7. D

56 Blood Transfusions

1. Blood transfusion involves the injection of blood into a person usually from one person to another.
2. Incorrect matching of blood groups for both the ABO system and RhD group will lead to complications. In an emergency situation when the patient's blood group is unknown 'O Rh negative is given.
3. Blood transfusions can be used to treat many different diseases and conditions, e.g. for cancers, for operations, for anaemia, iron deficiency, bleeding disorders, pregnancy and childbirth.
4. Blood group A has alleles I^AI^A or I^Ai. Blood group B has alleles I^BI^Bor I^Bi. Blood group AB has alleles I^AI^B. Blood group O has alleles ii.
5. The surface of red blood cells can have antigen A and/ or antigen B, or no antigens.
6. Blood group is determined genetically which causes the formation of antigens on the surface of red blood cells.
7. D-antigen is the most important antigen of the Rh blood groups system and if present the person is Rh+.
8. Around 17% of the human population is Rh−.
9. If a Rh− female has a Rh+ child and then falls pregnant and is carrying another Rh+ child maternal anti-D antibodies can cross the placenta to act against the foetus causing haemolytic disease. To stop the next child being potentially a 'rhesus baby' a Rh− mother who gives birth to a Rh+ baby is given an injection of anti-D antibodies to destroy the Rh+ cells that have entered her circulation. This destroys the cells before her own immune system can be sensitised to Rh+ blood.
10. O− has been called the universal donor as patients of all blood types can receive these transfusions without agglutination occurring.
11. Agglutination is the clumping of red blood cells due to the combination of an agglutinogen and its corresponding agglutinin (antigen with antibody).
12.

Type of transfusion	Descriptions
Red cell transfusion	For low haemoglobin or red cell count if a person has anaemia.
Platelet transfusion	If the person has a low platelet count.
Plasma	Fresh frozen plasma (FFP), e.g. to replace low or missing blood proteins or blood volume in a range of medical situations.
Cryodepleted plasma	Is plasma with some blood clotting proteins removed and is used for thrombotic thrombocytopenic purpure (TTP).
Cryoprecipitate	Has clotting proteins and fibrinogen and is used in massive transfusions where large numbers of blood components are needed to help with clotting.

13. The main reasons for blood transfusions are – 1. Blood loss – excessive bleeding, e.g. haemorrhage can lead to large loss of blood that affects blood volume and circulation. 2. Thrombocytopenia – spontaneous bleeding due to insufficient numbers of platelets. 3. Severe anaemia – insufficient numbers of functioning red blood cells reduces the amount of oxygen being supplied to cell threatening whole body metabolism. 4. Operations – many types of surgery require blood transfusions.
14.

Complication	Descriptions
Haemolytic reaction	If blood types are not correctly matched the patient's body will destroy and break down the transfused red blood cells.
High fluid volume	The introduction of blood too quickly can overload the circulatory system. Increased blood volume can distend blood vessels and reduce their resistance and they reduce venous return.
Allergic reaction	The immune system of the recipient may be activated causing itchiness, dizziness, headaches and breathing difficulties.
Transfusion related acute lung injury (TRALI)	Agglutination and other reactions to transfused blood can cause blockages in blood vessels in the lungs. This can affect breathing capacity leading to low oxygen levels which in turn affects metabolic processes and lead to life threatening situations.

15. B
16. C

57 The Digestive System

1. The digestive system consists of the alimentary canal and its associated organs.
2. The alimentary canal is a long tube running from the mouth to the anus.
3.

Food group	Enzyme	Product of digestion
Carbohydrate	Amylase	Glucose
Protein	Protease	Amino acid
Lipid	Lipase	Glycerol and fatty acids

4. Mechanical digestion occurs in the mouth where the teeth bite and chew food into smaller pieces and in the stomach where the muscles churn the food mixing the food with the gastric juice.
5. Amino acids, carbohydrates and minerals are absorbed into the bloodstream, while fatty acids and glycerol enter the lymphatic system.
6. Features that aid the absorption of digested food include: 1. The long length of small intestine. 2. Villi, covered with microvilli increase the surface area for absorption. 3. The thinness of the epithelial lining. 4. A rich blood supply in each villus.
7. In the bloodstream, glucose is carried to the liver by the hepatic portal vein where it may be stored or altered as needed by the body. If there is excess glucose it is stored as glycogen and when blood sugar levels fall, the glycogen is turned back into glucose and released so that blood sugar levels remain fairly constant.

8. Once in the lacteal, the fatty acids are recombined to form fats once again. The fats are often coated with special proteins, becoming lipoproteins for transportation. The lymphatic system transports the lipids from the digestive system to the circulatory system, draining into veins at two locations near the shoulders.
9. A
10. B
11. C
12. D
13. A
14. C
15. C
16. D
17. C
18. D
19. A

58 Teeth

1. The four types of teeth are incisors, canines, premolars and molars. Incisors are chisel shaped teeth at the front and are used for biting. Canines have a sharp point and are used for tearing meat. Premolars are used for crushing food and molars are used for grinding food.
2. The total number of teeth in a fully grown adult is 32 teeth. However, the last molars to grow (the wisdom teeth) do not usually appear until 18 years old or older. Since many people look after their teeth, there is often no room in their mouth for the wisdom teeth and they have them removed. Thus when comparing the number of teeth present in a person's mouth and the diagram, age and dental history will have to be considered.
3. Saliva dissolves some of the food, helps to lubricate the food and makes some small pieces stick together.
4. The enzyme in saliva is salivary amylase. Its function is to split starch into smaller units of the double sugar maltose.
5. Temporary teeth include the incisors, canines and premolars; they do not include the molars. The benefit of temporary teeth is that the first set of teeth can be smaller, softer and fewer in number and do not have to last a whole lifetime. As the child grows, the second set replaces this first set and each tooth can be larger in size and more teeth can fit into the mouth.
6. In the mouth mechanical digestion occurs when the teeth break the food into smaller pieces so that they can be swallowed or enzyme action can be increased. Chemical digestion is the action of salivary amylase which breaks down starch into maltose.
7. Digestion of starch stops when food reaches the stomach as the stomach is quite acidic and the salivary amylase is inactivated by low pH.
8. C
9. A
10. D

59 Play – Amino and Lipidet, A Most Lamentable Tragedy

Scene 1 – The Mouth

1. Food groups in fish and chips are carbohydrates, proteins and lipids.
2. Carbohydrate – Prince Carbohydrate and Glucose
   Protein – Protein, Amino, Glycine
   Lipid – Lipid, Lipidet, Glycerol, Phospholipid, Glycerol, Steroid
   Minerals and Ions – Vitamin A
3. Fat stores 4000 kJ energy per 100 grams while protein only stores 1720 kJ energy per 100 grams.
4. Glycine is the smallest of the 20 amino acids found in proteins. It is a precursor to proteins.
5. Salivary amylase breaks down starch in the mouth beginning the digestion of carbohydrates.
6. Peptide bonds hold amino acids together in a polypeptide.
7. Proteins make up more than 50% dry mass of most cells.
8. It is dangerous to be a protein in the stomach as protein digestion begins in the stomach. The pH of the stomach is 2 due to the presence of hydrochloric acid (HCl) which activates the enzyme pepsin which begins the breakdown of proteins by breaking the peptide bonds. Lipid digestion does not begin till the small intestine and it is thus safe for lipids to be in the stomach.

Scene 2 – The Small Intestine

1. Gastric acid kills bacteria and other micro-organisms that are ingested. The acid also denatures many of the proteins destroying their tertiary structure which allows enzymes to attack the bonds.
2. An amino acid has a central carbon atom with an amino group and a carboxyl group attached.
3. Carrots can be broken down to produce vitamin A as carotene can be split to form two molecules of vitamin A.
4. Vitamins were named as 'vital amines' and the two words were joined.
5. Vitamin A as a fat-soluble vitamin is absorbed with the fats across the wall of the small intestine.
6. Enzymes are proteins.
7. In the small intestine there are pancreatic amylases to break down the polysaccharides, pancreatic trypsin to break down the polypeptides and bile salts and pancreatic lipases to break down fat globules.

Scene 3 – Epithelial Cell in the Small Intestine

1. A monoglyceride can enter an epithelial cell of the small intestine by diffusing across the membrane.
2. Free amino acids can enter an epithelial cell by the sodium-dependent cotransport proteins.
3. Chylomicrons are triglycerides with cholesterol and phospholipids surrounded with a protein coat.
4. The function of a chylomicron is to transport lipids from the intestines around the body.
5. Chylomicrons are packed into a secretory vesicle and leave the epithelial cell by exocytosis as the size of their vesicle is too large to cross the basement membrane.
6. The vesicle will be absorbed into the lacteal and pass through the lymphatic system. It will enter the venous blood just before it reaches the heart.
7. There are a range of lipoprotein complexes as various ratios of protein, triglyceride, phospholipids and cholesterol lead to the formation of very low-density lipoprotein (VLDL), a low-density lipoprotein (LDL) or a high-density lipoprotein (HDL). The higher the amount of protein present, the heavier the weight.
8. 'Good cholesterol' is high-density lipoprotein and 'bad cholesterol' is low-density lipoprotein.

Scene 4 – The Bloodstream

1. Lipoprotein lipase is an enzyme found bound to the capillary endothelium of muscles and adipose tissue. Its function is to convert triglycerides in a chylomicron to fatty acids and monoglycerides.
2. Deamination is a process that removes an amino group from an amino acid to form ammonia.
3. Ammonia is highly toxic to cells and thus it needs to be quickly converted to urea.
4. The nitrogenous waste from proteins is urea. Urea is filtered out of the blood by the kidneys and excreted as part of urine.
5. An adipose cell is a fat cell.
6. Excess cholesterol is incorporated into bile and leaves the body from the digestive system.

60 Mechanical Digestion

1. Mechanical digestion involves the physical breakdown of food into smaller pieces. It involves teeth, bile and peristalsis.
2. Peristalsis is waves of muscular contraction of smooth muscle that moves food along the length of the digestive tract.
3. The gastrointestinal wall has four layers – the mucosa, the submucosa the muscular layer and the serosa.
4. Peristalsis involves relaxation of smooth muscle then contraction behind the chewed material to keep it from moving backward and then longitudinal contraction to push it forward.
5. (a) Bile is a solution secreted by the liver composed mainly of bile acids, bile pigments and cholesterol. It does not contain any digestive enzymes.
   (b) Bile salts are conjugated bile acids. This means the bile acid produced in the liver has been joined with an amino acid – either glycine or taurine.
   (c) Bile is stored in the gall bladder.

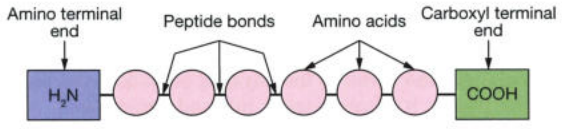
6. Bile salts emulsify fats acting like a detergent that breaks large droplets into smaller droplets.
7. Bile salts break lipid droplets into smaller droplets and do not change the chemical composition of the lipid. Bile salts do not contain an enzyme that will chemically react with the lipid.
8. Sphincters are ring-like valves that allow/prevent the movement of material from one section into the next.
9. Bile is needed in digestion to break large lipid droplets into small lipid droplets so that there is a greater surface area for the action of the lipase enzymes involved in chemical digestion of lipids.

61 Chemical Digestion

1. Chemical digestion involves the breakdown of food into simpler substances. It involves enzymes with distinctive operating conditions and function that are located in different sections of the digestive system.
2. Enzymes are organic catalysts that control the rate of reactions.
3. Digestive enzymes speed up the breakdown of large organic compounds into smaller molecules and thus required nutrients can be supplied at a faster rate for use in the body after ingestion.
4. (a) Two carbohydrate enzymes are amylases and disaccharidases.
   (b) Amylases break down polysaccharides into disaccharides, e.g. maltose and disaccharidases break down disaccharides into monosaccharides, e.g. glucose.
5. Carbohydrate digestion begins in the mouth with salivary amylase.
6. Gastric juice is a thin, almost colourless liquid secreted by glands in the lining of the stomach and contains hydrochloric acid (HCl) and the enzymes pepsin and rennin and mucus.
7. The main layers of the stomach wall are the epithelial layer, the connective tissue layer with nerves and blood vessels, the smooth muscle layer and the connective tissue and epithelial tissue layer.
8. An endopeptidase breaks apart long protein chains acting on the peptide bonds linking particular amino acids while an exopeptidase acts only on the end of a polypeptide chain splitting off dipeptides and single amino acids.
9.

Cell type in stomach wall	Function
Parietal cell	Secretes hydrochloric acid.
Chief cell	Secretes pepsinogen. Pepsinogens are activated to pepsins by hydrochloric acid and pepsin already in the stomach.
G cells	Secrete the hormone gastrin that passes into the bloodstream.

10. Dietary lipids are often in the form of triglycerides.
11. Chemical digestion of lipids involves the enzyme lipase which hydrolyses the triglycerides acting on the surface of the fat droplet forming free fatty acids and monoglycerides.
12. Colipase is a cofactor secreted by the pancreas. It displaces some bile salts that coat a lipid droplet allowing lipase access to fats inside the bile salt coating.
13.



14. (a) Pepsin needs an acidic environment.
    (b) Salivary amylase breaks carbohydrates into disaccharides.
    (c) Pancreatic lipase enters the alimentary canal at the beginning of the small intestine.
15. C
16. A
17. B

62 Accessory Digestive Organs

1. Accessory digestive organs are organs that aid digestion but are not a part of the alimentary canal and digestive tract, e.g. salivary glands, pancreas, liver, gall bladder and tongue.
2. (a) There are taste receptors for sweet, sour, bitter and salty.
   (b) For a substance to be tasted it must first dissolve in the saliva and mucus in the mouth. A tablet that is quickly gulped down has not dissolved and is thus not tasted.
3. The parotid salivary glands are on each side of the face. The submandibular salivary glands are on each side of the face under the angle of the jaw. The sublingual salivary glands lie under the floor of the mouth.
4.

Action of saliva	How it works
Chemical digestion	Salivary amylase begins digestion of ingested carbohydrates.
Lubrication	Ingested food is made into a bolus for swallowing.
Taste	Dry substances are dissolved in saliva and mucus which can then stimulate taste receptors in the tongue.
Cleansing the mouth	Cleanses and lubricates the mouth keeping tissues moist, soft and pliable.
Non-specific defence	Lysozymes, clotting factors and antibodies act against microbes in ingested food and water.

5. The pancreas is a pale grey gland that weighs about 60 grams and has both endocrine and exocrine tissues. It lies parallel to and just beneath the stomach.
6. The exocrine part of the pancreas has secretory cells that produce digestive enzymes and an alkaline solution that drain into small ducts.
7. (a) Pancreatic amylase is needed to break down carbohydrates into disaccharides.
   (b) Pancreatic lipase is needed to break down fats into fatty acids and monoglycerides.
   (c) Trypsin and chymotrypsin are needed to break peptide bonds next to certain amino acids and carboxypeptidase breaks peptides into amino acids.
8. The liver produces bile which is needed to emulsify fat droplets into smaller droplets.
9. (a) Hepatocytes are liver cells.
   (b) Hepatocytes synthesise the components of bile, e.g. bile acids, cholesterol and other organic components.
10. (a) The gall bladder is attached to the posterior surface of the liver by connective tissue.
    (b) The gall bladder stores bile.
11. The bicarbonate ions are needed to neutralise the acidity of the chime that comes from the stomach.
12. From the diagram the hormone CCK causes the contraction of the gall bladder and the relaxation of the sphincter at the mouth of the bile duct. This means that bile will leave the gall bladder and move into the duodenum to mix with the food mass to emulsify the fats eaten in the fatty meal.
13. B
14. D
15. A
16. C
17. A

63 Absorption

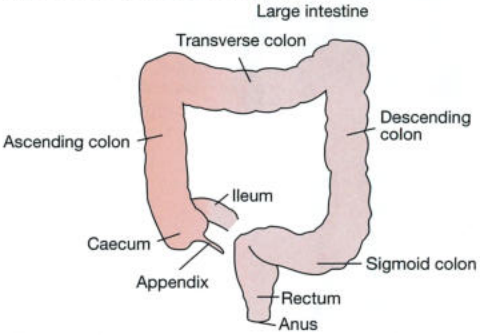
1. Absorption is the movement of small nutrients into the body across a membrane, e.g. from the small intestine into blood capillaries.
2. Assimilation is the use of absorbed nutrients to form parts of an organism.

3.

Method for absorption	Explanation of process
Simple diffusion	Lipids are absorbed into the cell by simple diffusion across the apical membrane. At the brush border surface, fatty acids and monoglycerides move out of micelles and into the epithelial cell. It is passive transport with substances moving down a concentration gradient. Water uptake often follows the osmotic gradient of sodium, potassium and ATPase.
Facilitated diffusion	Is passive transport with substances moving down a concentration gradient with the assistance of membrane proteins, e.g. glucose uses a glucose transporter. Amino acid structure is so variable there are multiple amino acid transport systems – many use facilitated diffusion.
Active transport	Requires energy in the form of ATP. Substances are often moved against a concentration gradient. Ion uptake, e.g. sodium, calcium, magnesium and iron are taken up by active transport. The sodium/potassium pump is highly important in maintaining the electrochemical gradient across the epithelial cell boundary of the lumen.
Endocytosis	Endocytosis is a form of active transport requiring energy input. Larger peptides are carried into the cell in vesicles. Immunoglobulins, e.g. IgA from plasma cells in the mucosa is absorbed through receptor mediated endocytosis and released into the intestinal lumen.

4. Amino acids, carbohydrates and minerals are absorbed into the bloodstream while fat soluble vitamins, fatty acids and glycerol enter the lacteal and the lymphatic system.
5. Some proteins are absorbed unchanged, e.g. immunoglobulins in breast milk and oral vaccines.
6. Tight junctions are cell to cell junctions in epithelia that do not allow much movement of material between the cells.
7. (a) Poor absorption is due to the absence of villi and the junctions between the epithelial cells are tight junctions.
   (b) Only small quantities of some highly lipid soluble substances, e.g. alcohol and some drugs, e.g. aspirin are absorbed through stomach wall.
8. Villi and microvilli increase surface area which increases the efficiency of absorption.
9. The middle section of the small intestine has many plicae each with many villi which increases the surface area for absorption of the products of digestion. Whereas the last section has fewer plicae, villi become smaller and there is less absorption of digested products.
10. Humans lack the necessary enzyme, cellulase to digest cellulose. The cellulose in plant matter in a human diet becomes fibre (roughage). Cooking breaks down cellulose so it can be digested.
11. Vitamins A, D, E and K are fat soluble and are absorbed in the same way as lipids. Whereas vitamins B and C are water soluble and can be absorbed directly into the blood from the small intestine.
12. Peyer's patches are sections of lymphoid tissue in the last section of the small intestine that help prevent the spread of potentially harmful bacteria that inhabit the large intestine.
13. Maltase completes the digestion of maltose to two glucose molecules. Sucrase completes the digestion of sucrose to a glucose and a fructose molecules. Lactose completes the digestion of lactose to a glucose and a galactose molecule.
14. Mechanical digestion in the small intestine involves peristalsis and the movement of food along the tract. Chemical digestion is completed in the small intestine with the large surface area providing increased contact for food substances and digestive juices containing enzymes. The large surface area of villi and microvilli with the close proximity of the capillaries and lymph vessels inside the villi aid absorption of digested food products into the bloodstream and lymphatic system.
15. C
16. A

64 Elimination

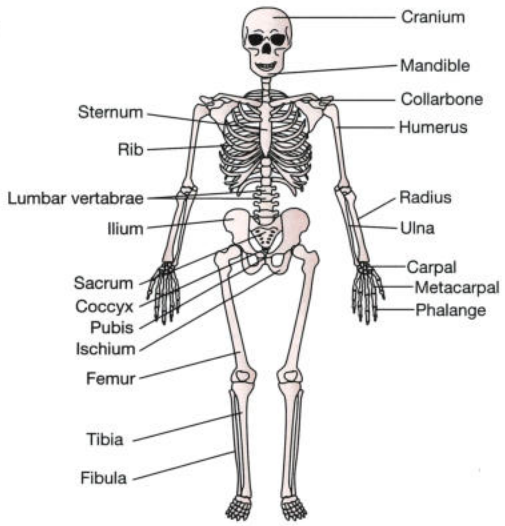
1. Elimination is the passing of undigested material out of the alimentary canal.
2. The human large intestine consists of the caecum, appendix, colon and rectum.
3. The movement of material into the large intestine is controlled by a sphincter muscle and valve.
4. 

5. The small intestine prevents migration of bacteria from the large intestine to the small intestine by Peyer's patches in submucosa layer of the wall of the small intestine which contains lymphoid nodules. The patches are most abundant at the end of the ileum and are part of the immune system.
6. Materials that are not absorbed and pass through the large intestine to be eliminated include: cellulose, lignin, bile pigments, bacteria, mucus and dead mucosal intestinal cells.
7. Differences in the walls of the small intestine and large intestine relate to their different functions. The wall of the large intestine is much thinner than the wall of the small intestine and the diameter of the colon is approximately three times the diameter of the small intestine. The large intestine lacks villi. The small intestine is concerned with completing digestion and the absorption of nutrients while the large intestine is concerned with the compaction of intestinal contents into faeces and the reabsorption of water and electrolytes.
8. Colonic bacteria produce significant amounts of absorbable vitamins, especially vitamin K and also biotin, folic acid and several B vitamins. Other by-products produced by the bacteria include gastrointestinal gases, e.g. hydrogen sulfide and methane. Faeces contains masses of bacteria.
9. Mass movement occurs when a strong peristaltic movement occurs (about twice an hour) that forces the contents along all sections of the colon.
10. A

65 The Musculoskeletal System

1. The musculoskeletal system provides support, maintains posture, gives stability and produces movement for the body.
2. It consists of muscles, cartilage, tendons, ligaments, joints and connective tissue.
3. The axial skeleton consists of the skull and vertebral column. The appendicular skeleton consists of the limb girdles and limbs.
4. The skull consists of 22 bones.
5. A suture is an immovable joint that has interlocking edges held together by a small amount of fibrous tissue.
6. (a) Ligaments are fibrous connective tissue that joins bones together at joints.
   (b) In the axial skeleton ligaments hold vertebrae together.
7. Two girdles are the shoulder girdle and pelvic girdle.
8. The pelvis is examined as the female pelvis is more shallow and rounded than the male pelvis to allow for the passage of the baby during childbirth.
9. The pentadactyl limb is the 5-digit limb.
10. In the uterus, the foetus curls up and the primary curve develops. At around 3 months old when the child holds up their head the secondary cervical curve develops and at around 12 to 15 months old when the child stands upright the secondary lumbar curve develops.

11.



12. D
13. A
14. B
15. C
16. D
17. B

66 Muscles

1. Three types are skeletal muscle, smooth muscle and cardiac muscle.
2. Limb muscles consist of skeletal muscle fibres, connective tissue, blood vessels and nerves.
3. Skeletal muscle appears as dark and light bands because of the arrangement of the filaments and is thus often called striated muscle.
4. Skeletal muscle consists of muscle fibres which consist of myofibrils which consist of myofilaments which consist of repeating units called sarcomeres.
5.

Component	Description
Striated muscle	Consists of a series of parallel muscle fibres held together.
Muscle fibres	Are very large and multinucleated with dark and light bands. The sarcolemma (cell membrane) surrounds the sarcoplasm (cytoplasm) and the sarcoplasm contains many myofibrils.
Myofibril	Myofibrils are cylinders (1 to 2 μm diameter) and as long as the entire cell. Myofibrils can shorten and cause contraction. Each myofibril is attached to the sarcolemma at each end of the cell thus contraction shortens the cell. Myofilaments are surrounded by sarcoplasmic reticulum which forms a network with transverse tubules.
Myofilament	Are protein filaments consisting mainly of actin and myosin. Actin is in thin filaments and myosin is in thick filaments. The actin and myosin are organised in sarcomeres which repeat along the myofilament. The arrangement of the thin and thick filaments gives the dark and light banded appearance of a myofibril.

6. A sarcomere is the fundamental repeating unit of a skeletal muscle myofilament and is the segment between two Z lines.
7. The I band is the area near the edge of the sarcomere where there is only thin filaments. The thin filaments are attached to the Z line. The A band is the central region determined by the length of the thick filaments where the thin and thick filaments overlap including the H zone where there are only thick filaments.
8. The sliding filament theory provides a mechanism for muscle contraction. During a contraction neither thin (actin) filaments nor thick (myosin) filaments change in length, but slide past each other when the myosin heads of the thick filaments bind to the active sites on the thin filaments.
9.

Electron micrograph X	Electron micrograph Y
Contracted form of sarcomere – the thin filaments and Z line have been pulled towards the M line and the I band has reduced width.	Relaxed form of sarcomere as the I band is full width.

10. The thin filaments consist of the protein actin whereas the thick filaments consist of the protein myosin.
11. Striated muscle is capable of rapid contraction though the contraction cannot be held as long as a contraction of smooth muscle.
12. Muscle fibres contain a large number of mitochondria to supply the energy needed for contraction.
13. (a) There are more muscle fibres in the middle of a muscle than at the ends of the muscle.
    (b) A group of muscle fibres is surrounded by connective tissue.
14. D
15. A
16. C
17. B
18. D

67 Musculoskeletal Terminology

1. Many anatomical terms have Latin or Greek roots.
2. Body is divided into head, neck, trunk and limbs.
3. Superior means towards the head while inferior means towards the feet.
4. (a) Capitis refers to the head.
   (b) Thorax refers to the chest.
   (c) Abdomen refers to the lower trunk region.
5. (a) Flexion means bending that decreases the angle between two parts and its opposite is extension which usually straightens out a bent part.
   (b) Flexion is the bending of the forearm upwards at the elbow.
6. Circumduction is a combination of flexion, extension, abduction and adduction. All four movements bring about the hand being able to move in a circle and the leg being able to move in a circle. An individual finger is also capable of circumduction.
7. Ventral means towards the front of the body while dorsal means towards the back of the body.
8. Proximal means closer to the trunk or point of origin while distal means farther from the trunk or point of origin.
9. Articulation is a joint between two or more bones and an articulating surface is the part of the bone that enters into the formation of a joint.
10. The diagram shows that atlas/axis articulation forms a swivel joint that allows rotation of the head from side to side and also allows a nodding movement.
11. Ligaments are dense bands of connective tissue that join bones to bones.
12. Tendons are connective tissue fibres that connect muscle to bone.
13. Motor neurons take messages from CNS to muscles for voluntary movements or reflex arc reactions. If the nerve supply is cut then the muscle cannot contract, stopping movement. Over time his can lead to atrophy of the muscle unless the nerve regenerates.
14. D
15. A
16. A

68 Joints

1. The three types of joints are: 1. Fixed or fibrous joints. 2. Cartilaginous or slightly movable joints. 3. Synovial or freely moving joints.
2. Main types of synovial joints are: 1. Ball and socket joint. 2. Hinge joint. 3. Gliding joint. 4. Pivot joint 5. Saddle joints.
3. Fixed joints have fibrous tissue between the bones, e.g. collagen fibres.
4. Sutures are fixed joints between bones of the cranium.
5. A cartilaginous joint has a pad of fibrocartilage that forms a disc between the bones, e.g. between vertebrae in the vertebral column.

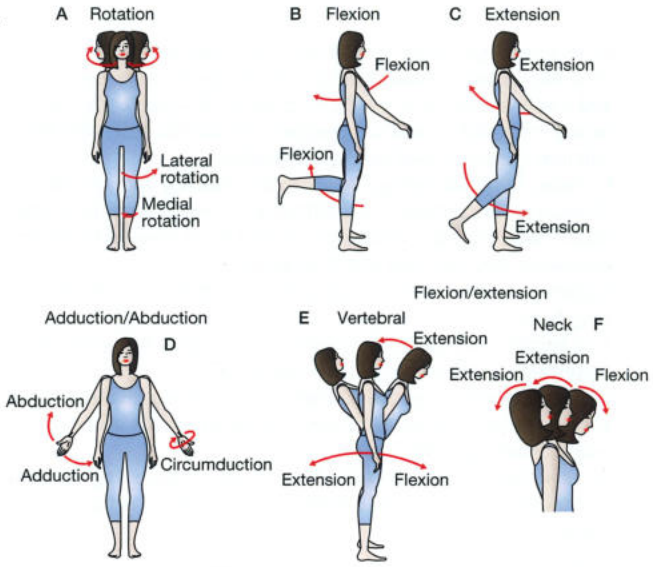
6. (a) Synovial joints are freely movable joints with a synovial cavity separating the bones.
   (b) There is a capsule (capsular ligament) surrounding the joint holding the bones together and an inner synovial membrane. The capsule consists of collagen fibres and dense connective tissue. The synovial membrane consisting of epithelial cells lines the capsule and secretes synovial fluid into the synovial cavity.
7. The synovial fluid – 1. Provides nutrients for the components of the joint cavity. 2. Acts as a lubricant. 3. Helps provide stability for the joint. 4. Has phagocytes for defence and removing cellular debris caused by the wear and tear of joint use.
8. Articular cartilage is at the ends of the bones where the bones are in contact and provides a smooth interface which can withstand compression during movement.
9. Ligaments are a type of connective tissue with fibres that join bones together at joints and prevent bones moving apart.
10. Tendons are a type of connective tissue made of fibres that connect muscle to bone so that when the muscle contracts the tendon pulls on the bone and the bone moves.
11. A dislocated joint occurs when the bone is displaced from its socket with accompanying damage to ligaments and sometimes tendons.
12.

Movement	Description
Flexion and extension	Changes the angle between body parts. Flexion decreases the angle, e.g. bending the knee and extension increases the angle, e.g. straightening the leg at the knee.
Adduction and abduction	Moves a structure with respect to the centre of the body. Abduction pulls a structure away from the middle of the body, e.g. moving the leg to the side while adduction pulls a structure towards the midline of the body, e.g. returning the leg to the body.
Rotation	Movement that spins a structure on an axis, e.g. rotating the tibia and fibula of the lower leg. Medial rotation (also called internal rotation) is towards the midline while lateral rotation is away from the midline.

13.

Feature	Ball and socket joint	Hinge joint
Example	Hip joint.	Knee joint.
Location	Found at the end of a long bone and bony surfaces do not contact each other due to articular cartilages.	Found at the end of a long bone and bony surfaces do not contact each other due to articular cartilages.
Function	Involved in walking.	Involved in walking.
Axes of movement	Multiaxial.	Movement in only one axis.
Kinds of movement	1. Flexion/ extension – flexion reduces the angle between the bones and extension increases the angle between the bones in anterior-posterior plane. 2. Abduction/adduction – abduction is movement away from the longitudinal axis of the body in frontal plane, adduction is moving it back. 3. Rotation.	1. Flexion/ extension – flexion reduces the angle between the bones and extension increases the angle between the bones in anterior-posterior plane. 2. Small amount of rotation possible.
Common synovial cavity	Present.	Not present.
Fat pad	Fat pad covers furrow in pelvis that holds the head of the femur.	Prominent fat pads provide padding around margins of the joint and assist reducing friction between patella and other tissues.
Capsule	Has large strong dense capsule.	No single unified capsule in the knee.
Stability	Extremely strong and stable joint partly due to massive capsule of hip joint.	Less stable than other hinge joints and the hip joint.

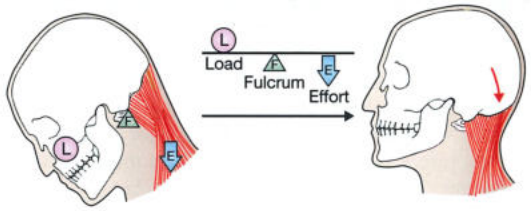
14.



15. A
16. C

69 Paired Muscles

1. Antagonistic muscles are a pair of muscles that oppose each other; one must relax while the other contracts, e.g. biceps and triceps.
2. A prime mover is the muscle that carries out an action, e.g. protagonist or agonist.
3. A synergist is the muscle that contracts at the same time as the prime mover and the antagonist opposes the prime mover.
4. A lever system is an arrangement of load, effort and pivot point that acts as a simple machine.
5. First class lever – the fulcrum is between the load and the effort. Second class lever – the load is between the effort and fulcrum. Third class lever – the effort is between the load and the fulcrum.
6. The third class lever system is the most common in the human body.
7. (a) Raising your head is an example of a first class lever system.
   (b)



8. When paired muscles are causing movement one muscle needs to contract and the other muscle needs to relax. To coordinate these two actions, nerves need to stimulate/inhibit the paired muscles to ensure smooth movement.
9. Motor neurons will stimulate a muscle to contract.
10. The diagram shows how the thigh muscle (muscle X) and the calf muscle (muscle Y) are part of pairs of antagonistic muscles that are involved in running. To straighten the leg and foot the buttock muscle, the thigh muscle and the calf muscle must all contract.
11. Proprioceptive receptors detect the positions of bones, joints and muscles monitoring the stretch of contraction of a muscle and the tension within a tendon.
12. Muscle tone is the resting tension in a skeletal muscle.
13. A muscle spindle is a sensory end organ in a muscle with proprioceptors that monitor the length of skeletal muscles.
14. Cellular respiration provides the energy for muscle contraction.
15. A
16. B
17. C
18. D

70 Bone

1. Bone makes up about 15% of the mass of an adult.
2. Hydroxyapatite is bone mineral and the main component of bone and tooth enamel. It is a form of calcium phosphate and can incorporate ions of sodium, magnesium and fluoride with the calcium carbonate and calcium phosphate
3. Compact bone and spongy bone can be identified with the naked eye.
4. Osteocytes are mature bone cells that secrete and maintain the matrix with both the release of calcium from bone to blood and the deposit of calcium salts into the matrix.
5. Epiphysis is the end of a long bone and diaphysis is the shaft of a long bone.
6. The epiphyseal plate is the cartilaginous region between the epiphysis and diaphysis of growing bone.
7. Lamellae are the concentric rings of bone matrix that surround a Haversian canal whereas lacunae are small spaces in the lamellae, each containing a bone cell.
8. A Haversian system is a cylinder of bone with circling layers of mineralised matrix called lamellae surrounding a central Haversian canal.
9.

Type of bone cell	Description
Osteocyte	Mature bone cells that secrete and maintain the matrix with both the release of calcium from bone to blood and the deposit of calcium salts into the matrix.
Osteoblast	Osteoblasts are bone-forming cells that secrete collagen and other parts of bone tissue. They are actively dividing bone cells.
Osteoclast	Osteoclasts are giant cells with many nuclei that help maintain optimum bone shape. They secrete acids by exocytosis from lysosomes. The acids dissolve the bony matrix and releases stored calcium and phosphate minerals in osteolysis.

10. Ossification is the process of laying down the inorganic parts of bone.
11. There needs to be a balance between osteoblasts which are producing matrix that binds minerals and osteoclasts which removing matrix and releasing minerals. If an imbalance occurs with the osteoclasts removing calcium salts at a faster rate than the action of osteoblasts the bone will weaken. If the imbalance has increased activity of the osteoblasts, then the bone will become more massive and stronger.
12. The medullary cavity is the space within the bone that contains marrow.
13. (a) Structure X is canaliculi which are little channels that radiate through the matrix from lacuna to lacuna and are the main transport route for supplies to and from the bone.
    (b) Structure Y is an osteocyte which is a mature bone cell in the lacuna.
14.

Compact bone	Spongy bone
Has Haversian systems.	No Haversian systems.
Relatively dense and solid.	Open network of plates and struts.
Forms walls.	Forms internal layer that surrounds the marrow.
Has concentric lamellae that form rings around the central canal.	Parallel lamellae form the struts and branching plates (trabeculae).
Heavier than spongy bone.	Lighter than compact bone.

15. (a) Periosteum is the outer covering on the bone surface. It consists of a fibrous outer layer and a cellular inner layer.
    (b) Functions include – 1. It protects the bone from surrounding tissue. 2. It is involved in bone growth and repair. 3. It is a place of attachment for blood vessels and nerves. 4. Near joints become continuous with the connective tissues that surround the joint and help stabilise the joint.
16. Red bone marrow tissue is found at the head of long bones and is the site of red blood cell formation. It is very soft tissue and less dense than bone whereas yellow bone marrow tissue is found in the shaft of the long bone and consists mainly of fat.
17. C
18. D
19. A
20. B

71 Cartilage

1. Cartilage is a type of flexible connective tissue consisting of intracellular material with collagen fibres in chondroitin sulfate and scattered cells.
2. Cartilage is a fibrous connective tissue as it consists of scattered cells in an intercellular material that is a mass of fibrous tissue.
3. (a) Chondroitin sulfates are complex polysaccharides that form complexes with protein to form proteoglycans.
   (b) Chondroitin sulfates make the matrix tougher and harder and give cartilage the ability to resist compression.
   (c) Osteoarthritis is often caused by loss of chondroitin sulfate and thus supplements of chondroitin sulfate are used in treatments for the disease.
4. Chondrocytes are cartilage cells and the only cells in cartilage matrix.
5. Cartilage is firm and can resist strain but also has flexibility with some degree of elasticity making it suitable to form framework structures in the body.
6. (a) Cartilage is avascular with no blood vessels as chondrocytes produce a chemical that stops the formation of blood vessels.
   (b) Being avascular means that the supply of oxygen and nutrients to chondrocytes is by diffusion and the removal of wastes also occurs by diffusion.
7. Since cartilage is avascular and relies on diffusion for the supply of nutrients and oxygen, damaged cartilage heals slowly as diffusion depends on the concentration gradient and is usually a slow process. Skin has a good blood supply and heals relatively quickly.
8. The perichondrium surrounds a chondrocyte and consists of an outer fibrous layer which gives support and protection and attaches to other structures and an inner cellular layer which helps growth and maintenance of the cartilage.
9.

Hyaline cartilage	Fibrocartilage	Elastic fibrocartilage
Has chondrocytes in small groups in a smooth and solid matrix and appears as a smooth blue-white tissue with no prominent fibres.	Has dense masses of white collagen fibres with widely spaced cells and a matrix similar to hyaline cartilage.	Has yellow elastic fibres in a solid matrix.
It provides stiff, slightly flexible support and reduces friction between bony surfaces.	It resists compression, limits relative movement and prevents bone to bone contact.	It provides support and can distort without damage and return to original shape.
It is found on the surface of parts of bones that form synovial joints. It forms part of the larynx, trachea and bronchi, part of the nasal septum and forms the costal cartilages that attach the ribs to the sternum. The entire skeleton of an embryo is made of hyaline cartilage.	It is found as ligaments joining bones, as intervertebral discs, on the rim of bony sockets such as the hip and shoulder joints, allowing the cavity to be deeper without restricting movement and forms the semilunar cartilages which are in the knee joint between articulating bone surfaces.	It forms the pinna (ear lobe), the external auditory canal, the epiglottis and part of the tunica media of blood vessel walls.

10. Cartilage is used to – 1. Support softer tissues, e.g. rings of cartilage in the trachea support the walls of the windpipe and prevent it from collapsing during inhalation and exhalation. 2. Form flexible connections, e.g. between sternum and ribs. 3. Act as a shock absorber, e.g. intervertebral discs. 4. Reduce friction between bony joints, e.g. bone surfaces at synovial joints,
11. With most staining techniques collagen fibres of hyaline cartilage are not usually visible using a light microscope, although the matrix is closely packed with collagen fibres. This shows the weakness of hyaline cartilage.

12.

Cartilage	Bone
Chondrocytes in lacunae.	Osteocytes in lacunae.
Matrix contains chondroitin sulfates forming proteoglycans.	Matrix contains crystals of hydroxyapatite with ions of sodium, magnesium and fluoride with calcium carbonate and calcium phosphate.
Can contain collagen, elastic fibres in different proportions.	Mainly collagen fibres.
Has covering of perichondrium with two layers.	Has covering of peristeum with two layers.
Has limited repair ability.	Has extensive repair ability.
Has limited strength, bend easily and hard to break.	Has strength, resists bending until breaking point reached.
Avascular.	Vascular.

13. D
14. A
15. C
16. B
17. B

72 Osteoporosis

1. Osteoporosis is a disease, primarily of ageing, that causes disability due to a reduction in bone mass and strength that compromises normal bone function.
2. Osteoporosis occurs when the amount of deposition is less than the reabsorption.
3. Spongy bone is first affected and thins.
4. Trabeculae are the small beams of osteocytes and matrix that develop along the lines of stress within bone. Trabuculae are found at the end of long bones, e.g. femur and provide strength and help withstand stress.
5. Decreased bone mass increases chance of bone fractures, e.g. wrist, hip, vertebrae.
6. Osteoporosis can also cause decreased height with age due to compressed vertebrae.
7. In both sexes peak bone mass occurs at around 30 to 35 years old.
8. After menopause lowered oestrogen levels can cause accelerated loss of bone mass in women. This means that females after menopause have lower bone density than men.
9. (a) Steroid hormones are lipids that can control the expression of specific genes and thus affect growth and development.
   (b) Steroid hormones include the corticosteroids (e.g. made in the adrenal cortex) and the sex steroids (e.g. made in the gonads or placenta).
   (c) The change in balance of steroid sex hormones (oestrogen and androgens) and corticosteroid hormones (e.g. glucocorticoids) affects bone deposition/reabsorption.
10. Risk factors that can cause decreased bone mass include lifestyle choices such as smoking, excessive alcohol intake, immobility, diet low in calcium, lack of exercise and drugs such as corticosteroids.
11. Prevention of osteoporosis involves having sufficient calcium and magnesium in the diet, adequate vitamin D to aid the absorption of calcium and regular weight bearing exercise that strengthens muscles and increases resistance.
12. Treatment of osteoporosis involves increased calcium and magnesium in the diet, increased vitamin D, regular exercise, no smoking and medication for some patients.
13. B
14. C

73 Osteoarthritis

1. Osteoarthritis is a degenerative non-inflammatory disease caused by the breakdown of joint cartilage and the underlying bone.
2. Osteoarthritis is caused when articular cartilage thins with age and wear and tear with the rate of renewal slower than the rate of thinning and cartilage breakdown.
3. Joints can become immobile when bones come into contact with each other and the ends of the bones begin to degenerate and the articular surfaces become misshaped.
4. Factors that affect the risk of osteoarthritis – 1. Genetics – family history. 2. Gender – females have a higher incidence than males. 3. Injury to joints and overuse.
5. Osteoarthritis causes joints to become swollen which leads to restricted movement of the affected joints and pain. Symptoms range in pain severity from stiffness to disabling pain.
6. Osteoarthritis causes joints to become swollen which leads to restricted movement of the affected joints and pain so that the key indicators are swelling and pain.
7. Common sites of swelling and pain are the hands, spine, hips and knees.
8. Diagnosis involves collection of data, e.g. about personal and family history, medical history, and a physical examination to test the range of motion of the joints and the position and alignment of the neck and spine.
9. (a) X-rays are used to show bone damage.
   (b) MRI (magnetic resonance imaging) shows images of the cartilage and other structures.
10. Treatment involves managing symptoms, e.g. pain management, reducing swelling and aiding stiffness. There is no cure. Some people have joint replacement. People are advised to maintain a healthy weight and keep active with exercise.

74 Excretion, Respiration and Water

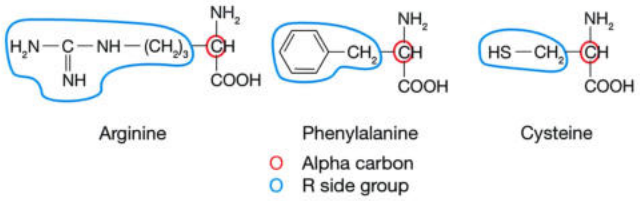
1. (a) The respiratory system ensures that gas exchange occurs so that oxygen enters the body and carbon dioxide leaves the body. The excretory system ensures the removal of metabolic wastes.
   (b) The two systems are highly integrated as the lungs act as both a respiratory organ and an excretory organ, e.g. gas exchange for respiration occurs in the lungs, but at the same time carbon dioxide and water are metabolic wastes and are excreted from the lungs.
2. (a) Water is an excellent solvent which allows many organic and inorganic substances to dissolve or be suspended thus allowing reactions to occur in solution and maintaining osmotic balance. Water has a high specific heat which means that water changes its temperature less when heat is either absorbed or released compared to many other materials which helps organisms to resist change in their body temperature. Water is highly cohesive, which means water molecules bond to their neighbours which assists transport of water in xylem against gravity.
   (b) Water must be kept constant as any change disrupts homeostasis. For example, a change in the concentration of a substrate will change the rate of reaction, or osmosis will occur causing oedema or dehydration.
   (c) The veins are carrying too much fluid which is not being excreted. The increased pressure causes the fluid to leak into the lungs which then reduces gas exchange so that cells receive insufficient oxygen and there is too much carbon dioxide in the bloodstream.
3. Respiratory surfaces of multicellular organisms need to have a large surface area for gas exchange, a concentration gradient across the membrane so diffusion will occur, a good blood supply to transport oxygen, carbon dioxide and water to and from the respiratory surface and need to be kept moist to assist the dissolving of gases so they pass across the membrane.

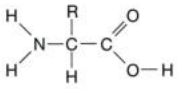
4. Respiratory systems maintain homeostasis by: 1. Their concentration gradient which moves air to the exchange surfaces of the lungs. 2. Protecting the lungs from drying out. 3. Removing carbon dioxide which assists in regulating blood pH.
5. Nearly every chemical reaction in cells involves water and the concentration of water must be kept relatively constant to maintain pH, maintain substrate concentration and the other conditions necessary for metabolic reactions. For example, because water acts as a solvent and many substances, e.g. nutrients, oxygen and wastes are carried to and from cells in water, the concentration of water determines the amount of substances being carried and their concentration in the blood or tissues. Water helps maintain a constant body temperature as water has a very high heat capacity which means it can absorb and retain heat.
6. C
7. A
8. B
9. D

75 Excretory Organs

1. Excretion is the removal of metabolic wastes from the body.
2. Metabolic wastes are excreted from the lungs, the skin, the digestive system and the kidneys.
3. The lungs excrete most of the carbon dioxide and a lot of water from the body.
4. Sweat glands are involved in excretion as they remove metabolic wastes from the body, e.g. some urea, salt (sodium chloride), wastes from carbohydrate metabolism and water (though the loss of water from sweat glands in the skin is usually associated with thermoregulation rather than excretion).
5. The liver is the site of the breakdown of red blood cells which leads to the formation of bile pigments. These pigments are passed into the intestine where they are excreted from the body with faeces. The elimination of undigested food is not excretion.
6. The three forms of nitrogenous wastes are ammonia, urea and uric acid.
7. In humans the kidney is the main organ for the excretion of nitrogenous waste.
8. The human kidney is involved in: 1. Excretion of nitrogenous waste. 2. Osmoregulation maintaining the water potential of the blood. 3. Ionic regulation maintaining ions levels, e.g. potassium (K^+) and sodium (Na^+). 4. Regulation of blood pH. 5. Regulation of blood pressure long term.
9. Osmoregulation is the regulation of the water potential and osmotic pressure of body fluids so that they do not become too dilute or concentrated.
10. Ammonia is highly toxic and very soluble in water. Humans need to convert ammonia to less toxic forms, e.g. uric acid or urea to protect the cells of their body while still being able to excrete nitrogenous waste.
11. C

76 Deamination

1. Deamination is the process when an amino group is removed from an amino acid or other compound.
2. Proteins contain the elements carbon, hydrogen, oxygen and nitrogen and often sulfur atoms.
3. The primary structure of a protein is a series of linked amino acids. The secondary structure of a protein are the coils and folds due to hydrogen bonds of the backbone, The tertiary structure is the overall shape due to the interactions between the side chains (R groups) of the various amino acids. The quaternary structure is the overall shape and structure due to two or polypeptide chains being present in one large macromolecule.
4. A peptide bond is a covalent bond that joins amino acids together in a polypeptide chain.
5. (a) An amino group is $-NH_2$
   (b) A carboxyl group is $-COOH$
6. 

7. The composition and structure of the variable side group (R) causes an amino acid to be polar, non-polar or electrically charged.
8. Formula for an amino acid is:

9. Transamination removes the amino group from an amino acid and attaches it to another carbohydrate molecule to form a new non-essential amino acid.
10. (a) Essential amino acids are the amino acids that must be included in the diet and cannot be synthesised by the body whereas a non-essential amino acid can be synthesised by the body.
    (b)

Essential amino acids	Non-essential amino acids
Lysine, tryptophan, threonine, methionine, phenylalanine, leucine, valine and isoleucine	Alanine, arginine, asparagine, aspartic acid, cysteine, glutamic acid, glutamine, glycine, proline, serine, tyrosine

11. Deamination occurs to the additional amino acids that are present after the body has stored sufficient protein or if there is insufficient protein in the diet the body proteins will be broken down and these amino acids will be deaminated to provide fuel for respiration.
12. (a) Deamination forms ammonia and a keto acid.
    (b) The ammonia is converted into urea and the keto acid is usually oxidised to release energy for metabolic processes.
13. Obligatory loss of protein means that there is breakdown of body proteins with the deamination of the amino acids as the person has insufficient protein intake in their diet.
14. Ammonia is highly soluble and can easily pass through membranes and since it is highly toxic it needs to be converted into urea which is less toxic to prevent brain damage and the person going into a coma.
15. Urea travels in the blood to the kidneys where it is excreted.
16. Uric acid is produced from ammonia and by the breakdown of nucleotides from nucleic acids, e.g. the breakdown of the genetic material of worn out cells.
17. (a) Part X = nitrogenous waste, e.g. ammonia then forming urea.
    Part Y = remainder of amino acid after the removal of the amino group, e.g. forming keto acid.
    (b) Any of the non-essential amino acids.
18. C
19. B

77 The Kidney

1. Nephron – functional unit of the kidney.
   Filtration – process which allows small particles to pass through filter but retains larger sized particles.
   Reabsorption – process in the kidney where particles which have been filtered out of the blood are returned to blood.
   Secretion – process where materials are released from one area to another.
   Filtrate – is the fraction that passes through the filter.
   Osmoregulation – the control of water and salt levels in the body to maintain homeostasis.
   Solute – the material that is dissolved in the solvent.
   Ureter – tube passing from the kidney to the bladder.
   Urethra – the tube passing from the bladder to the outside of the body.

2. Active transport uses energy to move materials across membranes, such as during the secretion of some ions, e.g. H^+ into the convoluted tubules of the kidney. Passive transport does not require an energy input, such as during the osmosis of water from the convoluted tubules back into the capillaries.
3. (a) Homeostasis is the maintenance of a constant internal environment.
   (b) Renal artery → glomerulus → Bowman's capsule → convoluted tubule → collecting duct → kidney pelvis → ureter → urinary bladder → urethra → outside body.
   (c) On a hot day with little fluid intake a person could dehydrate. The kidney will respond by increasing the amount of water reabsorbed in the renal tubules so that a concentrated urine is formed.
   (d) Red blood cells should not be present in urine as they are too large to pass through the pores in the epithelium of Bowman's capsule. If RBCs are present in the urine it indicates damage to the nephron and the person should seek immediate medical attention.
4. (a) Brown region is the renal cortex which consists of glomerulus and Bowman's capsule which is where the blood is filtered. Pinkish-red region is the medulla which consists of renal tubules and collecting tubules and is where reabsorption occurs. White region is the pelvis and consists of collecting ducts and is where urine collects to be passed to the ureter. The thin tube is the ureter.
   (b) Students would need to wear gloves to protect their hands and prevent possible infection. Students would have to correctly dispose of the dissected kidneys, making sure to remove all pieces of kidney from the lab as the tissue will putrefy and could transmit disease if it became infected.
5. (a) Filtration is the movement of small molecules such as glucose, amino acids, water, salts and urea from the glomerulus into Bowman's capsule. The movement is mainly due to increased blood pressure. The filtrate is the same as blood except it does not contain red blood cells or proteins. Reabsorption is the movement of needed nutrients back into the bloodstream from the renal tubules. All of the glucose and amino acids are reabsorbed and some salts and water is reabsorbed, depending on the needs of the body. Urea is not reabsorbed. In this way the kidney is regulating the amount of salt and water in the body, as well as excreting urea.
   (b) Reducing the intake of proteins will help to reduce the pressure within the kidneys and hence protects the kidneys – if the blood has a high concentration of protein compared to the lack of protein in the filtrate, there will be a high 'suction pressure' drawing water by osmosis from the blood. Proteins are needed in the diet to provide amino acids which are the building blocks for new cells and enzymes. If the protein intake is insufficient the patient will have difficulty maintaining cell growth, repair and maintenance.
6. Many of the waste products, especially the nitrogenous wastes such as ammonia and urea are toxic and therefore the body needs a method to quickly remove these products so that normal chemical reactions can function correctly, catalysed by enzymes working at optimum efficiency.
7. Diffusion is slow and is effective over short distances where there is a large concentration gradient. Mammals are complex multicellular animals and the processes of diffusion and osmosis would not be able to remove waste products, especially the toxic nitrogenous waste products, quickly enough to prevent toxic poisoning. Therefore a circulatory system is needed to deliver waste products to the excretory system which in turn removes the waste products from the blood.
8. D
9. A
10. A
11. C
12. D
13. B
14. C
15. B

78 Osmosis, Diffusion and Active Transport

1. Oxygen enters the blood from the lungs by diffusion. Glucose enters the blood from the small intestine by diffusion.
2. You would expect osmosis to occur and the water molecules will move into the cell. Since it is a human cell with only a thin cell membrane for support, the cell will swell and will most likely burst due to the amount of water entering the cell.
3. You would expect water to leave the distal tubule by osmosis and passive transport because of the concentration gradient created by the movement of the salt out of the nephron.
4. By changing the permeability of the collecting duct, the kidney can produce either concentrated or dilute urine by osmosis.
5. The kidney needs the high blood flow as it needs to filter the blood to regulate the composition and volume of body fluids. The kidney has a high energy requirement as most of the reabsorption is by active transport, e.g. maintaining the sodium pump and osmotic gradients.
6. (a) If the plasma becomes more dilute than the intracellular fluid within red blood cells then water will move its concentration gradient and move across the cell membranes into the red blood cells. The red blood cells will swell and could burst.
   (b) If the plasma becomes more concentrated than the intracellular fluid within red blood cells then water will move down the concentration gradient from the red blood cell into the plasma. The red blood cells will shrink.
7. A

79 Renal Dialysis

1. (a) Oedema occurs when tissues are swollen with fluid.
   (b) Pulmonary oedema is when fluid accumulates in the lungs.
2. Dialysis is used in emergency situations: when potassium levels are high and there is fluid overload and pulmonary oedema; if there is a need to remove medications or toxins from the blood; or if there is long-term renal failure.
3. (a) Dialysis is a process which passes blood over a semipermeable membrane so the solutes, e.g. waste products, diffuse through the membrane and are removed from the blood.
   (b) One advantage of haemodialysis is that it prolongs life when the kidneys have failed. One disadvantage is that each treatment requires 2 needle sticks and for 3 times a week each patient must spend 4 hours at a dialysis session.
   (c) Both the kidney and renal dialysis filter urea from the bloodstream. The kidney filters all substances from the blood and then needed substances are reabsorbed. Thus not only is urea removed, but the kidney also regulates water levels and controls the concentration of many other substances, e.g. sodium, potassium, phosphate levels. The dialysis machine only filters substances with no reabsorption, so people using dialysis have to watch their diet, restricting their intake of high potassium foods such as leafy vegetables and fruits and high phosphate foods such as beans and peas, as dialysis does not control the level of these substances.
   (d) For blood to be entering his urine, the glomerulus and Bowman's capsule could be damaged so that red blood cells are now filtering across to the filtrate instead of staying in the bloodstream.
4. Heparin, an anticoagulant, is administered to stop the blood clotting in the dialysis tubing.
5. The body's buffer system, which maintains homeostasis, is maintained used a dialysate, which is a solution of all the extracellular ions such as bicarbonate and acetate.
6. Urea, uric acid and the potassium, phosphate and sulfate ions are considered wastes and need to be removed from the blood. The hydrogen carbonate is needed to maintain correct pH levels and the glucose is needed for energy requirements.
7. A
8. D

80 Treatment of Conditions

1. Treatment of conditions due to system or organ dysfunction has changed through improvements in early diagnosis and appropriate use of drugs, physical therapy, radiation therapy and removal and/or replacement of affected parts.
2. Diagnostic testing involves gathering clinical information about a person's health.
3. Early and accurate diagnosis is highly important – 1. In managing many diseases and disorders. 2. If symptoms are reversible then early diagnosis may be able to halt, perhaps reverse symptoms or reduce the period of illness. 3. If symptoms are not reversible early treatment can stop the rate of onset and further decline due to the condition. 4. Early diagnosis gives a person some sense of empowerment. 5. Allows the patient to make plans, e.g. controlling lifestyle choices such as diet, smoking, alcohol and caffeine consumption, amount of exercise, managing finances, preparing living arrangements if mobility and flexibility are affected and arranging care and support.
4. Preliminary tests check vital signs, e.g. measuring blood pressure, breathing rate, heart rate, respiratory rate and temperature.
5. Laboratory tests check samples of blood, urine or body tissues.
6. Prenatal testing checks the health of an embryo or foetus.
7. Diagnostic imaging looks inside the body for indications of system or organ dysfunction and other medical conditions. It includes X-rays, CT scans, nuclear medicine scans, MRI scans and ultrasound. Computed tomography (CT) scans are a type of imaging using X-rays to look for broken bones, cancers, blood clots and signs of heart disease or internal bleeding.
8. (a) Genetic testing looks for indications of genetic disorders. There are more than 1000 genetic tests available from testing laboratories.
   (b) The Guthrie test or HPLC test is used to test newborn infants for phenylketonuria (PKU) as part of the national screening program for babies. Early detection means the child can be given a special diet so that the phenylalanine will not accumulate causing progressive mental retardation.
9. Pharmacology is the study of drugs looking at the possible uses, effects on the body and the interactions with other drugs. Pharmacogenomics investigates the effects of a patient's genome on the way the person will respond to a particular drug and the likely effectiveness of a particular drug for a particular genetic make-up. Pharmacogenetics is the study of the influence of inherited genetic differences of one or a few genes on the response of a drug.
10. Physiotherapy is a branch of rehabilitative health that uses specially designed exercises and equipment to help enhance or restore movement and physical function.
11. If a person has osteoporosis a physical therapist will design a specific program for the individual needs of the patient with specific exercises, e.g. weight bearing and/or resistance exercises. The person will also be advised on posture, how to improve balance to reduce the risk of falling and ways to improve muscle strength and mobility.
12. Radiotherapy or radiation therapy is a type of cancer treatment that uses beams of intense energy to kill cancer cells. Radiotherapy uses ionising radiation which consists of particles, X-rays or gamma rays with sufficient energy to cause ionisation in the medium through which it passes.
13. Chemotherapy is the use of anti-cancer drugs to destroy cancer cells.
14. (a) 'PET" stands for Positron Emission Tomography.
    (b) PET is an imaging technique which is used to detect radioisotopes in the diagnosis and treatment of diseases.
15. Gallium-67 is used to find cancers in lymph nodes and bronchi and malignant melanomas.
16. A prosthesis is an artificial body part, e.g. limb, joint or silicone breast implant that replaces removed disease-affected parts.
17. C
18. A

81 Chemical Monitoring

1. Chemical monitoring involves a range of technologies and different areas of scientific knowledge and understanding to look at medical conditions, collecting data which can be used in the diagnosis and treatment of different medical conditions.
2. Medical monitoring is often classified by the target of interest, e.g. cardiac monitoring, haemodynamic monitoring, respiratory monitoring, neurological monitoring.
3. (a) Pulse oximetry identifies oxygen levels in blood.
   (b) Pulse oximeters are non-invasive and allow continuous monitoring of the oxygen levels, e.g. for patients undergoing general anaesthesia, critical care units and is easily applied to fingers for adults and children.
   (c) Pulse oximeters cannot distinguish between different forms of haemoglobin, e.g. haemoglobin combined with carbon monoxide registers as 90% oxygenated haemoglobin and therefore overestimates the saturation.
4. (a) A capnometer is a device that measures carbon dioxide concentrations in respired gases.
   (b) A capnometer projects an infra-red beam through a sample of respired gas and measures the amount of light absorbed. The amount of light absorbed depends on the concentration of carbon dioxide present.
   (c) Capnometers could be useful in intensive care units, e.g. to show restoration of circulation after cardiopulmonary arrest and resuscitation.
5. A.

82 Experiment – Carbon Dioxide and pH

1. Exhaled air is richer in carbon dioxide than atmospheric air.
2. Universal indicator will show acidity/alkalinity by colour change. Adding this indicator to the distilled water gave the students another visual means of identifying the effect of dissolved carbon dioxide on pH besides watching the data logger plot a change on the graph.
3. Green universal indicator indicates neutrality while yellow indicates a more acidic level, but not highly acidic, which is red.
4. Distilled water should have pH = 7. There are a couple of possibilities. The data logger may not have been zeroed before beginning the experiment, or some of the dishwasher detergent contaminated the distilled water making the water very slightly alkaline.
5. In the experiment the independent variable was time and the dependent variable was the pH of the water.
6. Both the pH sensor and universal indicator show a change in pH of the water as carbon dioxide is breathed through the straw. They have different levels of accuracy. The universal indicator gives a rough indication using the accompanying colour chart of a change from an initial reading of approximately pH = 7 to a final reading of approximately pH = 5.5. The pH sensor connected to the data logger is more accurate showing a precise change over time with numerical values to 0.01.
7. Carbon dioxide makes the water more acidic.
8. Carbon dioxide is a waste product from cellular respiration.
   Glucose + oxygen → carbon dioxide + water + energy
9. To show exhaled air contains carbon dioxide, students should breathe through the straw into a test tube of limewater which turns milky with carbon dioxide.
10. Repetition would improve the reliability and show that these results were not an anomaly.
11. (a) Homeostasis is maintaining a constant internal environment.
    (b) Carbon dioxide is a waste product of respiration and is transported in the blood. Carbon dioxide levels affect pH with higher carbon dioxide levels raising the acidity of the blood. Homeostasis needs to monitor carbon dioxide levels to ensure constant pH for efficient metabolism.

12. (a) Exercise and strenuous physical activity rapidly increases the carbon dioxide level in the blood.
    (b) Stimulus (high CO_2 level) → chemoreceptor (aortic and carotid bodies and medulla oblongata) → messenger (electrochemical impulse along nerve) → effector (intercostal muscle and diaphragm) → response (faster and deeper breathing rate).
    (c) Response is negative feedback
13. C
14. B

83 CAT and MRI Scans

1. X-ray computed tomography uses X-rays to view inside a body giving virtual slices of the part being viewed.
2. A CAT scan is a computed axial tomography – one type of computed tomography now available to view inside bodies.
3. MRI scan means a magnetic resonance imaging scan which uses strong magnetic fields and radio waves to form images of the body.
4. In 2003 Paul Lauterbur and Sir Peter Mansfield were awarded the Nobel Prize in Physiology or Medicine for 'discoveries concerning magnetic resonance imaging'.
5.

Advantages of CT scans	Disadvantages of CT scans
Used in preventive medicine and the screening of diseases. Allows multiplanar reformatted imaging with images viewed on either axial, coronal or sagittal planes.	Uses ionising radiation which can cause cancers. Some people have allergic reactions to the contrast medium that is used or may feel sick for a while afterwards.

6.

Advantages of MRI scans	Disadvantages of MRI scans
It is non-invasive. Does not involve ionising radiation. It is used for medical diagnosis and to follow the progress of certain diseases in people. It is the most sensitive examination for small brain tumours.	Implanted medical devices that contain metal may malfunction or cause problems due to the strong magnetic field of MRI. Not all conditions can be clearly diagnosed by MRI.

84 Lifestyle Choices – Exercise and Diet

1. Lifestyle choices that can affect health include – 1. The amount and type of exercise done per day/week/month. 2. The use of drugs. 3. The type of diet.
2. Muscle fibres are as long as they need to be. If a person does not stretch the muscle it will contract to a reduced length to what is now needed. The shortening of muscle can occur if the muscle is in a flexed position in a splint or if it is not being used for some other reason.
3. A healthy diet needs to contain all the essential nutrients that cannot be synthesised by the body. This includes amino acids, fatty acids, minerals, vitamins and water.
4. Proteins are needed in the diet as they contain amino acids which are needed to build body structures, e.g. around two thirds of the total dry mass of a cell is protein and for cell metabolism, e.g. to form enzymes, for protection, e.g. form antibodies, and to maintain homeostasis, e.g. soluble proteins act as buffers to stabilise pH.
5. Since all metabolic pathways are not fully developed in children, there are additional essential amino acids that need to be given in the diet of infants and children, e.g. histidine.
6. Metabolic fuel is supplied by dietary lipids and carbohydrates.
7. The amount of energy needed by a person depends on their basal metabolism, sex, age, body size and activity.
8. (a) Obesity is often not defined as an absolute number but by looking at historical data and using a BMI greater than the 95th percentile.
   (b) Obesity in children is difficult to define as height varies with age during growth.
   (c) Obesity increases the risk of many diseases, e.g. coronary disease and shortens life expectancy.
9.

Factor	How it contributes to obesity
Cheap high energy foods	Many cheap and easily available foods, e.g. fast foods (hamburgers, chips, pizza) and packaged 'snack' foods have high levels of sugars and/or fats. They often do not give the 'full' sensation which means people relying on these foods often increase their daily calorie intake and eat additional food.
Large portion sizes	Some restaurants and fast food outlets serve larger portions or allow an 'upgrade'. This leads to increased intake of food and increased calorie intake beyond what is needed for age, gender and activity.
Vehicles for transport	Urban development and modern lifestyles have reduced the number of people walking or cycling. An increased fear of crime has reduced the number of children playing in parks and in local streets and many children mostly travel in vehicles to destinations.
Sedentary occupation	Many jobs rely on technology, e.g. computers forcing people to sit at desks and email has reduced the need to physically deliver messages to other workers. Development of labour saving devices, e.g. the control of mechanical equipment and housework has reduced physical activity, e.g. washing machines, vacuum cleaners, central heating. Buildings have elevators and escalators to reduce the need to walk up stairs.

10. Two of all the fatty acids cannot be made by the human body – omega-6 fatty acid, e.g. linoleic acid (LA) and omega-3 fatty acids, e.g. alpha-linolenic acid (ALA).
11. Humans need large amounts of calcium and phosphorus to maintain bones. Calcium is also needed for muscle and nerve functioning and phosphorus for ATP and nucleic acids. Iron is essential for haemoglobin and RBCs and in cellular respiration. Iodine is needed to produce thyroid hormones.
12. Food containing vitamin A include egg yolk, green or yellow vegetables, fruits, liver and butter.

85 Lifestyle Choices – Drugs

1. (a) A drug is any chemical substance that is taken into the body and causes modification of one or more functions of the body.
   (b) Many drugs are taken to relieve pain or treat an illness. Some drugs cause addiction and are habit forming, e.g. alcohol, nicotine and narcotics and these are taken as lifestyle choices.
2. An addiction refers to a physical or psychological dependence on a compulsive behaviour that negatively impacts the quality of life of the person and that the person feels powerless to stop.
3. Substance addiction involves the consumption of substance(s), e.g. alcohol, tobacco or psychoactive drugs such as narcotics, stimulants or sedatives.
4. The term 'substance dependence' is often used rather than 'addiction' as there is debate as to whether some substances, e.g. tobacco are 'addictive' making the person physically dependent on the substance with increasing tolerance for the substance. Substance dependence shows the reliance on the substance and the inability of the person to quit.
5. (a) Psychoactive drugs are chemical substances that alter mood, behaviour, perception or mental functioning.
   (b) Psychoactive drugs modify biochemical or physiological processes in the brain. They can either block or enhance the effect of neurotransmitters.
6. Many drugs, e.g. cocaine and amphetamines alter dopamine functioning and dopamine is thus related to addiction. Dopamine is a neurotransmitter that affects brain processes involved in behaviour and cognition, controlling voluntary movement, emotional and mood responses and the ability to experience pleasure and pain.
7. Excitatory drugs can increase dopamine release (amphetamines) whereas inhibitory drugs inhibit dopamine reuptake (cocaine and amphetamines).

8. Excitatory psychoactive drugs.

Name of drug	Brief description of drug
Cocaine	Blocks reuptake of neurotransmitter causing more neurotransmitter to be present in the synaptic cleft. The dopamine stimulates neurons which causes prolonged feelings of pleasure and excitement. Habitual use and prolonged exposure makes the brain adapt to it and leads to a decrease in the synthesis of dopamine by neurons. Cocaine stimulates the CNS and is an appetite suppressant. It enhances endurances, increases alertness and reduces feelings of fatigue. Cocaine gives feelings of pleasure and is associated with the 'reward pathway' Prolonged use causes itching, hallucinations and paranoid delusions.
Nicotine	Binds to the presynaptic receptors exciting the neuron to fire more action potentials and increases the release of dopamine into the synapse and can bind to receptors on the presynaptic neuron. Stimulates receptors that usually respond to acetylcholine causing an increase in adrenaline and increasing heart rate, blood pressure and blood sugar levels.
Amphetamines	Stimulate the CNS and suppress the appetite. Blocks reuptake of neurotransmitter causing more neurotransmitter to be present in the synaptic cleft.

9. Inhibitory psychoactive drugs.

Name of drug	Brief description of drug
THC (tetra hydrocannabinol)	Main psychoactive substance in the cannabis plant. THC binds to cannabinoid receptors (e.g. CB1 receptors in the brain) and causes relaxation, drowsiness and reduces pain. When acting on CB1 receptors in the hunger centres in the hypothalamus it increases the palatability of food and enhances appetite. There is conflicting evidence and debate about whether THC is addictive/non-addictive. THC reduces concentration, impairs decision making and affects short term memory. It has moderate analgesic effects, stimulates the appetite, inhibits vomiting and is used to treat pain. In high doses distorts perception.
Benzodiazepines	Enhance the effect of GABA and hyperpolarise the postsynaptic membrane so that impulses are not transmitted and the person experiences a sedative, hypnotic effect. Used to treat anxiety, insomnia and as a muscle relaxant to reduce convulsions. If taken with other CNS depressants, e.g. alcohol and opiates the result can be toxic.

10. Behaviours that indicate substance dependence or addiction include – 1. Increasing tolerance to substance with increased doses needed to achieve same result or with reduced response to same dose. 2. Withdrawal signs or symptoms if the substance is abruptly discontinued. 3. A continual desire or unsuccessful attempts to cut down or stop using the substance. 4. Craves use of the substance. 5. A large amount of effort and/or time is spent obtaining the substance. 6. Other activities, e.g. occupational, social or recreational requirements are negatively affected. 7. Despite arising physical and psychological problems, the substance use is maintained.
11. D

86 Lifestyle Choices – Alcohol

1. Most alcoholic drinks contain between 2% to 40% ethanol.
2. Around 25% can be absorbed straight into the bloodstream from the stomach.
3. (a) Alcohol dehydrogenase is an enzyme found in the lining of the stomach and in the liver and allows the consumption of alcoholic drinks.
   (b) There are at least six slightly different alcohol dehydrogenases in humans and the differences can be related to the ability of different people to process alcohol, e.g. male/female, old/young, or populations from different parts of the world.
4. Alcohol depresses the CNS and there are short term effects on the brain, producing a sedative, pleasant, relaxed effect. Intoxication impairs judgement and motor skills and chronic use causes liver and brain cell damage. Alcohol can be physiologically addictive with dangerous withdrawal symptoms. Alcoholics often suffer from nutrient deficiencies as they lose interest in food. High alcohol intake can also cause weight gain and obesity due to the high energy value of alcohol.
5. Long excessive use of alcohol increases the chances of several diseases, e.g. oesophageal and pancreatic cancer, damage to the immune system with higher susceptibility to infection, weaker and less regular heartbeat and most importantly liver damage.
6. The three primary types of alcohol induced liver disease are – 1. Alcoholic hepatitis which is an inflammation of the liver with destruction of individual liver cells and scarring. 2. Fatty liver which involves an excessive accumulation of fat inside the hepatocytes causing the liver to become enlarged. 3. Alcoholic cirrhosis which involves the destruction of normal liver tissue and permanent scarring making the cells non-functional.
7. Signs of liver damage include – 1. Pale stools – this is caused by a lack of sterobilin which is a brown pigment. Liver damage, e.g. in cirrhosis of the liver makes cells non-functional and affects the normal breakdown of haemoglobin. 2. Dark urine – due to larger than normal amounts of bilirubin mixes with urine caused by impaired functioning of liver cells. 3. Jaundice – a yellowing of the skin and sclera (white) of the eyes is due to high bilirubin levels. 4. Bruising and easy bleeding – liver damage can disrupt the manufacture of clotting factors and the absence of these factors leads to bleeding and bruising. 5. Excessive fatigue – disruption of normal metabolic processes in the liver, e.g. carbohydrate and protein metabolism reduces energy levels and the supply of nutrients, minerals and vitamins.
8. C

Topic Test

Section A – Multiple Choice (35 marks)

1. C	2. A	3. B	4. A	5. D
6. B	7. C	8. A	9. B	10. D
11. C	12. A	13. A	14. A	15. B
16. D	17. A	18. C	19. B	20. C
21. C	22. A	23. D	24. A	25. B
26. A	27. C	28. B	29. A	30. D
31. B	32. C	33. D	34. C	35. A

Section B – Written Response (65 marks)

36. (a) Marking Guidelines

Criteria	Marks
Correctly defines coenzyme	1

Sample answer:
Coenzymes are non-protein molecules which need to be present before an enzyme can function. They take part in the reaction and are changed by it.

(b) Marking Guidelines

Criteria	Marks
Names a coenzyme	1

Sample answer:
Coenzymes include vitamins and ATP.

(c) Marking Guidelines

Criteria	Marks
Distinguishes between a coenzyme and a cofactor	2

Sample answer:
Unlike coenzymes, cofactors are mineral ions or non-protein molecules that are needed by enzymes before they will catalyse a reaction and are not changed by the reaction, e.g. potassium ions and magnesium ions. Coenzymes are changed by the reaction.

37. Marking Guidelines

Criteria	Marks
Relates surface area to volume ratio with efficiency of diffusion AND Division of cell increasing surface area to volume ratio	3
Recognises size of surface area to volume ratio with size of cell AND Identifies relationship of surface area to volume ratio with efficiency of diffusion	2
Recognises size of surface area to volume ratio with size of cell OR Identifies relationship of surface area to volume ratio with efficiency of diffusion	1

Sample answer:
A large cell has a small surface area to volume ratio which makes diffusion of nutrients into the cell and diffusion of wastes out of the cell inefficient. A large cell will divide to make two smaller cells as this will increase the surface area to volume ratio and hence increase the efficiency of diffusion.

38. (a) Marking Guidelines

Criteria	Marks
Correctly identifies organelle	1

Sample answer:
Organelle is a Golgi body.

(b) Marking Guidelines

Criteria	Marks
Correctly describes function of organelle	2
Gives simple statement of function	1

Sample answer:
Golgi bodies receive, sort, store and secrete materials, e.g. proteins. Reactions occur on their large surface membrane and the materials are packaged into vesicles which transport the substance to the cell membrane for secretion or the substance is used to form lysosomes or other cytoplasmic components.

39. (a) Marking Guidelines

Criteria	Marks
Explains osmosis, states direction of movement of water and states what happens to potato	4
States water will leave potato, the potato shrivels due to osmosis	3
States water will leave the potato and the potato will shrivel OR States water will leave the potato or potato will shrivel and it is due to osmosis	2
States water will leave potato or potato will shrivel OR States osmosis occurs	1

Sample answer:
When cube Y is placed in a very strong salt solution, the concentration of water inside the potato cells is higher than the water concentration outside. This means that water will move out of the potato into the salt solution by osmosis. This will cause the potato to become shrivelled in appearance.

(b) Marking Guidelines

Criteria	Marks
Correctly explains why difference occurs	2
States correct difference	1

Sample answer:
If cube X was placed in a salt solution of similar concentration, the same situation will occur and osmosis will occur at the same rate. However, cube X is smaller than cube Y and has a larger surface area to volume ratio. This means that the cells in the centre of cube X will be affected sooner than the cells in the centre of cube Y and hence cube X will shrivel more quickly.

40. Marking Guidelines

Criteria	Marks
Correctly identifies three features and correctly explains why each feature is needed	3
Correctly identifies two features and correctly explains why each of these is needed OR Correctly identifies three features and correctly explains one of these features	2
Correctly identifies three features OR Correctly identifies one feature and correctly explains why it is needed	1

Sample answer:

Feature of respiratory system	Why it is needed
Large surface area	A large surface area is needed to provide an increased number of places for diffusion to occur. The larger the surface area the faster the rate of diffusion.
Rich blood supply	Gas exchange needs to supply oxygen to cells and remove carbon dioxide and wastes from cells. The blood system carries these wastes and thus a respiratory surface needs a rich blood supply to quickly load oxygen into the blood and to unload carbon dioxide from the blood.
Surface needs to be thin	A thin surface reduces the amount of diffusion needed allowing a quicker gas exchange across the surface.

41. Marking Guidelines

Criteria	Marks
Correctly distinguishes between osmosis, diffusion and active transport giving the concentration gradient present in each and the energy requirements for the movement of substances	6
Correctly provides some data about osmosis, diffusion and active transport	1 to 5

Sample answer:

Feature	Diffusion	Osmosis	Active transport
Definition	Is the movement of particles from an area of high concentration of particles to an area of low concentration of particles.	Is the movement of water across a semipermeable membrane from an area of high water concentration to an area of low water concentration.	The movement of a substance across a membrane against a concentration or electrochemical gradient.
Concentration gradient	High to low.	High to low.	Low to high.
Energy requirement	Passive movement – no energy input.	Passive movement – no energy input.	Requires an input of energy, e.g. ATP.

42. (a) Marking Guidelines

Criteria	Marks
Procedure outlined in steps in correct sequence Lists appropriate equipment used	3
Procedure outlined in steps in correct sequence OR Procedure outlined in steps out of sequence/only some steps and lists appropriate equipment	2
Procedure outlined in steps out of sequence/only some steps OR Lists appropriate equipment	1

Sample answer:
Exhaled air contains carbon dioxide and hence breathing through a straw will provide the source of carbon dioxide. To show the effect of dissolved carbon dioxide on the pH of water you need two test tubes half filled with water, each with two drops of universal indicator. One is to be left as a control. A student needs to exhale through a straw into the other test tube and note any colour change of the water.

(b) Marking Guidelines

Criteria	Marks
Gives correct equation to show chemical composition of substances involved Explains the meaning of results States expected results	3
Gives TWO of the following: Gives equation to show chemical composition of substances involved Explains the meaning of results States expected results	2
Gives ONE of the following: Gives equation to show chemical composition of substances involved Explains the meaning of results States expected results	1

Sample answer:
The indicator should stay green in the control and change to a yellowish-red in the one with carbon dioxide. This shows that the pH of water without carbon dioxide is neutral pH = 7, while the water with dissolved carbon dioxide has a pH = 4.5, making it acidic. Carbon dioxide reacts with water to form carbonic acid which in turn dissociates to form hydrogen carbonate ions and hydrogen ions.
$CO_2 + H_2O \rightarrow H_2CO_3 \rightarrow HCO_3^- + H^+$
The hydrogen ions make the solution acidic.

(c) Marking Guidelines

Criteria	Marks
Predicts an observable symptom Predicts change in amount of carbon dioxide present/ change in pH Predicts change of rate of reaction	3
Outlines any TWO of the following: Predicts an observable symptom Predicts change in amount of carbon dioxide present/ change in pH Predicts change of rate of reaction	2
Outlines any ONE of the following: Predicts an observable symptom Predicts change in amount of carbon dioxide present/ change in pH Predicts change of rate of reaction	1

Sample answer:
Enzymes catalyse reactions. Without carbonic anhydrase the reaction of carbon dioxide and water to form carbonic acid would be much slower, which would reduce the amount of carbon dioxide carried in the plasma. Since breathing rate and pulse rate are affected by carbon dioxide in the blood, these would slow down and since carbon dioxide also affects pH, the acidity of the plasma would change. The accumulation of these factors will disrupt normal body functioning, perhaps leading to coma and then death.

43. (a) Marking Guidelines

Criteria	Marks
Correctly identifies inhalation and exhalation	1

Sample answer:
Diagram R shows inhalation and diagram S shows exhalation.

(b) Marking Guidelines

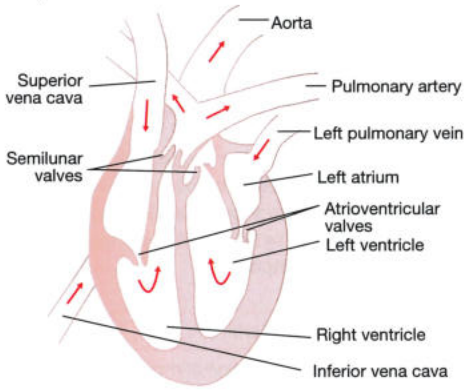
Criteria	Marks
Correctly uses information from the diagram to discuss the mechanism of inhalation and exhalation	3
Correctly uses information from the diagram and/or discusses the mechanism of inhalation and/or exhalation	1 to 2

Sample answer:
Diagram R shows inhalation with the diaphragm contracting and moving downwards and the expansion of the chest cavity with the ribs and sternum moving up and out. The pressure change caused by the change in volume of the thorax causes inhalation. Diagram S shows the diaphragm relaxing with the ribs and sternum moving down and in. This changes the pressure and volume of the thorax causing exhalation.

44. (a) Marking Guidelines

Criteria	Marks
Correctly labels all 9 parts on the diagram	3
Correctly labels 6 to 8 parts on the diagram	2
Correctly labels 3 to 5 parts on the diagram	1

Sample answer:



(b) Marking Guidelines

Criteria	Marks
Correctly shows direction of blood flow through the heart	2
Correctly shows direction of blood flow on one side of the heart	1

Sample answer:
See diagram for sample answer Question 44 (a).

(c) Marking Guidelines

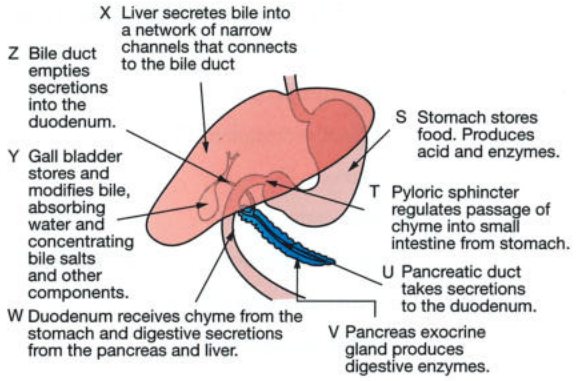
Criteria	Marks
Correctly compares the function of atrioventricular and semilunar valves	2
Describes the function of a valve	1

Sample answer:
Valves prevent the backflow of blood. The atrioventricular valves prevent backflow of blood from the ventricle into the atrium – the right atrioventricular valve is the tricuspid valve with 3 cusps and the left ventricular valve is the mitral valve with 2 cusps. The right semilunar valve or pulmonary valve has 3 semilunar cusps and prevents backflow back into the heart from the pulmonary artery. The left semilunar valve or aortic valve also has 3 semilunar cusps and prevents backflow from the aorta into the heart.

45. Marking Guidelines

Criteria	Marks
Correctly matches each part and description with diagram	8

Sample answer:



46. Marking Guidelines

Criteria	Marks
Correctly describes each level of muscle structure	4
Correctly describes some levels of muscle structure	1 to 3

Sample answer:

Diagram	Description
Level of structure = striated muscle	Striated muscle consisted of a series of parallel muscle fibres held together.
Level of structure = fibre	Skeletal muscle fibres are very large and multinucleated. Fibres have dark and light bands. The sarcolemma (cell membrane) surrounds the sarcoplasm (cytoplasm). Sarcoplasm contains many myofibrils.
Level of structure = myofibril	Myofibrils are cylinders (1 to 2 μm diameter) and as long as the entire cell. Myofibrils can shorten and cause contraction. Each myofibril is attached to the sarcolemma at each end of the cell thus contraction shortens the cell. Myofilaments are surrounded by sarcoplasmic reticulum (like smooth ER) which forms a network with transverse tubules.
Level of structure = myofilament	Myofilaments are protein filaments consisting mainly of actin and myosin. Actin is in thin filaments and myosin is in thick filaments. The actin and myosin are organised in sarcomeres which repeat along the myofilament. The arrangement of the thin and thick filaments gives the dark and light banded appearance of a myofibril.

47. (a) Marking Guidelines

Criteria	Marks
Correctly identifies process occurring at point X	1

Sample answer:
Process B.

(b) Marking Guidelines

Criteria	Marks
Correctly identifies both processes occurring due to active transport	1

Sample answer:
Process C and process D.

(c) Marking Guidelines

Criteria	Marks
Correctly outlines what happens in deamination	1

Sample answer:
Deamination is a reaction which removes an amino group from an amino acid or other compound. Deamination of amino acids in the liver leads to the formation of urea.

48. Marking Guidelines

Criteria	Marks
States why this future research is needed Identifies area for future research Gives a value or judgement on this impact States one impact on society of this advance showing its advantage States one impact on society of this advance showing a disadvantage Identifies one major advance	6
Identifies one major advance AND Identifies area for future research AND Up to THREE of the following: Gives a value or judgement on this impact States one impact on society of this advance showing its advantage States one impact on society of this advance showing a disadvantage States why this future research is needed	3 to 5
Identifies one major advance and its impact OR Identifies area for future research and why this research is needed OR Identifies one major advance and its impact and identifies area for future research	2
Identifies one major advance OR Identifies area for future research	1

Sample answer:
Answers may include:

- Measurement of oxygen saturation.
- Measurement of carbon dioxide concentrations.
- Donated blood and the use of its products.
- Artificial blood.
- Kidney dialysis.

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